

MR Imaging System

AIRIS Elite

TECHNICAL GUIDE

--- Check and Service Program ---

(V5.0N~)

HITACHI MEDICAL CORPORATION

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Section 1 Basics of Software

1.1 Composition of MOD Cartridge / DVD-RAM Cartridge / CD-ROM

The MOD cartridges, DVD-RAM cartridges, and CD-ROM supplied together with the system consist of the followings.

But, MOD is used when operation table is UltraSPARC-IIi-Engine, DVD-RAM and CD-ROM is used when operation table is Sun-Blade-2500 :

1.1.1 In the case of UltraSPARC-IIi-Engine

Table 1.1-1 Composition of MOD Cartridge 1

Side	Label	Description
A	SYSTEM SOFTWARE	MOD boot program, System program
B	Blank	This side cannot be used for storing image data, service backup and raw data.

Table 1.1-2 Composition of MOD Cartridge 2

Side	Label	Description
A	SERVICE BACKUP	For backup of adjustment values and other data
B	TEST MOD	For backup of image and library at the time of adjustment, a recommended protocol, SAMPLE PATTERN IMAGE DATA

NOTE: DVD-RAM and CD-R is available when DVD-RAM/CD-R option is installed.

1.1 Composition of MOD Cartridge / DVD-RAM Cartridge / CD-ROM

1.1.2 In the case of Sun-Blade-2500

Table 1.1-3 Composition of CD-ROM

Label	Description
SYSTEM SOFTWARE	CD-ROM boot program, System program

Table 1.1-4 Composition of DVD-RAM Cartridge

Side	Label	Description
A	SERVICE BACKUP	For backup of adjustment values and other data
B	TEST DVD	For backup of image and library at the time of adjustment, a recommended protocol, SAMPLE PATTERN IMAGE DATA

1.2 System Composition

The software configuration is shown in Fig. 1.2-1.

The software is largely divided into the front-end group and back-end group. The front-end includes scan and image reconstruction processes which are operated on real-time OS (OS-9), while the back-end plays a part of GUI (Graphical User Interface) processes related to post processes of image data and setting scan parameters which are operated on UNIX OS. The software is configured as Client/Server model to maximize the utilization of the multi-task function.

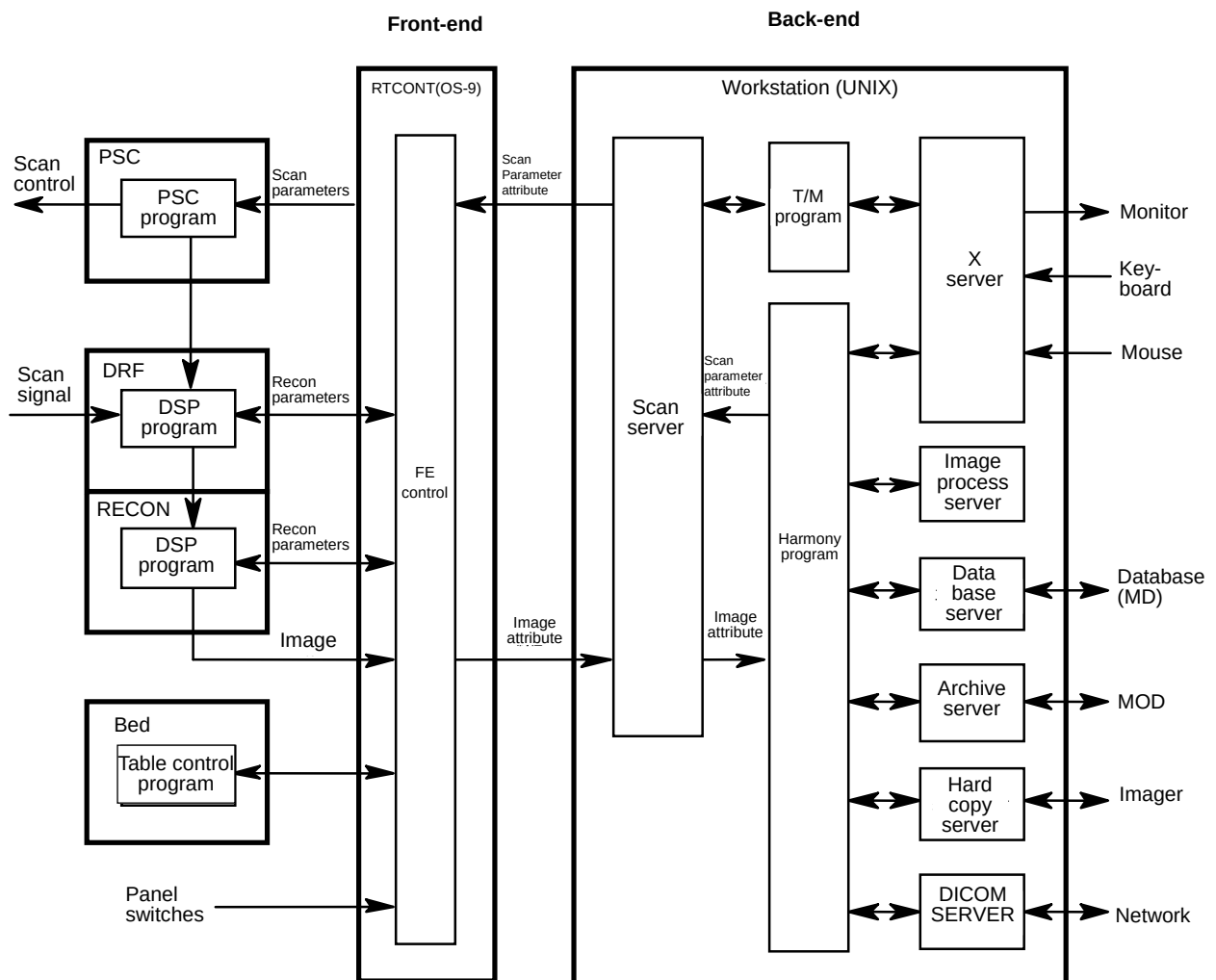


Fig. 1.2-1 System Configuration

1.3 Start-up Procedure of Check/Service Program

The Check/Service program can be started up by the following 7 methods.

The start-up procedures from 1.3.3 to 1.3.7 partially require registration of the license by the Key MOD or Key DVD.(Key DVD is necessary in the case of Sun-Blade-2500.)

Registration of the license adds an item in the menu.

For registration procedure of the license, refer to Section 3.

1.3.1 Start-up from System Tool Menu

1.3.2 Start-up from HMC Diagnostics Menu

1.3.3 Start-up from HMC Service Diagnostics Menu

1.3.4 Start-up from Maintenance Card

1.3.5 Start-up from Scan Card

1.3.6 Start-up from T/M Manager

1.3.7 Start-up from XTerm

1.3.1 Start-up from System Tool Menu

Click System Tool and make a selection from the Menu.

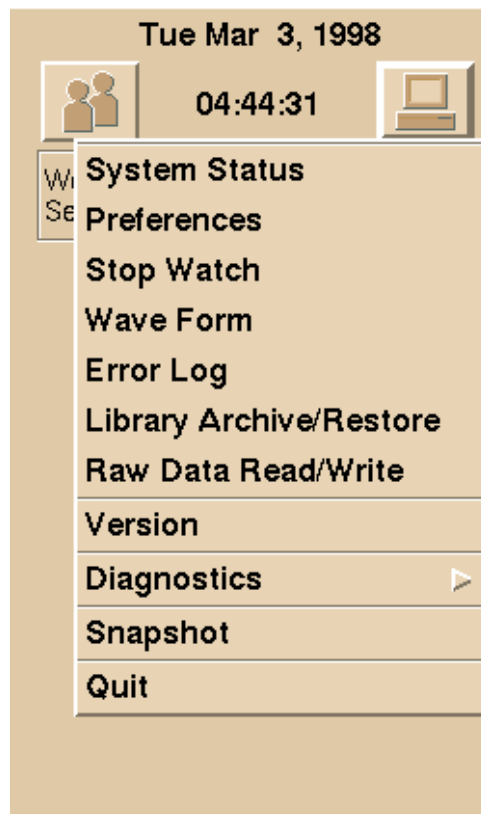


Fig. 1.3-1 Start-up from System Menu.

1.3.2 Start-up from HMC Diagnostics Menu

Make a selection from HMC Diagnostics Menu in the System Tool.

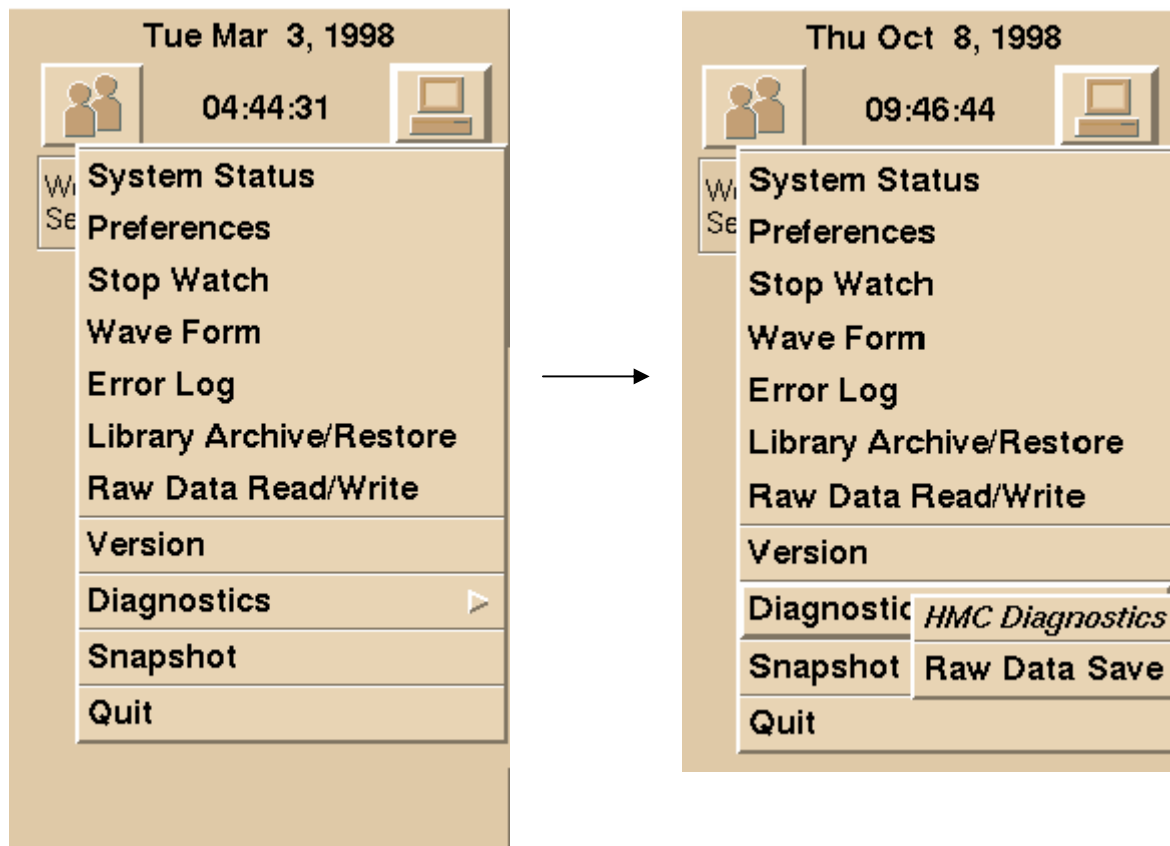


Fig. 1.3-2 Start-up from HMC Diagnostics Menu

1.3.3 Start-up from HMC Service Diagnostics Menu

Register the license by Key MOD/DVD, and make a selection from Service Diagnostics Menu.

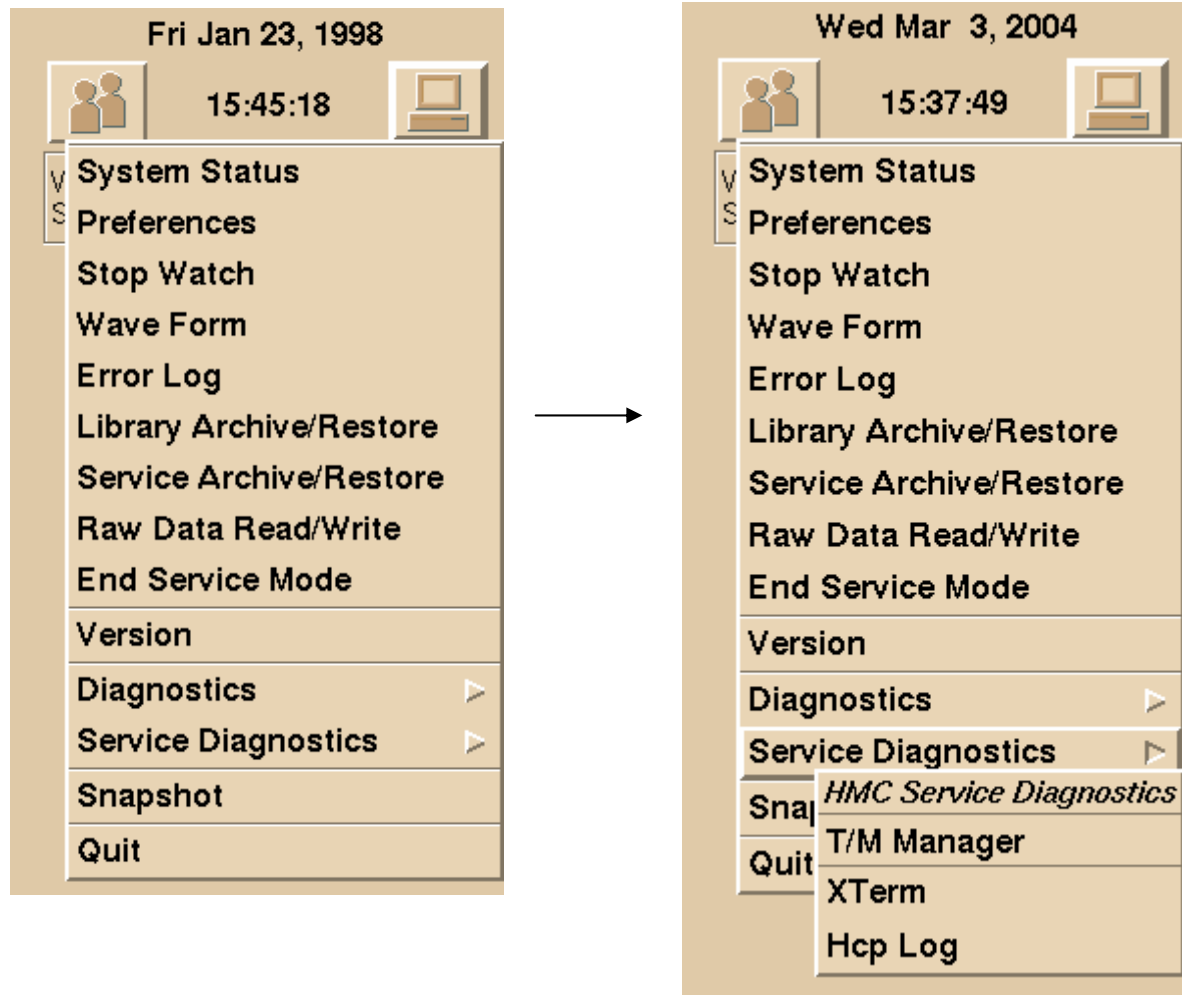


Fig. 1.3-3 Start-up from HMC Service Diagnostics Menu

1.3.4 Start-up from Maintenance Card

Open the Maintenance Card and click the desired one in the tags displaying check items at the lower left of the screen.

SNR	Special Text	SNR-NEMA	FOV	Uniformity	Thickness
Signal-to-Noise Ratio					
SNR_CNR_A			SNR_CNR_D		
X	W.	Mn	X	W.	Mn
Y	H.	SD	Y	H.	SD
SNR_CNR_B			SNR_CNR_C		
X	W.	Mn	X	W.	Mn
Y	H.	SD	Y	H.	SD
SNR_CTR_E					
X	W.	Mn	X	W.	Mn
Y	H.	SD	Y	H.	SD
defaults-1					
mm					

Fig. 1.3-4 Start-up from Maintenance Card

1.3.5 Start-up from Scan Card

Open the Scan Card to display Others Group. Clicking Others tag in the tags displaying group names at the lower left of the screen changes the window. Click Maint. Menu on the scan parameter setting area to open the menu.

Basic	Technique	Particular	Matrix	Gating	Dynamic	Filter	Fatsup	Others
Carry Offset	Mag..	Slice Ord. T	H→F	RCoil Mode	A	SAR [W/kg]	0.1	Maint.
Line/Box	Line	Slice Ord. S	R→L	Prescan	Auto	dB/dt [T/s]	35.7	off
Inf.FOV	Yes	Slice Ord. C	A→P	VolShim	Auto			F-Search
Stack.Mode	Normal	Plane Order	TRS					H1 Adjust
G.Intvl	ON	Sync.						M-Scan
Oblique	Full							Sgnl.Gen.
								B0 Search
								ECC Adjust
								Image Quality
								Limit Graphs

Fig. 1.3-5 Start-up from Scan Card

1.3.6 Start-up from T/M Manager

Select T/M Manager on the HMC Service Diagnostics Menu in the System Tool to open a menu. Items of menu are arranged as [Service], [Check] and [System] tags and clicking each tag changes over the item of menu.

By clicking [Cancel] button, the window is closed.

When starting the Check/Service program by clicking Screen re-display toggle button and setting it ON, after ending the program, T/M Manager is opened again.

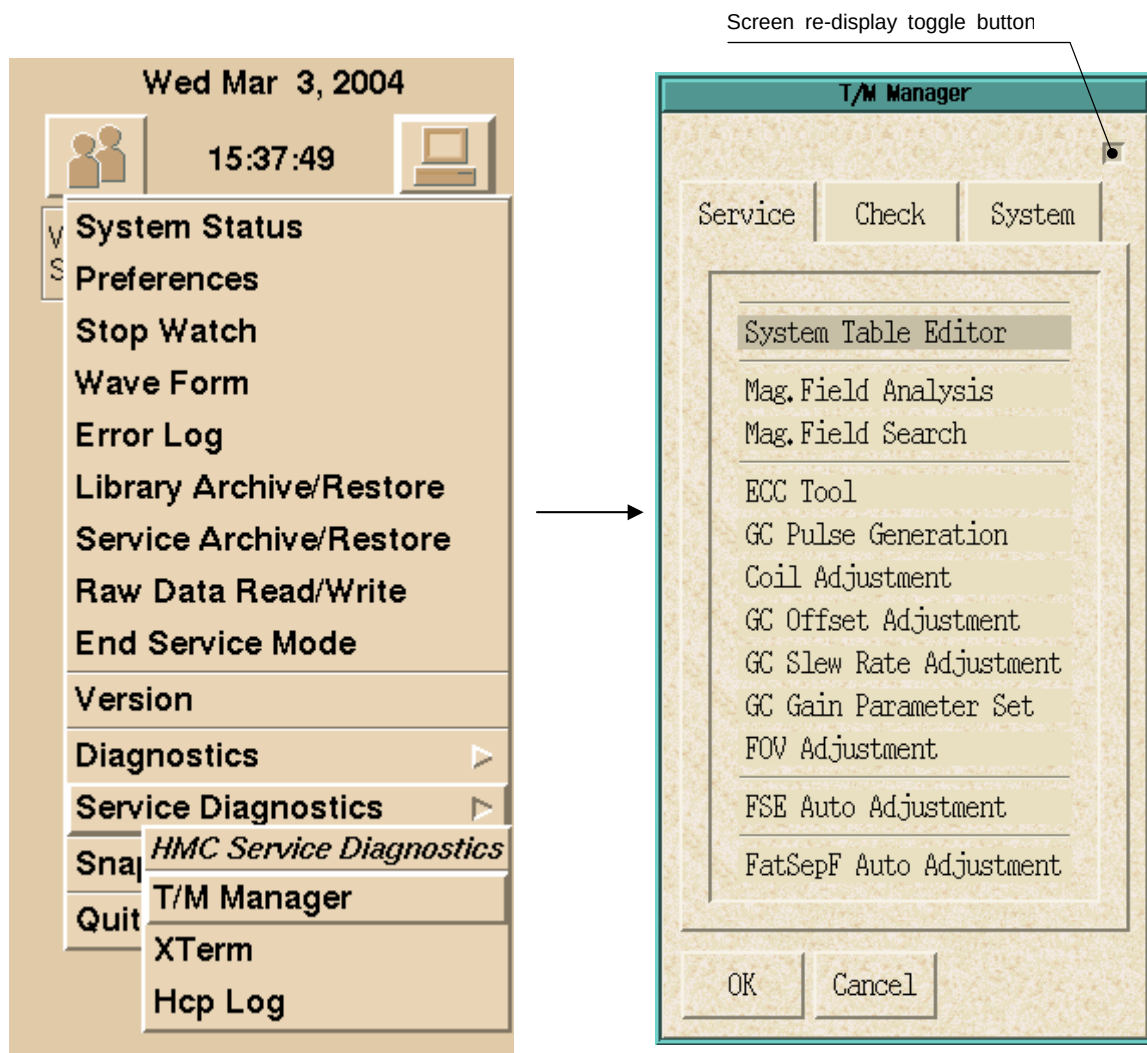


Fig. 1.3-6 Start-up from T/M Manager

1.3.7 Start-up from XTerm

Select XTerm on the HMC Service Diagnostics Menu in the System Tool to start XTerm.
By entering the command specified here, the program starts.

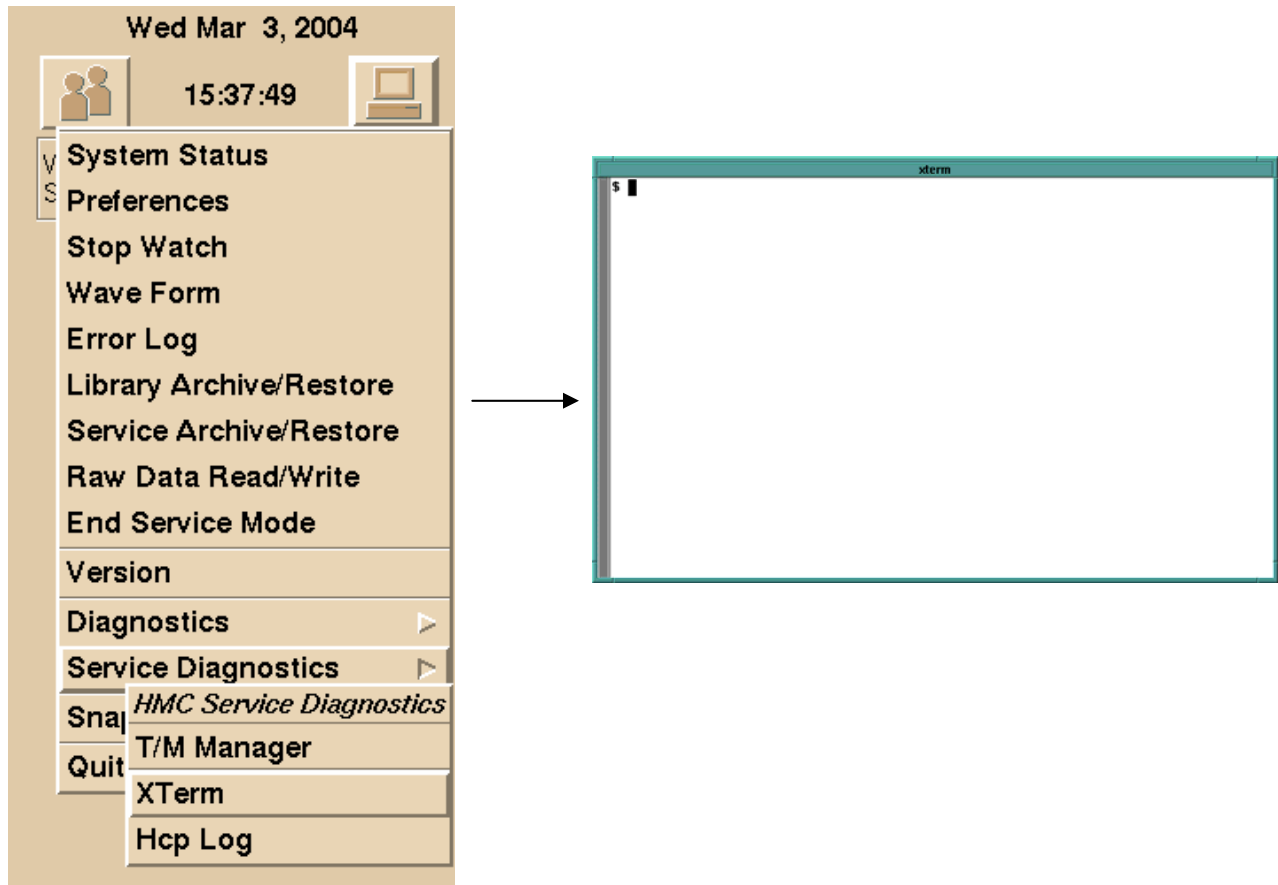


Fig. 1.3-7 Start-up from XTerm

1.4 Check/Service Program List

Check/Service programs can be started by six starting methods shown below.

The following is the program list according to each starting method. The figures at the top of each line show the Paragraph numbers in which each description is made.

1.4.1 Programs to be started from System Tool Menu

1.4.2 Programs to be started from HMC Diagnostics Menu

1.4.3 Programs to be started from HMC Service Diagnostics Menu

1.4.4 Programs to be started from Maintenance Card

1.4.5 Programs to be started from Scan Card

1.4.6 Programs to be started from T/M Manager

1.4.1 Programs to be started from System Tool Menu

These are programs mainly related to the system.

Some of the programs require license registration.

Table 1.4-1 Programs to be started from System Tool Menu

Item Name	Section & Subsection	Remarks
Registration of Name	Section 6 6.1	
Setting Display	Section 6 6.2	
Setting Film	Section 6 6.3	
Setting Task Card	Section 6 6.4	
Setting System	Section 6 6.5	
Setting Message	Section 6 6.6	
Setting Date and Time	Section 6 6.7	License registration required
Other Setting	Section 6 6.8	License registration required
Error LOG Display	Section 6 6.9	
Service Backup	Section 6 6.10	License registration required
Raw Data Read/Write	Section 6 6.11	
Software Version Display	Section 6 6.12	

1.4.2 Programs to be started from HMC Diagnostics Menu

These are programs to be mainly supplied to the user.

Table 1.4-2 Programs to be started from HMC Diagnostics Menu

Item Name	Section & Subsection
Easy Raw Data Save	Section 6 6.14

1.4.3 Programs to be started from HMC Service Diagnostics Menu

These are Test/Maintenance programs to be mainly used by serviceman. License registration is necessary to start up the program.

Table 1.4-3 Program List to be started from HMC Service Diagnostics Menu

Item Name	Section & Subsection
T/M Manager	Section 1 1.4.6
Hcp Log Display	Section 6 6.13

1.4.4 Programs to be started from Maintenance Card

These are programs mainly related to image quality check.

Some of the programs requires license registration for start-up.

Table 1.4-4 Programs to be started from Maintenance Card

Item Name	Section & Subsection	Remarks
S/N Check	Section 5 5.1	
Special Text	Section 5 5.2	
S/N-NEMA Check	Section 5 5.3	License registration required
FOV Check	Section 5 5.4	License registration required
Uniformity Check	Section 5 5.5	License registration required
Slice Thickness Check	Section 5 5.6	License registration required

1.4.5 Programs to be started from Scan Card

These are programs mainly used for adjustment.

License registration is necessary to start up the programs.

Table 1.4-5 Programs to be started from Scan Card

Item Name	Section & Subsection
Frequency Search	Section 6 6.28
H1 Adjustment	Section 6 6.29
Manual Scan	Section 6 6.30
Signal Generation Check	Section 6 6.31
ECC Adjustment	Section 6 6.37
B0 Search	Section 6 6.38

1.4.6 Programs to be started from T/M Manager

These are programs mainly related to adjustment and log display. License registration is necessary to start up the programs.

Table 1.4-6 Programs to be started from T/M Manager

Tag Name	Item Name	Section & Subsection
Service	System Table Editor	Section 6 6.15
	Magnetic Field Analysis	Section 6 6.16
	Magnetic Field Search	Section 6 6.17
	ECC Tool	Section 6 6.35
	GC Pulse Generation	Section 6 6.18
	Coil Adjustment	Section 6 6.32
	GC Offset Adjustment	Section 6 6.19
	GC Slew Rate Adjustment	Section 6 6.20
	GC Gain Parameter Setting	Section 6 6.21
	FOV Adjustment	Section 6 6.22
	FSE Auto Adjustment	Section 6 6.23
	FatSepF Auto Adjustment	Section 6 6.39
Check	Easy Raw Data Analysis	Section 6 6.24
	Prescan Data Display	Section 6 6.25
	Frequency Log Display	Section 6 6.26
	Check Sum Display	Section 6 6.27
	Status Check	Section 4 4.1
	B0 Current Check	Section 4 4.6
	Shim Current Check	Section 4 4.7
	Check Program	Section 4 4.2
	Patient Table Tool	Section 4 4.3
	WS board Check	Section 4 4.4
	NVRAM Check	Section 4 4.5
System	UPS Setup	Section 6 6.33
	UPS View	Section 6 6.34
	Language Setup	Section 6 6.36

1.5 Caution

When ending the system, make sure to end T/M program first. If the system ending procedure is taken without ending T/M program, such a case may occur that the system is not shut down normally.

Section 2 Registration of Software

2.1 General

The procedures which must be followed before and after software registration are described below.

2.1.1 General flow chart of operation procedures

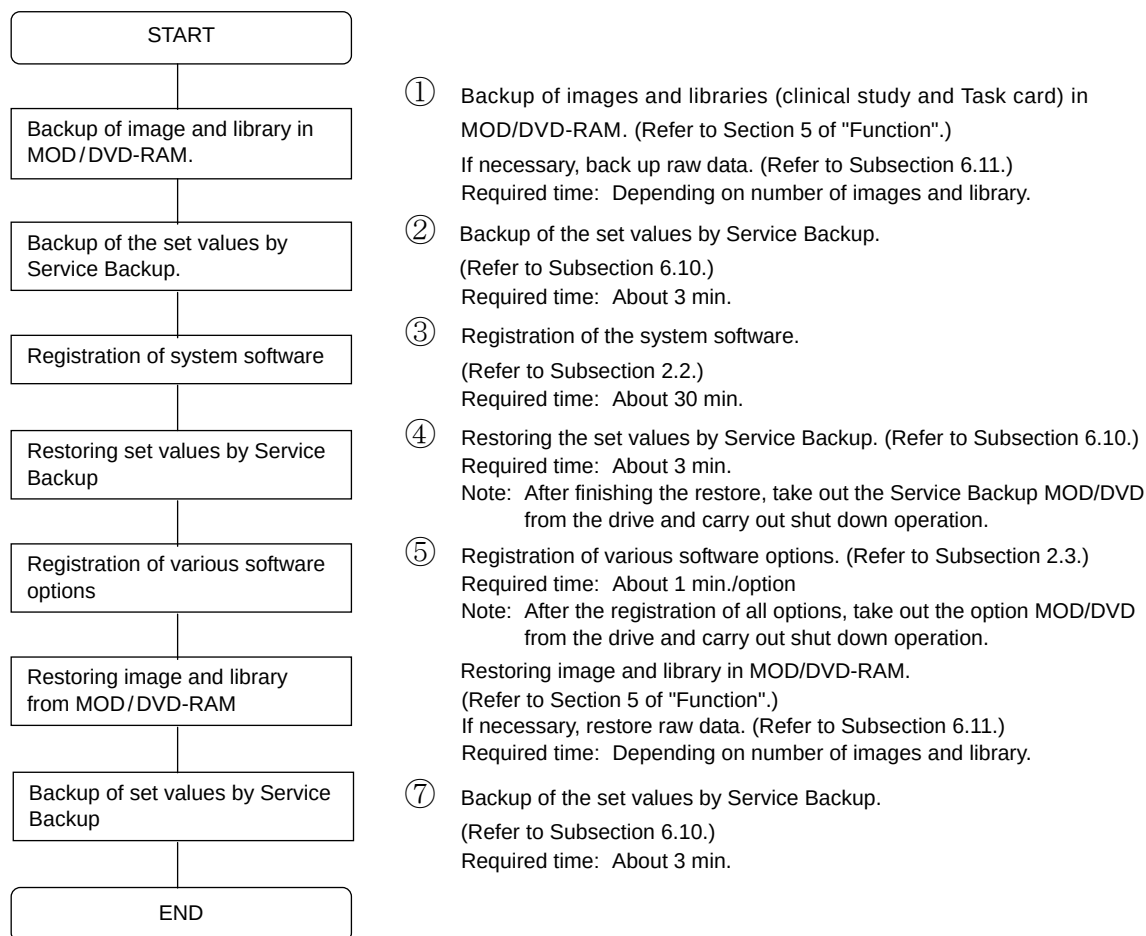


Fig. 2.1-1 General Flow Chart of Operation Procedures

2.1.2 Operation procedures for various operation works

< Operation works >	< Required procedures >
(a) Software version-up	: ① ② ③ ④ ⑤ ⑥ ⑦
(b) System recovery operation	: — — ③ ④ ⑤ ⑥ ⑦
(c) Addition of option software	: — — — — ⑤ — ⑦

2.2 Registration of System Software

2.2.1 General

Register the system software into the magnetic disk to make the system ready to operate.

2.2.2 Start-up

2.2.2.1 In the case of UltraSPARC-III-Engine:

- (1) If the system has already powered up, press CPU POWER button to shut down the equipment.
- (2) Set the MOD drive switch (referring to Fig. 2.2-1) to the right position (HDD mode).

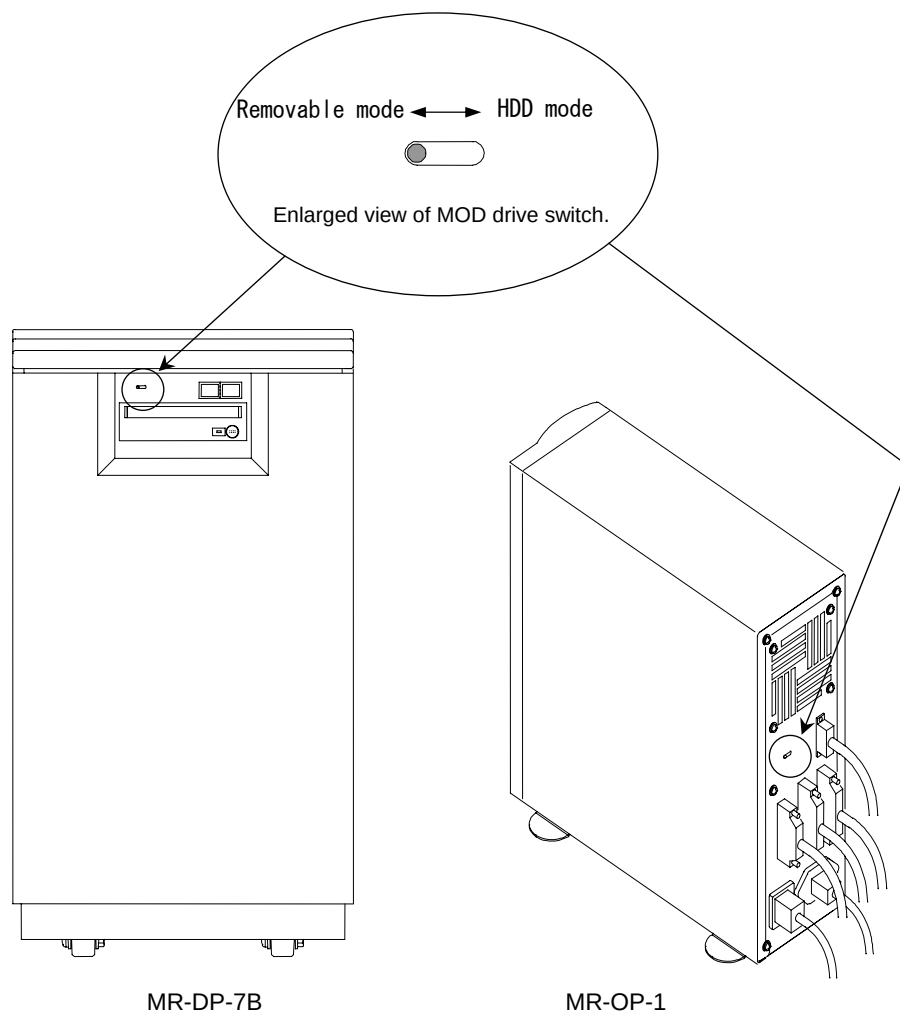
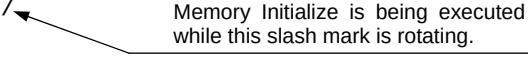


Fig. 2.2-1 Position of MOD Drive Switch

- (3) Press CPU POWER button to power up the equipment.
- (4) Immediately after the screen (white screen) is displayed, the Memory Initialize process starts. During this Memory Initialize process, press <Stop + A>. Then, the following ok prompt is displayed on the screen.

NOTE: Because the Memory Initialize process will be soon completed (in about 2 seconds), quickly proceed the above operation. If you miss this timing, follow the procedure described in Paragraph 2.2.6.1.

```
Initializing Memory /  
ok
```



Memory Initialize is being executed while this slash mark is rotating.

- (5) Enter the following command to confirm the HDD mode is selected.

```
probe-scsi-all
```

- (6) If the following underlined message is displayed, the HDD mode has been selected.

NOTE: If the HDD mode has not been selected, follow the operation described in Paragraph 2.2.6.2. In case the following message is not displayed after some time has passed, follow the operation described in Paragraph 2.2.6.4.

```
Target 1  
Unit 0  Removable Disk  HITACHI OD172-11  B178  
.  
.  
.  
ok
```

- (7) Insert the system software MOD into the MOD drive with the A-side faced up. However, when a console is MR-OP-1 type, insert the MOD where the A-side is made into the left.

NOTE: Do not apply write protect to the system software MOD.

- (8) After the LED on the MOD drive lights up, input the following command to boot up the system by the MOD cartridge.

```
boot disk1
```

- (9) Upon completion of the system bootup, the button shown in Fig. 2.2-2 is displayed.

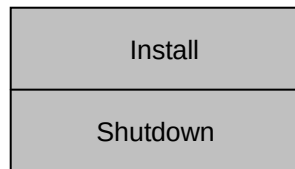


Fig. 2.2-2 System Software Registration Button

2.2.2.2 In the case of Sun-Blade-2500:

- (1) If the system has already powered up, choose Quit from System Tool to shut down the equipment.

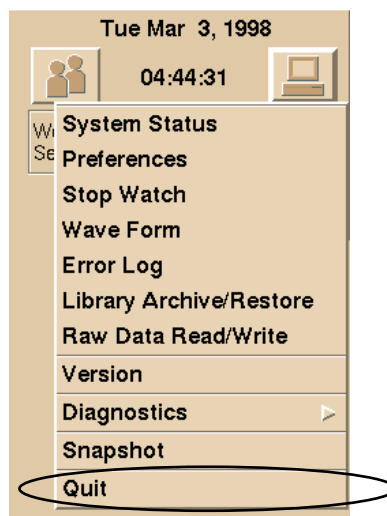


Fig. 2.2-3 How to system shut down by the system tool

- (2) If the MOD drive of the outside connection exists, turn off the power of that drive.
- (3) Press CPU POWER button to power up the equipment. (If the MOD drive of the outside connection exists, keep turning off the power of that drive.)

- (4) Just before the screen (white screen) is displayed, the Memory Initialize process starts. Repeat press <Stop + A> until the screen (white screen) is displayed after CPU POWER is pressed to be pressed <Stop + A> during this Memory Initialize process. Then, the following ok prompt is displayed on the screen.

NOTE: Because the Memory Initialize process will be soon completed (in about 2 seconds), quickly proceed the above operation. If you miss this timing, follow the procedure described in Paragraph 2.2.6.1.

Initializing Memory /
ok

Memory Initialize is being executed while this slash mark is rotating.

- (5) Insert the system software CD-ROM into the DVD-RAM drive.
- (6) After the color of LED of the DVD-RAM drive is turned off from green, input the following command to boot up the system by the CD-ROM.

```
boot cdrom
```

- (7) Upon completion of the system bootup, the button shown in Fig. 2.2-4 is displayed.

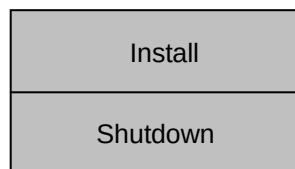


Fig. 2.2-4 System Software Registration Button

2.2.3 Installation

Click [Install]. After clicking the button, the window for registration of the system software is displayed to start the installation (about 30 minutes).

NOTE: Do not touch any buttons on the keyboard while the installation being executed. "Warning" is displayed during registration, but it does not mean any trouble.

2.2.4 Ending process

After finish of the installation, the dialog shown in Fig. 2.2-5 is displayed. By clicking [OK] button, the installation process has been completed.

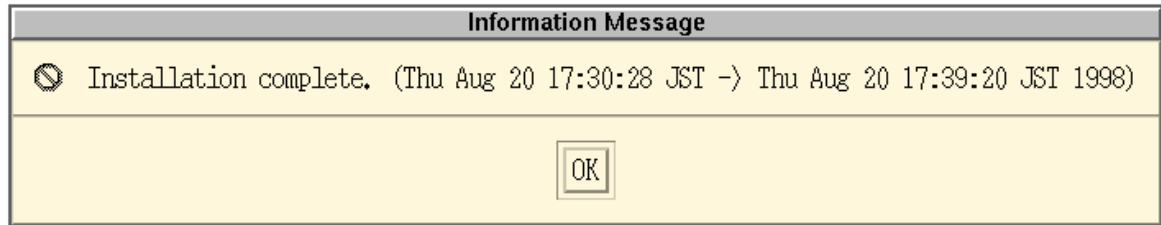


Fig. 2.2-5 Dialog to indicate the finish of installation

2.2.5 Checking registration of system software

The way of checking registration of system software is a little different at the case of the operation table UltraSPARC-III-Engine and Sun-Blade-2500.

2.2.5.1 In the case of UltraSPARC-III-Engine:

- (1) Set the MOD drive switch to the left position (Removable mode).
- (2) Press CPU POWER button and MR POWER button to power up the system.
- (3) Immediately after the screen (white screen) appears, Memory Initialize process starts. During this Memory Initialize process, press <Stop + A>. Then, the following ok prompt is displayed on the screen.

NOTE: Because the Memory Initialize process will be soon completed (in about 2 seconds), quickly proceed the above operation. If you miss this timing, follow the procedure described in Paragraph 2.2.6.1.

Initializing Memory /
ok

Memory Initialize is being executed while this slash mark is rotating.

- (4) Press the Eject button and take out the system software MOD.
- (5) To check that the Removable mode has been set, enter the following command.

```
probe-scsi-all
```

- (6) If the following underlined message is displayed, the Removable mode has been selected.

NOTE: If the Removable mode has not been selected, follow the operation described in Paragraph 2.2.6.3. In case the following message is not displayed after some time has passed, follow the operation described in Paragraph 2.2.6.4.

```
Target 1
Unit 0 Removable Device type 7 HITACHI OD172-11 B178
      .
      .
      .
ok
```

- (7) To reboot the system, boot it by using the magnetic disk of which installation has been completed. Enter the following command.

```
boot -r
```

- (8) The initial MR screen is displayed on the monitor. If this screen is displayed normally, it indicates that the installation has been successfully completed.

2.2.5.2 In the case of Sun-Blade-2500:

- (1) Press CPU POWER button and MR POWER button to power up the system.
- (2) Press the Eject button and take out the system software CD-ROM.
- (3) The initial MR screen is displayed on the monitor. If this screen is displayed normally, it indicates that the installation has been successfully completed.

2.2.6 Operation procedure for misoperation

2.2.6.1 In case timing for Memory Initialize is missed:

- (1) In case <Stop + A> is pressed before Memory Initialize

For the ok prompt, enter the command below to turn off the power supply. Then, press CPU POWER button to turn on the power supply of the system to retry the operation.

```
power-off
```

- (2) In case <Stop + A> is pressed after Memory Initialize

By entering the following command, the system is booted and Memory Initialize process is executed again.

```
sync
```

2.2.6.2 In case HDD mode has not been set:

For the ok prompt, enter the following command and power down the system. Upon checking the MOD drive switch set to the right position, restart the operation from the beginning.

```
power-off
```

2.2.6.3 In case Removable mode has not been set:

For the ok prompt, enter the following command and power down the system. Upon checking the MOD drive switch set to the left position, restart the operation from the beginning.

```
power-off
```

2.2.6.4 In case of no response to probe-scsi-all:

In case any message is not displayed even if more the 5 minutes has passed after entry of the command "probe-scsi-all", power down the system by turning off the switch on the rear of the control console. Press CPU POWER button again to power up the system and retry the operation.

2.3 Registration of Option Software

2.3.1 General

Option software is registered in the system to make the option software ready to use.

2.3.2 Start-up

Follow the next steps to start the registration process for option software.

Option software is MOD in the case of UltraSPARC-III-Engine ,and it is DVD in the case of Sun-Blade-2500.

2.3.2.1 In the case of UltraSPARC-III-Engine:

- (1) When the system has been powered up, press CPU POWER button to power down the system.
- (2) Make sure that write-protection is not applied to the option MOD.
- (3) Insert the option MOD into the drive with the A-side faced up. However, when a console is MR-OP-1 type, insert the MOD where the A-side is made into the left. The MOD cannot be inserted completely because the system is not powered up at this time, but go to the next step.
- (4) Press CPU POWER button to power up the system.
- (5) The option software MOD is automatically inserted into the drive and registration of the option software starts.

2.3.2.2 In the case of Sun-Blade-2500:

- (1) When the system has been powered up, choose Quit from System Tool menu to power down the system.

- (2) Make sure that write-protection is not applied to the option DVD.
- (3) Press CPU POWER button to power up the system.
- (4) Quickly insert the option DVD into the drive with the A-side faced up after turn on the system.
- (5) The option software DVD is automatically inserted into the drive and registration of the option software starts.

2.3.3 Display screen

The display screen and its function names are shown in Fig. 2.3-1

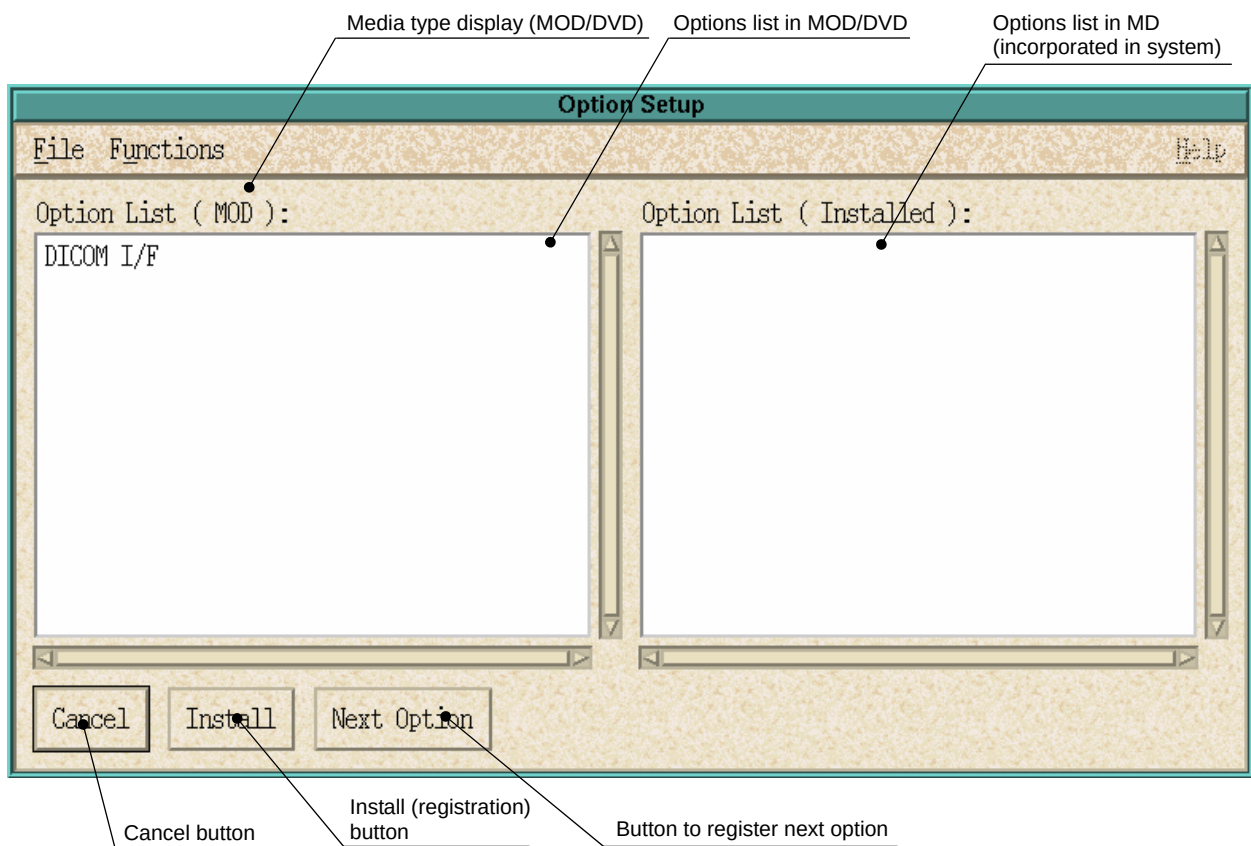


Fig. 2.3-1 Option Software Registration Screen

2.3.4 Operation procedure

Follow the next steps for the registration operation.

2.3.4.1 Installation of option software

Follow the next steps to register option software.

- (1) Make sure that the option software MOD/DVD is set in the drive and the options list in the MOD/DVD is displayed in the left area on the option software registration screen (Fig. 2.3-1). If the list is not displayed, insert the option software MOD/DVD again into the drive and click [Next Option] button to check the display.
- (2) Click [Install] button to start the installation.

NOTE: When UPS option newly installed, click [Install] button to display UPS setting tool (Refer to Item 6.33.) for setting type of UPS. Select the type of UPS and then click [OK] button. In this step, make sure to execute the setting the type of UPS without clicking [Cancel] button in UPS setting tool. If not, the UPS option is not installed correctly.

- (3) When the installation is completed, the dialog shown in Fig. 2.3-2 is displayed. Click [OK] button to confirm that the list in the left area on the option software registration screen has been copied into the list in the right area. If confirmed, the registration has been completed.



Fig. 2.3-2 Installation Completion Dialog

If more than one option software MOD/DVD are to be registered, follow the next steps.

- (4) Replace the option software MOD/DVD with other MOD/DVD and check to see that the LED on the drive lights up.
- (5) Click [Next Option] button. The option software list in the MOD/DVD is displayed in the left area on the option software registration screen.
- (6) Repeat the operations from (2) to (5).

NOTE: In case DICOM I/F has been registered, set DICOM by using the DICOM setting program. (Refer to Technical Guide for DICOM Net Interface.)

2.3.5 End

Take out the option software MOD/DVD from the drive and click [Cancel] button to display the screen shown in Fig. 2.3-3. Clicking [Yes] button on this screen closes this application and the system is automatically shut down.

Clicking [No] button returns to the previous screen without closing the application.

NOTE: After the system is shut down, press CPU POWER button.



Fig. 2.3-3 Exit Confirmation Dialog

2.3.6 Error specifications

Error specifications are shown in the Table 2.3-1.

Table 2.3-1 Error Specifications

Error Message	Cause	Solution
Option not entry.	Improper option software MOD/DVD is used.	Use the proper option software MOD/DVD.
This is not an Option MOD/DVD.	The MOD/DVD that is not the option MOD/DVD is in the drive unit.	Insert the option software MOD/DVD in the drive.
Can't access the file.	Internal error	
Serial number doesn't match.	Improper MOD/DVD is in use.	Use the proper option software MOD/DVD.
This option has already been installed.	The already installed option is set for reinstallation.	The installed option cannot be installed again.
This MOD/DVD is write-protected.	The MOD/DVD is write-protected.	Release write-protect of the MOD/DVD to set it to Rewritable.
Please insert MOD/DVD.	No MOD/DVD is in the drive unit.	Insert the option software MOD/DVD into the drive unit.
This MD is not supported.	The MD which is not supported is used as Second MD.	Use the MD supported by this system.

Section 3 Effective Setting of Check/Service Program

3.1 Registration of License

3.1.1 General

With the Key MOD/DVD, enter the password. If the password is correct, execute the License registration.

By executing the license registration, the Check/Service program is made effective and the Service Menu is added to System Tool, Scan Card and Maintenance Card.

3.1.2 Start-up

After starting-up the system, inserting the Key MOD/DVD into the MOD/DVD-RAM drive unit displays the password entry screen.

3.1.3 Display screen

3.1.3.1 In the case of the system for the overseas general purpose

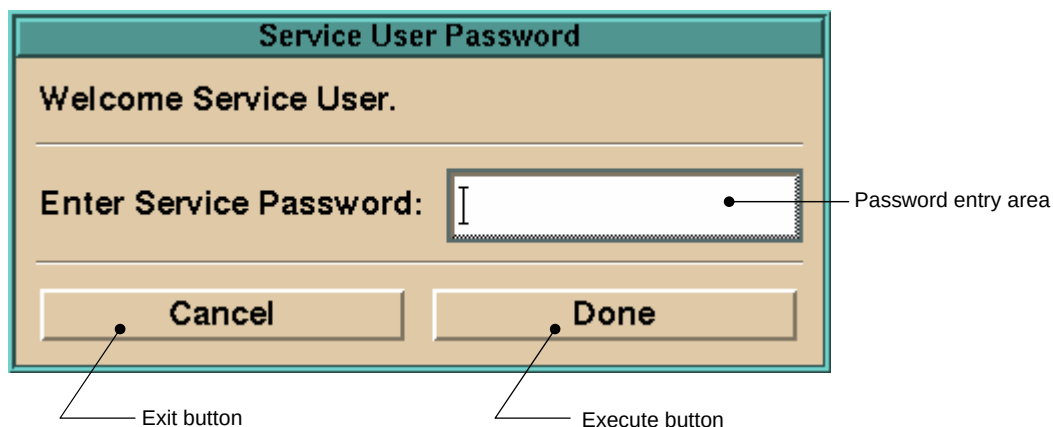


Fig. 3.1-1 Password Entry Screen (for overseas general)

3.1.3.2 In the case of the system for HMSA (When it is ratified in HIPAA.)

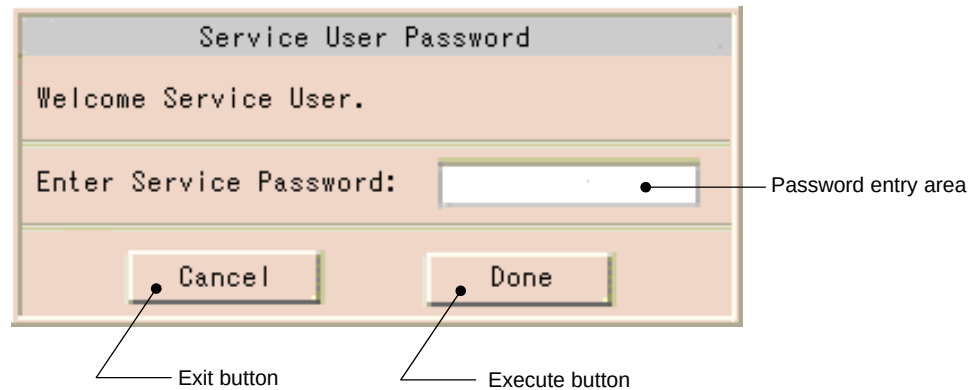


Fig. 3.1-2 Password Entry Screen (for HMSA)

3.1.4 Operation procedure

3.1.4.1 Entry of password

Click the password entry screen and enter the password.

3.1.4.2 Execution

After entry of the password, click [Done] button.

If the entered password is correct, the application is exited after the license registration is completed.

(1) In the case of the system for the overseas general purpose

If an incorrect password is entered, the dialog shown in Fig. 3.1-3 is displayed. After confirmation, click [Yes] button to return to the previous Password Entry screen.

To finish the program without doing anything, click [No] button. Clicking [Help] button displays the dialog shown in Fig. 3.1-4. After confirmation, click [Done] button. The previous Password Entry screen is displayed.

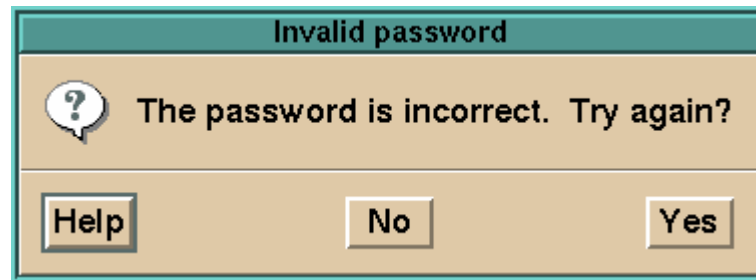


Fig. 3.1-3 Dialog displayed by Entry of Incorrect Password (for overseas general)

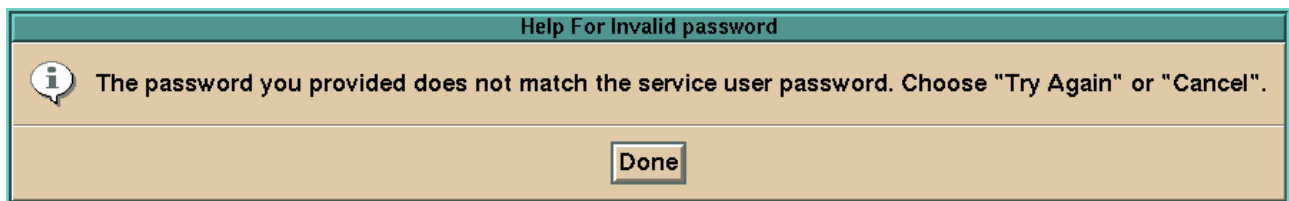


Fig. 3.1-4 Dialog displayed by Selection of [Help] (for overseas general)

(2) In the case of the system for HMSA (When it is ratified in HIPAA.)

If an incorrect password is entered, the dialog shown in Fig. 3.1-5 is displayed. After confirmation, click [Done] button to return to the previous Password Entry screen.

To finish the program without doing anything, click [Cancel] button.

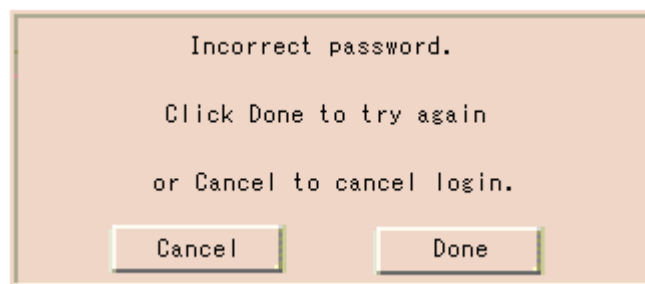


Fig. 3.1-5 Dialog displayed by Entry of Incorrect Password (for HMSA)

3.1.4.3 Unloading Key MOD/DVD

Press the Eject button on the MOD/DVD-RAM drive unit and take out the Key MOD/DVD.

3.1.5 End

Click [Cancel] button to exit from the application. In this case, the License registration is not executed.

3.2 Deletion of License

3.2.1 General

The License is deleted. By deleting the License, the Service Menu is deleted from System Tool, Scan Card and Maintenance Card.

3.2.2 Operation procedure

Select End Service Mode in the System Tool. Then, the License is deleted. This operation is, however, not executed unless the License has been registered.

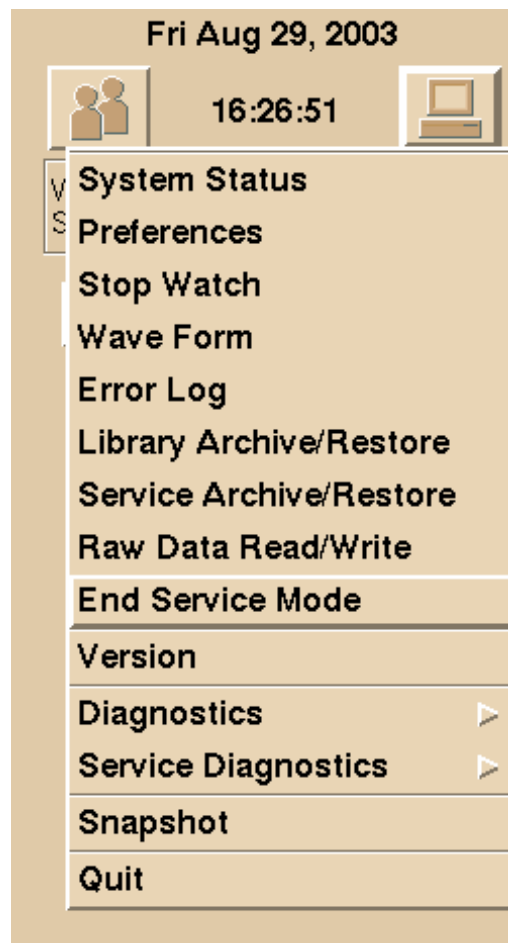


Fig. 3.2-1 Selection of End Service Mode

Section 4 System Check Program

4.1 Status Check

4.1.1 General

This is a tool to display the status signal on the status PCB. Change in status is always refreshed.

4.1.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Status Check in the displayed items and click [OK] or double-click the selected item.

4.1.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.1-1.

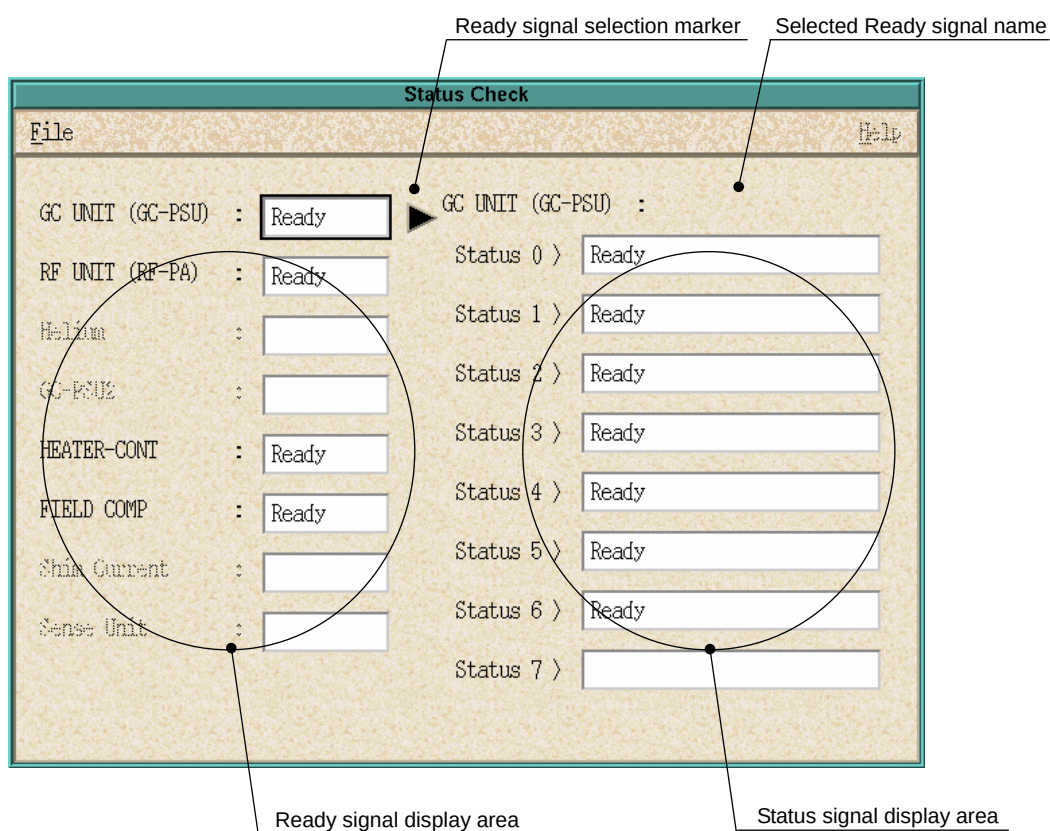


Fig. 4.1-1 Status Check Display Screen

4.1.4 Operation procedure

Follow the next steps to execute the process.

4.1.4.1 Selection of Ready signal

(1) Ready signals display

Status of the status signal of the selected Ready signal is displayed in the Ready signal display area. If all of the defined status signals are ready, **Ready** is displayed, while even only one of them is not ready, **Not Ready** is displayed. A list of Ready signals is shown in Table 4.1-1. Blank indicates the signal not been defined.

Table 4.1-1 Ready Signals List

Ready signal	Name	Details of status signal
Ready0	GC UNIT	Table 4.1-2
Ready1	RF-PA	Table 4.1-3
Ready2	-----	Table 4.1-4
Ready3	-----	Table 4.1-5
Ready4	HEATER-CONT	Table 4.1-6
Ready5	FIELD COMP	Table 4.1-7
Ready6	-----	Table 4.1-8
Ready7	-----	Table 4.1-9

(2) How to see ready signals display

The Ready signal marker indicates the Ready signal of the selected Status currently selected and displayed.

When the system is started up, the Ready signal at the following position is selected to indicate the status.

When no error occursGC UNIT status

If an error occurs.....Status of the Ready signal positioned at the highest part out of errors

To change Ready signal to be displayed, click the area where the status of the Ready signal is displayed. Pressing <Tab> key displays the status of the Ready signal in the next field, and pressing Shift + <Tab> keys displays the status of the Ready signal in the previous field.

4.1.4.2 Status signals display

Content to be displayed differs depending on model of the equipment and Ready signal. Displayed contents depend upon Ready signals. Table 4.1-2 ~ Table 4.1-9 show messages to be displayed when an error is found in each status signal. In case status signal is normal, Ready is displayed. Blank represents the signal which is not defined.

Table 4.1-2 GC UNIT (GC-PSU)

Ready signal	Error Status Display
Ready0	Xch Amp. Fault
Ready1	Ych Amp. Fault
Ready2	Zch Amp. Fault
Ready3	Xch Amp. Overheat
Ready4	Ych Amp. Overheat
Ready5	Zch Amp. Overheat
Ready6	Primary Power Fault
Ready7	-----

Table 4.1-3 RF UNIT (RF-PA)

Ready signal	Error Status Display
Ready0	Not Ready
Ready1	Shutdown
Ready2	-----
Ready3	-----
Ready4	-----
Ready5	-----
Ready6	-----
Ready7	-----

Table 4.1-4 Helium (non-displaying)

Ready signal	Error Status Display
Ready0	-----
Ready1	-----
Ready2	-----
Ready3	-----
Ready4	-----
Ready5	-----
Ready6	-----
Ready7	-----

Table 4.1-5 GC-PSU2 (non-displaying)

Ready signal	Error Status Display
Ready0	-----
Ready1	-----
Ready2	-----
Ready3	-----
Ready4	-----
Ready5	-----
Ready6	-----
Ready7	-----

Table 4.1-6 HEATER-CONT

Ready signal	Error Status Display
Ready0	Upper heater Overheat
Ready1	-----
Ready2	Lower heater Overheat
Ready3	-----
Ready4	Upper heater Modeset error
Ready5	-----
Ready6	Lower heater Modeset error
Ready7	-----

Table 4.1-7 FIELD COMP

Ready signal	Error Status Display
Ready0	Fault
Ready1	-----
Ready2	-----
Ready3	-----
Ready4	-----
Ready5	-----
Ready6	-----
Ready7	-----

Table 4.1-8 Shim Current (non-displaying)

Ready signal	Error Status Display
Ready0	-----
Ready1	-----
Ready2	-----
Ready3	-----
Ready4	-----
Ready5	-----
Ready6	-----
Ready7	-----

Table 4.1-9 Sense Unit (non-displaying)

Ready signal	Error Status Display
Ready0	-----
Ready1	-----
Ready2	-----
Ready3	-----
Ready4	-----
Ready5	-----
Ready6	-----
Ready7	-----

4.1.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 4.1-3 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.



Fig. 4.1-2 Quit Button

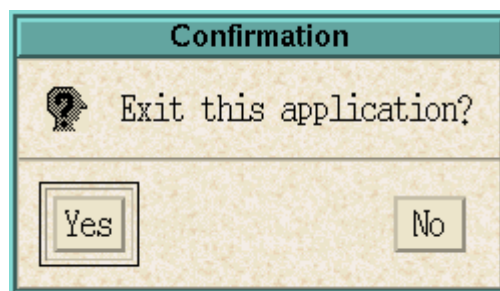


Fig. 4.1-3 Exit Confirmation Dialog

4.1.6 Error specifications

Error specifications are shown in Table 4.1-10.

Table 4.1-10 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading the SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application

4.2 Check Program

4.2.1 General

This tool checks hardware on each PCB. It is applicable to RCN board, PSC board and RT-CONT board.

4.2.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Check Program in the displayed items and click [OK] or double-click the selected item.

4.2.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.2-1.

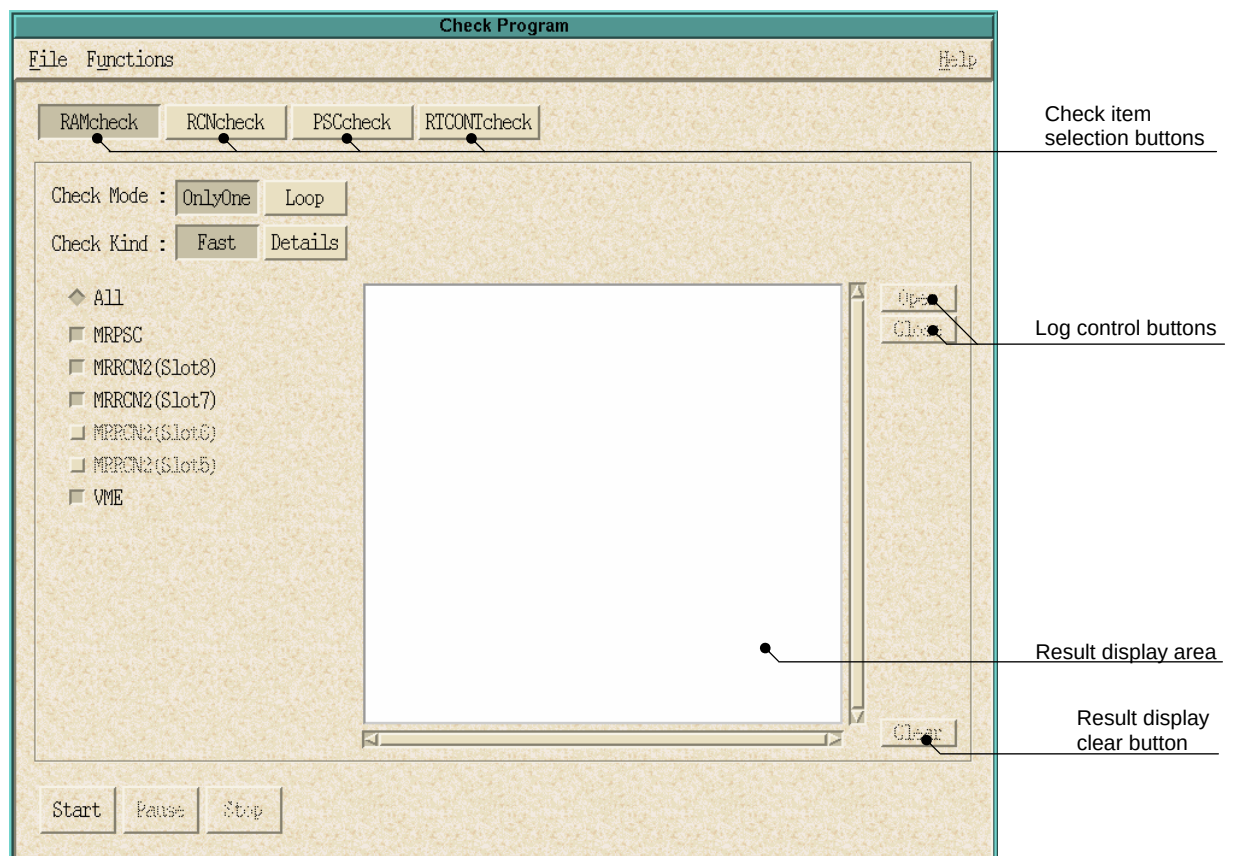


Fig. 4.2-1 Check Program Display Screen

4.2.4 Operation procedure

Follow the next steps to execute the process.

4.2.4.1 Updating parameters

Use the keyboard or mouse cursor to update each parameter. The following methods are available.

4.2.4.1.1 Selecting an item with pushbutton

Pushbuttons displayed in line are for single selection. Clicking with the mouse cursor updates any parameter.

4.2.4.1.2 Entering string into entry field

Place the mouse cursor in any entry field of parameter to be updated to select the entry field. Use the keyboard to enter a numerical value or string, and press <Enter> or remove the focus from the entry field to update the parameter.

4.2.4.1.3 Selecting an item with toggle button

Clicking any button (☐, ☐) or label with the mouse cursor to select it.

4.2.4.2 Selecting a function

This tool includes six functions as shown in the table below. Each function will be described in separate sections.

Table 4.2-1 Function Table

Item No.	Function name	Contents
4.2.4.3	RAMcheck	Checking MRPSC, MRRCN2 (Slot 8~5) and VME memories.
4.2.4.4	RCNcheck	Link check
		Flag check
4.2.4.5	PSCcheck	Timer check
		Data check
4.2.4.6	RTCONTcheck	Panel switch check
		Panel LED check
		Buzzer check
4.2.4.7	Log display	RAMcheck
		RCNcheck
		PSCcheck
		RTCONTcheck
4.2.4.8	System information display	Information display such as: ▪ Kind of RCN board ▪ Number of RCN boards ▪ Max. number of coil channels possible to scan ▪ LM memory capacity

4.2.4.3 RAMcheck

Operate as described below.

4.2.4.3.1 Operation procedure

- (1) Select [RAMcheck] button from the check item selection buttons on the startup screen.

If RAMcheck has been already selected, proceed to (3). When started up, RAMcheck has been selected by default.

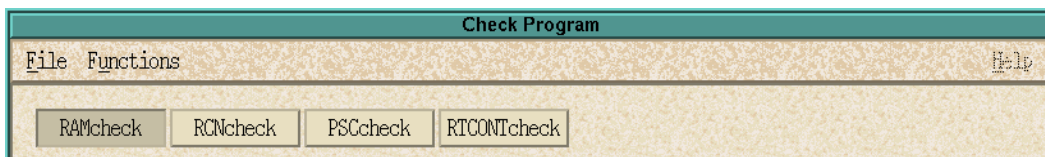


Fig. 4.2-2 Check Item Selection Buttons

- (2) RAMcheck screen is displayed.
Any board not installed is grayed out.

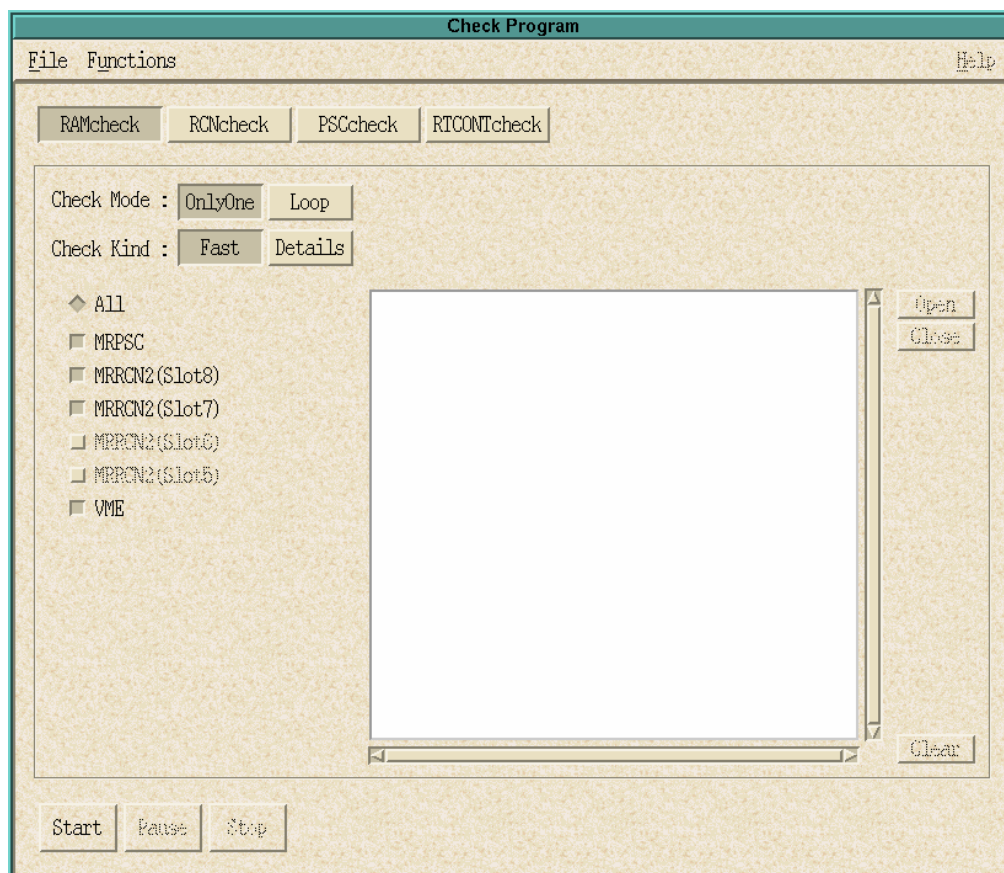


Fig. 4.2-3 RAMcheck Screen

- (3) Select a check mode (either of [OnlyOne]/[Loop] buttons).

When started up, OnlyOne has been selected by default.

OnlyOne Check executes the designated check only once, while Loop Check repeats execution of the designated check. Selecting Loop Check allows the Error Stop Toggle button to be checked. If it is set to On, checking is suspended when an error occurred.

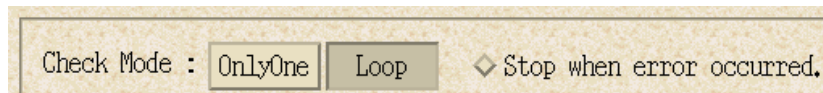


Fig. 4.2-4 Error Stop Toggle Button

- (4) Select the kind of check (either of [Fast]/[Details] buttons).

When starting up, Fast has been selected.

As the check of Fast checks the minimum necessary memory only, the checking is completed within short time. To the contrary, the check of Details checks all memories and long time is required to complete the check.

This function is effective for MRRCN2 board check and VME check.

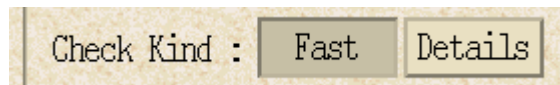


Fig. 4.2-5 Check Kind Selection Buttons

- (5) Use the board selection button to set any board to be checked On.

If clicking ☐ of each board, On/Off is switched over (see below). Multiple selections are allowed. ☒ All button switches over On/Off of all memories of all boards as a batch. When started up, all memories of all boards have been set to On by default.



Fig. 4.2-6 Board Selection Buttons

- (6) Click any individual board/memory selection button (see below) to open the individual board/memory selection screen (see (7)).

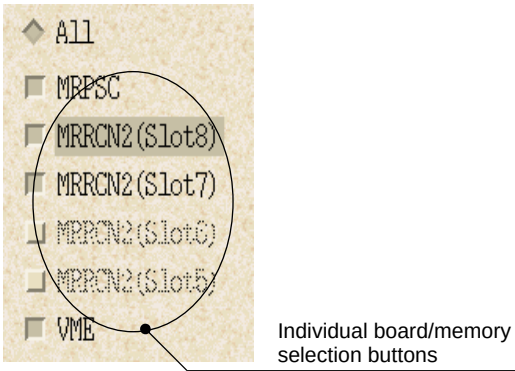


Fig. 4.2-7 Individual Board/Memory Selection Buttons

Table 4.2-2 Individual Memory Selection Screen

Button name	Figure No.	Figure title
MRPSC	4.2-8	MRPSC Memory Individual Selection Screen
MRRCN2 (Slot 8) MRRCN2 (Slot 7) MRRCN2 (Slot 6) MRRCN2 (Slot 5)	4.2-9	MRRCN2 (Slot 8~5) Memory Individual Selection Screen
VME	4.2-10	VME Memory Individual Selection Screen

(7) Operation of each individual selection screen is described below.

While each individual selection screen is displayed, however, the other operations are disabled.

- Operation of MRPSC memory individual selection screen

◇ All button switches over On/Off of all MRPSC memories as a batch.

The other buttons switch over On/Off individually.

Clicking [Apply] button or board selection button (MRPSC) returns to the previous screen.

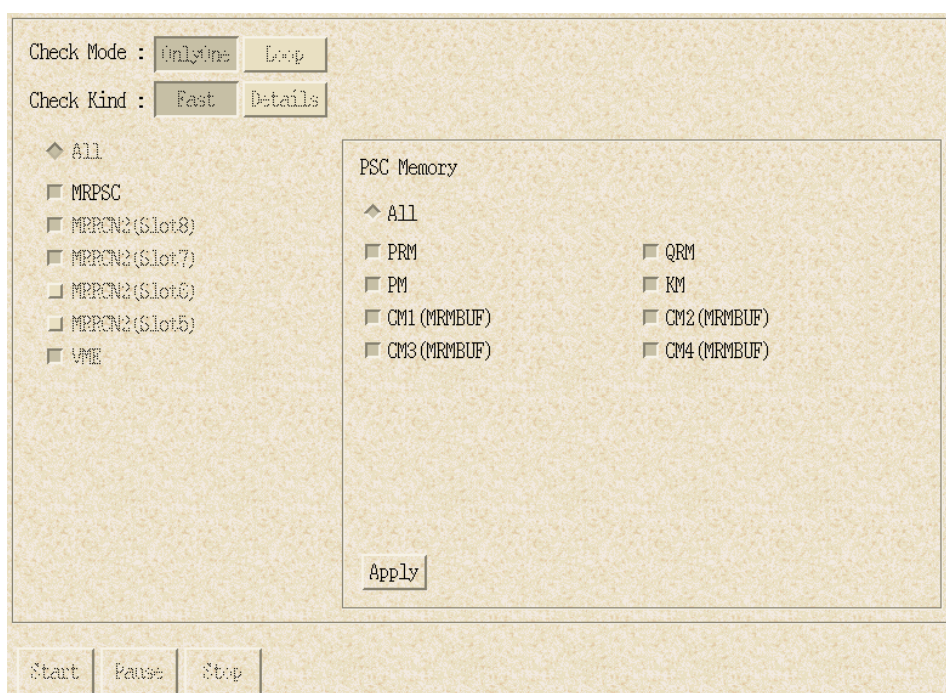


Fig. 4.2-8 MRPSC Memory Individual Selection Screen

- Operation of MRRCN2 (Slot 8~5) memory individual selection screen
 - ◇ All button switches over On/Off of all memories of relevant board as a batch.

The other buttons switch over On/Off individually.

Clicking [Apply] button or board selection button (MRRCN2 (Slot 8~5)) returns to the previous screen.



Fig. 4.2-9 MRRCN2 (Slot 8~5) Memory Individual Selection Screen

- Operation of VME memory individual selection screen
 - ◇ All button switches over On/Off of all VME memories as a batch.

Each button of MRRCN2 (Slot 8~5) switches over On/Off of EMA/EMB/VMA/VMB of each board as a batch.

Each button of EMA/EMB/VMA/VMB switches over On/Off individually.

Clicking [Apply] button or board selection button (VME) returns to the previous screen.

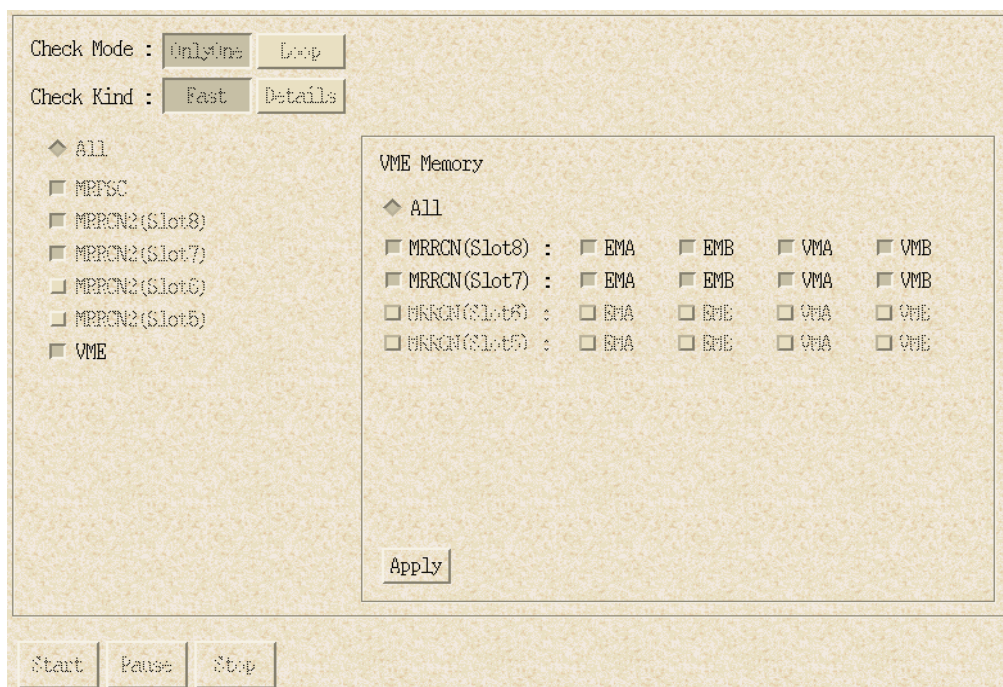


Fig. 4.2-10 VME Memory Individual Selection Screen

- (8) Clicking [Start] button starts checking.

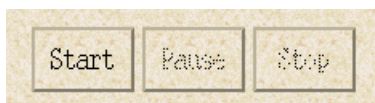


Fig. 4.2-11 Start Button

- (9) To stop checking, click [Pause] button.

Checking is suspended. If clicking [Stop] button, RAMcheck is terminated.

If clicking [Start] button, checking resumes from the point where the check has been stopped.

In [OnlyOne] check, however, when all selected items are completed, checking stops automatically with a bell sound.



Fig. 4.2-12 Buttons while Suspended

- (10) Checking progress is displayed in the result display area. Progress displayed is saved as a log as it is.

Clicking [Clear] button clears the result display area. This clears the screen only, but no log is cleared.

For displaying the log, refer to 4.2.4.7.



Fig. 4.2-13 Clear Button

4.2.4.3.2 Result display

Examples of result display are shown below.



Fig. 4.2-14 An Example of RAMcheck Result Display (OnlyOne Check)



Fig. 4.2-15 An Example of RAMcheck Result Display (Loop Check)

4.2.4.4 RCNcheck

Operate as follows.

4.2.4.4.1 Operation procedure

- (1) Use the check item selection button of the startup screen to select RCNcheck.
If RCNcheck has been already selected, proceed to (3).

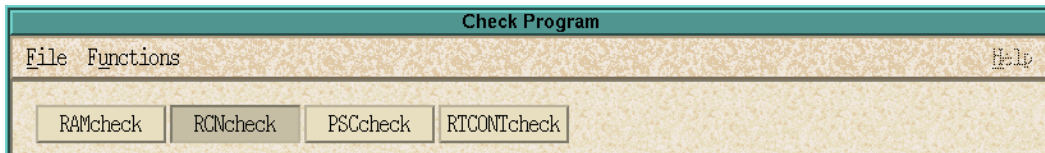


Fig. 4.2-16 Check Item Selection Button

- (2) RCNcheck screen is displayed.

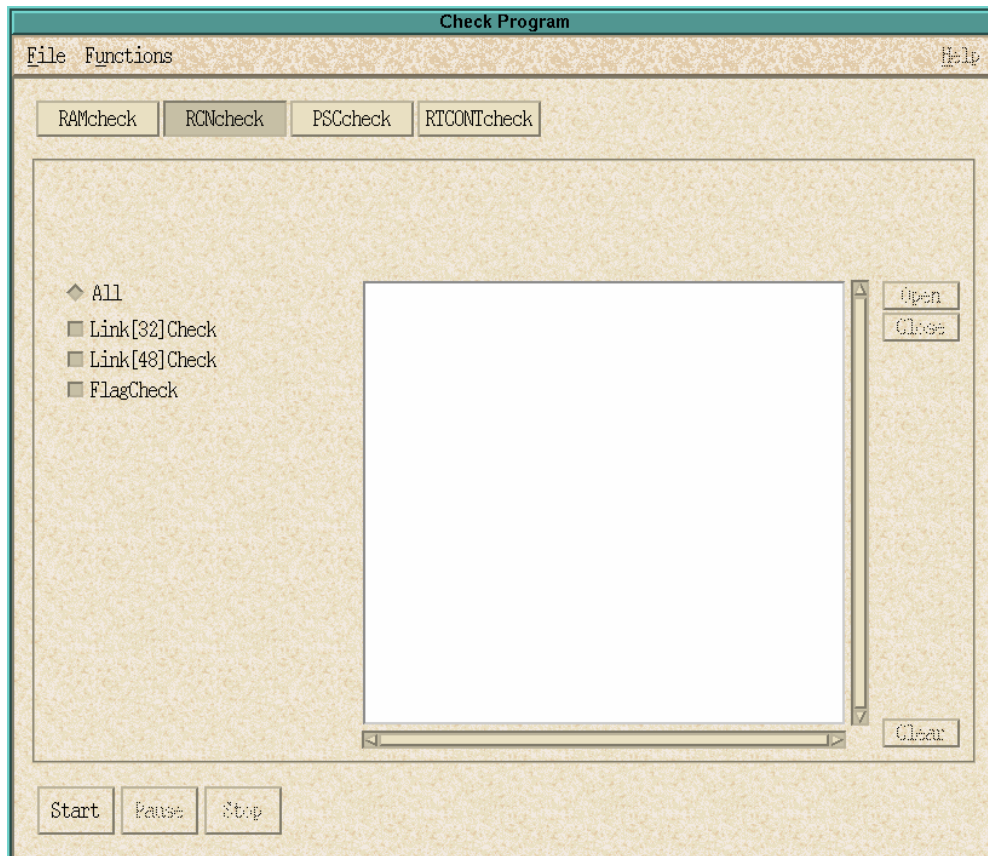


Fig. 4.2-17 RCNcheck Screen

- (3) RCNcheck includes items shown in Table 4.2-3. Set any item to be checked to On.

All items have been selected by default. Clicking  All button allows On/Off to be switched over as a batch.

Table 4.2-3 RCNcheck Items

Item
LINK [32] check
LINK [48] check
Flag check

- (4) Clicking [Start] button starts checking.



Fig. 4.2-18 Start Button

- (5) Check progress is displayed in the result display area. Progress displayed is saved as a log as it is.

Clicking [Clear] button clears the result display area. This clears the screen only, but no log is cleared.

For displaying the log, refer to Item 4.2.4.7.



Fig. 4.2-19 Clear Button

- (6) When completed, checking stops automatically with a bell sound.

4.2.4.4.2 Result display of RCNcheck

An example of RCNcheck is displayed below.



Fig. 4.2-20 An Example of RCNcheck Result Display

4.2.4.5 PSCcheck

Operate as follows.

4.2.4.5.1 Operation procedure

- (1) Select [PSCcheck] button from the check item selection buttons of the startup screen. If PSCcheck has been already selected, proceed to (3).

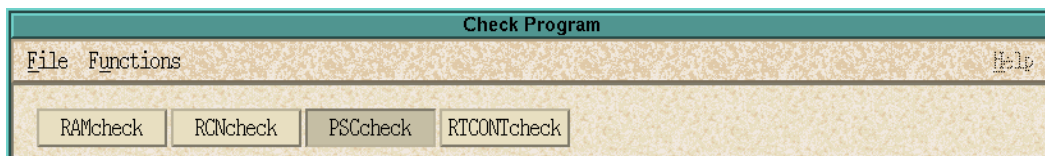


Fig. 4.2-21 Check Item Selection Buttons

- (2) PSCcheck screen is displayed.

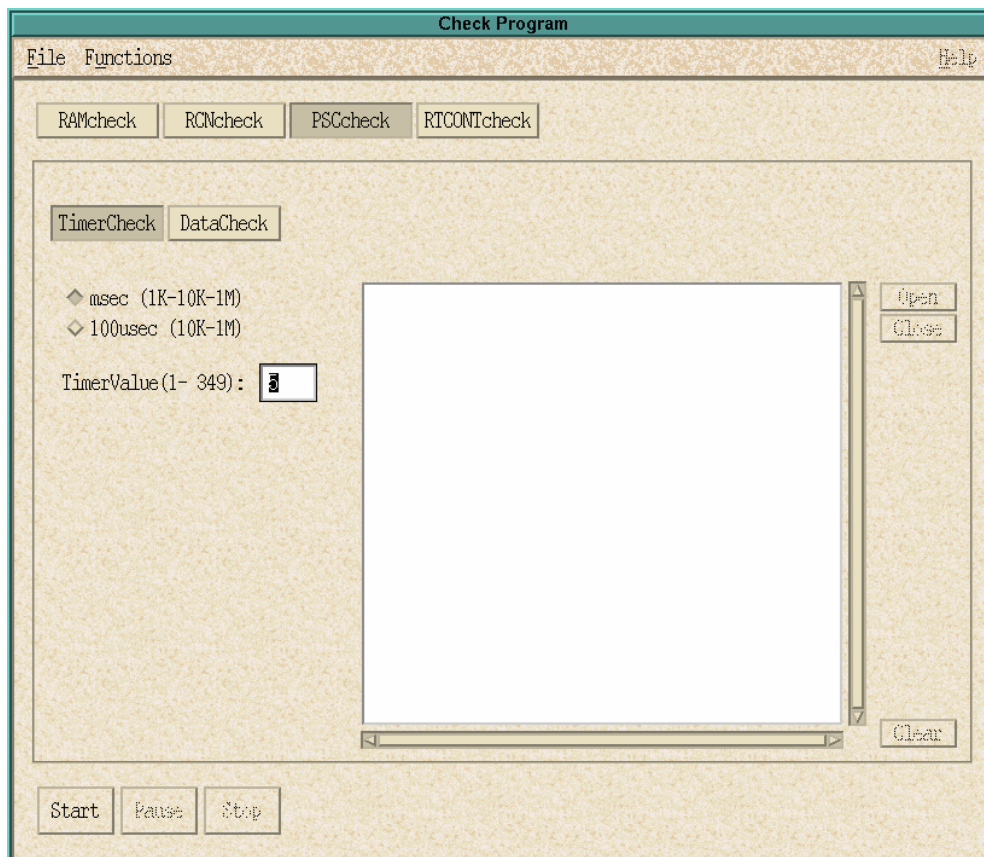


Fig. 4.2-22 PSCcheck Screen

- (3) PSCcheck includes items shown in the table below. Operation procedure will be described for each item.

Table 4.2-4 PSCcheck Items

Item No.	Item
4.2.4.5.1.1	Timer check procedure
4.2.4.5.1.2	Data check procedure

4.2.4.5.1.1 Timer check procedure

- (4) Click [TimerCheck] button.

This button has been selected by default when started up.

- (5) Select unit.

Click ☐ msec button for the timer in millisecond unit (Fig. 4.2-23), while

☐ 100 μ sec button for the timer in microsecond unit (Fig. 4.2-24). Timer entry field differs for each unit. The timer in millisecond unit has been selected by default when started up.

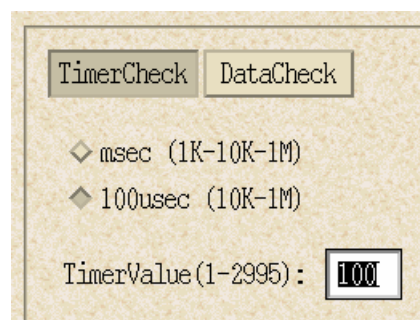
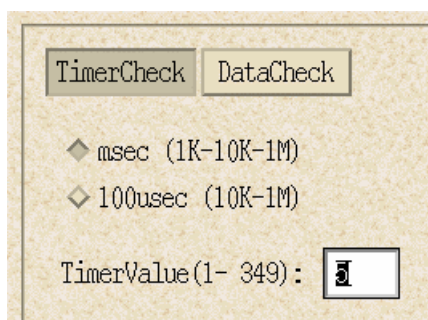


Fig. 4.2-23 Screen where msec Selected Fig. 4.2-24 Screen where μ sec Selected

- (6) Enter a timer value.

Allowed range for the timer is as shown in the table below.

Table 4.2-5 Range of Timer Value

Timer type	Range of timer value
msec	1 ~ 349
100 μ sec	1 ~ 2995

- (7) Clicking [Start] button starts checking.



Fig. 4.2-25 Start Button

- (8) Status of progress is displayed in the result display area (Fig. 4.2-30). Progress displayed is saved as a log as it is.

Clicking [Clear] button allows the result display area to be cleared. This clears the screen only, but no log is cleared..

For displaying log, refer to 4.2.4.7.



Fig. 4.2-26 Clear Button

- (9) When completed, checking stops automatically with a bell sound.

4.2.4.5.1.2 Data check procedure

- (4) Click the data check button.

The screen as shown below is displayed.

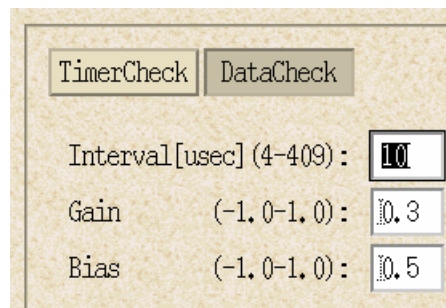


Fig. 4.2-27 Data Check Screen

- (5) Enter any parameter if necessary.

Range of parameters selectable is shown below.

Table 4.2-6 Range of Parameters Selectable

Parameter type	Range of parameter
Interval	4 ~ 409
Gain	-1.0 ~ 1.0
Bias	-1.0 ~ 1.0

- (6) Clicking [Start] button starts checking.



Fig. 4.2-28 Start Button

- (7) Status of progress is displayed in the result display area (Fig. 4.2-31). Progress displayed is saved as a log as it is.

Clicking [Clear] button allows the result display area to be cleared. This clears the screen only, but no log is cleared.

For displaying log, refer to 4.2.4.7.



Fig. 4.2-29 Clear Button

- (8) When completed, checking stops automatically with a bell sound.

4.2.4.5.2 Result display of PSCcheck

Examples of PSCcheck are shown below.



Fig. 4.2-30 An Example of Timer Check Result Display

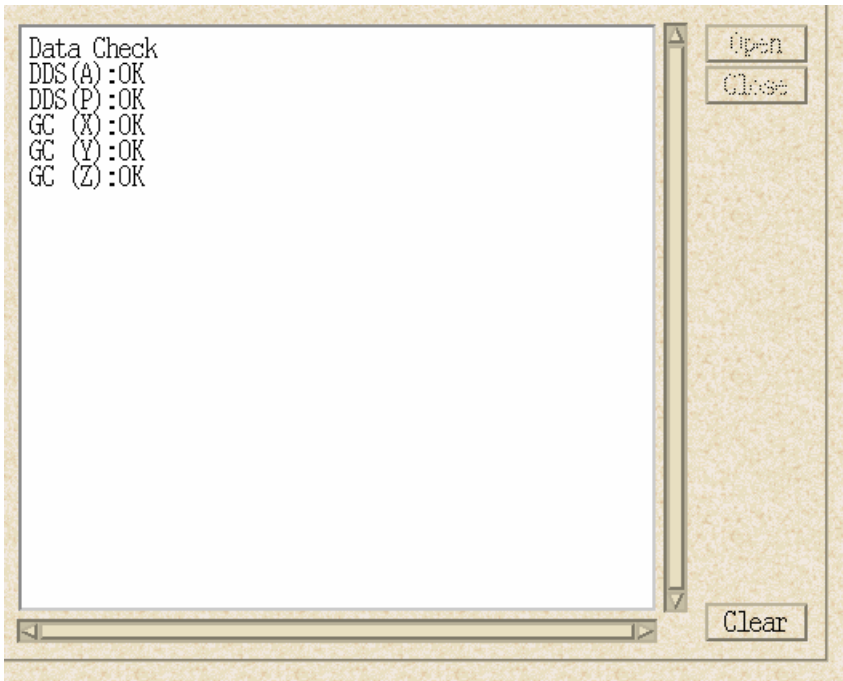


Fig. 4.2-31 An Example of Data Check Result Display

4.2.4.6 RTCONTcheck

Operate as follows.

4.2.4.6.1 Operation procedure

- (1) Select [RTCONTcheck] button from item selection buttons of startup screen
If RTCONTcheck has been already selected, proceed to (3).

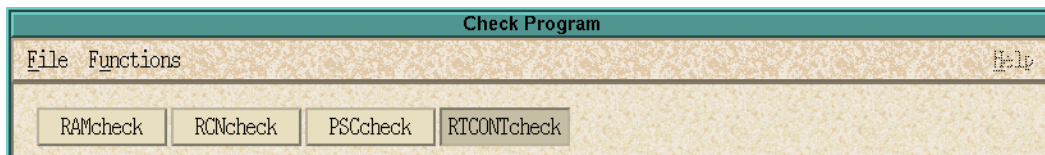


Fig. 4.2-32 Check Item Selection Buttons

- (2) RTCONTcheck screen is displayed.

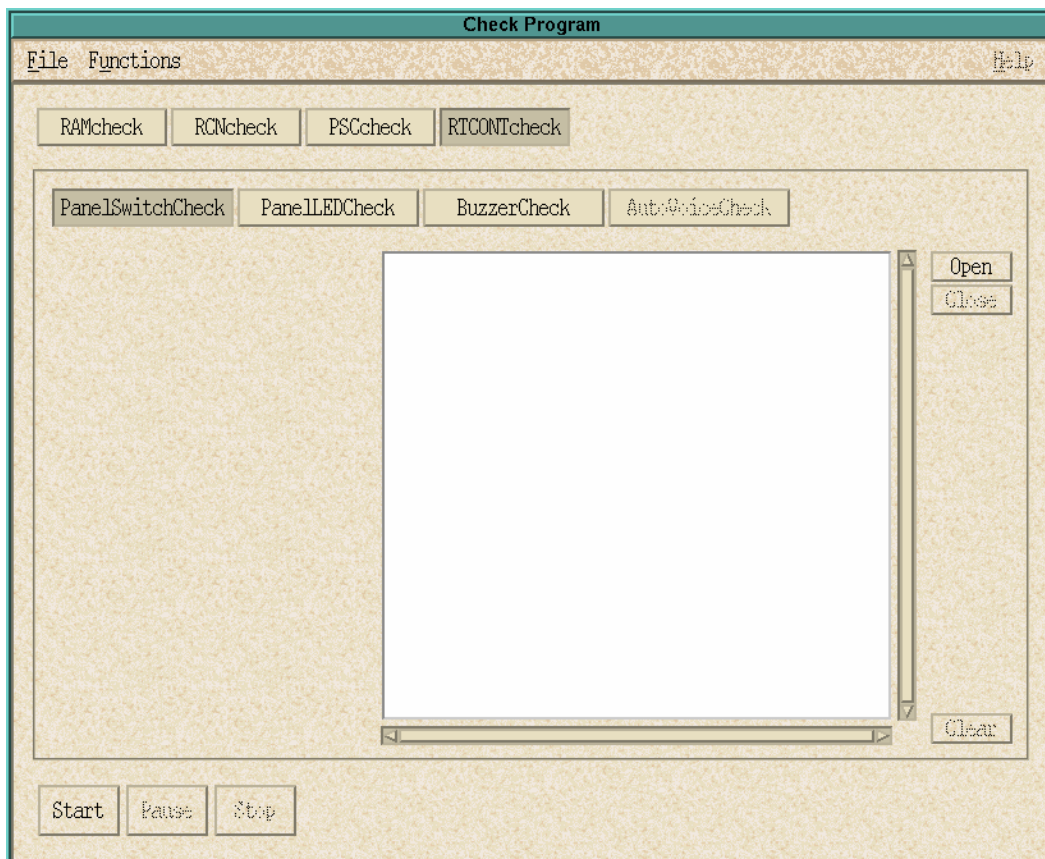


Fig. 4.2-33 RTCONTcheck Screen

- (3) RTCONTcheck includes items shown in the table below. Procedure will be described for each item.

Table 4.2-7 RTCONTcheck Items

Item No.	Item
4.2.4.6.1.1	Panel switch check
4.2.4.6.1.2	Panel LED check
4.2.4.6.1.3	Buzzer check

4.2.4.6.1.1 Panel switch check procedure

- (4) Click [PanelSwitchCheck] button.



Fig. 4.2-34 Panel Switch Check Buttons

- (5) Click [Start] button.



Fig. 4.2-35 Start Button

- (6) Pressing any panel switch displays the name and code No. of the switch thus pressed (Fig. 4.2-44).

Displayed data is saved as a log as it is. Clicking [Clear] button allows the result display area to be cleared. This clears the screen only, but no log is cleared.

For displaying the log, refer to 4.2.4.7.



Fig. 4.2-36 Clear Button

- (7) Clicking [Stop] button terminates checking the panel switch.



Fig. 4.2-37 Stop Button

4.2.4.6.1.2 Panel LED check procedure

- (4) Click [PanelLEDCheck] button.

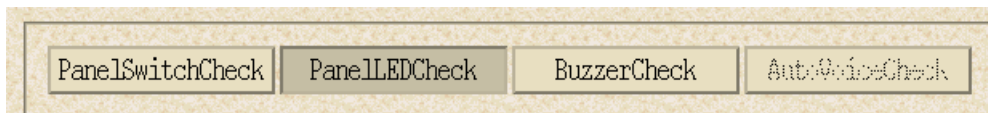


Fig. 4.2-38 Panel LED Check Buttons

- (5) Click [Start] button.

LED repeats illuminating, flashing and not illuminating. No display data is available.



Fig. 4.2-39 Start Button

- (6) Clicking [Stop] button to terminate panel LED check.



Fig. 4.2-40 Stop Button

4.2.4.6.1.3 Buzzer check procedure

- (4) Click [BuzzerCheck] button.

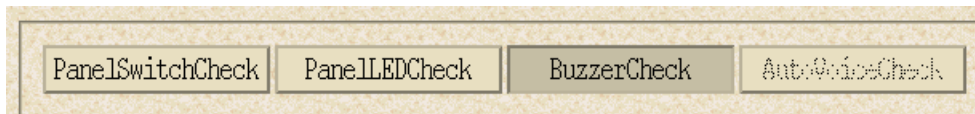


Fig. 4.2-41 Buzzer Check Button

- (5) Click [Start] button.

A buzzer sounds for one second and stops for two seconds and repeats this sequence. No display data is available.



Fig. 4.2-42 Start Button

- (6) Clicking [Stop] button to terminate buzzer check.



Fig. 4.2-43 Stop Button

4.2.4.6.2 Result display of RTCONTcheck

An example of RTCONTcheck is shown below.

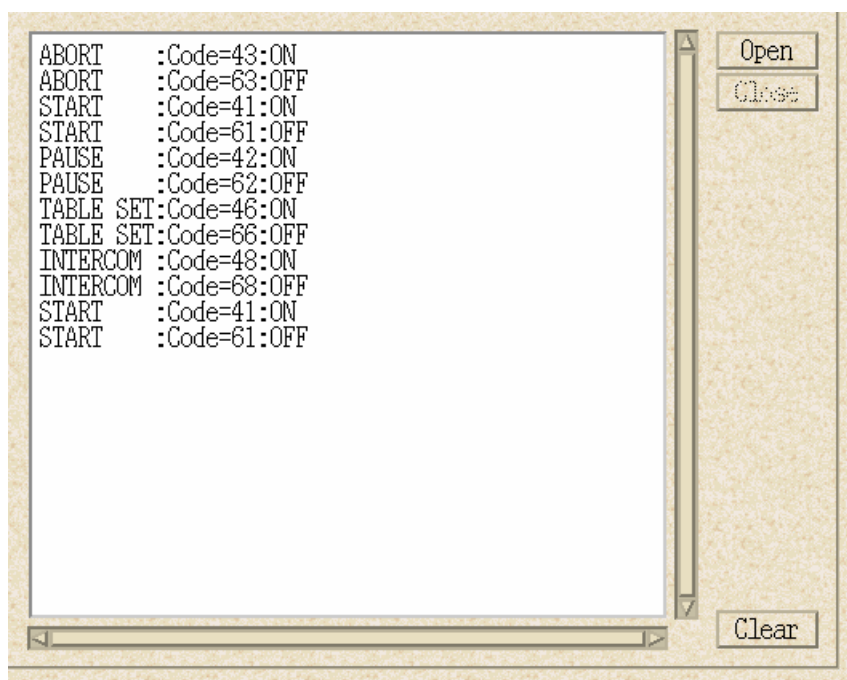


Fig. 4.2-44 An Example of Panel Switch Check Result Display

4.2.4.7 Log display

The log is that messages displayed in the result display area are saved in H/D. For every check item (RAMcheck/RCNcheck/PSCcheck/RTCONTcheck), one time data is saved. To display the log, click [Open] button on each check screen. Clicking [Close] button returns to the previous screen. However, the log display is not available while checking.



Fig. 4.2-45 Log Control Buttons

4.2.4.8 System information display

If selecting [MRRCN Information...] from "Functions Menu" of the Menu Bar, the system information screen of Fig. 4.2-47 is displayed.

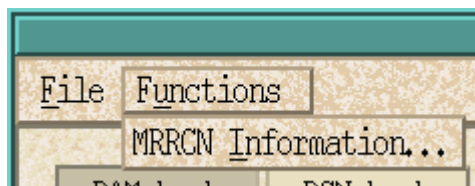


Fig. 4.2-46 System Information Display Button

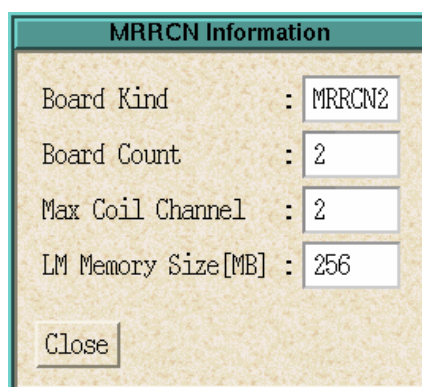


Fig. 4.2-47 System Information Screen

Table 4.2-8 Explanation of System Information

Name	Content	Display
Board Kind	Kind of RCN board	MRRCN2
Board Count	Number of RCN board	1 / 2 / 4
Max Coil Channel	Number of maximum coil channels possible to scan	1 / 2 / 4
LM Memory Size [MB]	LM memory capacity	256 / 512

4.2.5 End

If selecting [Quit] from “File Menu” of the menu bar, a dialogue box of Fig. 4.2-49 pops up. Clicking [No] here returns to the previous screen. Clicking [Yes] displays a reboot confirmation dialogue of Fig. 4.2-50. When no check has been executed, the reboot confirmation dialogue is not displayed.

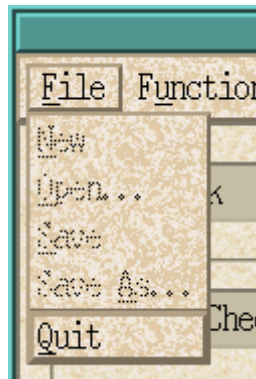


Fig. 4.2-48 Quit Button



Fig. 4.2-49 Exit Confirmation Dialogue

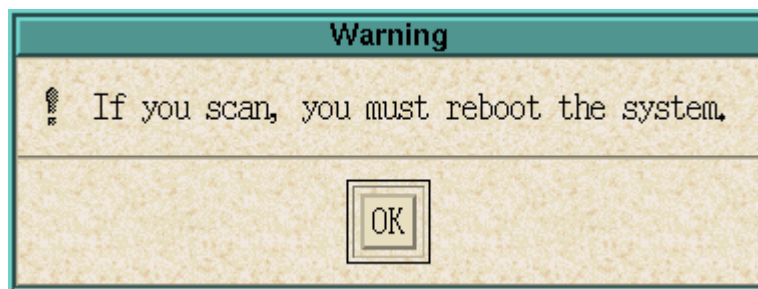


Fig. 4.2-50 Reboot Confirmation Dialogue

4.2.6 Error specifications

Error specifications are shown in Table 4.2-9.

Table 4.2-9 Error Specifications

Error message	Meaning	Action to be taken
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	End the task now running and retry in the same procedure.
Can't start task.	Failed to start application.	Check system status and retry in the same procedure.
Can't end task.	Failed to end application.	Check system status and retry in the same procedure.
The translation of parameter failed.	Failed to transfer parameter.	Check system status and retry in the same procedure.
Can't read the SYSTEM_TABLE.	Reading the SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application

4.2.7 Other cautions

If executing the check program even once (clicking [Start] button), operations such as scanning and a part of check service programs other than the check program cannot be started up. When the other operation such as measurement is to be operated, reboot the system. Starting up the check program needs no reboot.

4.3 Patient Table Tool

4.3.1 General

This tool communicates with firmware for the patient table to change status of the table, obtain various information of the table and display them.

4.3.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Patient Table Tool in the displayed items and click [OK] or double-click the selected item.

4.3.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.3-1.

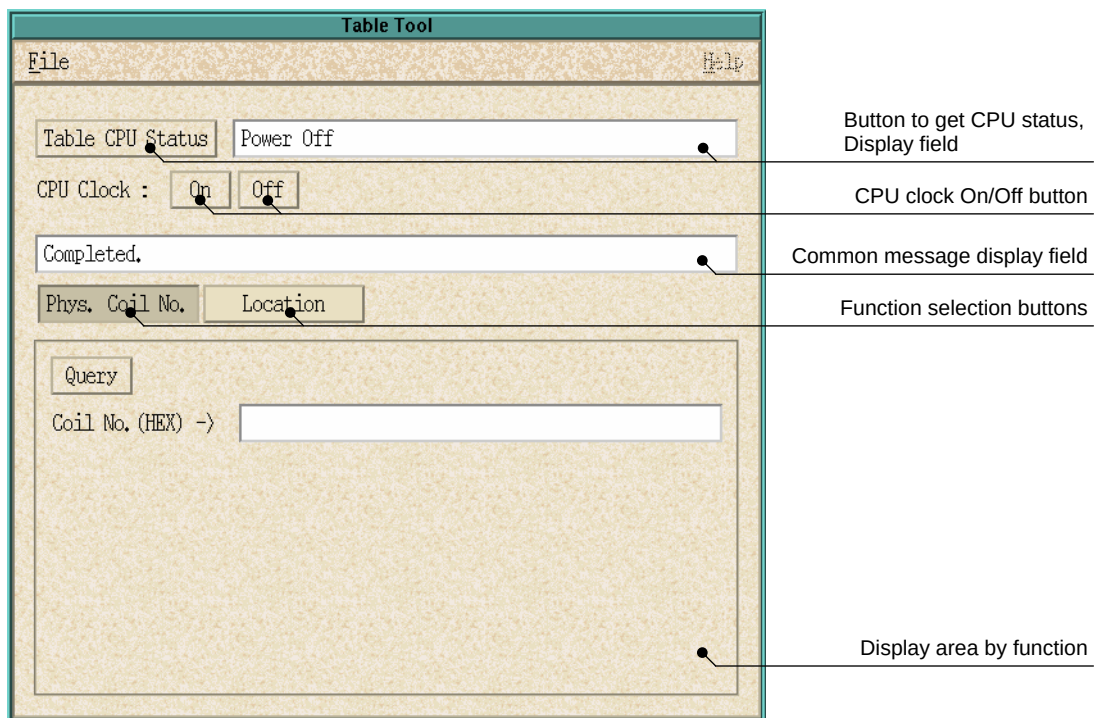


Fig. 4.3-1 Patient Table Tool Display Screen

4.3.4 Operation procedure

Follow the next steps to execute the process.

4.3.4.1 Update of parameters

Use the keyboard and mouse cursor to update each parameter. The following methods are available for updating parameters.

4.3.4.1.1 Selecting item with pushbutton

Pushbuttons displayed in line are for single selection. Use the mouse cursor to click and update any parameter.

4.3.4.1.2 Entering string into entry field

Place the mouse cursor in any entry field of parameter to be updated to select the entry field. Use the keyboard to enter a numerical value or string, and press <Enter> or remove the focus from the entry field to update the parameter.

4.3.4.2 Function selection

This tool includes functions shown in the table below. Procedure will be described for each item. Execution result of each function is displayed in the common message display field (Table 4.3-2).

Table 4.3-1 Function Table

Item No.	Item	Relevant button
4.3.4.3	Getting CPU status	Table CPU Status
4.3.4.4	Clock On/Off	On/Off
4.3.4.5	Getting Coil No.	Phys. Coil No.
4.3.4.6	Getting location information	Location

Table 4.3-2 Message Displayed in Common Message Display Field

Message	Meaning
Completed.	Normal end
Retry Time-out Error	Retry/Timeout error
Communication Error	Table CPU power off
Input Value Error	Parameter error
TB Control Error	Control error
Error	Other error

4.3.4.3 Getting CPU status

Operate as follows.

- (1) Click [Table CPU Status] button on the screen.
- (2) CPU status is displayed in the CPU Status Display field. A message displayed in the CPU Status Display field is shown below.
This function is also executed when starting up.



Fig. 4.3-2 Example Display of CPU Status

Table 4.3-3 Messages Displayed in CPU Status Display Field

Message	Meaning
Power On	CPU is running.
Power Off	CPU stops.

4.3.4.4 Clock On/Off

Operate as follows.

- (1) Click CPU Clock [On]/[Off] button on the screen (see the figure below).
- (2) Result is displayed in the CPU status display field.



Fig. 4.3-3 Clock On/Off

4.3.4.5 Getting Coil No.

Operate as follows.

- (1) Click [Phys. Coil No.] button among the function selection buttons.

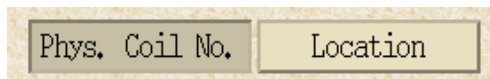


Fig. 4.3-4 Function Selection Buttons

- (2) Display area by function switches over to the coil No. display screen.
- (3) Clicking [Query] button on the screen displays coil No. in the coil No. display field.

This function uses the On/Off function of CPU Clock. Therefore, CPU status changes according to execution of this function.

Message displayed in the coil No. display field is shown below.

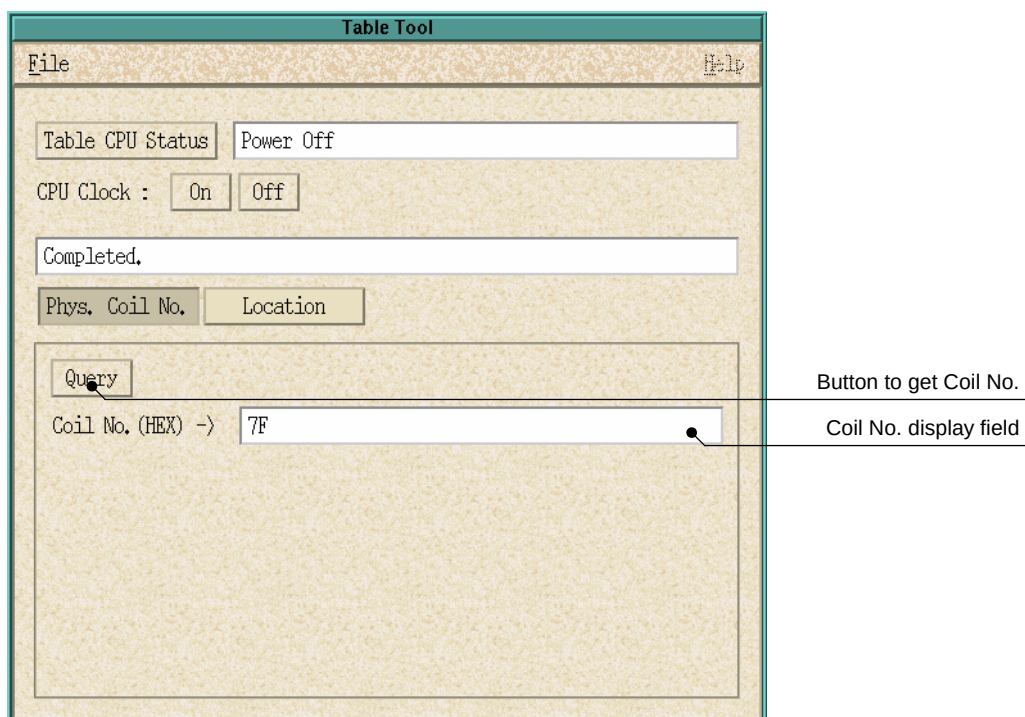


Fig. 4.3-5 Example Display of Result of Getting Coil No.

Table 4.3-4 Messages Displayed in Coil No. Display Field

Message	Meaning	Remarks
Numerical value	Coil No.	Hexadecimal
Coil Disconnection Error	Coil not connected	

4.3.4.6 Getting location information

Operate as follows.

- (1) Click [Location] button from the function selection buttons.

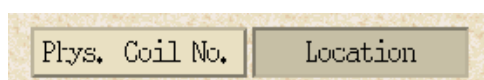


Fig. 4.3-6 Function Selection Buttons

- (2) Display area by function switches over to the screen for location information display.
- (3) Clicking [Query] button on the screen displays location information in the location information display area.

This function uses the On/Off function of CPU Clock. Therefore, CPU status changes according to execution of this function.

Messages displayed in the location information display field are shown in Table 4.3-5.

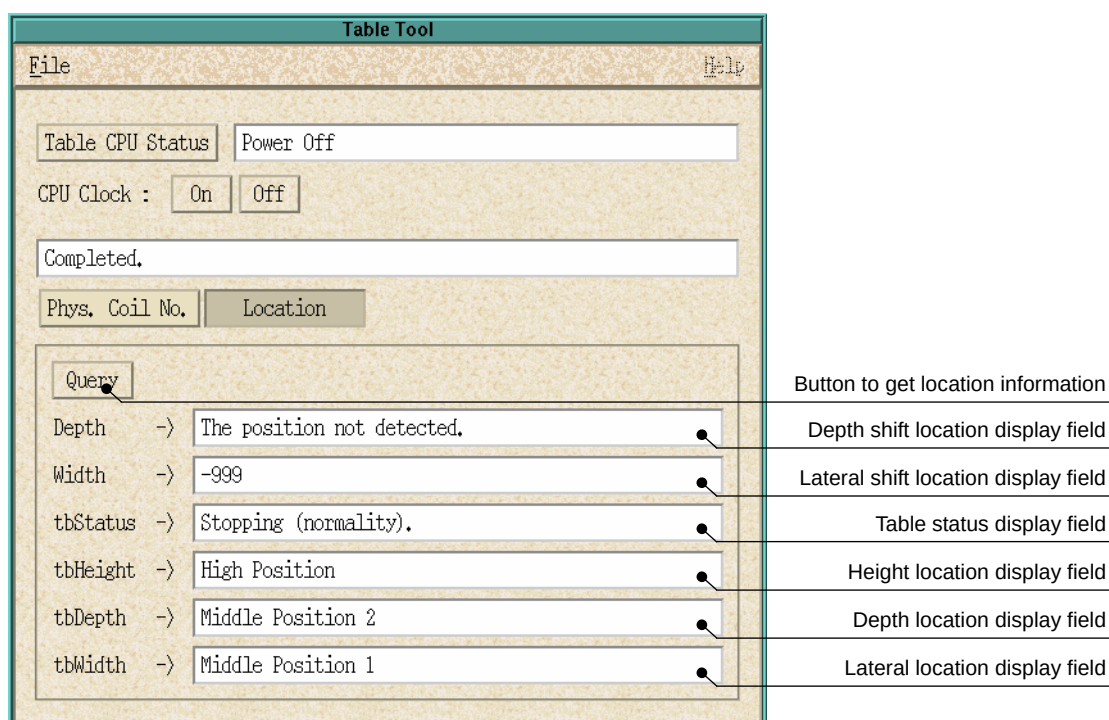


Fig. 4.3-7 Example Display of Result of Getting Location Information

Table 4.3-5 Message Displayed in Location Information Display Area

Item	Message	Meaning	Remarks
Depth	Numerical value (absolute position display)	Depth position	
	The position not detected.	No absolute position detectable.	
Width	Numerical value (absolute position display)	Width position	
	The position no detected.	No absolute position detectable.	
tbStatus	Stopping (normality).	Stopping normally.	
	Moving for the manual operation.	Moving by the manual operation.	
	Set sequence inside.	Sequence in setting	
	Stopping (Emergency).	Stopping in emergency	
tbHeight	Low Position	Lowest position	
	Middle Position	Middle position	
	High Position	Highest position	
tbDepth	Backward position	Most backward position	
	Middle Position1	Middle position 1	
	Middle Position2	Middle position 2	
	Forward position	Most forward position	
tbWidth	Pulse Position	Most plus position	
	Middle Position1	Middle position 1	
	Central position	Center position	
	Middle Position2	Middle position 2	
	Minus Position	Most minus position	

4.3.5 End

Selecting [Quit] from “File Menu” of the menu bar pops up a dialogue box shown in Fig. 4.3-9. Clicking [Yes] here terminates application, while [No] returns to the previous screen.



Fig. 4.3-8 Quit Button

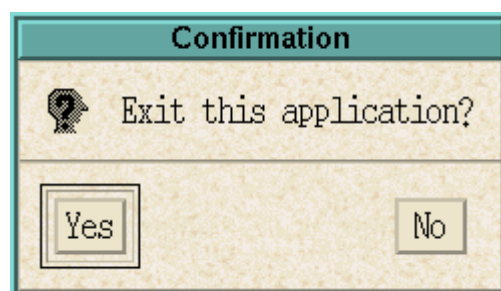


Fig. 4.3-9 Exit Confirmation Dialogue

4.3.6 Error specifications

Error specifications are shown in Table 4.3-6.

Table 4.3-6 Error Specifications

Error message	Meaning	Action to be taken
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	End the task now running and retry in the same procedure.
Can't start task.	Failed to start application.	Check system status and retry in the same procedure.
Can't end task.	Failed to end application.	Check system status and retry in the same procedure.
The translation of parameter failed.	Failed to transfer parameter.	Check system status and retry in the same procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button : Read again [Exit] button : Closing application

4.4 WS Board Check

4.4.1 General

The following boards in the workstation are checked up.

- CPU
- Memory
- Network card (In case the system is possible to DICOM communication.)
- Graphic card
- SCSI device
- USB device (Sun-Blade-2500 only)

4.4.2 Start-up

- (1) In the case of Ultra SPARC-III-Engine; Confirm that MOD drive switch is set at left side (Removable mode). If it is set at right side, refer to Paragraph 2.2.5 to changeover the switch to Removable mode.
- (2) Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then T/M Manager is started (refer to Paragraph 1.3.6).
Either click **WS Board Check** in the displayed items and click [OK] or double-click the selection item. The image shown in Fig. 4.4-1 is displayed.

4.4.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.4-1.

4.4.3.1 In the case of UltraSPARC-III-Engine:

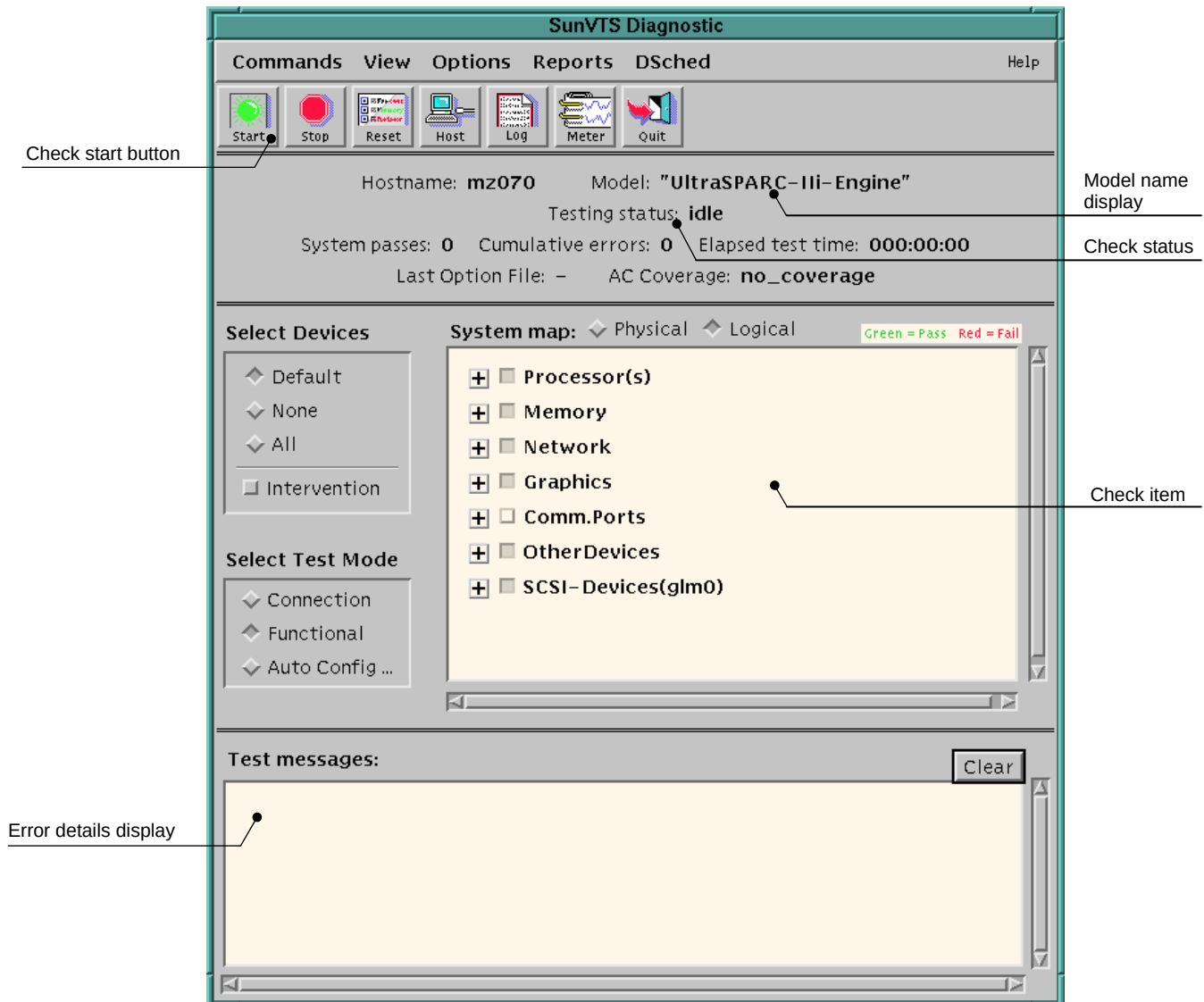


Fig. 4.4-1 WS Board Check Program Display Screen
(in the case of UltraSPARC-III-Engine)

4.4.3.2 In the case of Sun-Blade-2500:

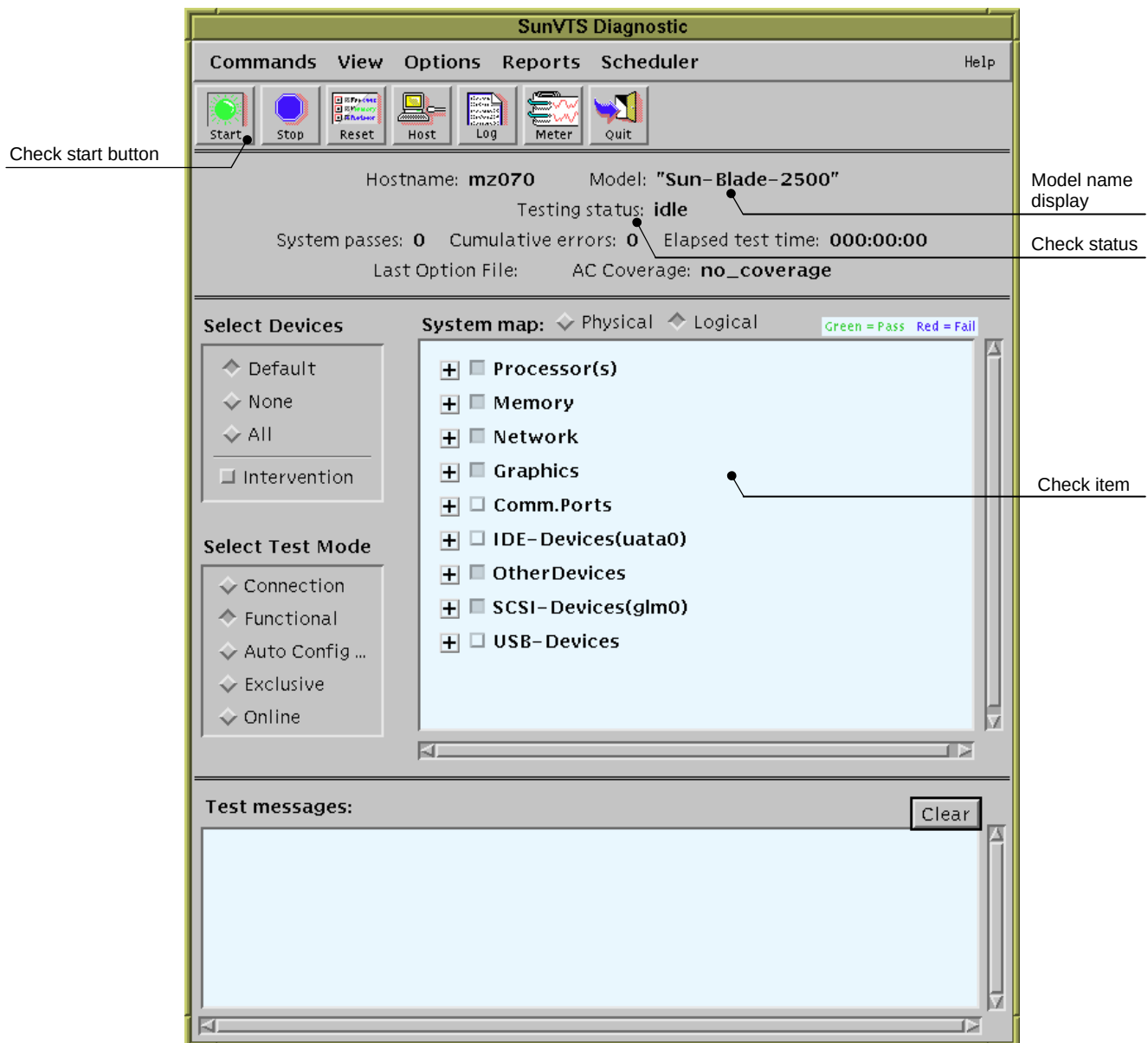


Fig. 4.4-2 WS Board Check Program Display Screen
(in the case of Sun-Blade-2500)

4.4.4 Operation procedure

Follow the next steps to execute the process.

4.4.4.1 Setting check items

- (1) When DICOM communication is possible with DICOM I/F provided, Network check can be made. Click [+] button of Network, and the following is displayed.

- (a) In the case of UltraSPARC-III-Engine, check hme1 only as shown in Fig. 4.4-3.

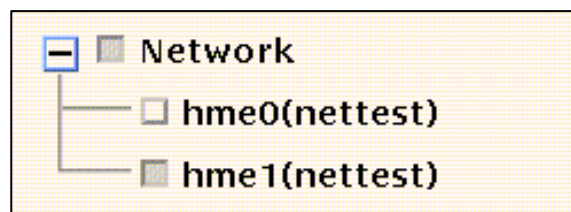


Fig. 4.4-3 Setting Network (In the case of UltraSPARC-III-Engine)

Note: When it is impossible to carry out DICOM communication, never check Network. Also, Do not check hme0 as an error occurs certainly.

- (b) In the case of Sun-Blade-2500, check hme1 only as shown in Fig. 4.4-4.

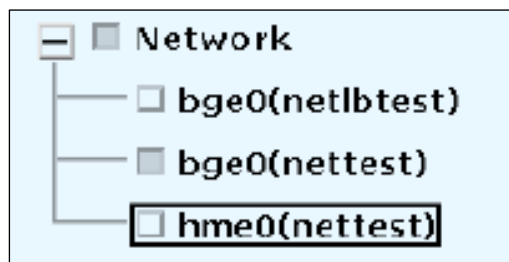


Fig. 4.4-4 Setting Network (In the case of Sun-Blade-2500)

Note: When it is impossible to carry out DICOM communication, never check Network. Also, Do not check hme0, bge0(betlbtest) as an error occurs certainly.

- (2) By selecting Threshold... from Options menu, the dialog shown in Fig. 4.4-5 is displayed. Here, click ▲ button of Max System Passes: to change 0 to 1 and click [Apply] button.

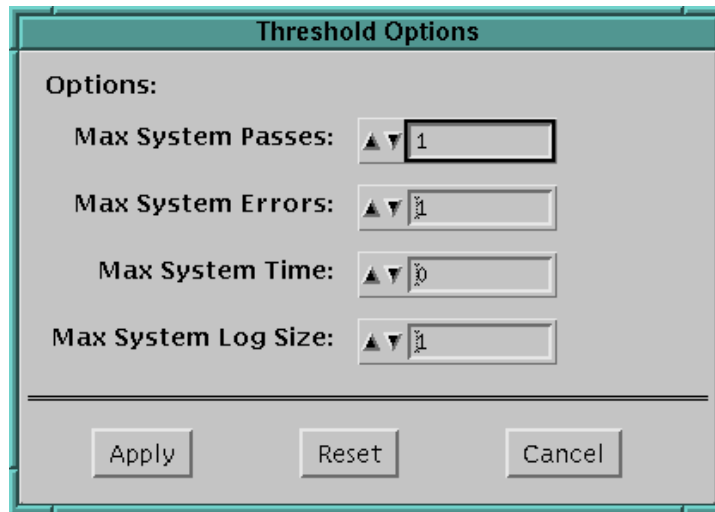


Fig. 4.4-5 WS Board Check Program Display Screen

4.4.4.2 Starting Test

Click [Start] Button to start WS Board Check program. The mark "*" is put at the upper right of the item being tested, while the item for which test is completed is displayed in green. If any abnormality is detected, the item is displayed in red and an error message is displayed in Test Message box. By clicking [+] button on System Map, details of each board can be displayed.

The Test is completed in about 6 minutes. When all test have been completed, Testing Status becomes idle.

Note: Do not touch the keyboard and mouse during the Test.

4.4.5 End

Select Quit SunVTS from [Commands Menu] and select UI and Kernal. WS Board Check Program ends.

4.4.6 Error specifications

Error specifications are shown in Table 4.4-1.

Table 4.4-1 Error Specifications

Error occurring item	Part supposed to be defective		Items to be checked
Processor(s)	Workstation board	_____	_____
Memory		Memory card	Check if memory card is correctly inserted.
Network		DICOM I/F board	Check if DICOM I/F board is correctly inserted.
Graphics		Graphic board	Check if graphic card is correctly inserted.
Other Device		_____	_____
SCSI-Devices		Magnetic disk	Check if SCSI cable is correctly connected.
IDE-Devices (Sun-Blade-2500 only)		DVD-RAM drive	Check if IDE cable is correctly connected.
USB-Devices (Sun-Blade-2500 only)		Keyboard, Mouse	Check if USB cable is correctly connected.

4.5 NVRAM Check

4.5.1 General

NVRAM used for the WS board holds board setting information by the built-in battery. Therefore, NVRAM needs to be replaced periodically in accordance with battery life. This system displays a dialogue of Fig. 4.5-1 when starting up the system if time before one month of NVRAM replacement is reached.



Fig. 4.5-1 Dialogue Warning NVRAM Replacement

This tool confirms time for NVRAM replacement and changes setting.

4.5.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click NVRAM Check in the displayed items and click [OK] or double-click the selected item.

4.5.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.5-2.

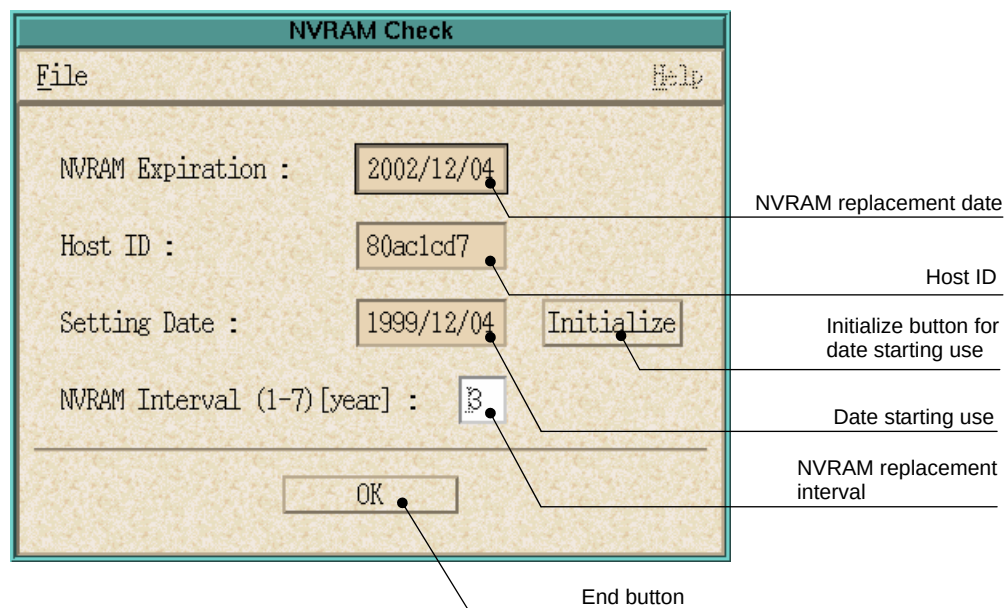


Fig. 4.5-2 NVRAM Check Display Screen

4.5.4 Operation procedure

Follow the next steps to execute the process.

4.5.4.1 Changing NVRAM replacement interval

Place the mouse cursor in the entry field of NVRAM Interval to select the entry field. Use the keyboard to enter a numerical value, and press <Enter> of the keyboard to update the parameter. Setting range for replacement interval (year) is 1~7.

4.5.4.2 Initialize of date to start using NVRAM

Click [Initialize] button to set the date of the day. Also, Host ID is renewed.

4.5.4.3 Saving setting change

To save setting changes, select [Save] of “File Menu” of the menu bar. Update confirmation screen of Fig. 4.5-4 is displayed. Clicking [Yes] reflects the set value. Clicking [No] reflects no set value.

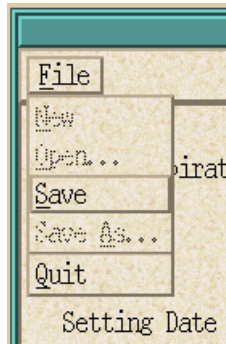


Fig. 4.5-3 Save Button

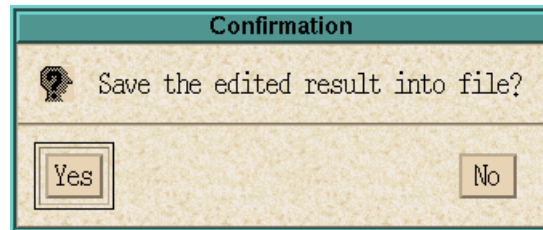


Fig. 4.5-4 Update Confirmation Screen

4.5.5 End

If clicking [OK] or selecting [Quit] from “File Menu” of the menu bar, a dialogue box of Fig. 4.5-6 pops up if no setting value has been changed. Click [Yes] to end application, or click [NO] to return to the previous screen.

If any setting has been changed but not yet saved, the update confirmation screen of Fig. 4.5-4 is displayed. After responding to it, a dialogue box of Fig. 4.5-6 is displayed. Operate as described above.

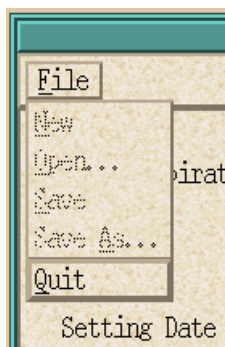


Fig. 4.5-5 Quit Button

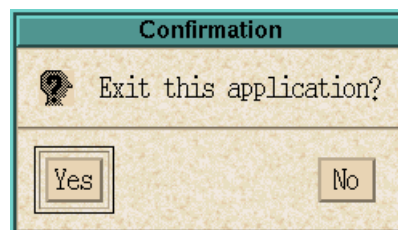


Fig. 4.5-6 End Confirmation Dialogue

4.5.6 Error specifications

Error specifications are shown in Table 4.5-1.

Table 4.5-1 Error Specifications

Error message	Meaning	Action to be taken
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
NVRAM Interval is 1 to 7.	NVRAM Interval is 1 to 7.	Reenter any parameter in a range of 1 to 7.
Host ID is different.	Host ID differs.	Displayed when NVRAM has been replaced. Replace the NVRAM of correct Host ID, or initialize by [Initialize] button.
Setting file is not found. Please do Service Restore now.	Setting file is not found. Execute Service Restore now.	Execute Service Restore.

4.6 B0 Current Check

4.6.1 General

This is a tool to check the B0 current by operating the B0 controller (DGBC) and outputting B0 waveform.

4.6.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click B0 Current Check in the displayed items and click [OK] or double-click the selected item.

4.6.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.6-1.

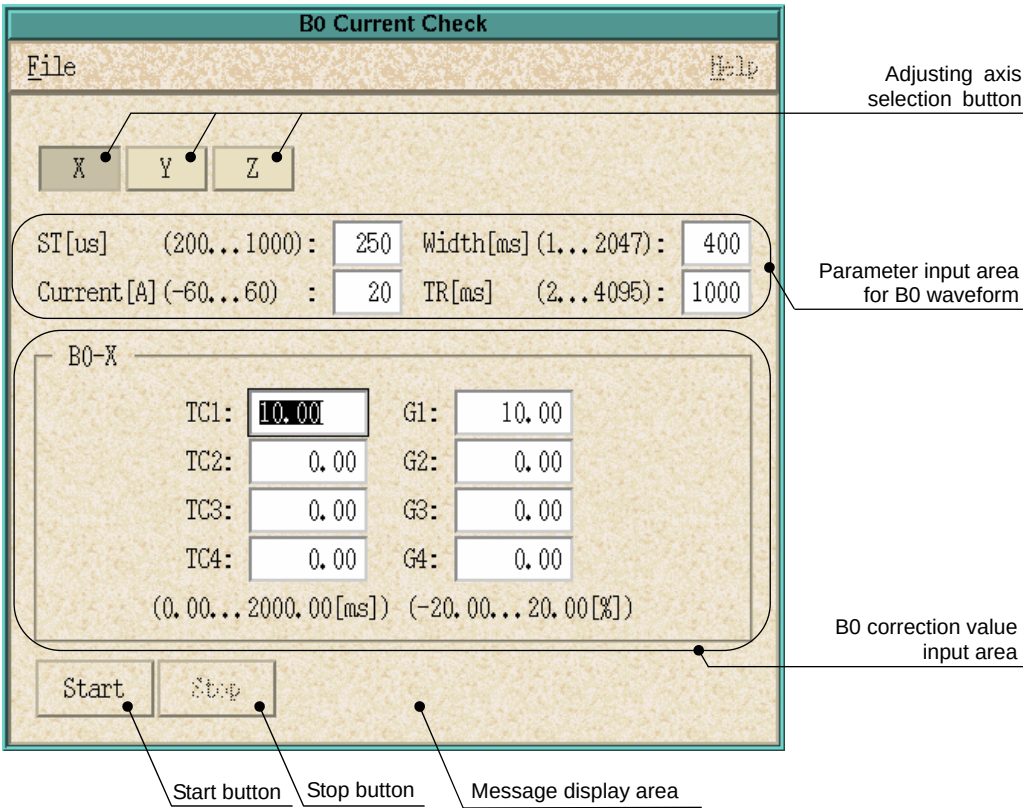


Fig. 4.6-1 B0 Current Check Display Screen

4.6.4 Operation procedure

Follow the next steps to execute the process.

4.6.4.1 Updating parameters

Using the keyboard and mouse cursor, change parameters. The following methods are available to change parameters.

4.6.4.1.1 Selection of parameter by pushbutton

The push-buttons arranged in array are of single-selection type, and by clicking one of them by the mouse cursor, the corresponding parameter is changed.

4.6.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press **<Enter>** on the keyboard to change the parameter.

NOTE: It takes some time until the window will be operable after pressing **<Enter>**. This is because the input parameter is set in the B0 controller (DGBC) from time to time.

Table 4.6-1 Parameter Setting Range

Parameter Name			Setting Range	Remarks
Adjustment axis			X, Y, Z	
B0 Pulse	ST[μs]	Slew Time	200 ~ 1000	
	Current[A]	Current Value	-60 ~ 60	
	Width	Pulse Width	1 ~ 2047※	
	TR	Repetition Time	2 ~ 4095※	
B0 Correction value	TC[ms]	Time Constant	0.00 ~ 2000.00	
	G	Gain	-20.00 ~ 20.00	× 0.03[%]

※ Note that because Duty ratio of 80% is set as the limit for input the setting is no possible unless Width (pulse width) and TR (repetition time) satisfy the following relation. If any erroneous input is made, the input area of Width and TR turns to yellow, and under this condition B0 waveform cannot be output.

$$(\text{Duty ratio } 80\%) \geq (\text{Width} \div \text{TR}) \times 100$$

4.6.4.2 Output of B0 waveform

Click on [Start] button. The B0 waveform of the currently displayed axis is output and the flashing axis name of the B0 being output is displayed in the message display area.

The B0 waveform parameter is the value common to all the axes. For input of B0 waveform parameter, refer to Paragraph 4.6.4.1.

While B0 waveform being output, any parameter for B0 waveform and B0 correction value other than B0 waveform output axis cannot be changed. To change any of these parameters, stop output of B0 waveform.

4.6.4.3 Stopping B0 waveform output

Click on [Stop] button. Output of B0 waveform is stopped.

4.6.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 4.6-3 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

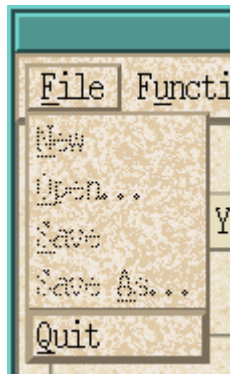


Fig. 4.6-2 Quit Button



Fig. 4.6-3 Exit Confirmation Dialog

4.6.6 Error specifications

Error specifications are shown in Table 4.6-2.

Table 4.6-2 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.

4.6.7 Other cautions

With the TC (time constant) set at 0, if the set value of G (Gain) is any value other than 0, there may be cases where a malfunction occurs on the hardware. Therefore, if TC (time constant) is set at 0, set 0 for G (Gain).

4.7 Shim Current Check

4.7.1 General

This is a tool to check the shim current by operating the Shim power supply and outputting waveform looks like stairs.

4.7.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Shim Current Check in the displayed items and click [OK] or double-click the selected item.

4.7.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 4.7-1.

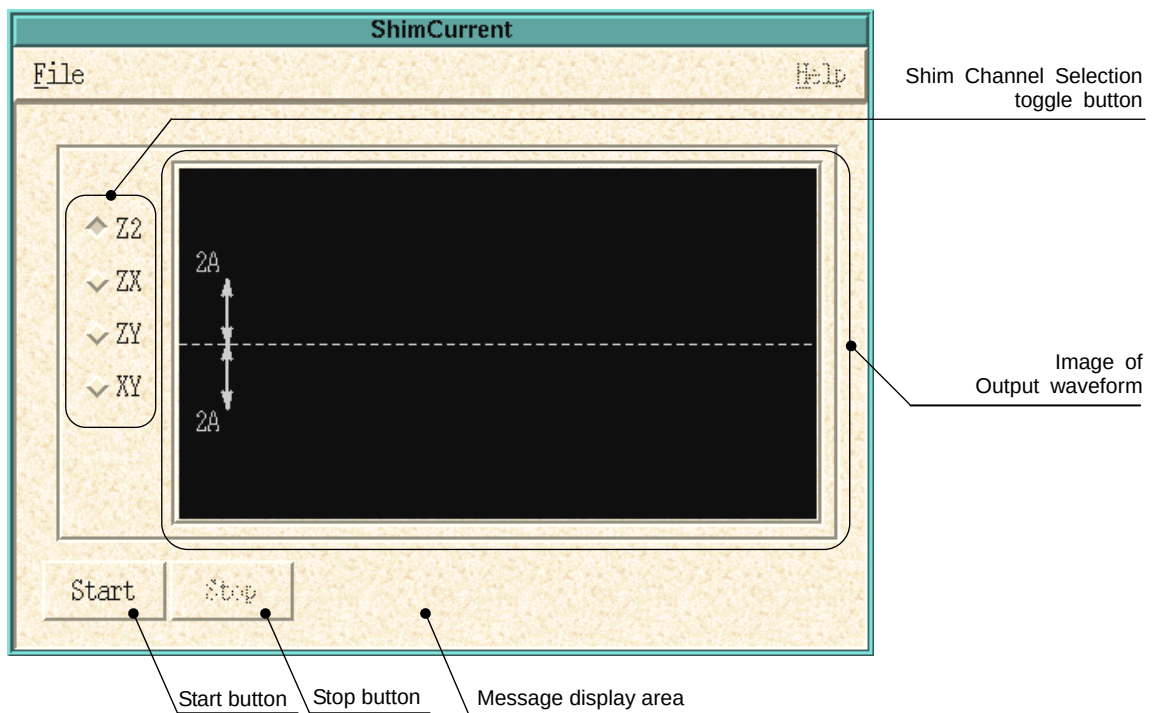


Fig. 4.7-1 Shim Current Check Display Screen

4.7.4 Operation procedure

Follow the next steps to execute the process.

4.7.4.1 Selection of shim channel

Using the shim channel selection toggle button, select [Z2] or [ZX] or [ZY] or [XY]

4.7.4.2 Output of waveform

Click on [Start] button. The waveform of the currently displayed channel is output and the flashing message is displayed in the message display area.

While waveform being output, the waveform is displayed in Fig4.7.2 and the shim channel cannot be changed. To change the shim channel, stop output of waveform.

Note : The updating interval of each electric current value in stair shape waveform may not be constant. This is for setting the current value to shim power supply upon occasion.

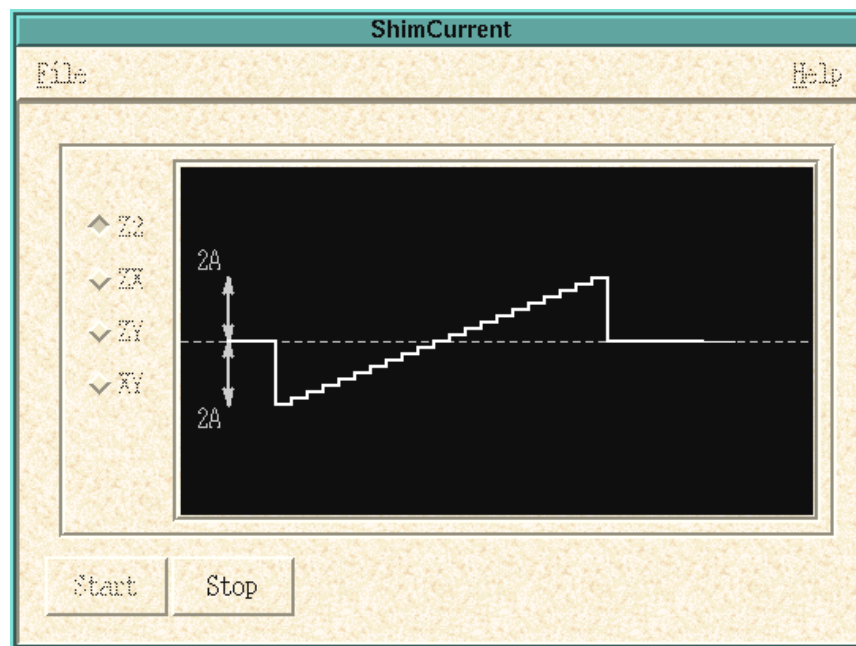


Fig. 4.7-2 Shim Current Check Display Screen(While waveform being output)

4.7.4.3 Stopping GC waveform output

Click on [Stop] button. Output of waveform is stopped.

4.7.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 4.7-4 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

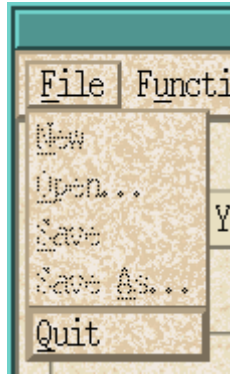


Fig. 4.7-3 Quit Button



Fig. 4.7-4 Exit Confirmation Dialog

4.7.6 Error specifications

Error specifications are shown in Table 4.7-2.

Table 4.7-2 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.

Setting ROIs at the center and four corners of the image, a mean value and standard deviation of each ROI are calculated and a S/N (SNR) is obtained by using the following formula.

$$\text{S/N (SNR)} = (\text{Mean value of ROI at center}) \div (\text{A mean value of standard deviation values of ROI at four corners})$$

As displaying on the most front of the screen either the scan card used to scan the image for S/N check or the data card in which the image is read for S/N check, click [Maint] button at the lower right of the screen. On the displayed Maintenance Card, click [SNR] tag on the Control face.

Display screen is shown below.

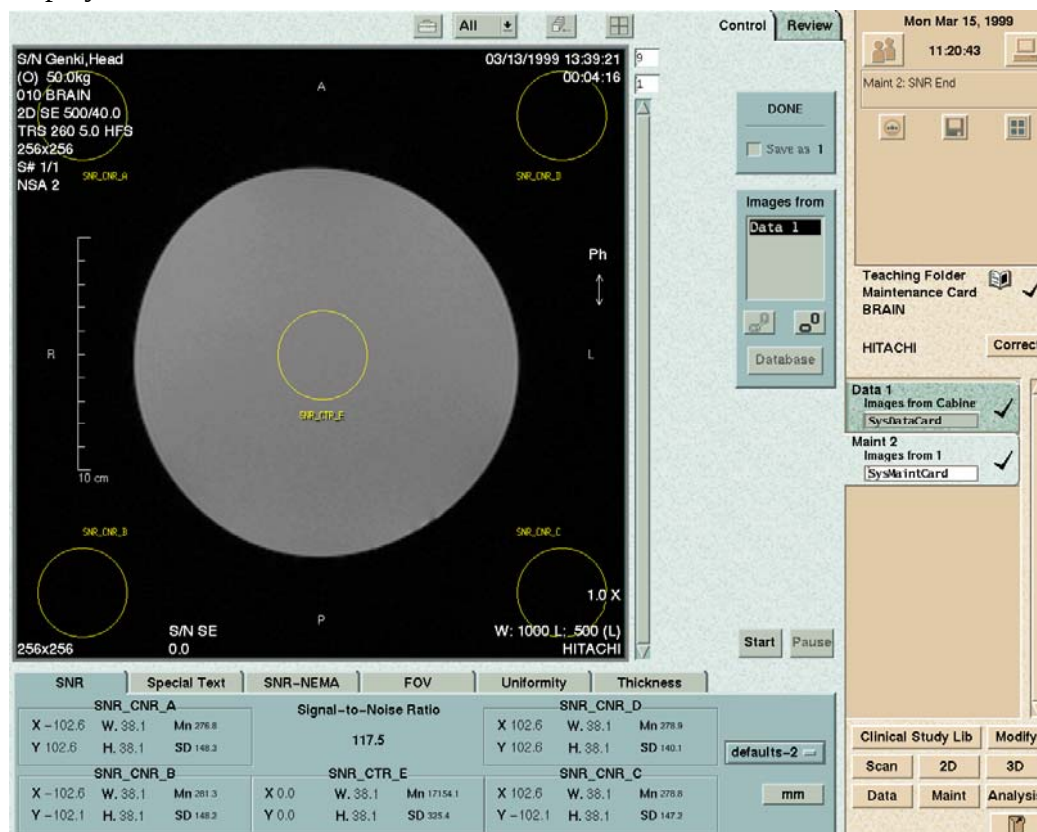


Fig. 5.1-1 S/N Check Display Screen

5.1.4 Operation procedure

Follow the next steps to execute the process.

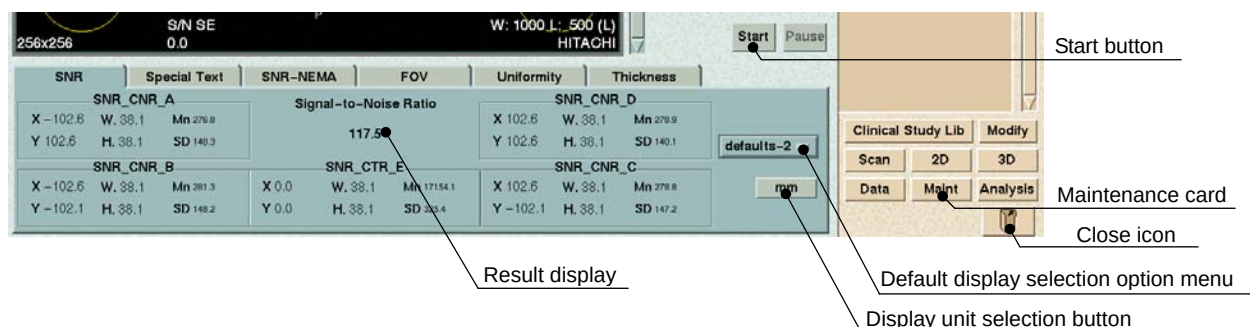


Fig. 5.1-2 S/N Check Function Buttons

5.1.4.1 ROI display selection

Two defaults are available for display of ROI. Use the [defaults-2]. Click the Default Selection Option Menu and select either [defaults-1] or [defaults-2] on the Menu.

5.1.4.2 Movement of ROI display position

Five ROIs are displayed on the image. Check the ROIs for their correct positions. If display position of any of ROIs at the four corners is not appropriate, click the ROI to make it movable and drag the center marker (move the marker with the left mouse button pressed) to move the ROI to the appropriate position.

5.1.4.3 Selection of display unit

Two units are available as display unit. Click the Display Unit Selection button and select the display unit of either [mm] or [Pixels].

5.1.4.4 Starting process

Click [Start] button on the control screen. Upon completion of the process, the result is displayed in Signal-to-Noise Ratio.

5.1.5 End

Click the Close icon on the lower right of the screen to close the folder.

5.2 Special Text

5.2.1 General

The detailed data of measurement and image reconstruction are displayed.

5.2.2 Start-up

Click [Maint] button on the lower right of the screen with displaying state of the scan card measured the image aiming at Special Text or the data card written the image aiming at Special Text on the most front screen.

Upon displaying the Maintenance Card, click [Special Text] tag on the control face.

5.2.3 Display screen

Display screen is shown below.

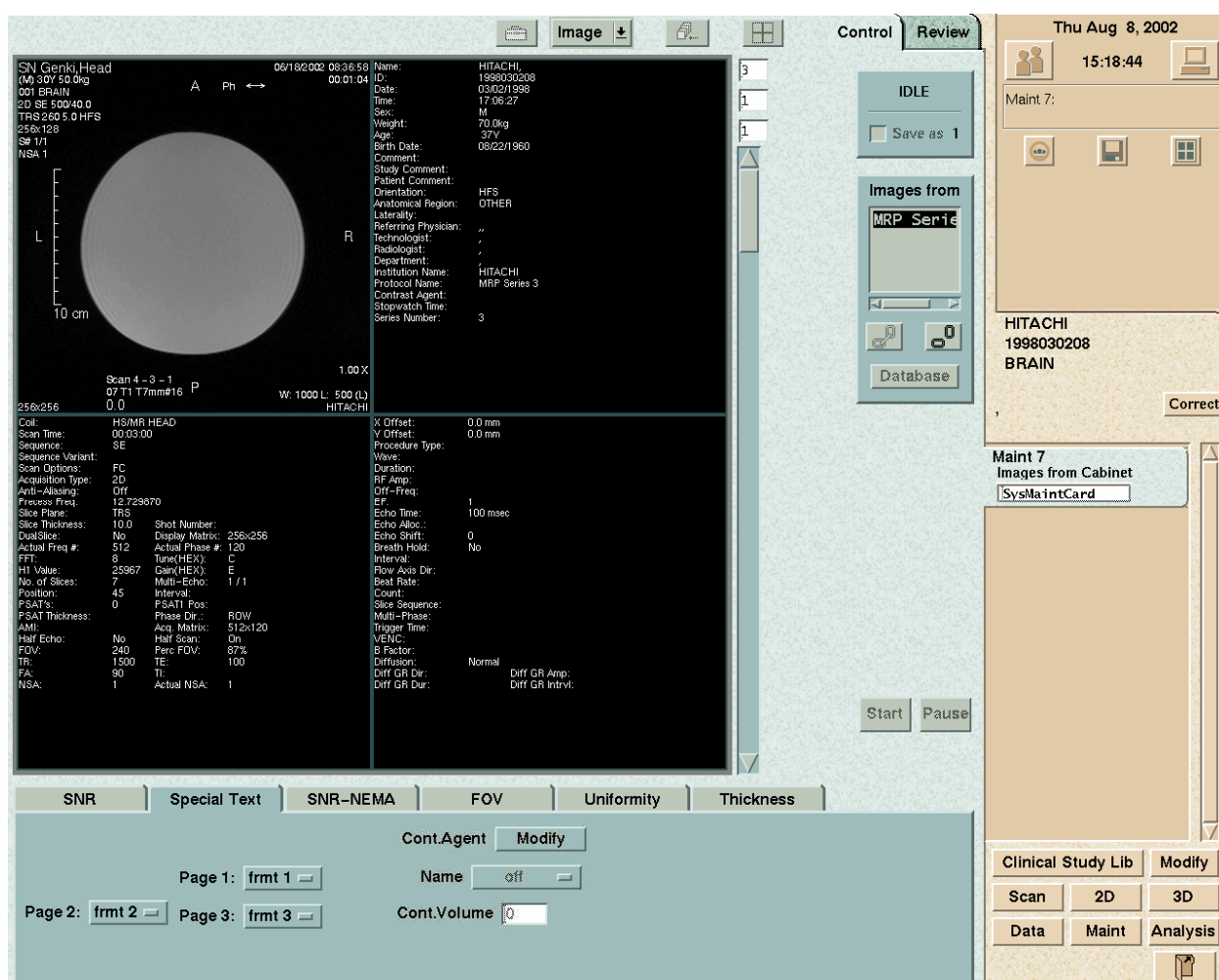


Fig. 5.2-1 Special Text Display Screen

5.2.4 Operation procedure

Follow the next steps to execute the process.

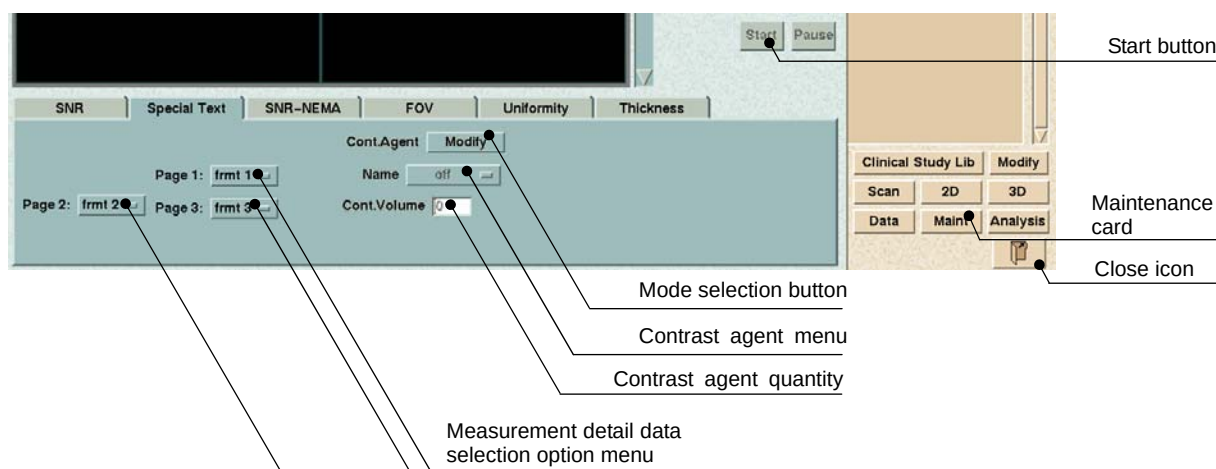


Fig. 5.2-2 Name of Special Text Function

5.2.4.1 Measurement detail data selection

Select measurement data displayed in the frame on the monitor.

Page 1 can select the measurement data from the upper right frame, page 2 can select it from the lower left frame and page 3 can select it from the lower right frame.

Measurement detail data selection option menu can be indicated from "frmt1" to "frmt5".

Refer to "INSTRUCTION MANUAL(Functions) Section4 4-6-2 Special Text Detailed Scan Data" for the details of the information to display.

5.2.4.2 Contrast agent name selection

If it is made contrast agent name selection mode with mode selection button, operation of contrast agent menu will be attained. It is stored in image data, if contrast agent name is selected and start button is pushed.

5.2.4.3 Contrast agent quantity change

Set the contrast agent name change mode by the mode selection button, and the contrast agent quantity can be input. Input a contrast agent quantity and push start button. The contrast agent quantity is saved into image data.

5.2.5 End

Click the Close icon on the lower right of the screen to close the folder.

5.3 S/N-NEMA Check

5.3.1 General

From the subtraction result between the two images acquired with the same scan conditions, a S/N is calculated and displayed.

5.3.2 Start-up

Register the T/M license with Key MOD/DVD (refer to Section 3).

Read the images of two series acquired with the same conditions into the data card, and using the Pick Image function select two images of which S/N-NEMA is to be checked.

With the data card displayed on the most front screen, click [Maint] button on the lower right of the screen. Upon displaying the Maintenance Card, click [SNR-NEMA] tag on the control face.

5.3.3 Display screen

Display screen is shown below.

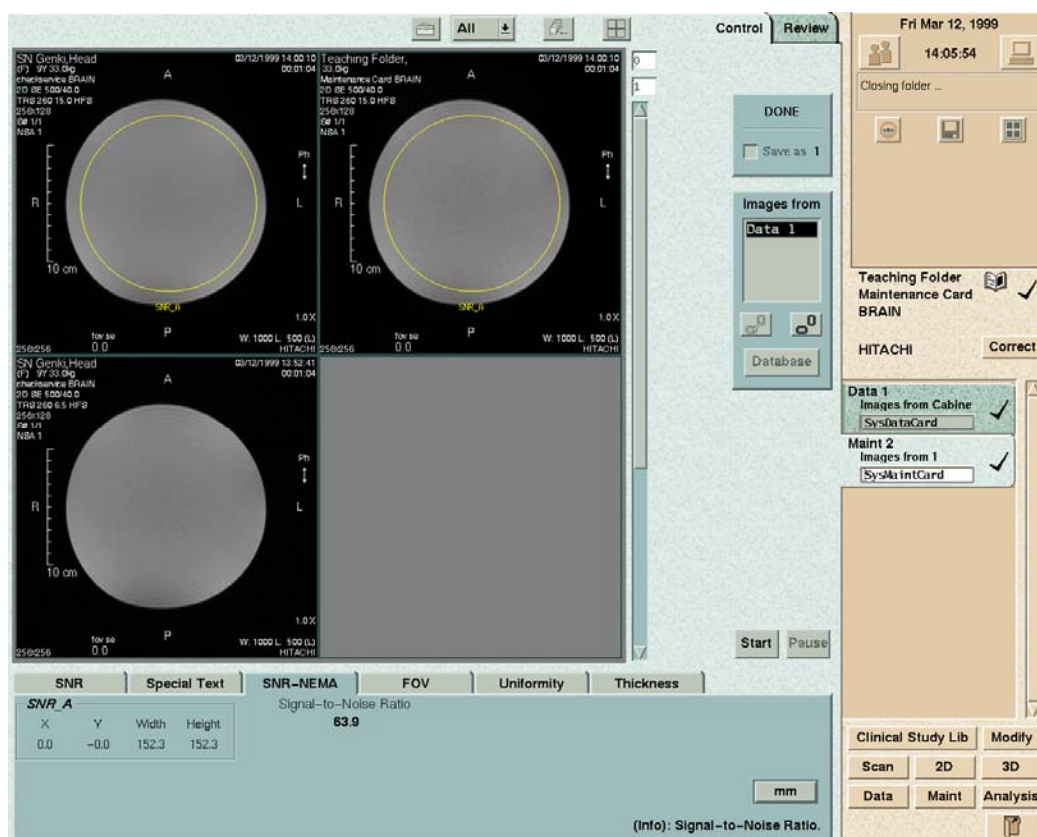


Fig. 5.3-1 S/N-NEMA Check Display Screen

5.3.4 Operation procedure

Follow the next steps to execute the process.

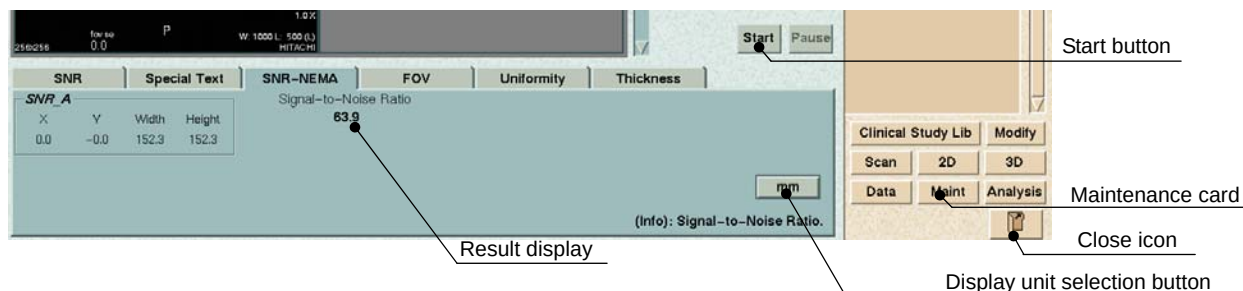


Fig. 5.3-2 S/N-NEMA Check Function Buttons

5.3.4.1 Displayed image

The images displayed on the left of the screen are the two selected images and the one displayed on the right is the image obtained by subtraction of the two images.

5.3.4.2 Movement of ROI display position

An ROI is displayed on the image. Check the ROI for its appropriate position. If display position of the ROI is not appropriate, click the ROI to make it movable and drag the center marker (move the marker with the left mouse button pressed) to move the ROI to the appropriate position.

5.3.4.3 Selection of display unit

Two units are available as display unit. Click the Display Unit Selection button and select the display unit of either [mm] or [Pixels].

5.3.4.4 Starting process

Click [Start] button on the control screen. Upon completion of the process, the result is displayed in Signal-to-Noise Ratio.

5.3.5 End

Click the Close icon on the lower right of the screen to close the folder.

5.4 FOV Check

5.4.1 General

The weight distances of the patterns of the pins located at the top-bottom and right-left on the FOV phantom image (the distance between pins of the FOV phantom is 100 mm) and the angle formed by the lines between the top-bottom and right-left pins are calculated and displayed.

5.4.2 Start-up

Register the T/M license with Key MOD/DVD (refer to Section 3).

As displaying the scan card scanned the image for FOV check or data card read the image for FOV check on the most front screen, click [Maint] button on the lower right of the screen. Upon displaying the Maintenance Card, click [FOV] tag on the control face.

5.4.3 Display screen

Display screen is shown below.

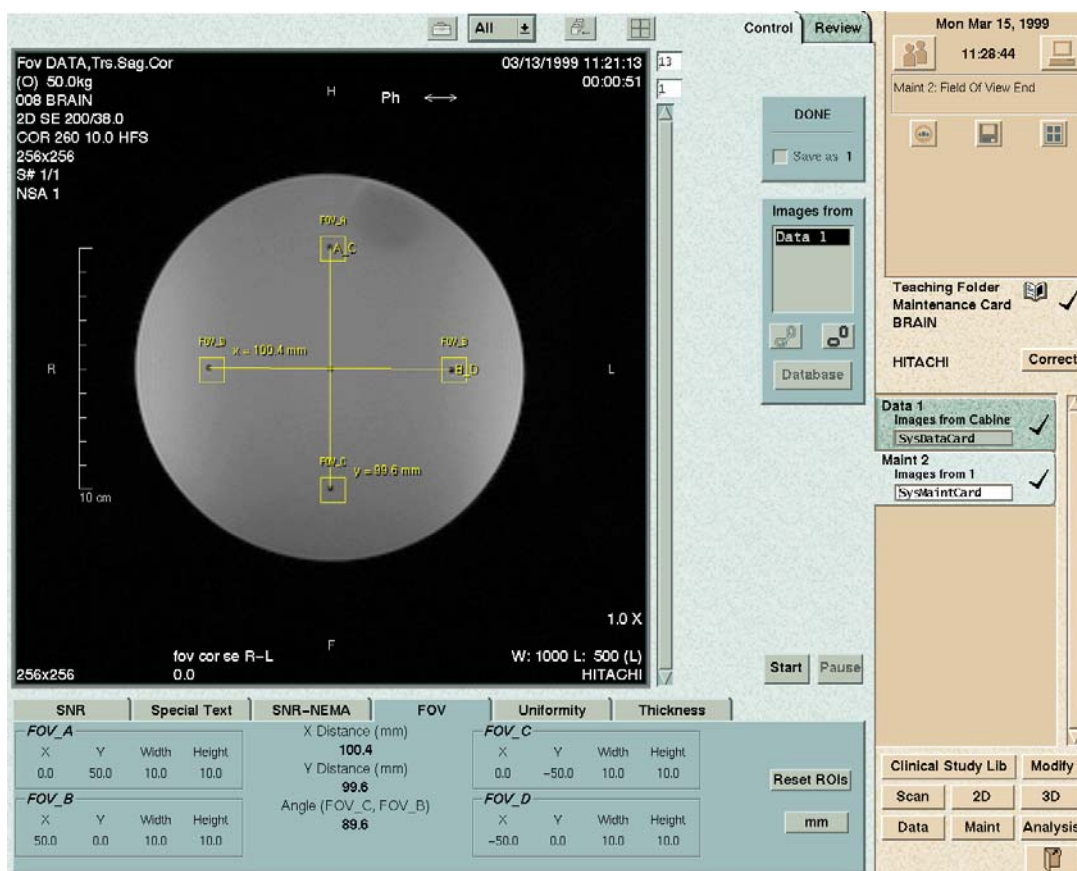


Fig. 5.4-1 FOV Check Display Screen

5.4.4 Operation procedure

Follow the next steps to execute the process.

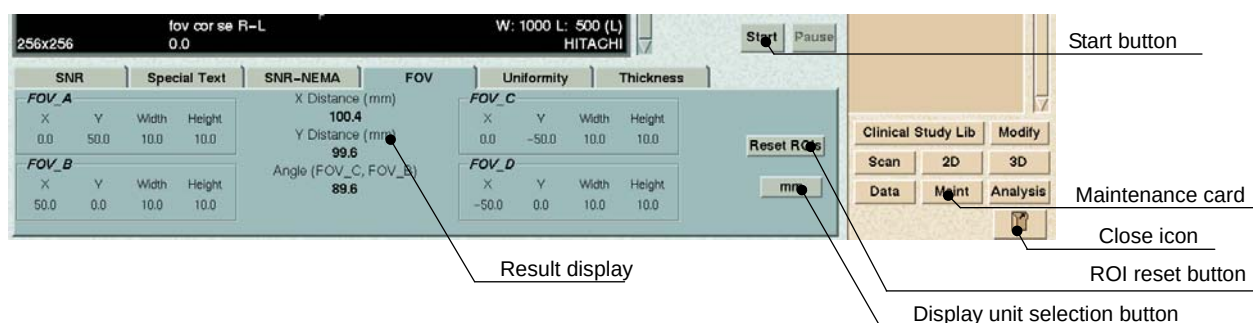


Fig. 5.4-2 FOV Check Function Buttons

5.4.4.1 Resetting ROI display position

Rest ROI display position to return to the default position.

5.4.4.2 Movement of ROI display position

Four ROIs are displayed on the image. Check the ROIs for their correct positions. If display position of any of ROIs is not appropriate, click the ROI to make it movable and drag the center marker (move the marker with the left mouse button pressed) to move the pin on the phantom to the center of the center marker in the ROI.

5.4.4.3 Selection of display unit

Two units are available as display unit. Click the Display Unit Selection button and select the display unit of either [mm] or [Pixels].

5.4.4.4 Starting process

Click [Start] button on the control screen. Upon completion of the process, the distances between pins in X and Y directions are displayed in X Distance and Y Distance, and the intersection angle is displayed in Angle.

5.4.5 End

Click the Close icon on the lower right of the screen to close the folder.

5.5 Uniformity Check

5.5.1 General

Uniformity of the acquired image is calculated and displayed.

5.5.2 Start-up

Register the T/M license with Key MOD/DVD (refer to Section 3).

As displaying the scan card scanned the image for uniformity check or data card read the image for uniformity check on the most front screen, click [Maint] button on the lower right of the screen.

Upon displaying the Maintenance Card, click [Uniformity] tag on the control face.

5.5.3 Display screen

Display screen is shown below.

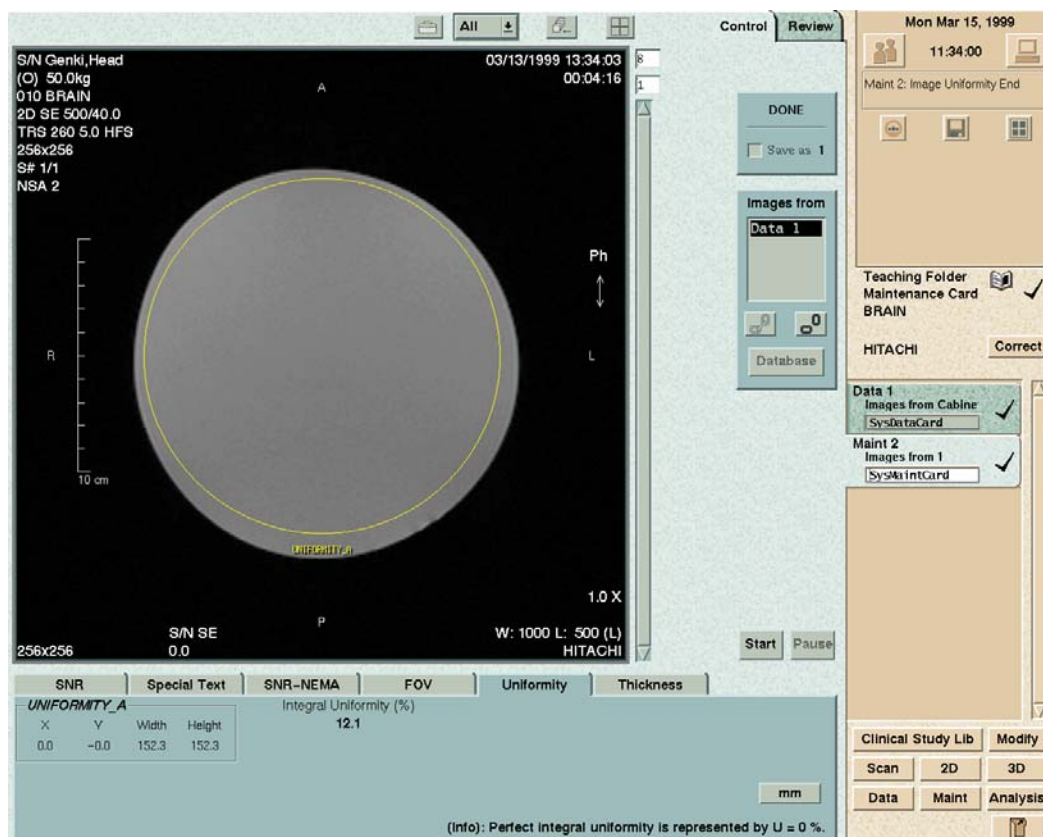


Fig. 5.5-1 Uniformity Check Display Screen

5.5.4 Operation procedure

Follow the next steps to execute the process.

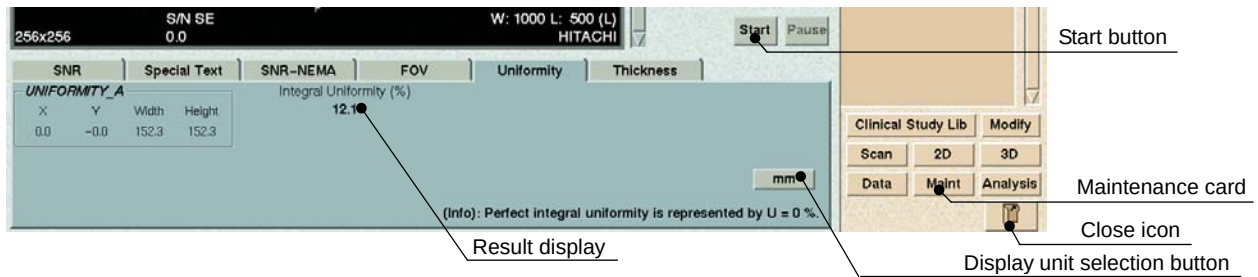


Fig. 5.5-2 Uniformity Check Function Buttons

5.5.4.1 Movement of ROI display position

An ROI is displayed on the image. Check the ROI for its correct positions. If display position of the ROI is not appropriate, click the ROI to make it movable and drag the center marker (move the marker with the left mouse button pressed) to move the ROI to the appropriate position.

5.5.4.2 Selection of display unit

Two units are available as display unit. Click the Display Unit Selection button and select the display unit of either [mm] or [Pixels].

5.5.4.3 Starting process

Click [Start] button on the control screen. Upon completion of the process, the result is displayed in Integral Uniformity.

5.5.5 End

Click the Close icon on the lower right of the screen to close the folder.

5.6 Slice Thickness Check

5.6.1 General

The slice thickness of the acquired image is calculated from its profile and displayed. The calculated results of each thickness at 1/4, 1/2 and 3/4 position in the depth direction of the obtained slice profile line are displayed in FW3QM, FWHM and FWQM respectively.

5.6.2 Start-up

Register the T/M license with Key MOD/DVD (refer to Section 3).

As displaying the scan card scanned the image for slice thickness check or data card read the image for slice thickness check on the most front screen, click [Maint] button on the lower right of the screen.

Upon displaying the Maintenance Card, click [Thickness] tag on the control face.

5.6.3 Display screen

Display screen is shown below.

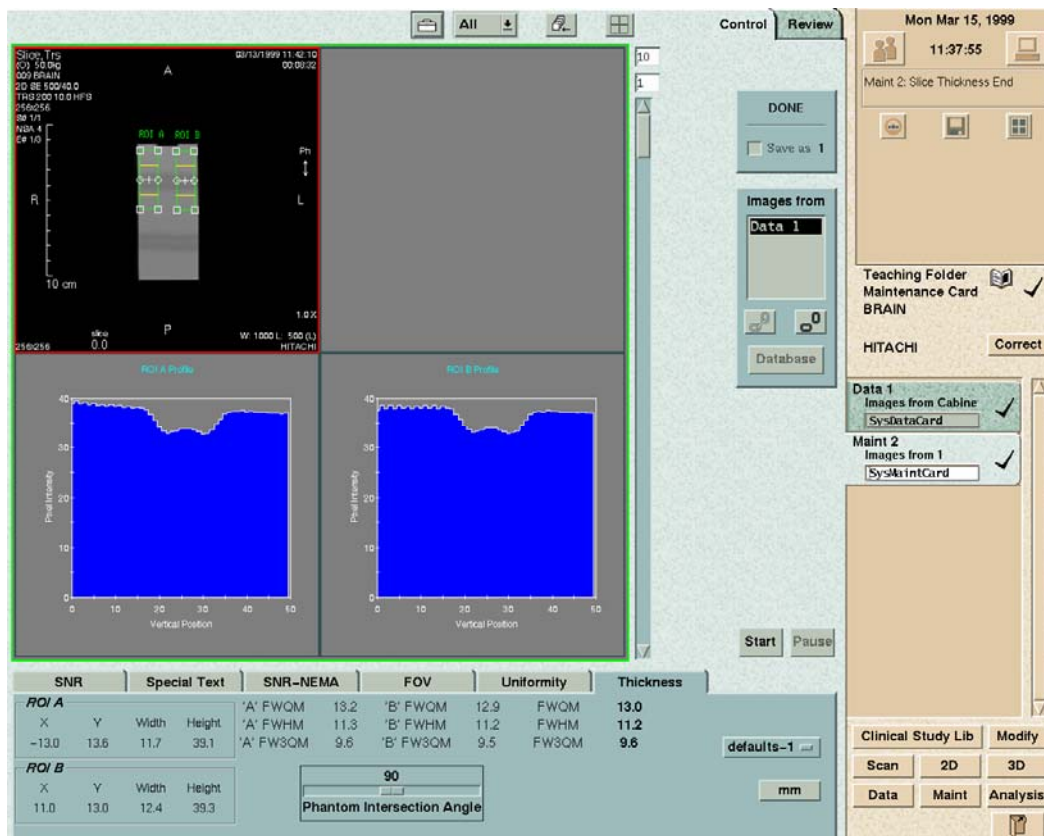


Fig. 5.6-1 Slice Thickness Check Display Screen

5.6.4 Operation procedure

Follow the next steps to execute the process.

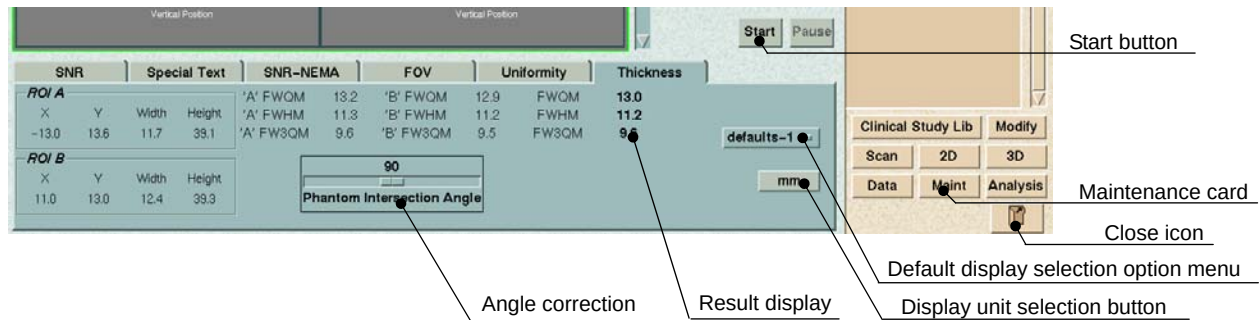


Fig. 5.6-2 Slice Thickness Check Function Buttons

5.6.4.1 ROI display selection

Two defaults are available for ROI display. Click the Default Selection Option Menu and select either [defaults-1] or [defaults-2] on the Menu.

5.6.4.2 Movement of ROI display position

Two ROIs are displayed on the image. Check the ROIs for their correct positions. If display position of any of ROIs is not appropriate, click the ROI to make it movable and drag the center marker (move the marker with the left mouse button pressed) to move the ROI to the appropriate position.

5.6.4.3 Selection of display unit

Two units are available as display unit. Click the Display Unit Selection button and select the display unit of either [mm] or [Pixels].

5.6.4.4 Angle correction

Change the angle correction parameter for the intersection angle with the phantom plate in Phantom Intersection Angle scale.

5.6.4.5 Starting process

Click [Start] button on the control screen. Upon completion of the process, the profile of each ROI and slice thickness are displayed on the bottom of the screen and result display area respectively.

5.6.5 End

Click the Close icon on the lower right of the screen to close the folder.

Section 6 Service Program

6.1 Registration of Name

6.1.1 General

Names of radiologist, physician, technologist, department and hospital name are registered.

To names of radiologist, physician, technologist and department which can be selected at patient registration, each registered name is added.

6.1.2 Start-up

Select Preferences in the System Tool. Click People/Names tag in the displayed items list.

6.1.3 Display screen

Display screen is shown below.

The screenshot shows the 'System Preferences' dialog box with the 'People/Names' tag selected. The dialog is titled 'System Preferences' and has a toolbar with icons for 'People/Names', 'Viewing', 'Film', 'Task Card', 'System', 'Messages', 'Date/Time', and 'Service'. Below the toolbar is the 'Site Preferences' section, which contains four panels: 'Radiologists', 'Technologists', 'Departments', and 'Physicians'. Each panel has an 'Add' button, a 'Delete' button, and a list box. The 'Radiologists' panel has a list box with 'HITACHI' and a vertical scrollbar. The 'Technologists' panel has a list box with 'HITACHI' and a vertical scrollbar. The 'Departments' panel has a list box with 'HITACHI' and a vertical scrollbar. The 'Physicians' panel has a list box with 'HITACHI' and a vertical scrollbar. At the bottom of the dialog is the 'Institution Name' field, which contains 'HITACHI'. There are 'Cancel' and 'Done' buttons at the bottom right.

Category	Family Name	Given Name
Radiologists	HITACHI	
Technologists	HITACHI	
Departments	HITACHI	
Physicians	HITACHI	

Institution Name: HITACHI

Fig. 6.1-1 People/Names Tag Display Screen

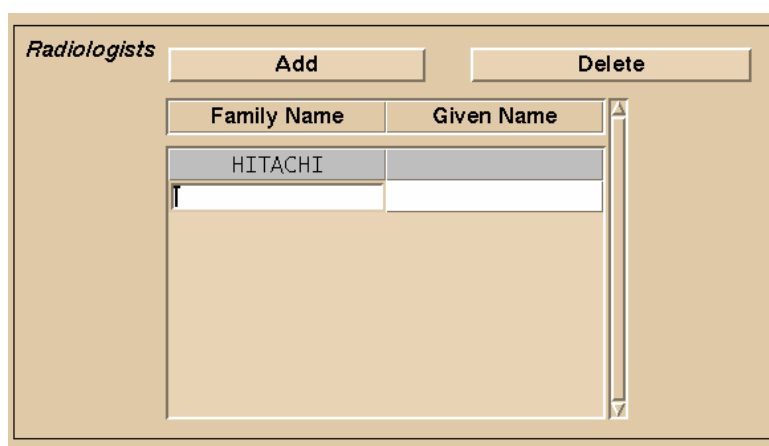
6.1.4 Operation procedure

Follow the next steps to execute the process.

6.1.4.1 Entry of radiologist name, physician, technologist name, and department name

<Addition>

Click [Add] button of the item to be added. Since one blank is added, click the entry area to make an entry and press <Enter> to set it.



The screenshot shows a window titled "Radiologists". At the top, there are two buttons: "Add" and "Delete". Below these buttons is a table with two columns: "Family Name" and "Given Name". The first row of the table has "HITACHI" in the "Family Name" column and is highlighted. Below this row, there is a large empty rectangular area, likely for additional entries or details. A vertical scrollbar is visible on the right side of the table area.

Fig. 6.1-2 Screen for clicking [Add] Button

<Deletion>

Select the name to be deleted and click [Delete] button of that item.

6.1.4.2 Setting hospital name

Enter the hospital name (or institute name) in *Institution Name*.

6.1.5 End

By pressing [Done] button, the name is registered and the process ends. To terminate the process without registering the name, click [Cancel] button.

6.2 Setting Display

6.2.1 General

Various parameters for display are set.

6.2.2 Start-up

Select Preferences in System Tool. Click Viewing tag in the displayed items list.

6.2.3 Display screen

Display screen is shown below.

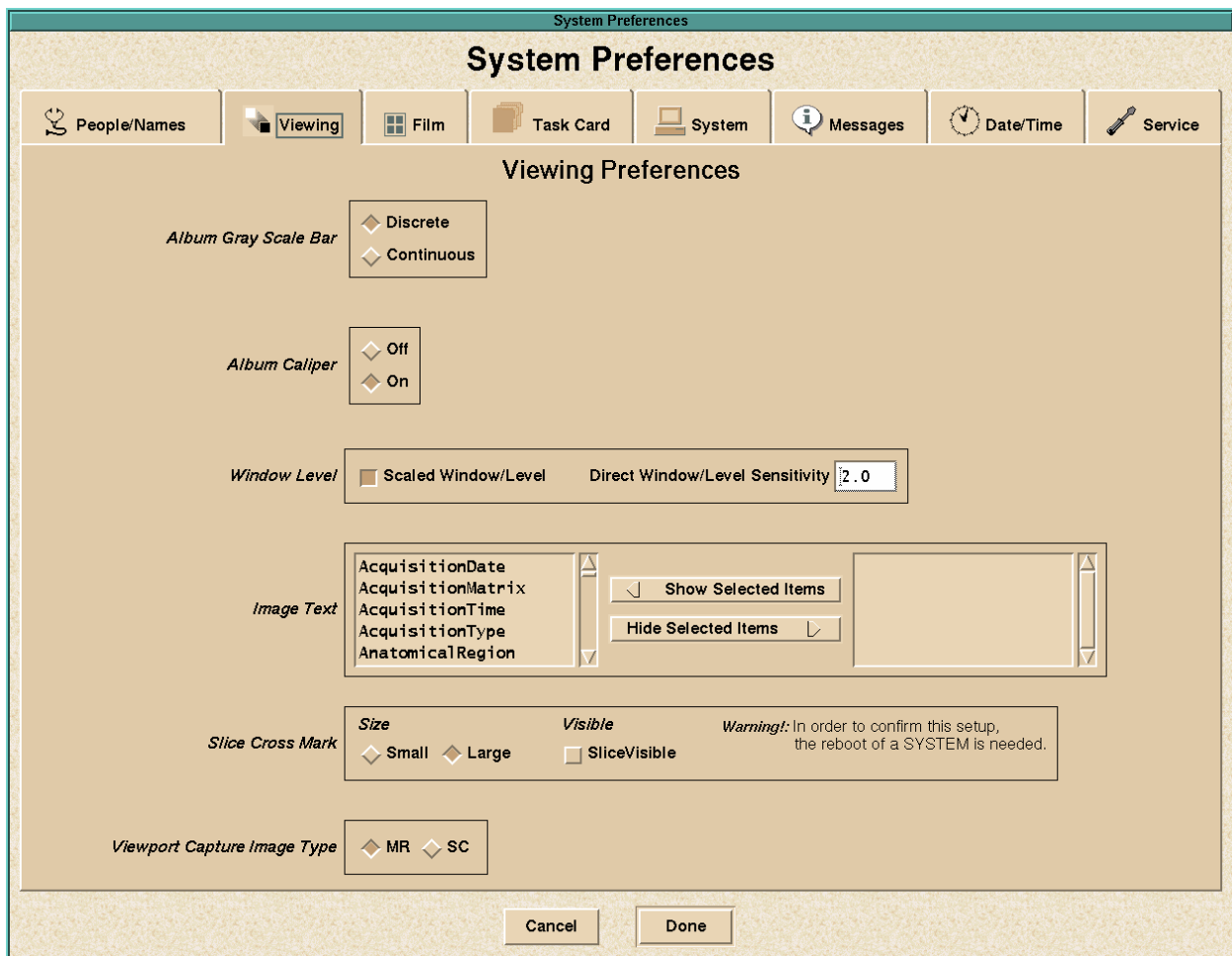


Fig. 6.2-1 Viewing Tag Display Screen

6.2.4 Operation procedure

Follow the next steps to execute the process.

6.2.4.1 Setting kind of gray scale

Select the desired gray scale by the **Album Gray Scale Bar**.

6.2.4.2 Setting line caliper

Select the display of line caliper when creating a new card by the **Album Caliper**.

6.2.4.3 Selection of function to normalize window width/level of display image

To execute the function to normalize window width and level of the display image, press and hold the [**Scaled Window/Level**] button.

6.2.4.4 Selection of mouse sensitivity to change window width/level

In changing window width and level of the image by the middle mouse button using **Direct Window/Level Sensitivity**, amount of change corresponding to the moving distance of the mouse is defined. The smaller the amount, the lower the window width/level sensitivity, while the larger the amount, the higher the sensitivity. Any value from 0.1 to 9.9 can be selected.

6.2.4.5 Selection of attendant data to be displayed on image

Attendant data to be displayed on the image can be selected by **Image Text**. To set the item being selected for display to non-display, select the item and press [**Hide Selected Items**] button. To set the item being selected for non-display to display, select the item and press [**Show Selected Items**] button.

6.2.4.6 Setting slice cross mark

Slice cross mark is defined in **Slice Cross Mark**. The size of cross mark contained in a slice line is set up by **Size**. The display of the slice line under mouse operation is set up by **Visible**.

6.2.4.7 Setting viewport capture image

An image format for saving images created by the capture tool is defined in **Viewport Capture Image Type**. Select **MR** when you perform additional processing on stored images, such as post-processing, changes in the Window Level/Width, magnification or minification, etc. Select **SC (Secondary Capture)** when you display images in other system. You can change only the Window Level/Width of the images stored in this format.

6.2.5 End

Clicking [**Done**] button ends the process for display setting. For cancellation of the process, click [**Cancel**] button.

6.3 Setting Film

6.3.1 General

Various parameters for film are set.

6.3.2 Start-up

Select Preferences in System Tool. Click Film tag in the displayed items list.

6.3.3 Display screen

Display screen is shown below.

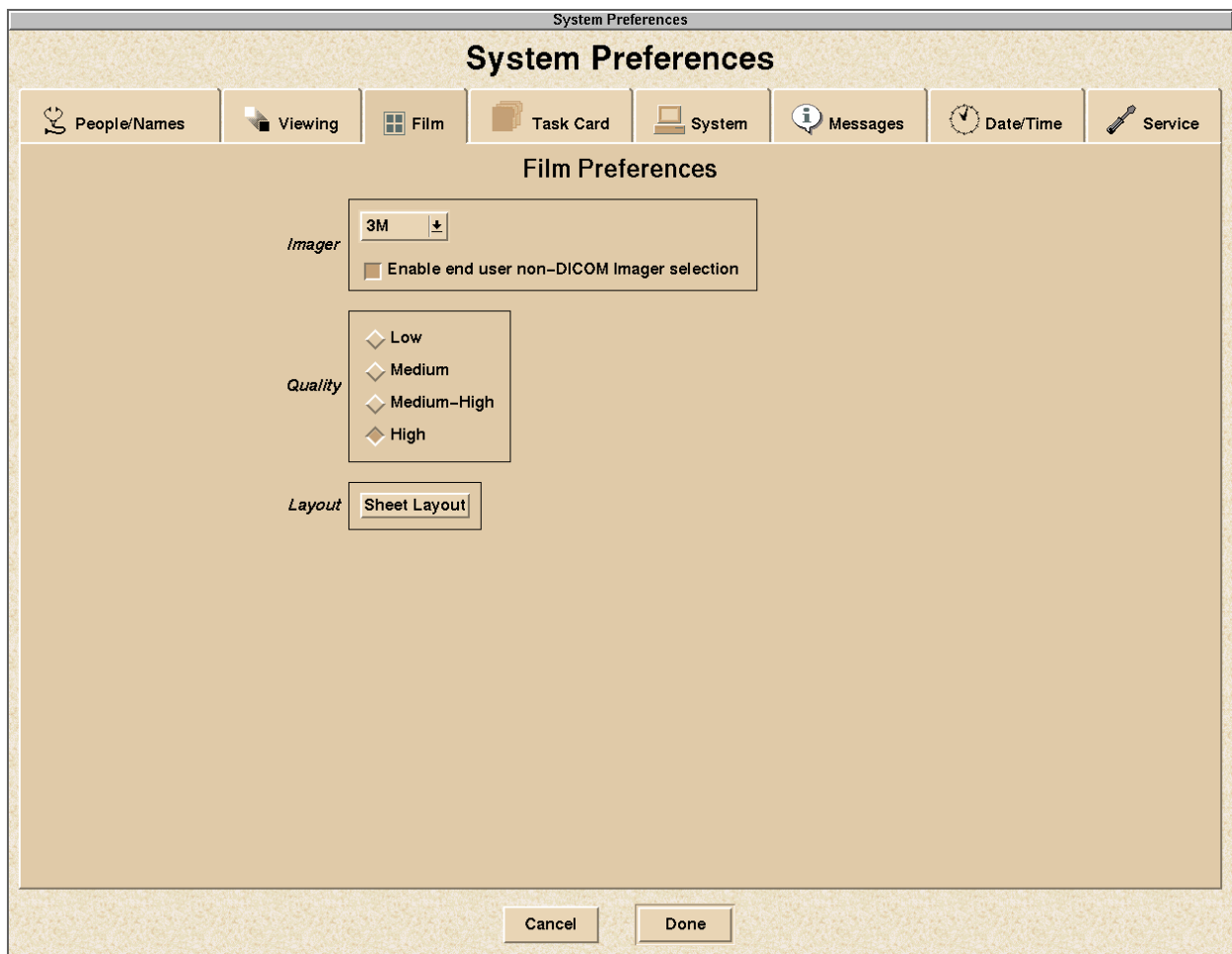


Fig. 6.3-1 Film Tag Display Screen

6.3.4 Operation procedure

Follow the next steps to execute the process.

6.3.4.1 Setting imager type to be connected

Select the imager to be connected by **Imager**. However, it is necessary to press and hold the [Enable end user non-DICOM Imager selection] button.

6.3.4.2 Selecting quality of image to be transferred to imager

Select the quality of the image to be transferred to the imager by **Quality**. Selecting [High] increases data volume after image interpolation by the system, resulting in longer time until filming.

6.3.4.3 Setting default of film layout

Select the default of film layout to be outputted to the imager by **Layout**.

6.3.5 End

Clicking [Done] button ends the process for display setting. For cancellation of the process, click [Cancel] button.

6.4 Setting Task Card

6.4.1 General

Various parameters for task card are set.

6.4.2 Start-up

Select Preferences in System Tool. Click Task Card tag in the displayed items list.

6.4.3 Display screen

Display screen is shown below.

The screenshot shows the 'System Preferences' dialog box with the 'Task Card' tab selected. The dialog has a title bar 'System Preferences' and a menu bar with icons for People/Names, Viewing, Film, Task Card, System, Messages, Date/Time, and Service. The main content area is titled 'Task Card Preferences' and contains three sections: 'MIP Default Settings' with a '3DAI' dropdown set to 'None' and an 'Enhanced Quality' checkbox; 'Contrast Agents' with an 'Add' button, a 'Delete' button, and a list box containing 'Gd-DTPA'; and 'TaskCard Auto Archive Setting' with a 'Clinical Study' checkbox and a 'Device' field set to 'MOD'. At the bottom are 'Cancel' and 'Done' buttons.

Section	Parameter	Value
MIP Default Settings	3DAI	None
	Enhanced Quality	☐
Contrast Agents	Add	Button
	Delete	Button
TaskCard Auto Archive Setting	Clinical Study	☐
	Device	MOD

Fig. 6.4-1 Task Card Tag Display Screen

6.4.4 Operation procedure

Follow the next steps to execute the process.

6.4.4.1 Setting MIP default

The default of MIP function is defined in MIP Default Settings.

3DAI will select a type of 3D Adaptive Imaging function used as the default (None/Sn.Ves/Lg.Ves). When a 3D card is created using the system default, the 3DAI which is set will be the default 3DAI of the 3D card.

Enhanced Quality will set the quality of MIP images. When it is set to ON, it creates high quality MIP images. In this case, creation speed will be decreased compared to the case that it is set to OFF.

6.4.4.2 Setting contrast agent name

Enter the initial value for names of the selectable Contrast Agents.

6.4.4.2.1 Addition of contrast agent name

Press [Add] button to display a new input area.

Enter the contrast agent name to be added.

6.4.4.2.2 Deletion of contrast agent name

Select the contrast agent name to be deleted, and press [Delete] button.

6.4.4.3 Setting auto archive

The auto archive function is defined in TaskCard Auto Archive Setting. When performing auto archive function and a card is saved with an image, set Clinical Study at selected status. Select external storage used as a preservation place for Device.

6.4.5 End

Clicking [Done] button ends the process for display setting. For cancellation of the process, click [Cancel] button.

6.5 Setting System

6.5.1 General

The following parameters are set.

- Unit of weight
- Selection of birth date or age entry
- Display/Non-display of Accession Number
- Automatic addition of patient ID No.
- Clock display style
- Screen Saver ON/OFF
- Size of mouse cursor
- Movement of mouse cursor

6.5.2 Start-up

Select Preferences in System Tool. Click System tag in the displayed items list.

6.5.3 Display screen

Display screen is shown below.

System Preferences

People/Names Viewing Film Task Card **System** Messages Date/Time Service

System General Preferences

Weight Units ☒ Kilograms ☐ Pounds

Preferred Registration Age Input ☒ Birth Date ☐ Age

Accession Number Availability ☐ Off ☒ On

Patient ID Auto Increment ☐ Off ☒ On

Clock Style 23:59:59

Inactive Screen Blanking Idle Time (Minutes): 10.0 ☐ Blanking Off **Warning!** Turning off blanking may cause damage to the screen phosphor.

Cursor ☒ Normal ☐ Medium ☐ Large

Mouse Movement Acceleration Fast Speed: 2.0 Distance Moved before Acceleration: 5

Cancel Done

Fig. 6.5-1 System Tag Display Screen

6.5.4 Operation procedure

Follow the next steps to execute the process.

6.5.4.1 Unit of weight

Select either kg unit or lb. unit for **Weight Units**.

6.5.4.2 Selection of birth date or age entry

Select either birth date or age entry for **Preferred Registration Age Input**.

6.5.4.3 Display or non-display of Accession Number

When Accession Number is used as the data for controlling the examination, select On in the items of **Accession Number Availability**. By selecting On, an entry field for Accession Number is displayed in Register New Patient UI or Correct Patient UI.

6.5.4.4 Automatic addition of patient ID No.

Set automatic addition display On/Off of patient ID No. in new patient registration screen in **Patient ID Auto Increment**.

In case of ON, patient ID No. is automatically displayed from the patient ID No. registered last in the new patient registration screen.

6.5.4.5 Clock display style

Select clock display mode for **Clock Style**. The digital display and analog display styles are available.

In the digital display style, either 24 hours display or 12 hours display and either display or non-display of the second unit can be selected.

6.5.4.6 Screen Saver ON/OFF

Screen Saver is set to either ON or OFF for **Inactive Screen Blanking**. In case of ON, set a start time of the Screen Saver in **Idle Time** in the unit of minute.

In case of OFF, click **[Blanking Off]** button.

6.5.4.7 Size of mouse cursor

Select the desired size of mouse cursor with **Cursor**.

6.5.4.8 Movement of mouse cursor

Operation of the mouse is defined in **Mouse Movement Acceleration**. **Fast Speed** represents a coefficient between the moving speed of the mouse itself and movement of the mouse cursor. The larger value the coefficient, the larger the cursor movement.

Distance Moved before Acceleration represents number of pixels of the moved mouse before accelerated. The smaller value the number of pixels, the less cursor movement before acceleration.

6.5.5 End

Clicking **[Done]** button ends the process after completing the setting. For cancellation of the process without executing the setting, click **[Cancel]** button.

6.6 Setting Message

6.6.1 General

The list of messages is displayed. Some of the messages can be suppressed not to be displayed.

6.6.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select Preferences in the System Tool. Click Message tag in the displayed items list.

6.6.3 Display screen

Display screen is shown below.

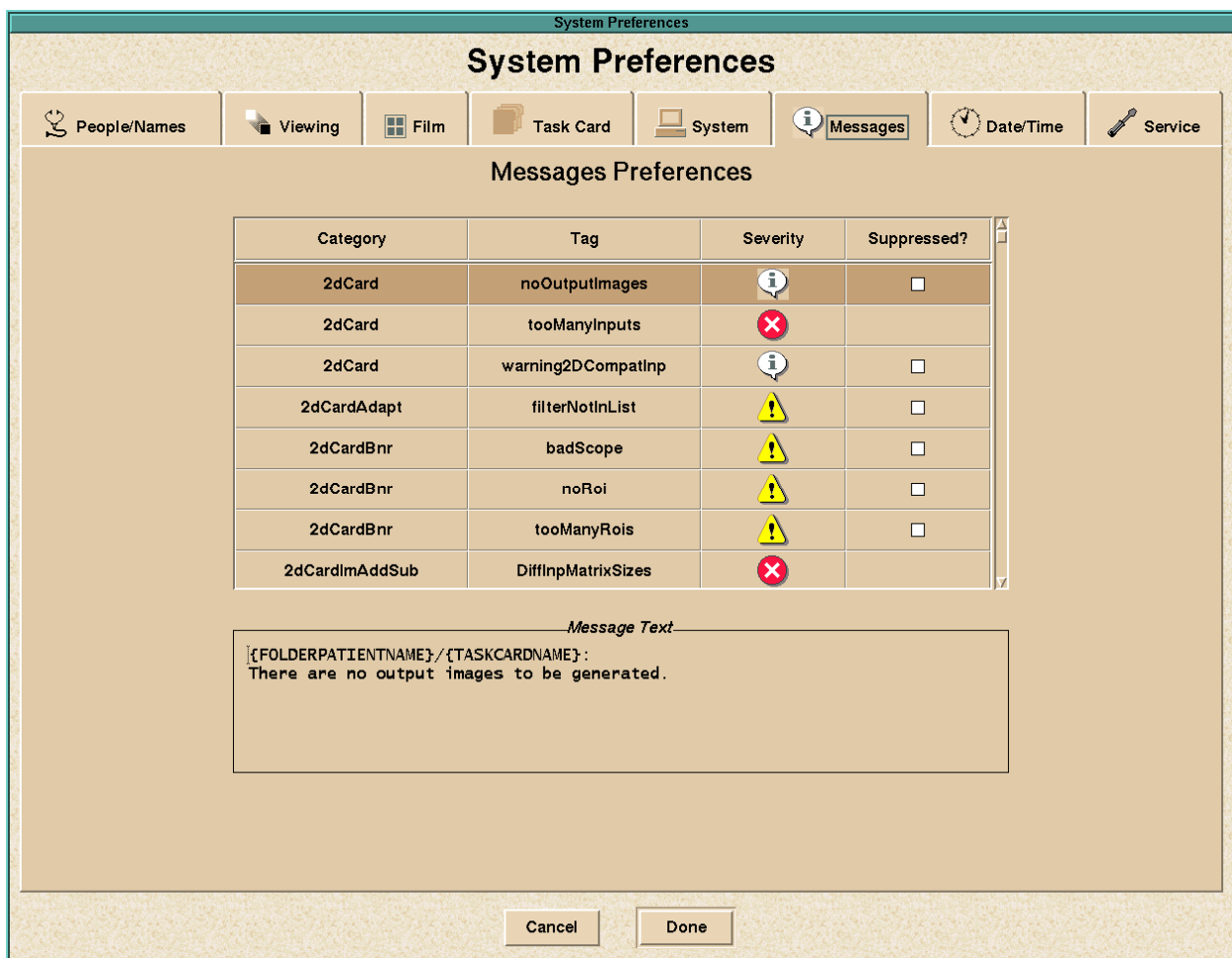


Fig. 6.6-1 Message Tag Display Screen

6.6.4 Operation procedure

Follow the next step to execute the process.

6.6.4.1 Suppression of display message

Display of the message of which Suppressed? item is attached with ☐ can be suppressed. Double-click ☐.

6.6.5 End

Clicking [Done] button ends the message setting process. For cancellation of the process, click [Cancel] button.

6.7 Setting Date and Time

6.7.1 General

Date and time of the system is set.

6.7.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select Preferences in the System Tool. Click Date/Time tag in the displayed items list.

6.7.3 Display screen

Display screen is shown below.

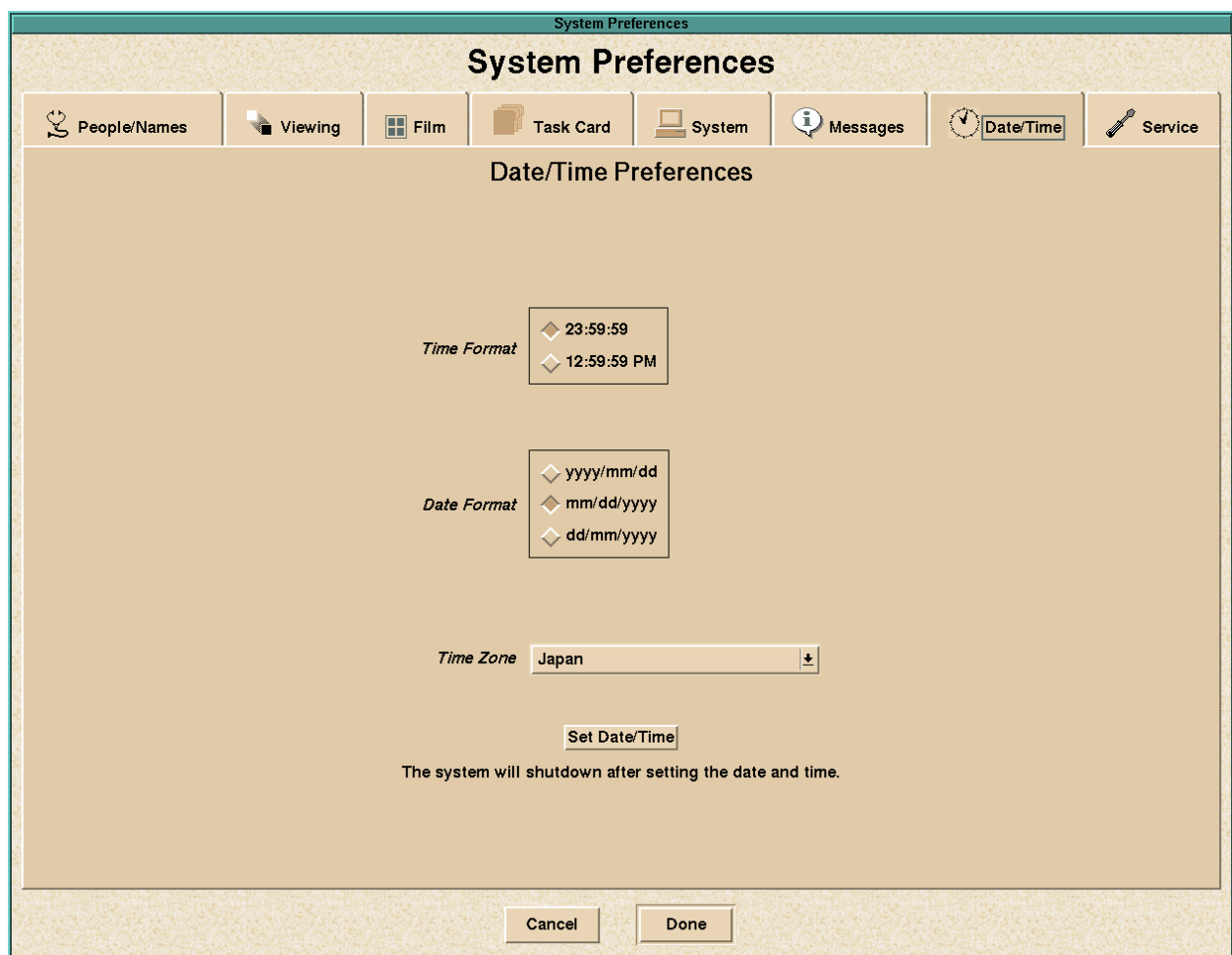


Fig. 6.7-1 Date/Time Tag Display Screen

6.7.4 Operation procedure

Follow the next steps to execute the process.

6.7.4.1 Setting time format

Select time format for Time Format item.

6.7.4.2 Setting date format

Select date format for Date Format item.

6.7.4.3 Setting time zone

Select time zone for Time Zone item.

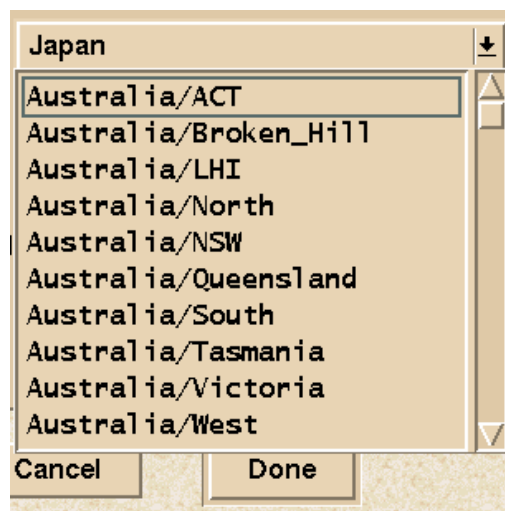


Fig. 6.7-2 Time Zone Setting Screen

6.7.4.4 Setting date and time

- (1) Clicking [Set Date/Time] button displays the dialog shown in Fig. 6.7-3. If the program can be terminated, click [Yes] button, otherwise click [No] button. Clicking [No] button returns to display of the previous screen.

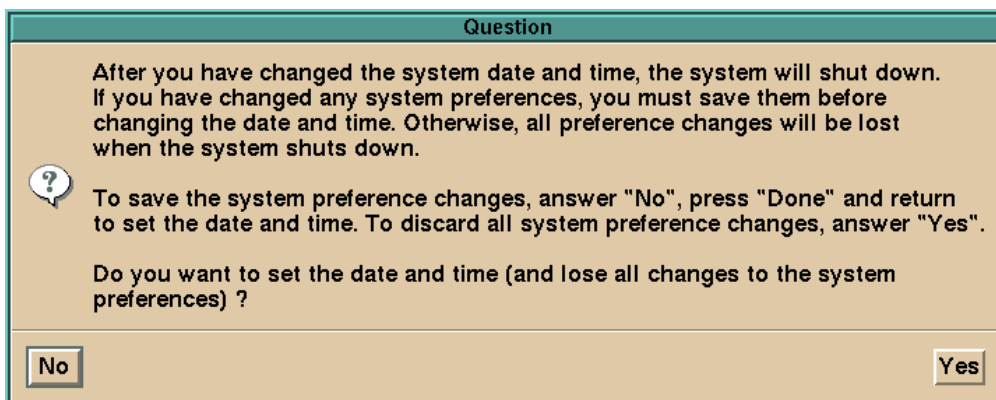


Fig. 6.7-3 Confirmation Dialog for Program Termination

- (2) Clicking [Yes] button displays the screen to set date and time. On this screen, enter date and time in the format designated in Paragraphs 6.7.4.1 and 6.7.4.2. Upon completion of the entry, press <Enter> key. If the input date and time are same as those of the specified, [Done] button becomes accessible. To execute the setting, click [Done] button.

To cancel execution of the setting, click [Cancel] button to return to display of the previous screen.

NOTE: Enter date and time in this order and place a space between them.

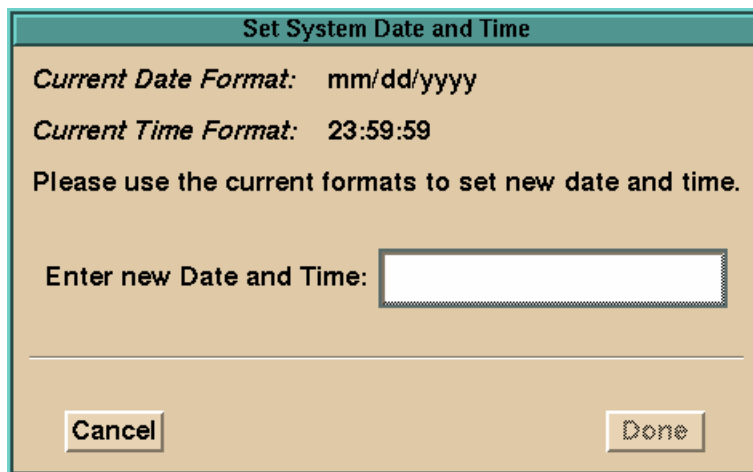
A dialog box titled "Set System Date and Time" with a teal header. The background is light beige. It contains the following text: "Current Date Format: mm/dd/yyyy", "Current Time Format: 23:59:59", and "Please use the current formats to set new date and time." Below this is a label "Enter new Date and Time:" followed by a white text input field with a thin black border. At the bottom, there are two buttons: "Cancel" on the left and "Done" on the right, both with a light beige background and a thin black border.

Fig. 6.7-4 Date and Time Entry Screen

- (3) Clicking [Done] button displays the dialog shown in Fig. 6.7-5. If the status allows to shut down the system, click [Yes] button, otherwise click [No] button. Clicking [Yes] button executes the setting and the system is automatically shut down. Clicking [No] button returns to the previous screen.

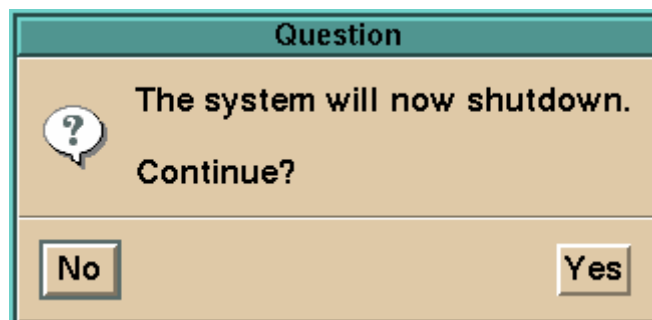
A dialog box titled "Question" with a teal header. The background is light beige. It contains a speech bubble icon with a question mark on the left. To the right of the icon, the text reads: "The system will now shutdown." followed by "Continue?". At the bottom, there are two buttons: "No" on the left and "Yes" on the right, both with a light beige background and a thin black border.

Fig. 6.7-5 System Shutdown Confirmation Dialog

6.7.5 End

Clicking [Done] button executes the setting and shuts down the system. To shut down the system without executing the setting, click [Cancel] button.

6.7.6 Other cautions

Executing the date and time setting will automatically shut down the system. Therefore, make the system ready to be shut down before executing the setting.

6.8 Other Setting

6.8.1 General

Set the items to be set at the time of installation.

6.8.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select Preferences in the System Tool. Click Service tag in the displayed items list.

6.8.3 Display screen

Display screen is shown below.

The screenshot shows the 'System Preferences' dialog box with the 'Service' tab selected. The 'Service Preferences' section contains the following settings:

- Image ID Settings:**
 - PatientName Size:** Radio buttons for 'Normal' (selected) and 'Large'.
 - BirthDate:** A checkbox for 'Display On' which is currently unchecked.
 - Adjustment ratio for Filming:** A text input field containing '1.00'.
- Wave Form:** A row of checkboxes for 'Enabled' (checked), 'ECG', 'Pulse', 'Resp', and 'JOMO'.
- Bed Orientation:** Radio buttons for 'Head/Foot' (selected) and 'Left/Right'.
- Display LookupTable:** A dropdown menu showing 'GREY.cm.EIZO'.
- Archive Verify:** A checkbox for 'Part10 Write Verify' which is unchecked.

Two warning messages are present:

- Next to the 'BirthDate' checkbox: *Warning!:* In order to confirm this setup, the reboot of a SYSTEM is needed.
- Next to the 'Part10 Write Verify' checkbox: *Warning!:* In order to confirm this setup, the reboot of a SYSTEM is needed.

At the bottom of the dialog are 'Cancel' and 'Done' buttons.

Fig. 6.8-1 Service Tag Display Screen

6.8.4 Operation procedure

Follow the next steps to execute the process.

6.8.4.1 Image ID display setting

Set PatientName Size and BirthDate in Image ID Settings.

For PatientName Size, set the patient name display size on the film. In case of Large, the patient name on the film is displayed to be larger than the other display items as shown in Fig. 6.8-2. In case of Normal, the patient name is displayed in the same size as in the other display items. However, the patient name display on the screen remains unchanged.

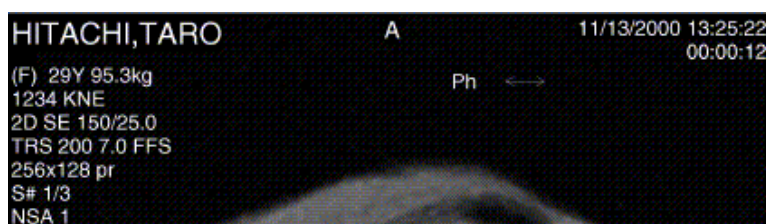


Fig. 6.8-2 Film output results

For BirthDate, set whether birth date (Born:MM/DD/YYYY) is displayed or not at the upper left corner of the image as shown in Fig. 6.8-3. Birth Date is displayed by turning on Display On.

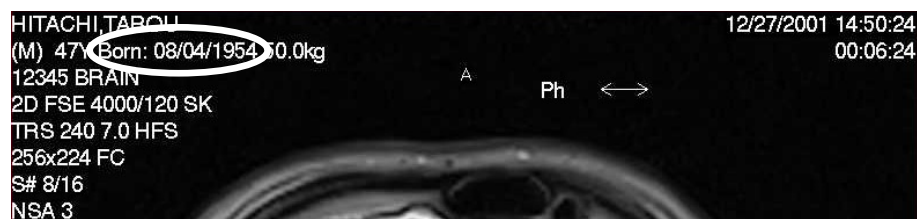


Fig. 6.8-3 Birth date display

For Adjustment Ratio for Filming, set the size ratio of text fonts displayed on image. The Image ID on Auto-Film Preview, on FilmTool, on Snapshot and on Film is displayed with the size ratio specified. However, the ratio of text fonts for text page and special text page remains unchanged.

6.8.4.2 Selection of displayable bio-signal

Items to be displayed such as ECG can be selected by user. Select the user-selectable items to be displayed. More than one item can be selected.

6.8.4.3 Selection of moving direction of patient table against gantry

Select the moving direction of the patient table against the gantry.

6.8.4.4 Selection of LookUp Table of display

Select LookUp Table of display from Display Lookup Table.

6.8.4.5 Selection of Archive write verify

Select the write verify feature when images are archived to media. When "Part10 Write Verify" is set ON, reliability of archive is improved, however, it takes more time to perform archive.

6.8.5 End

Clicking [Done] button ends the setting process of hospital name and contrast agent name. To cancel the process, click [Cancel] button.

6.9 Error LOG Display

6.9.1 General

The error logs stored in the magnetic disk are displayed. The error LOGs for the last 10 system bootings can be selected and displayed.

6.9.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select Error LOG in the System Tool.

6.9.3 Display screen

Display screen is shown below.

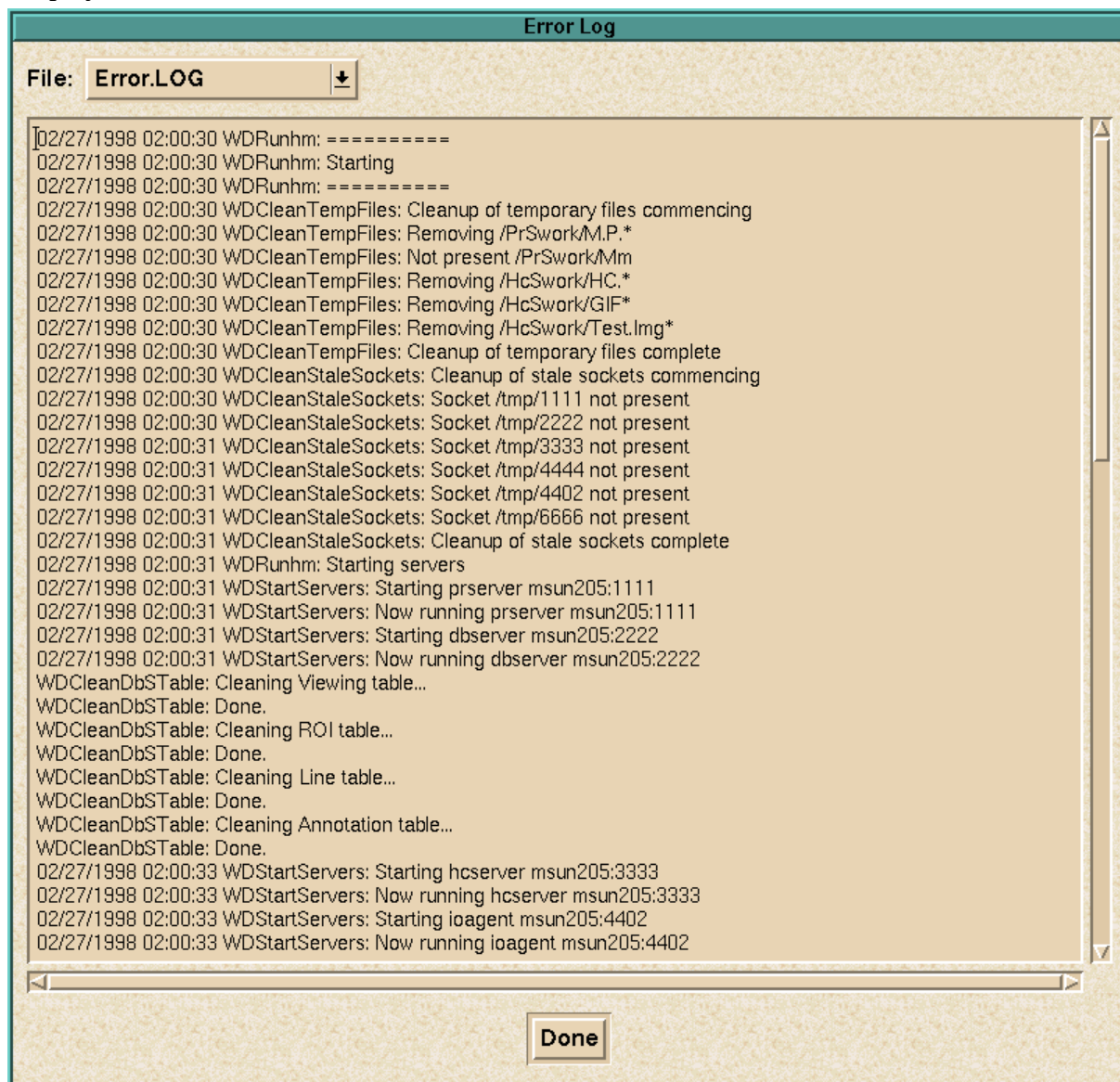


Fig. 6.9-1 Error LOG Display Screen

6.9.4 Operation procedure

Select the error LOG to be displayed.

Error LOG file name is given in such way that the currently activated system error LOG is Error.LOG, that of the system just before the current one is Error.LOG.0, the one before the last two systems is Error.LOG.1, and Error LOGs up to Error.LOG.9 can be selected.

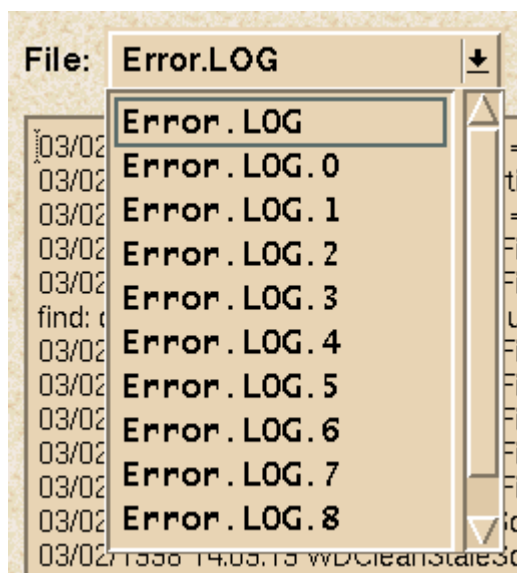


Fig. 6.9-2 Selection of Error LOG

6.9.5 End

Click [Done] button to finish the Error LOG display.

6.9.6 Other cautions

This process can be started even though the License registration has not been carried out. However, as detailed information cannot be obtained, make sure to start the process after executing the License registration.

6.10 Service Backup

6.10.1 General

The adjustment values and user-set values are stored in the MOD / DVD-RAM cartridge and read from the MOD / DVD-RAM cartridge to register in the magnetic disk.

The following data can be stored:

- System tables
- System parameters
- Pre-scan data
- DICOM settings
- User preset
- Data related to T/M

6.10.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select Service Archive/Restore in the System Tool.

6.10.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.10-1.

The screenshot shows the 'Service Archive and Restore' window. At the top, there's a title bar and a main title. Below the title, there's a 'Select an Operation' section with 'Archive' and 'Restore' options, and a 'MOD' label. To the right, there are input fields for 'File Set Name' and 'File Set Description'. Below these, there's a table titled 'Archived Service File Sets' with columns for 'File Set Name', 'Description', and 'Archive Date'. At the bottom left, there's a 'Volume Label' field showing 'Unknown'. At the bottom right, there's a 'Free Space' field showing 'Unknown'. Below these, there's an 'Initialization' section with an 'Enter label' field and an 'Initialize' button. At the very bottom, there are four buttons: 'Help...', 'Ok', 'Continue', and 'Done'.

① Operation selection area
 ② File Set Name input area
 ③ File Set Description input area
 ④ Display area of MOD/DVD contents
 ⑤ MOD/DVD volume label
 ⑥ Blank MOD/DVD capacity
 ⑦ [Help] button
 ⑧ Input area of MOD/DVD volume label
 ⑨ [Ok] button
 ⑩ [Continue] button
 ⑪ [Done] button
 ⑫ [Initialize] button
 ⑬ Media type display

- ① Operation selection area : Either Backup or Restore is selected.
- ② File Set Name input area : The file set name is entered.
- ③ File Set Description input area : The descriptions of the file set are entered.
- ④ Display area of MOD/DVD contents : The contents of the MOD/DVD are displayed.
- ⑤ MOD/DVD volume label : The volume label of the MOD/DVD is displayed.
- ⑥ Blank MOD/DVD capacity : The blank MOD/DVD capacity is displayed.
- ⑦ [Help] button : Help is displayed.
- ⑧ Input area of MOD/DVD volume label : The volume label of the MOD/DVD is input.
- ⑨ [Ok] button : Backup or Restore is executed.
- ⑩ [Continue] button : After the MOD/DVD is replaced, backup or restore process is continued.
- ⑪ [Done] button : The system exits from the application.
- ⑫ [Initialize] button : The MOD/DVD is initialized.
- ⑬ Media type display : MOD/DVD is displayed.

Select the media if option registration was done.

Fig. 6.10-1 Service Backup Display Screen

6.10.4 Operation procedure

Follow the next steps to execute the process.

6.10.4.1 Initialization of MOD/DVD

NOTE: If the MOD/DVD has been already initialized, do not initialize it. The content of backup will be lost.

- (1) Insert the MOD/DVD to be initialized into the drive unit.
- (2) Input the volume label to the MOD/DVD to be initialized in the volume label area.
- (3) Click [Initialize] button. The dialog shown in Fig. 6.10-2 is displayed. Click [Yes] button. Initialization of the MOD/DVD automatically starts.
- (4) Upon completion of initialization of the MOD/DVD, the dialog shown in Fig. 6.10-3 to exit from the initialization process is displayed. After confirmation of the initialization, click [Done] button.

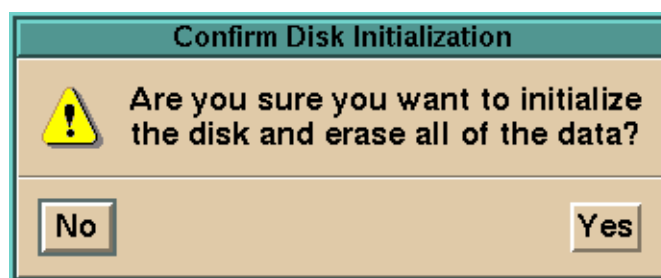


Fig. 6.10-2 Confirmation Dialog to exit from Initialization

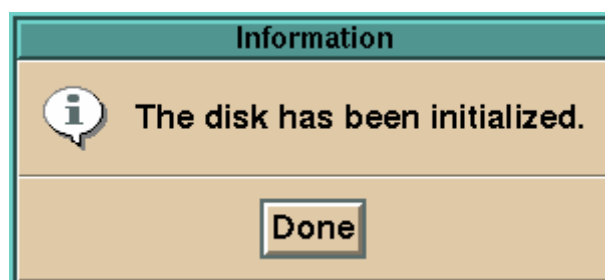


Fig. 6.10-3 Dialog to exit from Initialization

- (5) Upon completion of the initialization, the volume label input in MOD/DVD Volume Label is displayed.

6.10.4.2 Backup

NOTE: The system parameters and user-preset values in the backup are not saved in the magnetic disk unless the system is rebooted after setting.

Therefore, to set these values and take their backup, be sure to reboot the system and execute the backup operation.

- (1) Insert the initialized MOD/DVD for service backup.
- (2) Set Operation to Archive.
- (3) Input the file set name in File Set Name.

NOTE: Name with up to 50 characters can be input in File set Name, but note that the contents of the MOD/DVD can be displayed with only up to 36 characters.

- (4) Input the file set contents in File Set Description.

NOTE: The contents up to 50 characters can be input in File Set Description.

- (5) Click [Ok] button. The backup operation into the MOD/DVD automatically starts.
- (6) When no more file space is available in the MOD/DVD during backup operation, the dialog shown in Fig. 6.10-4 is displayed. After confirmation, click [Done] button and replace the MOD/DVD with a new one. After the MOD/DVD is identified, click [Continue] button. The backup operation is restarted.

NOTE: When a new MOD/DVD is used, the MOD/DVD must be initialized to use it for service backup. After inserting the new MOD/DVD, initialize the MOD/DVD according to the procedures described in Paragraph 6.10.4.1 before clicking [Continue] button. When an initialized MOD/DVD is used, the initialization operation is not required.

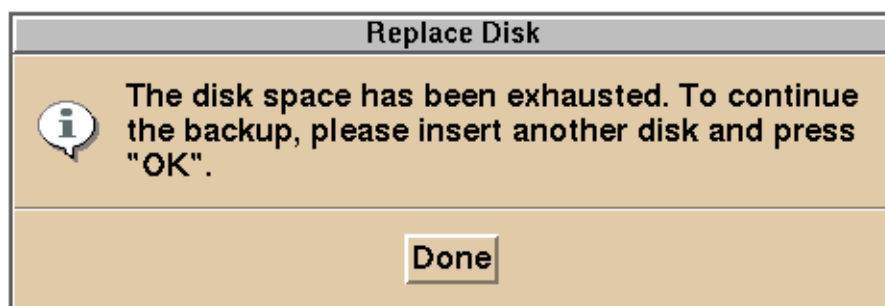


Fig. 6.10-4 Dialog requesting Replacement of MOD/DVD

- (7) Upon completion of the backup process, the dialog shown in Fig. 6.10-5 is displayed. After confirmation, click [Done] button.

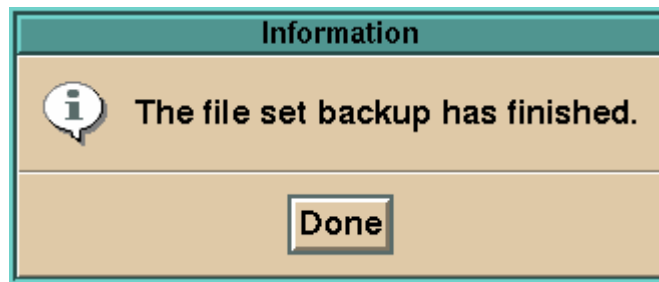


Fig. 6.10-5 Dialog to exit from Backup

- (8) The backup file contents are displayed in the MOD/DVD content display area.

6.10.4.3 Restore

NOTE: Make sure to carry out this work after CPU READY LED being turned on.

Do not begin the work during CPU READY LED being flickering as the system table is unable to renew while the LED is flickering.

- (1) Insert the MOD/DVD for service backup in which the backup file is stored.
- (2) Set Operation to Restore.
- (3) Select the file to be restored on the MOD/DVD contents display list.

NOTE: To restore a file, only the file backed up by the same system can be restored. When the file backed up by any other system is tried to be restored, the dialog shown in Fig. 6.10-6 is displayed. After confirmation, click [OK] button. After clicking [OK] button, the system is automatically powered down.

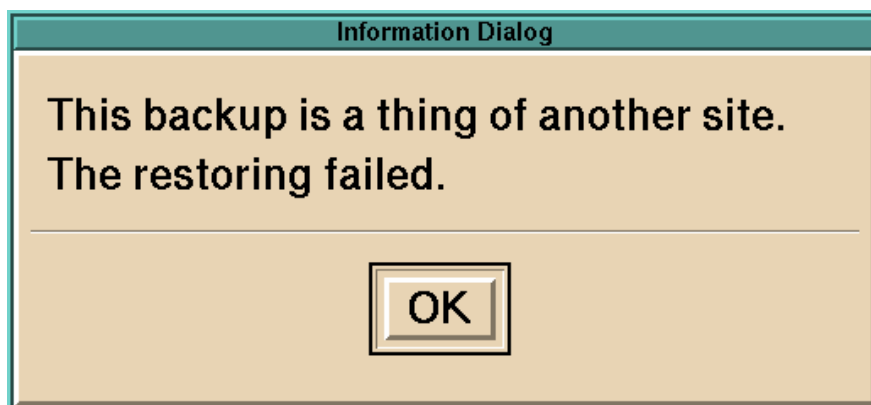


Fig. 6.10-6 Unsuccessful Restoring Dialog

- (4) Click [OK] button to start restoring data from the MOD/DVD to the magnetic disk.
- (5) In case the files to be restore have been backed up in two MODs/DVDs, when restoring process from the first MOD/DVD is completed, the dialog shown in fig. 6.10-7 is displayed. After confirmation, click [Done] button and replace the MOD/DVD with another MOD/DVD where the remaining backup data are stored. When the new MOD/DVD is identified, click [Continue] button. The restoring operation is continued.

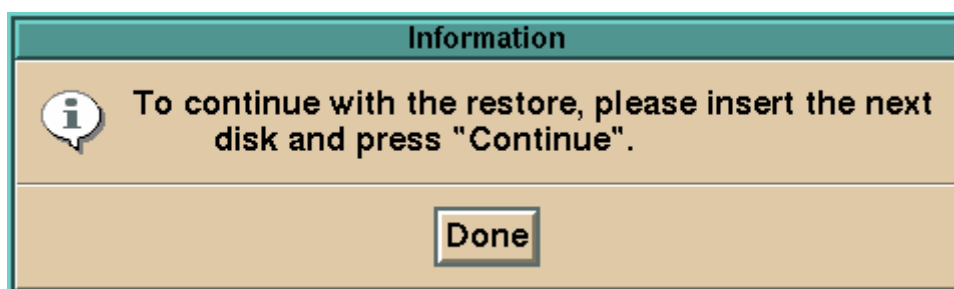


Fig. 6.10-7 Dialog requesting MOD/DVD Replacement

After completion of restoration, the dialog shown in Fig. 6.10-8 is displayed. Take out Service Backup MOD/DVD from the drive, and click [Done] button. At this time, the system ends automatically.

NOTE: After the end of system, press CPU POWER button.

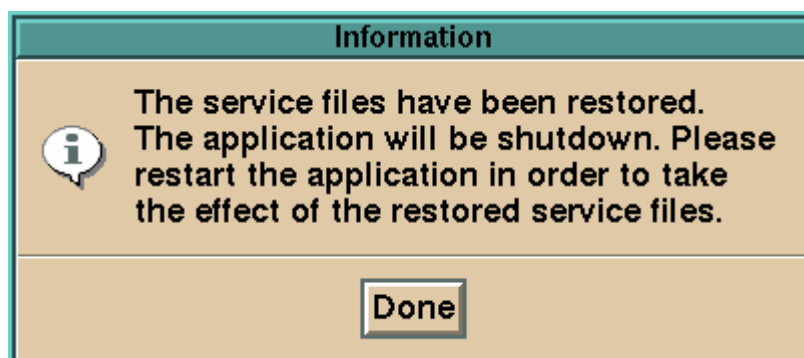


Fig. 6.10-8 Dialog to exit from Restoring Process

6.10.4.4 [Help] button display

Clicking [Help] button display the dialog shown in Fig. 6.10-9.



Fig. 6.10-9 [Help] Button Display

6.10.5 End

Clicking [Done] button ends the application.

6.10.6 Error specifications

If any error occurs, the dialog shown in Fig. 6.10-10 is displayed. In such case, work out a solution according to Table 6.10-1.

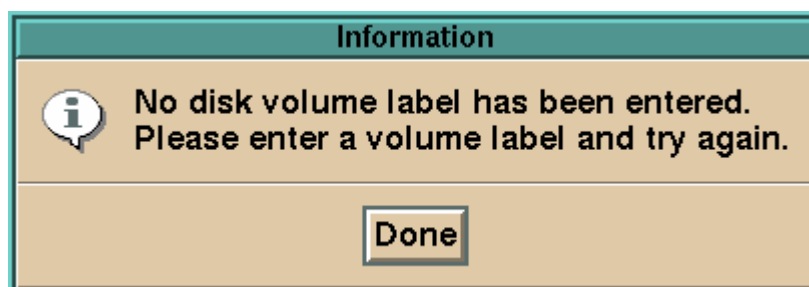


Fig. 6.10-10 Error Message

Table 6.10-1 Error Messages and Solutions

Message	Cause	Solution
No disk volume label has been entered. Please enter a volume label and try again.	Initialization was executed without entry of any volume label.	Input a volume label and try the operation again.
No file set name has been entered, or the name you entered already exists in the disk. Please enter a file set name, or use another name, and then try again.	The backup was executed without entry of File Set Name and File Set Description. Or the file with the same File Set Name exists.	Input each File Set Name and execute the backup. Input a different File Set Name.
The disk space has been exhausted. To continue the backup, please insert another disk and press "OK".	The backup was executed with the MOD/DVD without available file space.	Replace the MOD/DVD with another MOD/DVD having a file space.
No file set has been selected. Please select a file set and try again.	Restoring operation was tried without selecting any file set.	Select a file set and executed the backup.

6.10.7 Other cautions

- Backup of file can be taken only with an initialized MOD/DVD.
- Backup of file can be taken until the MOD/DVD Free Space shown on the lower right of the contents display area counts down to 0.
- If the file space of the MOD/DVD becomes zero during backup operation, the remaining data can be backed up in the second MOD/DVD.
- To restore the file, only the file which was backed up with the same system can be restored.
- After the restore operation is completed, the system is automatically shut down. Therefore, make the system ready to be shut down before starting the restore operation.

6.11 Raw Data Read/Write

6.11.1 General

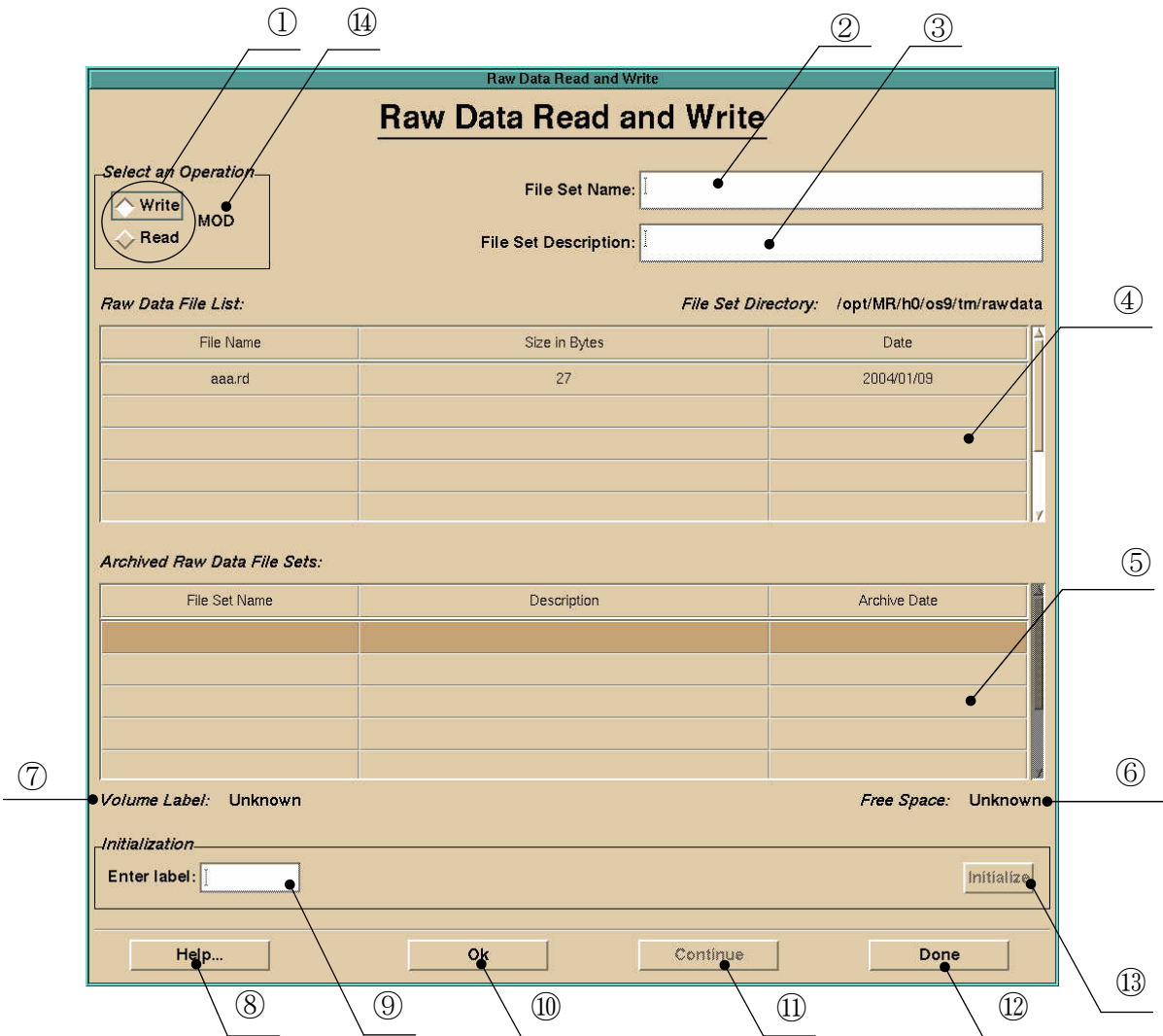
Raw data saved in the MD to be described in Sub-Section 6.14 is stored in the MOD / DVD-RAM cartridge. The data stored in the MOD / DVD-RAM cartridge is also read in the MD.

6.11.2 Start-up

Select Raw Data Read/Write in the System Tool.

6.11.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.11-1.



- | | |
|--------------------------------------|--|
| ① Operation selection area | : Either Write or Read is selected. |
| ② File Set Name input area | : The file set name is entered. |
| ③ File Set Description input area | : The descriptions of the file set are entered. |
| ④ Display area of MD contents | : The contents of the MD are displayed. |
| ⑤ Display area of MOD/DVD contents | : The contents of the MOD/DVD are displayed. |
| ⑥ Blank MOD/DVD capacity | : The blank MOD/DVD capacity is displayed. |
| ⑦ MOD/DVD volume label | : The volume label of the MOD/DVD is displayed. |
| ⑧ [Help] button | : Help is displayed. |
| ⑨ Input area of MOD/DVD volume label | : The volume label of the MOD/DVD is input. |
| ⑩ [Ok] button | : Write or Read is executed. |
| ⑪ [Continue] button | : After the MOD/DVD is replaced, Write or Read process is continued. |
| ⑫ [Done] button | : The system exits from the application. |
| ⑬ [Initialize] button | : The MOD/DVD is initialized. |
| ⑭ Media type display | : MOD/DVD is displayed. |

Select the media if option registration was done.

Fig. 6.11-1 Raw Data Read/Write Display Screen

6.11.4 Operation procedure

Follow the next steps to execute the process.

6.11.4.1 MOD/DVD initialization

Insert the MOD/DVD to be initialized into the MOD/DVD drive unit.

Input the MOD/DVD label name and click [Initialization] button.

6.11.4.2 Writing raw data

Write the raw data in the MD into the MOD/DVD.

Insert the MOD/DVD into the MOD/DVD drive.

Press [Write] button and select the raw data to be written from the raw data list in the MD.

Input the file set name and file description. Upon confirming the input items and selected raw data, click [OK] button.

In case the file space in the MOD/DVD has been exhausted during writing raw data the dialog shown in Fig. 6.11-2 is displayed. Upon confirmation, click [Done] button and replace the MOD/DVD with a new one. After identifying the MOD/DVD, click [Continue] button to continue the process.

NOTE: Before using a MOD/DVD, the MOD/DVD must have been initialized.

When a new MOD/DVD is inserted, initialize the MOD/DVD according to the procedure described in Paragraph 6.11.4.1 before clicking [Continue] button. If the initialized MOD/DVD is used, the initialization process is not required.

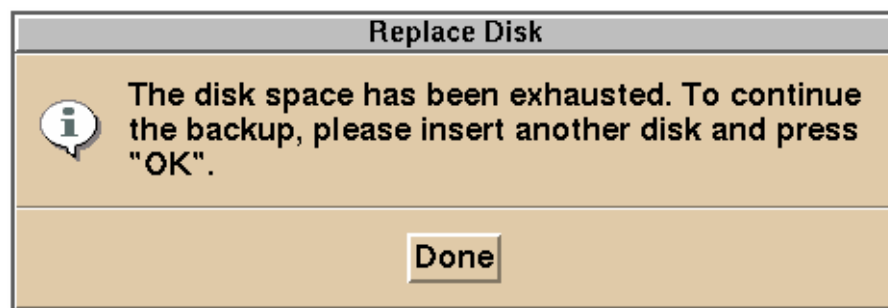


Fig. 6.11-2 Dialog to replace MOD/DVD

6.11.4.3 Reading raw data

Raw data in the MOD/DVD is read into the MD. Insert the MOD/DVD into the drive unit. Click [Read] button. Select the file set name to be read from the raw data list in the MOD/DVD, and upon confirmation of the selected data click [OK] button.

6.11.5 End

Clicking [Done] button ends the application.

6.11.6 Error specification

Error specification is shown in Table 6.11-1.

Table 6.11-1 Error Specification

Message	Cause	Solution
The disk space has been exhausted. To continue the backup, please insert another disk and press "OK".	Backup was executed on the MOD/DVD without no file space.	Replace the MOD/DVD with an MOD/DVD with a file space.

6.12 Software Version Display

6.12.1 General

Software version is displayed.

6.12.2 Start-up

Select the Version in the System Tool.

6.12.3 Display screen

The software version is displayed. The information of the service pack is also displayed, if it is installed.

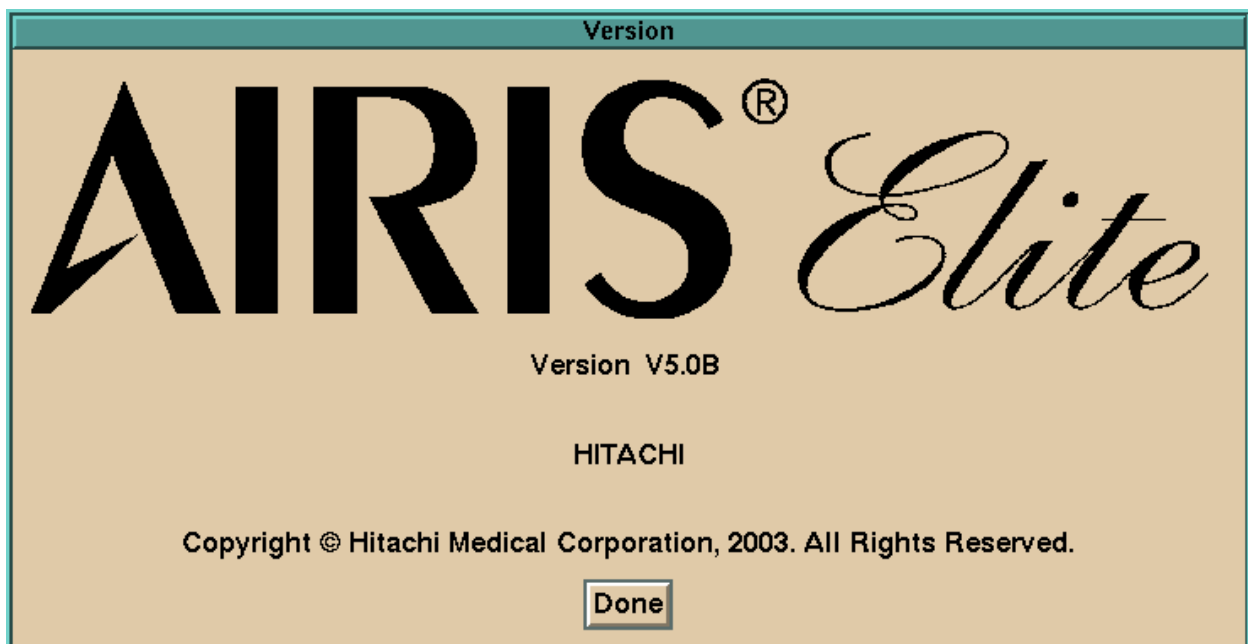


Fig. 6.12-1 Software Version Display Screen

6.12.4 End

Click [Done] button. The version display ends.

6.13 Hcp Log Display

6.13.1 General

The Hcp log stored in the magnetic disc are displayed. The Hcp log is a communication record with the digital connection imager.

6.13.2 Start-up

Register T/M license with Key MOD/DVD (refer to Section 3), and select Hcp Log from “HMC Service Diagnostics Menu” in the System Tool.

6.13.3 Display screen

Display screen is below.



Fig. 6.13-1 Hcp Log Display Screen

6.13.4 End

Click [Done] button to finish the Hcp Log Display.

6.14 Easy Raw Data Save

6.14.1 General

This is a tool to save raw data acquired by scan. Without exiting from the tool, the procedures of Scan → Save → Scan → Save can be repeated. The tool can also be applied to multi-slice, any slice can be saved.

6.14.2 Start-up

Select Raw Data Save from "HMC Diagnostics Menu" in the System Tool.

6.14.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.14-1.

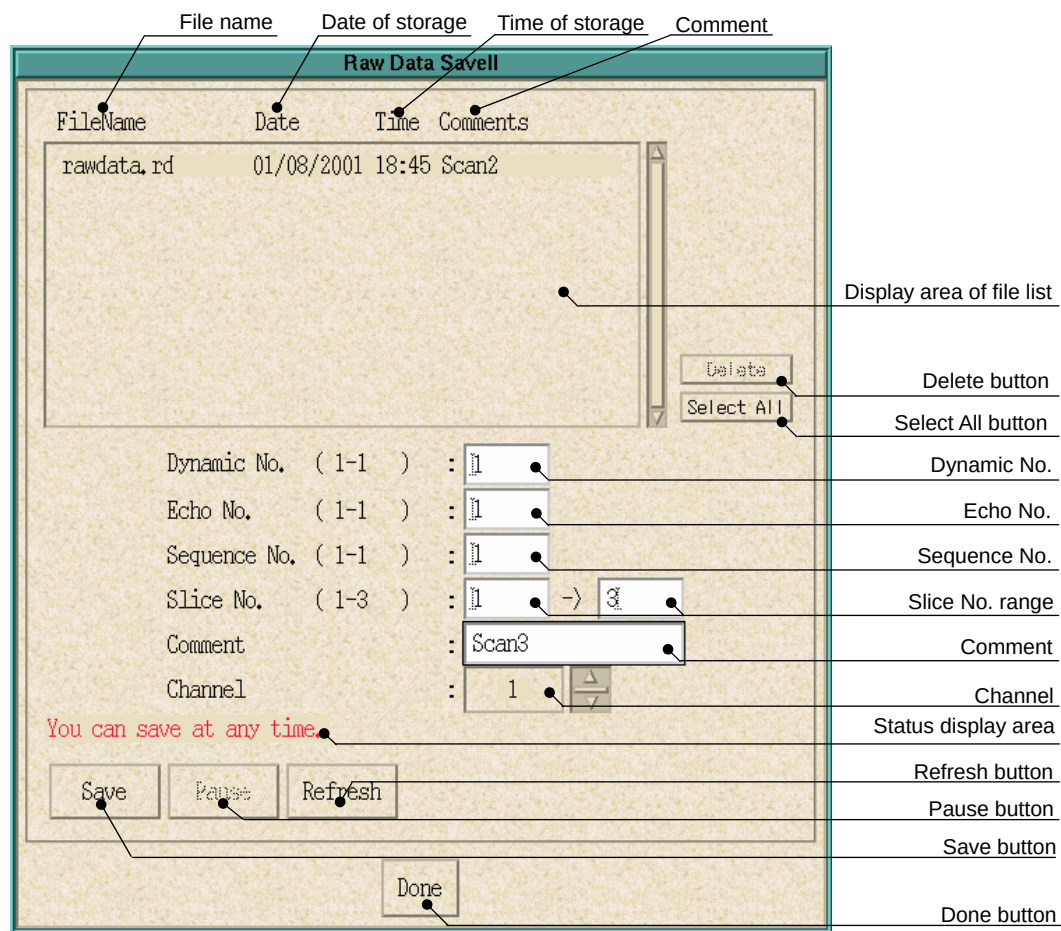


Fig. 6.14-1 Easy Raw Data Save Display Screen

6.14.4 Operation procedure

Follow the next steps to execute the process.

6.14.4.1 Updating parameters

Using the keyboard and mouse cursor, update each parameter. The updating can be done in the following procedures.

6.14.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

Table 6.14-1 Parameter Setting Range

Parameter name	Parameter Setting Range	Description
Dynamic No.	1 ~	Input measuring No. of the dynamic scan to be saved.
Echo No.	1 ~	Input echo No. of the multi-echoes to be saved.
Sequence No.	1 ~	Input sequence No of multi-sequences to be saved (PC measurement). No.1 is the original image sequence, while No.2~4 are R-L, A-P, and H-P axis image sequences, respectively.
Slice No. range	1 ~	Input the range of slice numbers of multi-slices to be saved.
Comment	Alphanumeric character within 16 characters	Input a comment. Original information can be added to a raw data file.
Channel	1 ~	Specify channel No. of MAC (Multiple Array Coil) to be saved. For the specifying method, refer to Paragraph 6.14. 4.1.2.

6.14.4.1.2 Channel selection

Select a desired channel by clicking [▲] or [▼] button on the right side of the channels list or by inputting <↑> or <↓> key after shifting the focus to the list. The settable range is restricted to only the channels where raw data are saved when scanning is done with MAC (Multiple Array Coil).

6.14.4.2 Refreshing the raw data information

If this tool was started before scanning or if scanning ~ saving process is repeated, click [Refresh] button without fail after scanning. If [Save] button is clicked without refreshing, raw data cannot be saved correctly. If this tool was started after scanning, the raw data information is already refreshed, and it is not necessary to click [Refresh] button again.

6.14.4.3 Saving raw data

Clicking [Save] button starts raw data saving. As executing the saving process, the progress status is displayed with number of slices. To repeat scan and save, click [Refresh] button every time a scan is made. File name is always suffixed with [rawdata. rd].

However, this application is provided with the storage limit by checking the available file size.

If the following alarm dialog shown in Fig. 6.14-2 is popped up, click [OK] button and follow the instruction message.

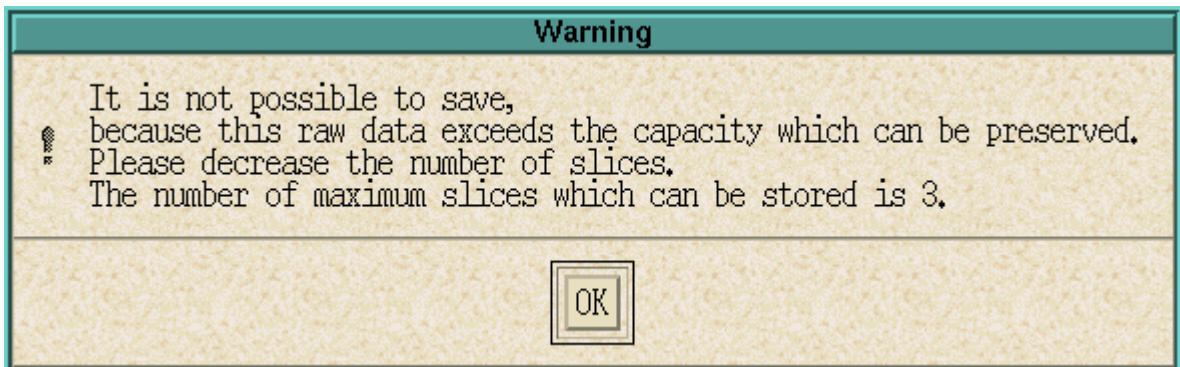


Fig. 6.14-2 Alarm Dialog

6.14.4.4 Suspension of raw data saving process

To suspend the raw data saving process being executed, click [Pause] button. The suspension dialog shown in Fig. 6.14-3 is popped up. Clicking [Stop] button here terminates the saving process, while clicking [Continue] button restart the saving process.

If the number of slice is one or while the last slice is being saved, this button cannot be pressed.

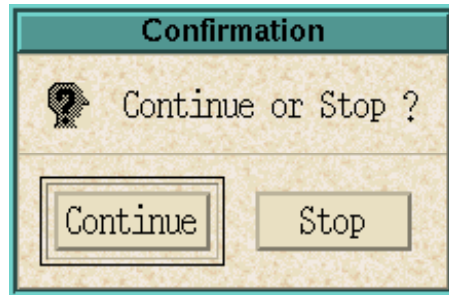


Fig. 6.14-3 Suspension Dialog

6.14.4.5 Deletion of raw data save file

Select a file to be deleted from the files list display area, and click [Delete] button. For the method of selecting a raw data save file, refer to the following description. The raw data save file which has been saved by this tool is displayed at all times. If date, time, and comments are described by a broken line, no file exists. In this case, this raw data save file cannot be selected.

6.14.4.5.1 Single and plural selection methods

By pressing the mouse button on the files list display area, the file name below the mouse button is reversed and the file name at the pressed position is selected. This reversal does not move even when shifting the mouse. By pressing the mouse button next, the reversal of the previously selected file name remains unchanged, so that plural file names can be selected.

When clicking a selected file name, the file name is reset as before from being reversed to cancel its selection.

6.14.4.5.2 All selection

If it is desired to select all items in the files list display area once, click [Select All] button. If this button is clicked once more, all items are reset to be non-selected.

6.14.4.6 Status display

Various operating conditions of this tool are displayed in the status display area.
The following table shows the messages and status.

Table 6.14-2 Description of status display

Messages	Status	Description
You can save at any time.	Savable	Indicates the savable status. Click [Refresh] button without fail when scanning was newly done.
Progress * / *	Being saved	Indicates saving in progress. This display is [Save slice No. / total save number]. For interrupting save, refer to 6.14.4.4.
Saving completed.	Saved completely	Indicates that raw data have been saved normally.
After scanning, push Refresh button.	Not savable	Indicates that no scanning has been done after the system has been started, so that no raw data are saved in the system memory. Click [Refresh] button after scanning.

6.14.5 End

Clicking [Done] button ends the application.

6.14.6 Error specifications

Error specifications are shown in Table 6.14-3.

Table 6.14-3 Error Specifications

Message	Status	Solution
Another task is running.	Another task is running.	Terminate the task being executed and retry the same operation procedure.
Can't start task.	Starting application failed.	Check the system status and retry the same operation procedure.
Can't end task.	Ending application failed.	Check the system status and retry the same operation procedure.
The translation of parameter failed.	Parameter transport failed.	Check the system status and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table failed.	[Retry] button : Retry the reading. [Exit] button : End the application.
Event Invalid.	Request is not accepted.	Check the system status and retry the same operation procedure.

6.14.7 Other cautions

In case this application is started without any scan, any data cannot be saved in this status. Make a scan and click [Refresh] button, then save the data. When this application is started after one or more scans are made, data can be immediately saved.

6.15 System Table Editor

6.15.1 General

This is a tool to change the system table. Using the keyboard or mouse, the system table can be changed by updating parameters and saving them into the H/D or memory. The parameters displayed after start-up are the ones read from the memory.

6.15.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click System Table Editor in the displayed items and click [OK] or double-click the selected item.

6.15.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.15-1. The parameters displayed after start-up are the ones read from the memory.

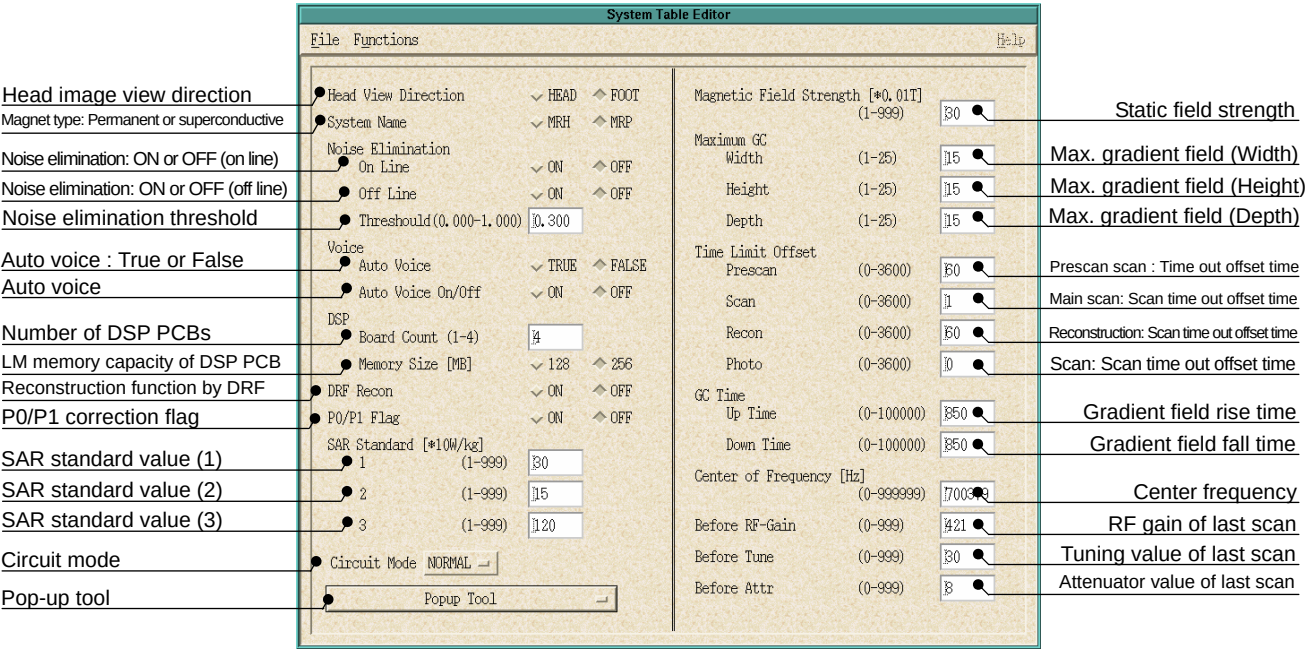


Fig. 6.15-1 System Table Editor Display Screen

6.15.4 Operation procedure

Follow the next steps to execute the process.

6.15.4.1 Updating parameters

Using the keyboard and mouse cursor, update parameters. The updating can be done in the following procedures.

6.15.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

Changed parameter can be confirmed by the color of the entry area changed to blue.

6.15.4.1.2 Selection of item in option menu

The parameters displayed in option menu can be changed by selecting the desired item from the menu displayed by clicking the button.

6.15.4.1.3 Selection of single selection button

Parameter displayed by a single selection button can be changed by clicking the deselected button.

Table 6.15-1 Parameter Setting Range

Parameter Name	Available Range for setting	Remarks
Head image view direction	HEAD, FOOT	Unchangeable item
Type of magnet: Permanent or Superconductive	MRH, MRP	Unchangeable item
Noise elimination selection (on line)	ON, OFF	
Noise elimination selection (off line)	ON, OFF	
Noise elimination threshold	0.000 ~ 1.000	
Auto voice type	TRUE, FALSE	Unchangeable item
Auto voice	ON, OFF	Unchangeable item
Number of DSP PCBs	1 ~ 4	
LM memory capacity of DSP PCB	128, 256	
Reconstruction function by DRF	ON, OFF	Unchangeable item
P0/P1 correction flag	ON, OFF	
SAR standard value	1 ~ 999	Unchangeable item
Circuit mode	NORMAL, EPI	Unchangeable item
Static field strength	1 ~ 999	Unchangeable item
Max. gradient field	1 ~ 25	Unchangeable item
Prescan scan time out offset time	0 ~ 3600	Unchangeable item
Main scan time out offset time	0 ~ 3600	Unchangeable item
Reconstruction scan time out offset time	0 ~ 3600	Unchangeable item
Scan time out offset time	0 ~ 3600	Unchangeable item
Gradient field rise time	0 ~ 100000	Unchangeable item
Gradient field fall time	0 ~ 100000	Unchangeable item
Center frequency	0 ~ 999999	Unchangeable item
RF gain of last scan	0 ~ 999	Unchangeable item
Tuning value of last scan	0 ~ 999	Unchangeable item
Attenuator value of last scan	0 ~ 999	Unchangeable item

6.15.4.2 Reading system table

Select [Read] from "Functions Menu" and select either [Memory] or [Harddisk] from the opened menu. [Memory] and [Harddisk] show data stored in the front end memory and H/D respectively.

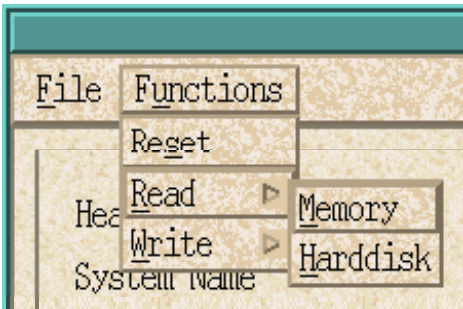


Fig. 6.15-2 Read Button

6.15.4.3 Writing system table

Select [Save] from "File Menu", and either click [H/D] or [Memory] on the dialog shown in Fig. 6.15-4 or select [Write] from "Functions Menu". Then, select either [Memory] or [Harddisk] on the opened menu. [Memory] and [Harddisk] show data stored in the front end memory and H/D respectively.

To exit from the process without saving the dialog shown in Fig. 6.15-4, click [Cancel].



Fig. 6.15-3 Save Button

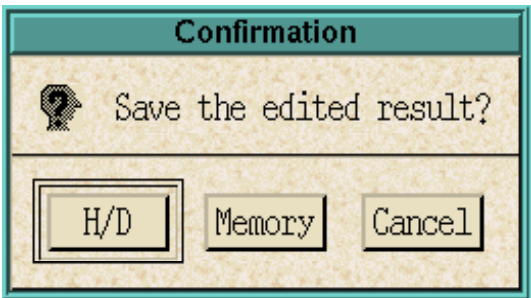


Fig. 6.15-4 Dialog for Saving

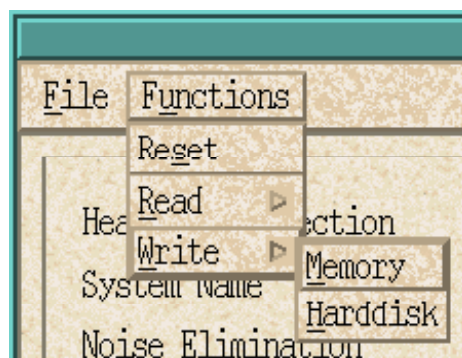


Fig. 6.15-5 Write Button

6.15.4.4 Resetting changed data

Select [Reset] from "Functions Menu". Doing so cancels change of parameter and the whole system table returns to the status where the last reading or writing was made.



Fig. 6.15-6 Reset Button

6.15.4.5 Starting pop-up tool

Click [Popup Tool] on the left bottom of the main screen. By selecting any item on the displayed menu, the selected tool is popped up. However, while the pop-up tool being displayed, editing operation on the main screen cannot be done.

Table 6.15-2 Pop-up Tool Menu

Pop-up Tool Name	Paragraph to be referred
Coil Data Entry Tool	6.15.4.6
FS GC Parameter Setting Tool	6.15.4.7
GC Parameter Setting Tool	6.15.4.8
Calibration Scan Setting Tool	6.15.4.9
EPI Setting Tool	6.15.4.10
MPG Pulse Coefficient Setting Tool	6.15.4.11
FatSep (2PT)Coefficient Setting Tool	6.15.4.12
FatSep (3PT)Coefficient Setting Tool	6.15.4.13
F0 Shift Correction Setting Tool	6.15.4.14

6.15.4.6 Coil data entry tool

This tool is used to input (enter), change, or delete the coil information.

6.15.4.6.1 Display screen

The display screen and function names are shown in Fig. 6.15-7~Fig. 6.15-10.

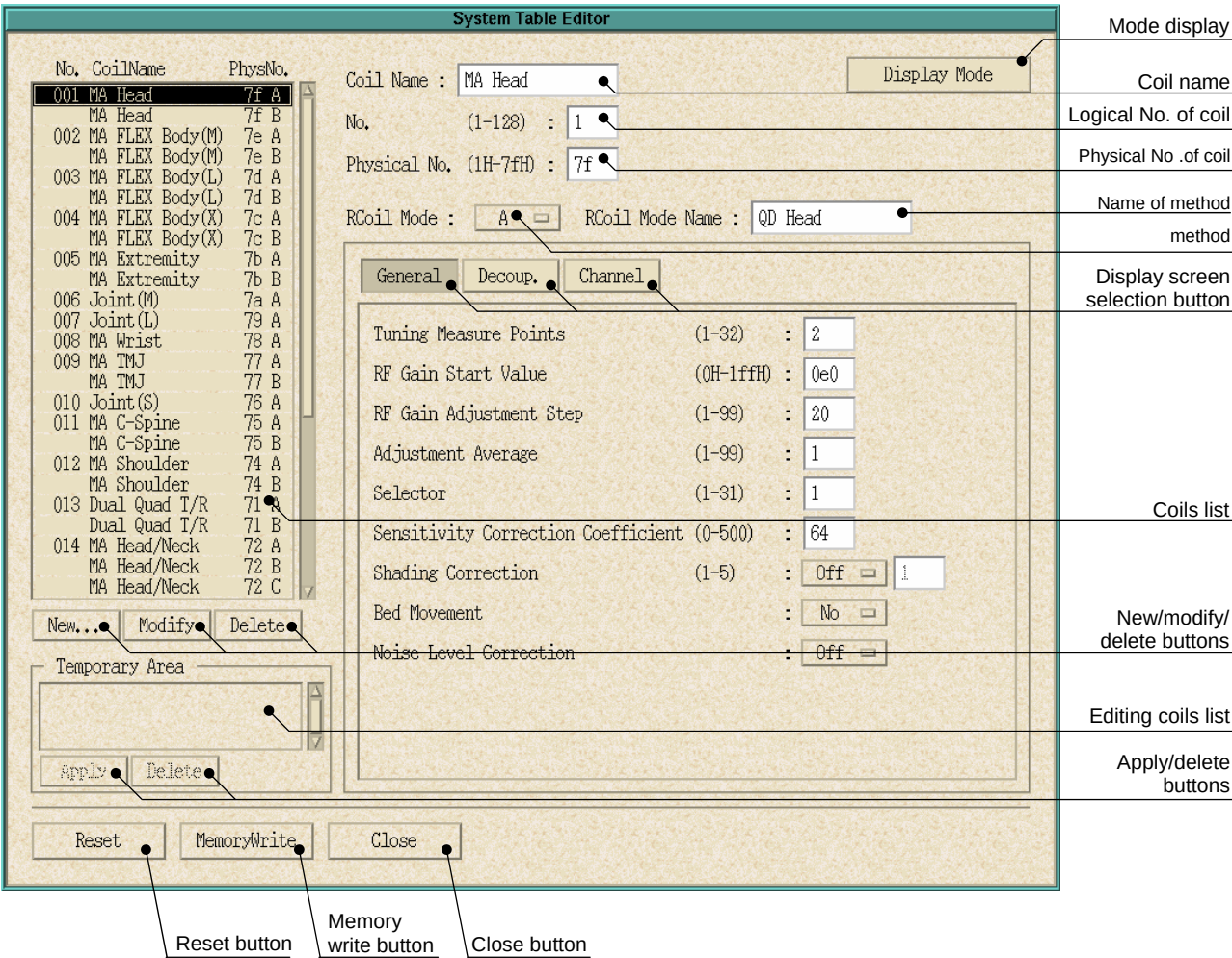


Fig. 6.15-7 Coil Data Entry Tool Display Screen

GeneralDecoup.Channel

Tuning Measure Points

(1-32)

:

2

No. of tuning measuring points

RF Gain Start Value

(0H-1ffH)

:

0e0

RF gain adjusting start value

RF Gain Adjustment Step

(1-99)

:

20

RF gain adjusting step value

Adjustment Average

(1-99)

:

1

NSA during RF gain adjustment

Selector

(1-31)

:

1

Coil selector mode

Sensitivity Correction Coefficient

(0-500)

:

64

MAC sensitivity correction coefficient

Shading Correction

(1-5)

:

Off

1

Shading correction

Bed Movement

:

No

Table move enable/disable

Noise Level Correction

:

Off

Noise level correction

Fig. 6.15-8 Coil Data Entry Tool Display Screen (General)

GeneralDecoup.Channel

No.	Current (0H-ffH)	
1	64	Receiver 1
2	64	Receiver 2
3	2f	Trans 1
4	00	Trans 2
5	00	Trans 3 / Receiver SW 1
6	00	Trans 4 / Receiver SW 2
7	00	Receiver 3
8	00	Receiver 4
9	00	Receiver 5
10	00	Receiver 6
11	00	Receiver 7
12	00	Receiver 8
13	00	Receiver 9
14	00	Receiver 10
15	00	Receiver 11
16	00	Receiver 12
17	00	Receiver 13
18	00	Receiver 14
19	00	Receiver 15
20	00	Receiver 16

Decoupler operation mode + current
(No. 1~20)

Fig. 6.15-9 Coil Data Entry Tool Display Screen (Decoup.)

General Decoup. **Channel**

Number of Receiver Channels : 2

RAPID Factor Interlock (1.0~8.0) : X: 1.0 Y: 1.0 Z: 1.0

Channel	Reception Line	QD Attenuation Value	MAC Amp. Correction (1~200)
1	3	0	100
2	0	0	100
3	0	0	100
4	0	0	100
5	0	0	100
6	0	0	100
7	0	0	100
8	0	0	100

No. of receiving channels for actual use

RAPID Factor Interlock

MAC amplitude correction coefficient (Channel 1~8)

QD attenuation value (Channel 1~8)

Receiving line division (Channel 1~8)

Fig. 6.15-10 Coil Data Entry Tool Display Screen (Channel)

6.15.4.6.2 Update of parameters

For the method of updating the parameters, refer to 6.15.4.1.

Table 6.15-3 Parameter setting ranges (Coil Data Entry Tool)

Display screens	Parameter names	Settable ranges
Common	Coil name	ASCII code
	Logical No. of coil	1 ~ 128
	Physical No. of coil	1H ~ 7fH
	Method	A ~ H, J, K
	Name of method	ASCII code
General	No. of tuning measuring points	1 ~ 32
	RF gain adjusting start value	0H ~ 1ffH
	RF gain adjusting step value	1 ~ 99
	NSA during RF gain adjustment	1 ~ 99
	Coil selector mode	1 ~ 31
	MAC sensitivity correction coefficient	0 ~ 500
	Shading correction	Off, On(1 ~ 5)
	Table move enable/disable	Yes, No
	Noise Level Correction	Off, On
Decoup.	Decoupler operation mode + current (No. 1 ~ 20)	0H ~ ffH
Channel	No. of receiving channels for actual use	1 ~ 8
	RAPID Factor Interlock (X, Y, Z)	1.0 ~ 8.0
	Receiving line division (Channel 1 ~ 8)	0, 1, 2, 3
	Attenuation value (Channel 1 ~ 8)	0, -2, -4, -6
	MAC amplitude correction coefficient (Channel 1 ~ 8)	1 ~ 200

6.15.4.6.3 Coil information display

Click the desired coil information to be displayed out of the coil list and the editing coils list. Each coil information is displayed in the display area. The display area comprises three screens consisting of [General], [Decoup.], and [Channel]. These screens can be switched by clicking a corresponding button, respectively.

6.15.4.6.4 New entry procedure of coil information

(1) Click [New...] button

A new preparation data selection screen is displayed as shown in Fig. 6.15-11.

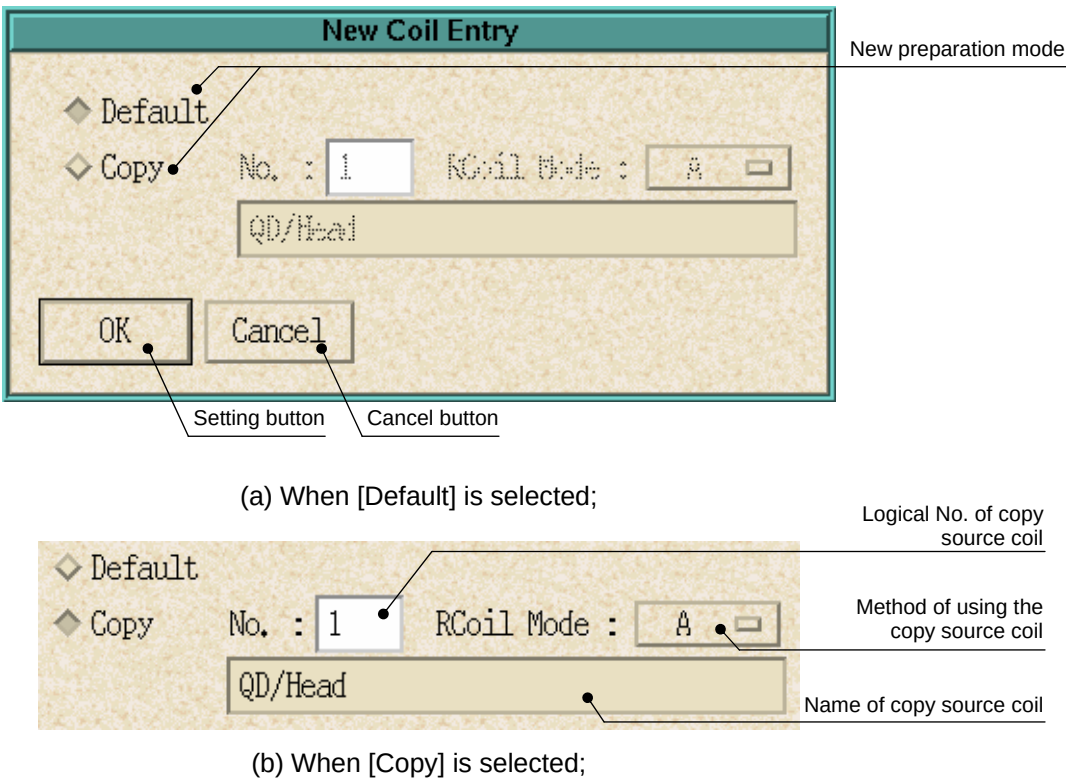


Fig. 6.15-11 New preparation data selection screen

(2) Select the new preparation mode.

Select the new preparation mode by clicking [Default] when data are prepared from the default coil information or click [Copy] when data are prepared by copying entry coil information. In case of [Copy], designate the logical number of coil of copy source data and method, and check the coil name. If the combination of the designated coil number and method of

coil is not entered yet, [***NOT FOUND***] is displayed in the coil name to cause [OK] button to be not operable.

When the new preparation data selection screen is displayed under the selected condition of the list of coils, the logical number and application method of the selected coil are set first. In other cases, logical No of coil [0] and method of coil [A] are set first.

(3) Click [OK] button

A new data item is added to the editing coils list, and the mode display is changed from [Display Mode] to [Edit Mode].

(4) Change the coil information parameters

Change each parameter according to Table 6.15-3.

(5) Click [Apply] button

The changed coil information is defined and entered in the coils list.

The new item is added to the coils list and the mode display is changed from [Edit Mode] to [Display Mode].

The following dialogues may be displayed during entry. In such a case, take necessary remedial measures.

Table 6.15-4 Warning dialogues during new entry

Messages	Meanings	Remedial methods
Overwrite the parameters of this coil?	Do you want to overwrite the parameters of this coil?	[Yes] button : Overwritten [No] button : Entry is canceled
This physical no. overlap.	Physical No overlaps.	This data cannot be entered. Check physical No. If not abnormal, it shows that a coil having this physical No. is already entered. Use the same value to logical No.
Unify physical no. of this coil?	Do you want to unify physical No. of this coil?	[Yes] button : Unified to new physical No. [No] button : Entry is canceled
Specify logical no. and physical no. of this coil.	Specify logical no. and physical no. of this coil.	Specify logical no. and physical no. of this coil.

6.15.4.6.5 Edition and entry procedures of coil information.

- (1) Select the coil information to be edited from the coils list and click [Modify] button.

An edition data item is added to the editing coils list, and the mode display is changed from [Display Mode] to [Edit Mode].

[Edit Mode] is also selected by directly changing parameters after selecting the coil information to be edited from the coils list.

- (2) Change the coil information parameters

Change each parameter according to Table 6.15-3.

- (3) Click [Apply] button

The changed coil information is defined and entered in the coils list. The item prepared in (1) is replaced with the edition source item and the mode display is changed from [Edit Mode] to [Display Mode].

The following dialogues may be displayed during entry. In such a case, take necessary remedial measures.

Table 6.15-5 Warning dialogues during new entry

Messages	Meanings	Remedial methods
Overwrite the parameters of this coil?	Do you want to overwrite the parameters of this coil?	[Yes] button : Overwritten [No] button : Entry is canceled
This physical no. overlap.	Physical No overlaps.	This data cannot be entered. Check physical No. If not abnormal, it shows that a coil having this physical No. is already entered. Use the same value to logical No.
Unify physical no. of this coil?	Do you want to unify physical No. of this coil?	[Yes] button : Unified to new physical No. [No] button : Entry is canceled

6.15.4.6.6 Deletion of coil information

Select the desired coil information to be deleted from the coils list and click [Delete] button of the coils list. A dialog is displayed as shown in Fig. 6.15-12. By clicking [Yes], the selected coil information is deleted. By clicking [No], the deletion is canceled.

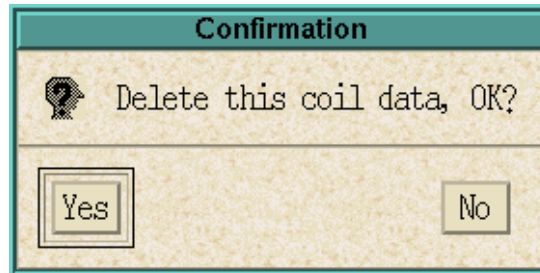


Fig. 6.15-12 Coil Information Deletion Check Dialog

6.15.4.6.7 Deletion of editing coil information

Select the editing coil information to be deleted from the editing coils list, and click [Delete] button in the editing coils list.

6.15.4.6.8 Resetting the coil information

Click [Reset] button in the tool. The coil information is reset to the data when these data were saved into memory or H/D last.

Note: All the coil information of editing coils list is discarded.

6.15.4.6.9 Writing into memory

Click [Memory Write] button in the tool.

6.15.4.6.10 Termination of coil data entry tool

Click [Close] button on the tool, and the coil data entry tool is terminated. The entered coil information is saved into memory even after this termination.

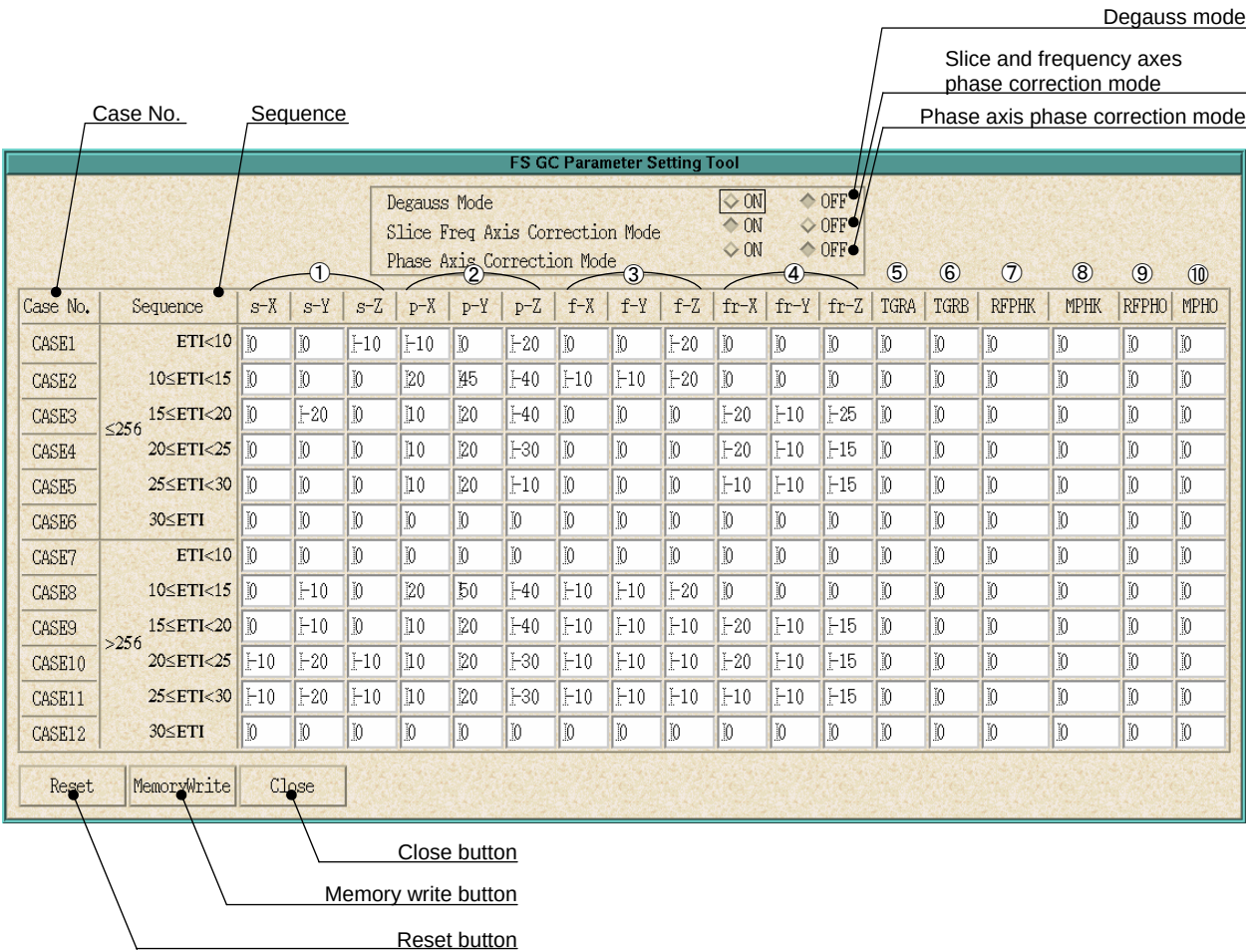
Note: All the coil information of editing coils list is discarded.

6.15.4.7 GC parameter setting tool for FS

This is a tool to set parameters for FS.

6.15.4.7.1 Display screen

The display screen and function names are shown in Fig. 6.15-13.



- ① Adjustment items in slice direction (X, Y, Z)
- ② Adjustment items in phase direction (X, Y, Z)
- ③ Adjustment items in frequency direction (X, Y, Z)
- ④ Adjustment items in frequency direction (Rephase)(X,Y,Z)
- ⑤ First echo correction value in frequency direction
- ⑥ Second echo correction value in frequency direction
- ⑦ Excitation phase coefficient
- ⑧ Reception phase coefficient
- ⑨ Excitation phase offset
- ⑩ Reception phase offset

Fig. 6.15-13 GC Parameter Setting Tool Screen for FS

6.15.4.7.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1. Some of the unchangeable parameters can, however, be changed by using FSE Auto Adjustment, and only confirmation is made in this Paragraph.

Table 6.15-6 Parameter Setting Range(GC parameter setting tool for FS)

Parameter Name	Available Range for setting	Remarks
Adjustment items in slice direction (X, Y, Z)	-999 ~ 999	Unchangeable item
Adjustment items in phase direction (X, Y, Z)	-999 ~ 999	Unchangeable item
Adjustment items in frequency direction (X, Y, Z)	-999 ~ 999	Unchangeable item
Adjustment items in frequency direction (X, Y, Z)	-999 ~ 999	Unchangeable item
First echo correction in frequency direction	-999 ~ 999	Unchangeable item
Second echo correction in frequency direction	-999 ~ 999	Unchangeable item
Degauss mode	ON, OFF	Unchangeable item
Slice and frequency axes phase correction mode	ON, OFF	Unchangeable item
Phase axis phase correction mode	ON, OFF	Unchangeable item
Excitation phase coefficient	-2000 ~ 2000	Unchangeable item
Reception phase coefficient	-2000 ~ 2000	Unchangeable item
Excitation phase offset	-180 ~ 180	Unchangeable item
Reception phase offset	-180 ~ 180	Unchangeable item

6.15.4.7.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.7.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.7.5 Exit from GC parameter setting tool for FS

Click [Close] button in the tool. The GC parameter setting tool for FS is closed.

At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.8 GC parameter setting tool

This is a tool to set GC parameters.

6.15.4.8.1 Display screen

The display screen and function names are shown in Fig. 6.15-14.

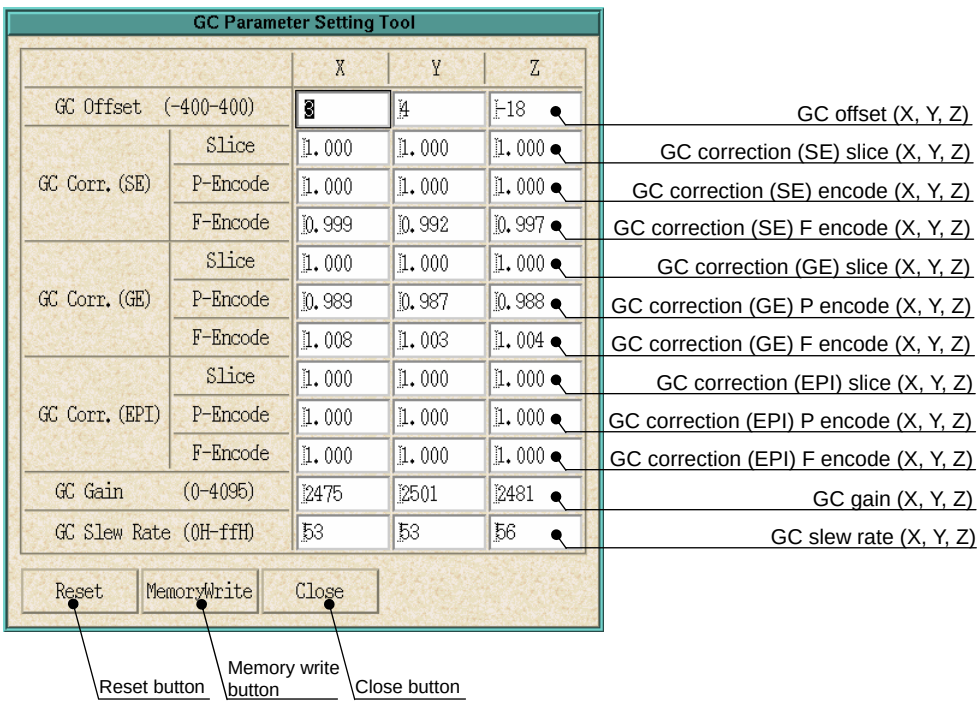


Fig. 6.15-14 GC Parameter Setting Tool Screen

6.15.4.8.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1. Some of the unchangeable parameters can be changed by using GC Offset Adjustment, GC Gain Parameter Setting, FOV Adjustment, GC Slew Rate Adjustment, etc., and only confirmation is made in this Paragraph.

Table 6.15-7 Parameter Setting Range (GC parameter setting tool)

Parameter Name	Available Range for setting	Remarks
GC offset (X, Y, Z)	-400 ~ 400	Unchangeable item
GC correction (SE, GE, EPI) slice (X, Y, Z)	-999.000 ~ 999.000	
GC correction (SE, GE, EPI) P encode (X, Y, Z)	-999.000 ~ 999.000	Unchangeable item
GC correction (SE, GE, EPI) F encode (X, Y, Z)	-999.000 ~ 999.000	Unchangeable item
GC gain (X, Y, Z)	0 ~ 4095	Unchangeable item
GC slew rate (X, Y, Z)	0H ~ ffH	Unchangeable item

6.15.4.8.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.8.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.8.5 Exit from GC parameter setting tool

Click [Close] button in the tool. The GC parameter setting tool is closed.

At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.9 Calibration Scan Setting Tool

This is a tool to set the execution flag for calibration scan.

6.15.4.9.1 Display screen.

The display screen and function names are shown in Fig. 6.15-15.

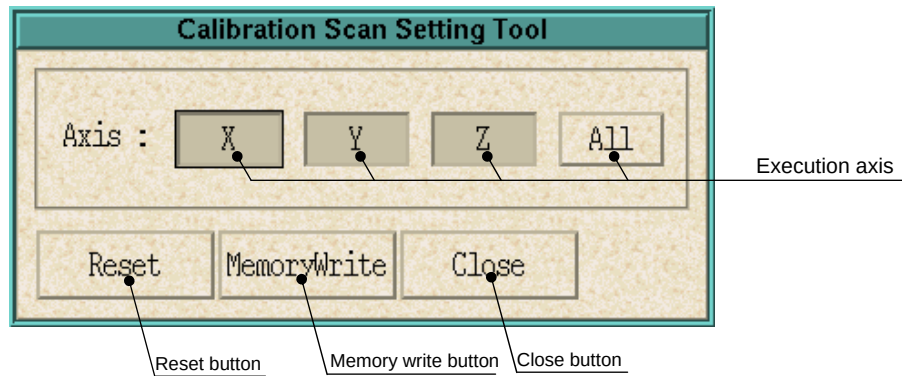


Fig. 6.15-15 Calibration Scan Setting Tool Screen

6.15.4.9.2 Selection of adjustment axis

Select an axis to execute the calibration scan in pre-scan. By clicking an axis indicating each axis, the axis to be adjusted is selected. More than one axis can be selected. Clicking [All] button sets adjustment of all axes.

Table 6.15-8 Parameter Setting Range (Calibration scan)

Parameter Name	Setting range
Calibration execution flag (X, Y, Z, All)	ON, OFF

6.15.4.9.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.9.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.9.5 Exit from calibration scan setting tool

Click [Close] button in the tool. The calibration scan setting tool is closed. At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.10 EPI Setting Tool

This is a tool to set System, Shim and EPI parameters.

6.15.4.10.1 Display screen

The display screen and function names are shown in Fig. 6.15-16.

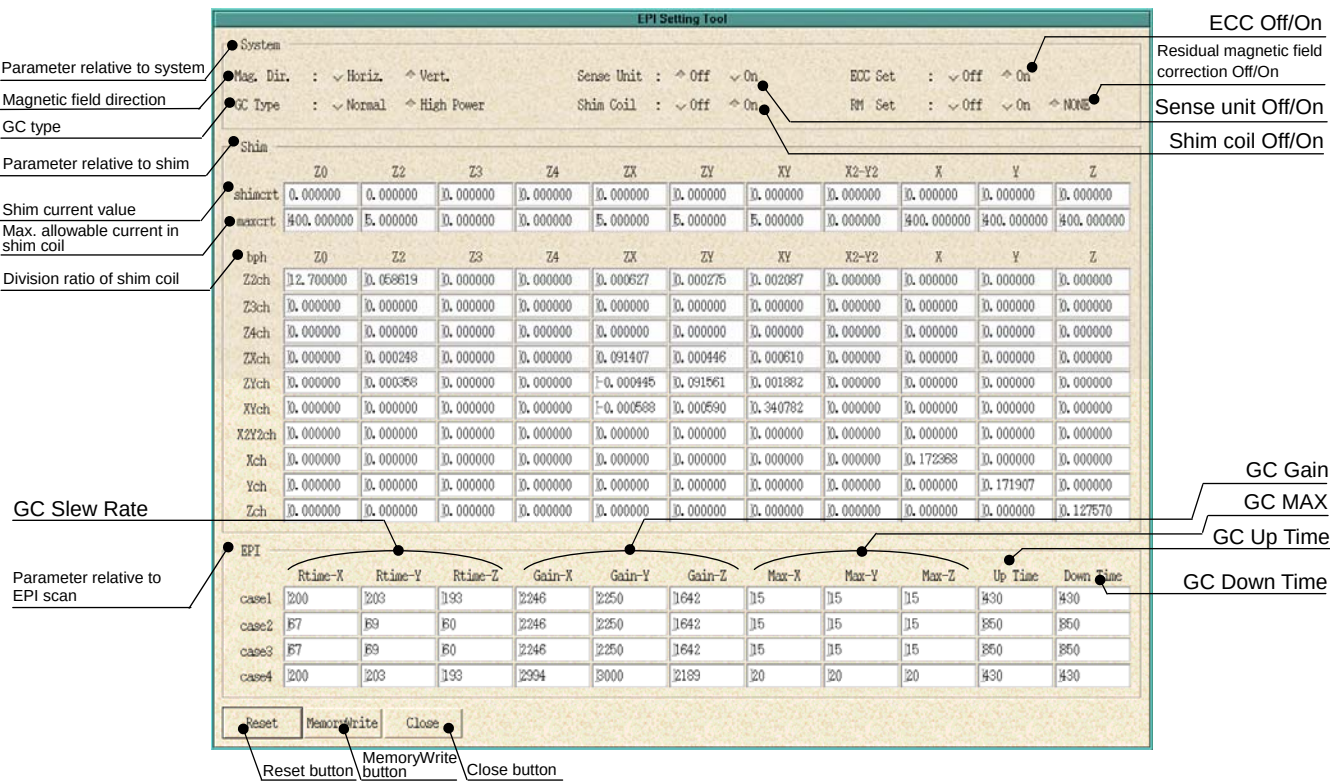


Fig. 6.15-16 EPI Setting Tool Screen

6.15.4.10.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1.

Table 6.15-9 Parameter Setting Range (EPI Setting Tool)

Parameter Name		Setting range	Remarks
System	Magnetic field direction	Horiz., Vert.	Unchangeable item
	GC type	Normal, High Power	Unchangeable item
	Sense unit Off/On	Off, On	Unchangeable item
	Shim coil Off/On	Off, On	Unchangeable item
	ECC Off/On	Off, On	Unchangeable item
	Residual magnetic field correction Off/On	Off, On, NONE	Unchangeable item
Shim	Shim current value	-999999999~999999999	Unchangeable item
	Max. allowable current in shim coil	0~400	Unchangeable item
	Division ratio of shim coil	-----	Unchangeable item
EPI	GC Slew Rate-X,Y,Z	0~255	Unchangeable item
	GC Gain-X,Y,Z	0~4095	
	GC Max-X,Y,Z	1~25	Unchangeable item
	Up time of GC	0~100000	Unchangeable item
	Down time of GC	0~100000	Unchangeable item

6.15.4.10.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.10.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.10.5 Exit from EPI Setting Tool

Click [Close] button in the tool. The EPI Setting Tool is closed.

At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.11 MPG Pulse Coefficient Setting Tool

This is a tool to set MPG pulse parameters.

6.15.4.11.1 Display screen

The display screen and function names are shown in Fig. 6.15-17.

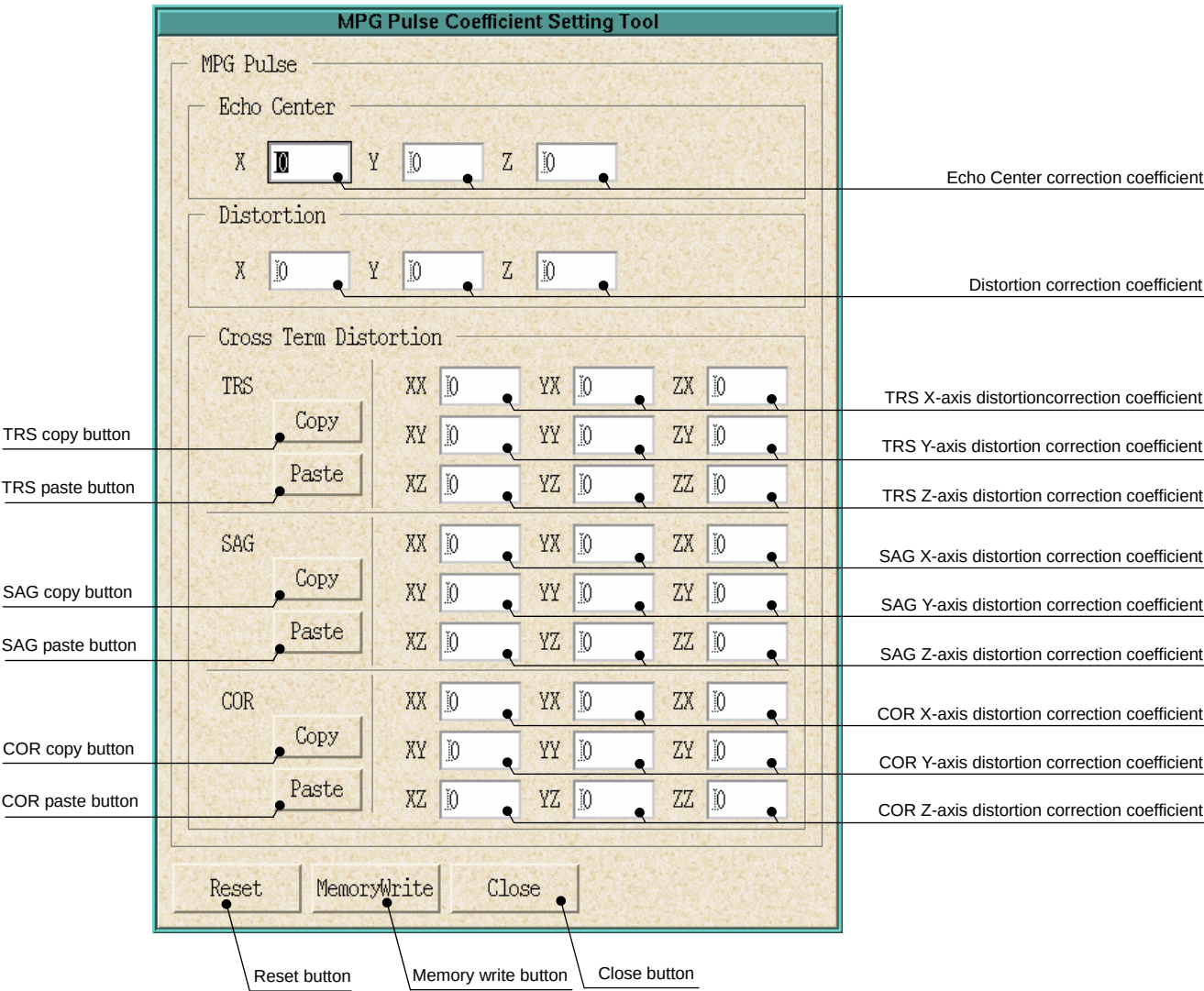


Fig. 6.15-17 MPG Pulse Coefficient Setting Tool Screen

6.15.4.11.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1.

For the cross term distortion correction coefficient, Copy button and Paste button are arranged in each section. The correction coefficient in a certain section can be transferred to the other section.

Table 6.15-10 Parameter Setting Range (MPG Pulse Coefficient Setting Tool)

Parameter Name	Setting range	Remarks
Echo Center correction coefficient -X,Y,Z	-200~200	
Distortion correction coefficient -X,Y,Z	-200~200	
TRS X-axis distortion correction coefficient-XX,YX,ZX	-200~200	Distortion correction coefficient to be applied to X-axis in case of TRS imaging
TRS Y-axis distortion correction coefficient-XY,YY,ZY	-200~200	Distortion correction coefficient to be applied to Y-axis in case of TRS imaging
TRS Z-axis distortion correction coefficient-XZ,YZ,ZZ	-200~200	Distortion correction coefficient to be applied to Z-axis in case of TRS imaging
SAG X-axis distortion correction coefficient-XX,YX,ZX	-200~200	Distortion correction coefficient to be applied to X-axis in case of SAG imaging
SAG Y-axis distortion correction coefficient-XY,YY,ZY	-200~200	Distortion correction coefficient to be applied to Y-axis in case of SAG imaging
SAG Z-axis distortion correction coefficient-XZ,YZ,ZZ	-200~200	Distortion correction coefficient to be applied to Z-axis in case of SAG imaging
COR X-axis distortion correction coefficient-XX,YX,ZX	-200~200	Distortion correction coefficient to be applied to X-axis in case of COR imaging
COR Y-axis distortion correction coefficient-XY,YY,ZY	-200~200	Distortion correction coefficient to be applied to Y-axis in case of COR imaging
COR Z-axis distortion correction coefficient-XZ,YZ,ZZ	-200~200	Distortion correction coefficient to be applied to Z-axis in case of COR imaging

6.15.4.11.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.11.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.11.5 Exit from MPG Pulse Coefficient Setting Tool

Click [Close] button in the tool. The MPG Pulse Coefficient Setting Tool is closed.

At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.12 FatSep(2PT) Coefficient Setting Tool

This is a tool to set the Fat/Water separation(2PT) correction coefficient.

6.15.4.12.1 Display screen

The display screen and function names are shown in Fig. 6.15-18.

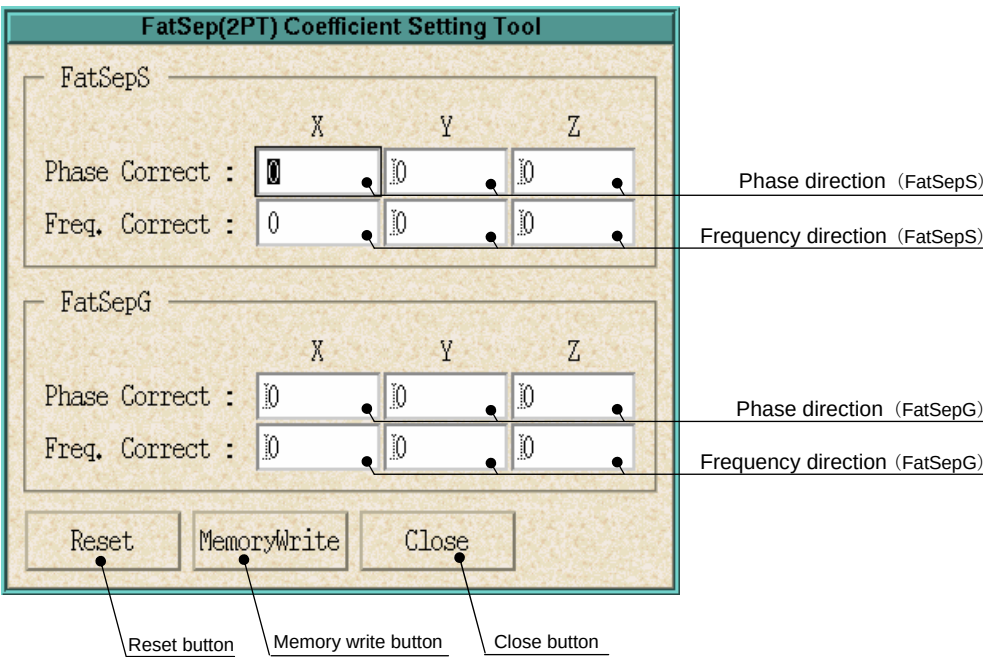


Fig. 6.15-18 FatSep(2PT) Coefficient Setting Tool Display Screen

6.15.4.12.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1.

Table 6.15-11 Parameter Setting Range (FatSep(2PT) Coefficient Setting Tool)

Parameter Name		Setting Range	Remark
FatSepS	Phase direction -X,Y,Z	-500~500	
	Freq. direction -X,Y,Z	-500~500	
FatSepG	Phase direction -X,Y,Z	-500~500	
	Freq. direction -X,Y,Z	-500~500	

6.15.4.12.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.12.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.12.5 Exit from FatSep(2PT) Coefficient Setting Tool

Click [Close] button in the tool. The FatSep(2PT) Coefficient Setting Tool is closed.

At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.13 FatSep(3PT) Coefficient Setting Tool

This is a tool to set the Fat/Water separation(3PT) correction coefficient.

6.15.4.13.1 Display screen

The display screen and function names are shown in Fig. 6.15-19.

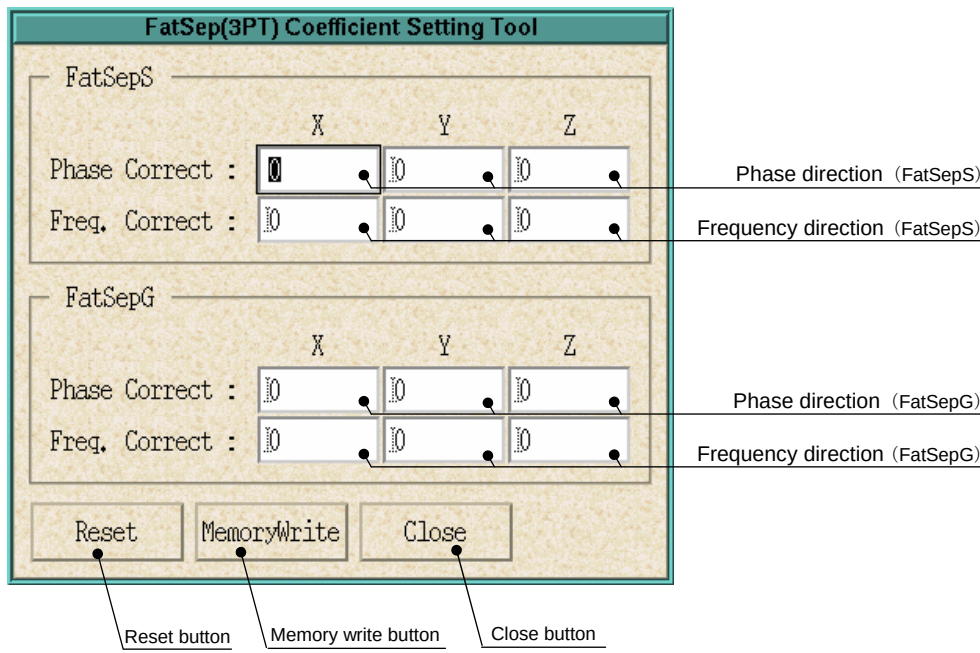


Fig. 6.15-19 FatSep(3PT) Coefficient Setting Tool Display Screen

6.15.4.13.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1.

Table 6.15-12 Parameter Setting Range (FatSep(3PT) Coefficient Setting Tool)

Parameter Name		Setting Range	Remark
FatSepS	Phase direction -X,Y,Z	-500~500	
	Freq. direction -X,Y,Z	-500~500	
FatSepG	Phase direction -X,Y,Z	-500~500	
	Freq. direction -X,Y,Z	-500~500	

6.15.4.13.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.13.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.13.5 Exit from FatSep(3PT) Coefficient Setting Tool

Click [Close] button in the tool. The FatSep(3PT) Coefficient Setting Tool is closed.

At this time, the changed parameter is kept memorized after closing the tool.

6.15.4.14 F0 Shift Correction Setting Tool

This is a tool to set the parameters for F0 Shift Correction.

6.15.4.14.1 Display screen

The display screen and function names are shown in Fig. 6.15-20.

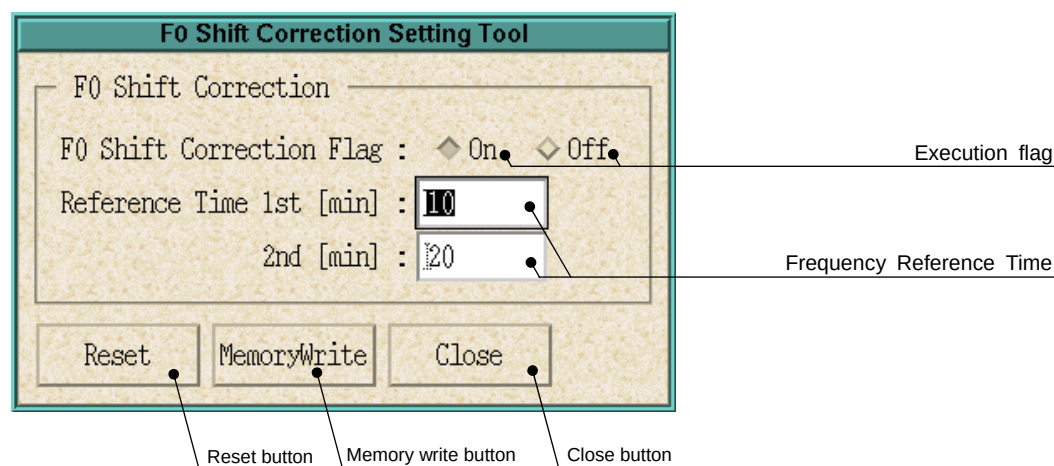


Fig. 6.15-20 F0 Shift Correction Setting Tool Display Screen

6.15.4.14.2 Parameter change

For parameter change procedure, refer to Paragraph 6.15.4.1.

Table 6.15-13 Parameter Setting Range (F0 Shift Correction Setting Tool)

Parameter Name	Setting Range	Remark
Execution flag	On, Off	Unchangeable item
Frequency Reference Time	-200~200	Unchangeable item

6.15.4.14.3 Recovery of parameter

Click [Reset] button in the tool. Parameters are reset to the data when these data were saved into memory or H/D last.

6.15.4.14.4 Writing into Memory

Click [Memory Write] button in the tool.

6.15.4.14.5 Exit from F0 Shift Correction Setting Tool

Click [Close] button in the tool. The F0 Shift Correction Setting Tool is closed. At this time, the changed parameter is kept memorized after closing the tool.

6.15.5 End

Selecting [Quit] button from "File Menu" in the menu bar pops up the dialog shown in Fig. 6.15-22. Clicking [Yes] button on this dialog ends the process, while clicking [No] button returns to the previous screen without ending the process.



Fig. 6.15-21 Quit Button



Fig. 6.15-22 Exit Confirmation Dialog

In case any parameter is changed and is not yet saved, the dialog shown in Fig. 6.15-23 is popped up. Clicking [Yes] button writes the changed parameter into the system table (H/D and memory) and the application is closed. To close the application without writing it into the system table, click [No] button. Clicking [Cancel] button returns to the previous screen without closing the application.

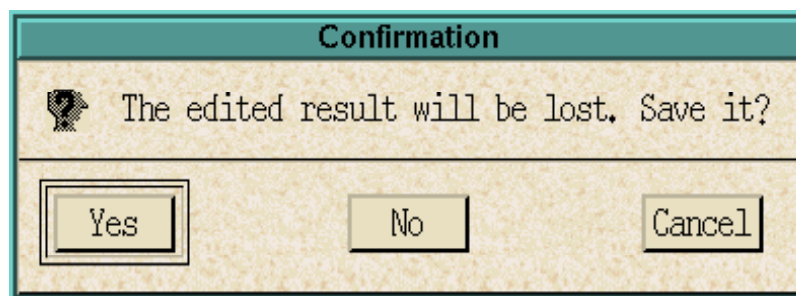


Fig. 6.15-23 Save Confirmation Dialog

6.15.6 Error specifications

Error specifications are shown in Table 6.15-14.

Table 6.15-14 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Can't write the SYSTEM_TABLE.	Writing the system table has failed.	Checking the system status, retry the same operation procedure.
Event Invalid.	Request is not accepted.	Checking the system status, retry the same operation procedure.

6.16 Easy Magnetic Field Analysis

6.16.1 General

This is a tool to measure the field strength at the specified position in the magnetic field and to perform magnetic analysis and estimated analysis from the obtained data.

6.16.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Mag. Field Analysis in the displayed items and click [OK] or double-click the selected item.

6.16.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.16-1.

Though it somewhat takes time to initialize at starting, that is not abnormal.

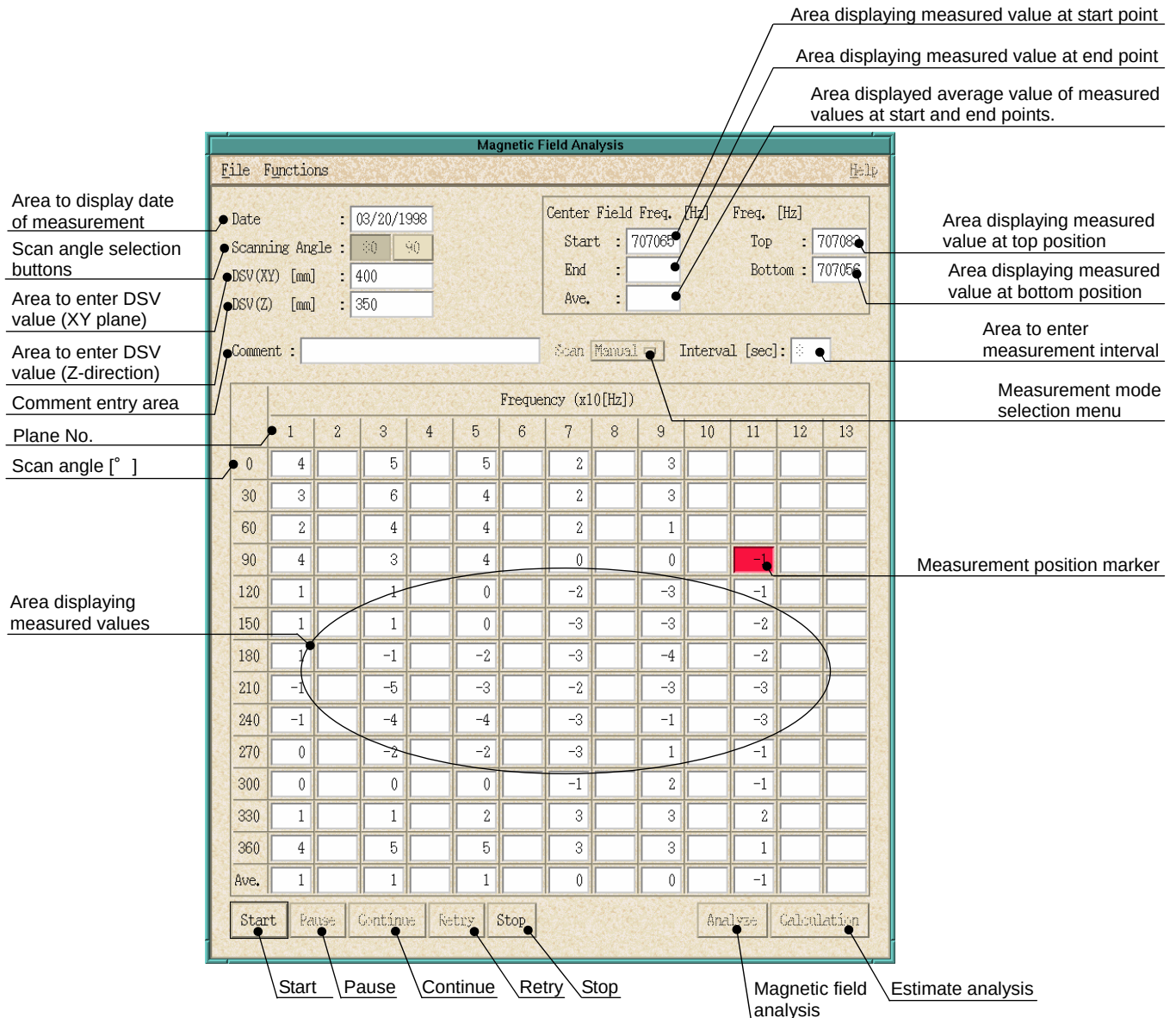


Fig. 6.16-1 Easy Magnetic Field Analysis Display Screen

6.16.4 Operation procedure

Follow the next steps to execute the process.

6.16.4.1 Updating parameters

Using the keyboard and mouse cursor, change parameters. The following methods are available to change parameters.

6.16.4.1.1 Selection of parameter by pushbutton

Move the mouse cursor to the button and click it to select the parameter to be changed.

6.16.4.1.2 Input of character string in entry area

Move the mouse cursor to the entry area of the parameter to be changed and select the entry area. Using the keyboard, input figures or a character string, and change the parameter by either pressing <Enter> on the keyboard or removing the focus.

Table 6.16-1 Parameter Setting Range

Parameter Name	Available Range for setting	Remarks
Scan angle	30° / 90°	
DSV (XY) [mm]	50 ~ 450	
DSV (Z) [mm]	50 ~ 450	
Comment	32 characters	
Measurement mode	Manual / Auto	
Measurement interval	3 ~ 120	Only for Auto measurement mode

6.16.4.2 Measurement start in Manual mode

The measurement in Manual mode is started by selecting Manual on "Scan Option Menu" and clicking [Start]. The measurement point during measurement is indicated with the measurement position marker.

To retry the measurement, move the mouse cursor to the previous measurement point before the current point and click the mouse button to move the measurement marker.

After the measurements up to the final measurement point (measurement of the center position of measurement end) are completed, click [Analyze] button to perform analysis. When the analysis process is completed normally, the result is automatically saved in Case:1. For details of the storage, refer to Paragraph 6.16.4.12.

To retry the measurement, click [Stop] button and select [Clear All] on "Functions Menu", and the window is cleared and [Start] button is enabled.

6.16.4.3 Measurement start in Auto mode

The measurement in Auto mode is started by selecting Auto on "Scan Option Menu" and clicking [Start] button. The measurement point during measurement is indicated with the measurement position marker, and measurements are automatically carried out in the specified measurement intervals until either the measurement is stopped or up to the final measurement point (measurement of the center position of measurement end).

Clicking [Start] button enables [Pause] button. To suspend the measurement, click this button. Any of [Continue], [Retry] and [Stop] buttons can be selected. For their functions, refer to the following paragraphs.

6.16.4.4 Continuation of measurement

Continuation of measurement means that in case the measurement being suspended the measurement is to be started from the point where the measurement was suspended.

In Manual mode, the measurement can be continued by clicking [Start] button. In Auto mode, with the [Pause] button clicked the measurement can be continued by clicking [Continue] button.

6.16.4.5 Retry of measurement

Retry of measurement means that if the measurement is required again as the result of measurement at a certain point the measurement is to be done from the point before the current position.

In Manual mode, the measurement can be retried by moving the measurement marker to the position from where the measurement is to be started and clicking [Start] button.

In Auto mode, with the [Pause] button clicked the measurement can be made again by clicking [Retry] button.

6.16.4.6 Termination of measurement

Termination of measurement means that the window is cleared and the measurement mode is canceled to carry out the measurement again from the start position.

In Manual mode, the measurement can be terminated by clicking [Stop] button.

In Auto mode, with the [Pause] button clicked the measurement can be terminated by clicking [Stop] button.

To carry out the measurement again, select [Clear All] on "Functions Menu" to return the window to the initial display. Note that data are not automatically saved when the measurement is terminated.

6.16.4.7 Magnetic field analysis

Only when magnetic fields are at all the specified positions are measured, [Analyze] button is enabled.

Because data are not saved unless the analysis process is executed, be sure to execute the analysis process. After executing the analysis process, clicking [Analyze] button displays the analyzed result.

6.16.4.7.1 Analysis result display screen

The window to display analysis result is shown in Fig. 6.16-2.

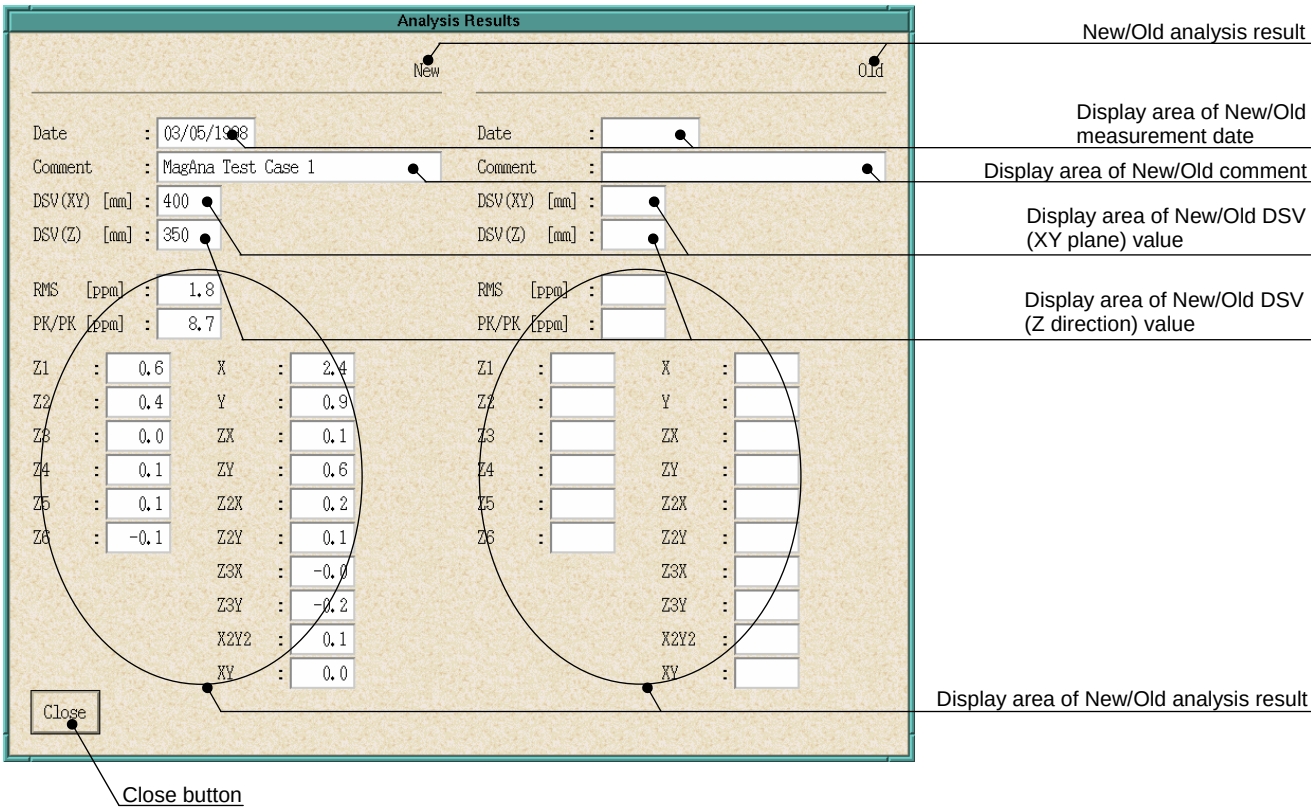


Fig. 6.16-2 Analysis Result Display Screen

6.16.4.7.2 Analysis result

The analysis result for the data being displayed is displayed at the left side of the screen, while the analysis result to the data of the previous measurement is displayed at right side of the screen. In case of the first measurement or reading the saved data after the system start-up, the analysis result display area at the right side of the screen is displayed as blanked.

6.16.4.7.3 Exit from analysis result display screen

Click [Close] button on the left bottom of the screen. This screen can be repeatedly displayed.

6.16.4.8 Calculation analysis

[Calculation] button is enabled only for the data to which analysis process is applied.

Calculation analysis means that the analysis process is executed with the DSV value different from the measured DSV value. Clicking [Calculation] button pops up the dialog shown in Fig. 6.16-3.

Input a DSV value. The input range is same as the DSV value at measurement. The default is the DSV value at measurement.

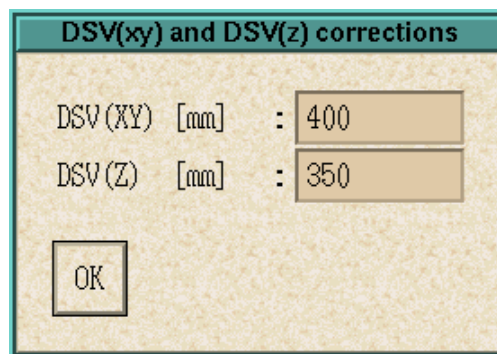


Fig. 6.16-3 Dialog for Entry of DSV Value

6.16.4.8.1 Calculation analysis result display screen

The window to display calculation analysis result is shown in Fig. 6.16-4.

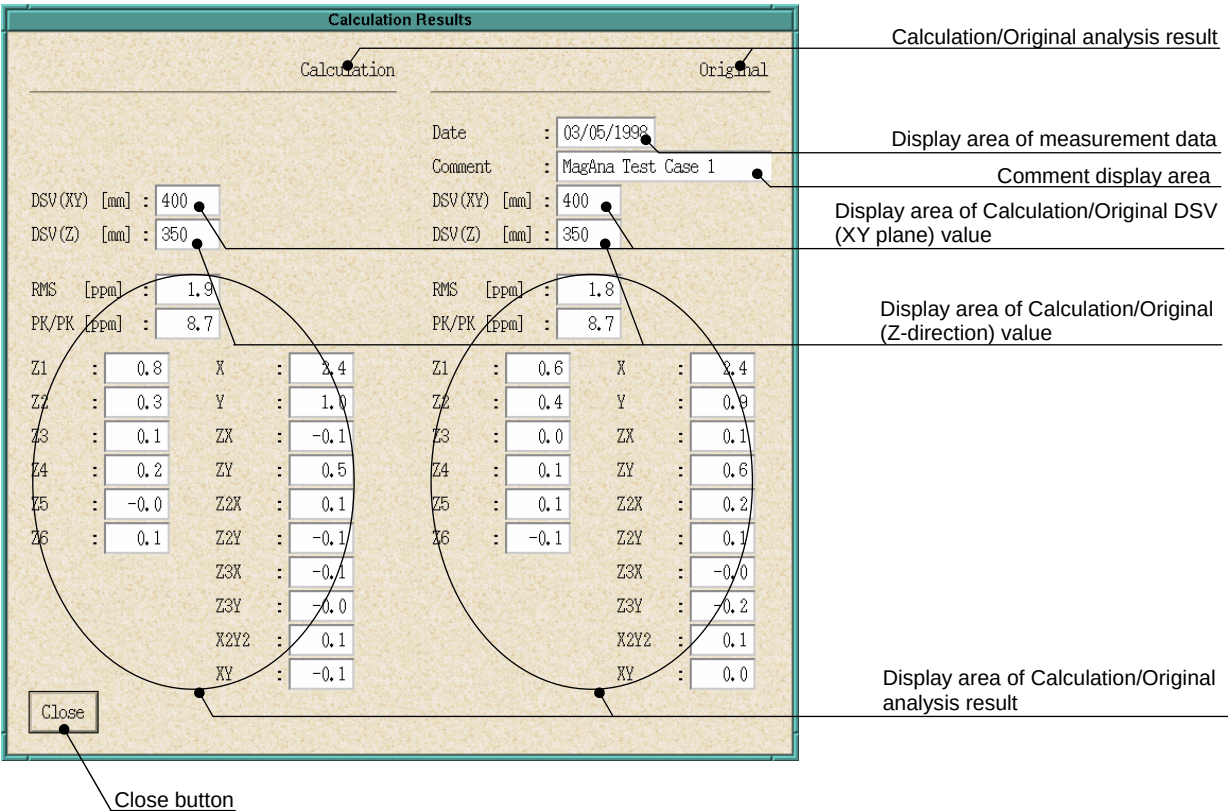


Fig. 6.16-4 Calculation Analysis Result Display Screen

6.16.4.8.2 Calculation analysis result

The calculation analysis result for the DSV value different from the displayed data is displayed at the left side of the screen and the analysis result for the displayed data is displayed at the right side.

6.16.4.8.3 Exit from calculation analysis result display screen

Click [Close] button on the left bottom of the screen. This screen can be repeatedly displayed.

6.16.4.9 Data storage (H/D)

The data displayed on the window is stored in the H/D. Data of up to three measurements can be stored. Click [Save] on "File Menu". The dialog for data storage is popped up as shown in Fig. 6.16-6.

Click the area to be stored in the storage data list on the dialog, and click [OK] to save the data, while click [Cancel] to cancel storage.

The comment displayed below the storage data list is used as comment for storage and the comment on the window is displayed as default. When the comment is changed on the dialog, the comment on the window is also changed. Comments up to 32 characters can be entered.



Fig. 6.16-5 Storage (H/D) Button

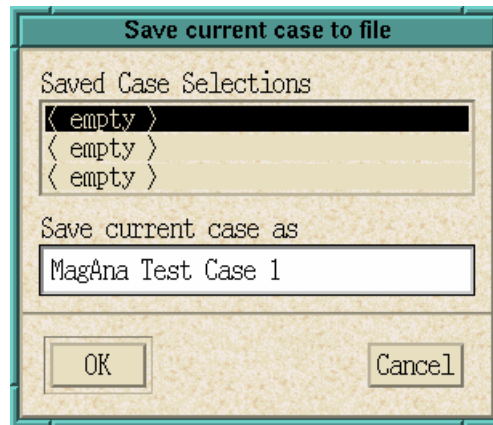


Fig. 6.16-6 Dialog for Storage

6.16.4.10 Opening stored data

Click [Open ...] on "File Menu". The dialog for opening data is displayed as shown in Fig. 6.16-8.

Click the data to be opened on the storage data list display and click [OK] to read the stored data, while click [Cancel] to cancel reading the data.

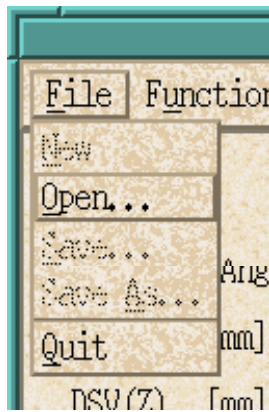


Fig. 6.16-7 Open (H/D) Button

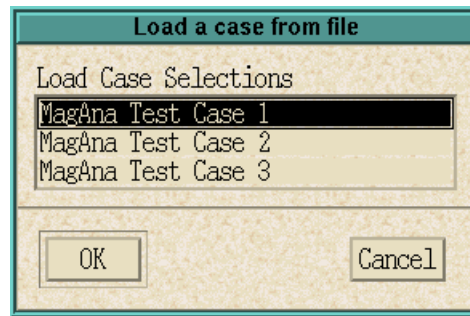


Fig. 6.16-8 Dialog for Opening Data

6.16.4.11 New measurement

To open the new window, select [Clear All] on "Functions Menu". The current window is cleared and a new measurement can be made.

6.16.4.12 Automatic data saving

In case [AutoSaveToggle] is set to ON, the measurement is made up to the last measurement position (measurement end center position), and only the data to which the analysis process was applied are automatically saved. The saved data can be maintained until the tool is exited and upon exiting from the tool, the data is lost. To store the data on the H/D, refer to Paragraph 6.16.4.9.

The latest data is saved in Case:1, and following new data up to 10 measurements are sequentially saved in Case:2, ... , Case:10. In case data are for more than 10 measurements, data are erased from the oldest data (Case:10).

In case [AutoSaveToggle] is set to OFF, data are not saved unless [Save Case] on "Functions Menu" is selected. For [AutoSaveToggle] refer to Paragraph 6.16.4.14.

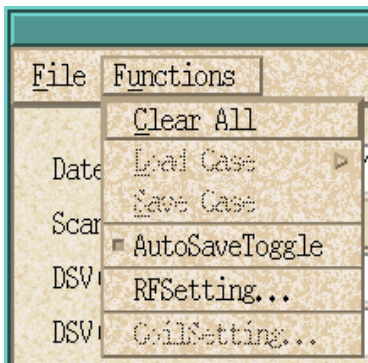


Fig. 6.16-9 Clear All Button

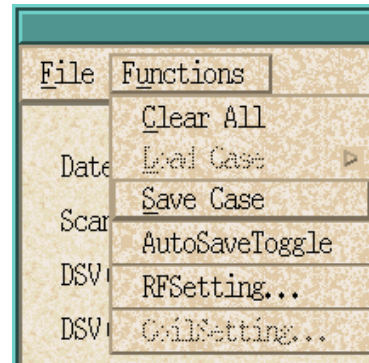


Fig. 6.16-10 Save Case Button

6.16.4.13 Loading stored data

Click [Load Case] on "Functions Menu" to display the menu for stored data.

Select the data to be displayed again on this menu. The data is displayed on the window.

The loaded data can also be stored in the H/D.

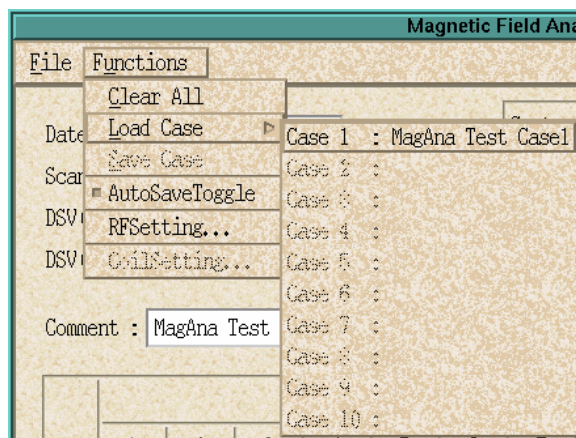


Fig. 6.16-11 Load Case Button

6.16.4.14 Permission of automatic storage

Permission of automatic storage means that measurement is made up to the last measurement position (measurement end center position) and a selection is made whether the data to which the analysis was applied is automatically stored or not. The default is ON for automatic storage.

To set permission of automatic storage, select [AutoSaveToggle] on "Functions Menu".

The pop-up button is OFF, while the push-down button is ON. Data up to 10 measurements can be stored.

6.16.4.15 RF setting

RF gain and receiver gain values are set.

Select [RFSetting ...] from "Functions Menu". RF Setting window will appear.

The value shown in the window is the current set value.

After changing each parameter to desired value, press [Set] button. To cancel the parameter changed in the procedure, press [Cancel] button.

For parameter change procedure, refer to paragraph 6.16.4.1.2.

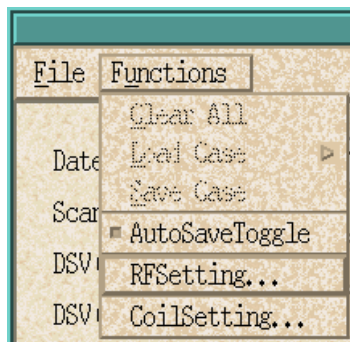


Fig. 6.16-12 RFSetting Button

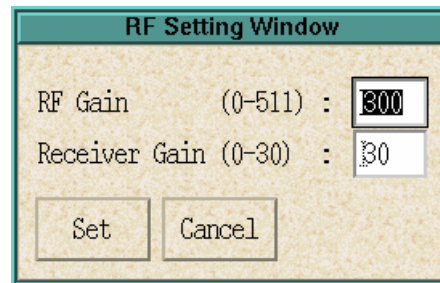


Fig. 6.16-13 RF Setting Window

Table 6.16-2 Parameter Setting Range(RF Setting Window)

Parameter Name	Available Range for setting	Remarks
RFGain	0 ~ 511	
Receiver Gain	0 ~ 30	Only even numbers

6.16.4.16 Coil setting

On the coil setting window, either automatic coil identification or manual setting can be selected.

6.16.4.16.1 Default

The default at start-up is Manual (coil No.: 32). This is the coil No. set for the easy magnetic field analysis.

6.16.4.16.2 Coil setting window display

Select [CoilSetting ...] on "Functions Menu". The coil setting window is popped up. The window at start-up shows the default value.

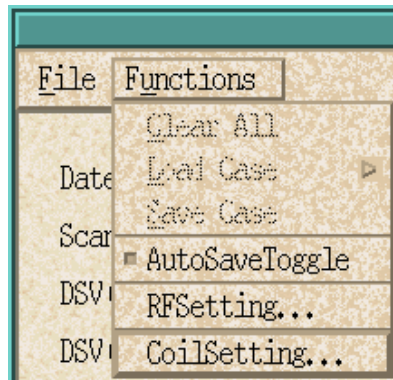


Fig. 6.16-14 Coil Setting Window Display Button

6.16.4.16.3 Display screen

The display screen and function names are shown in Fig. 6.16-15 and Fig. 6.16-16.

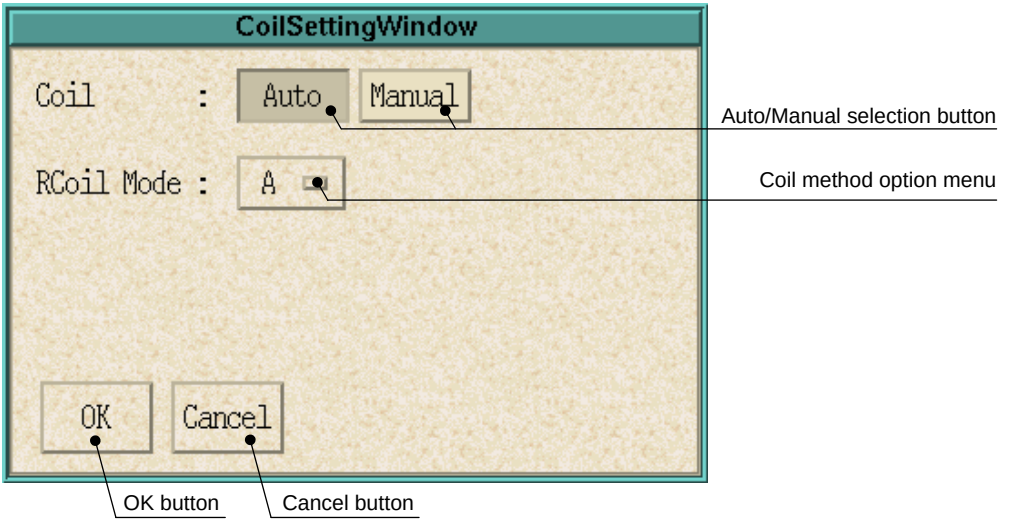


Fig. 6.16-15 Coil Setting Window Display Screen (Auto mode)

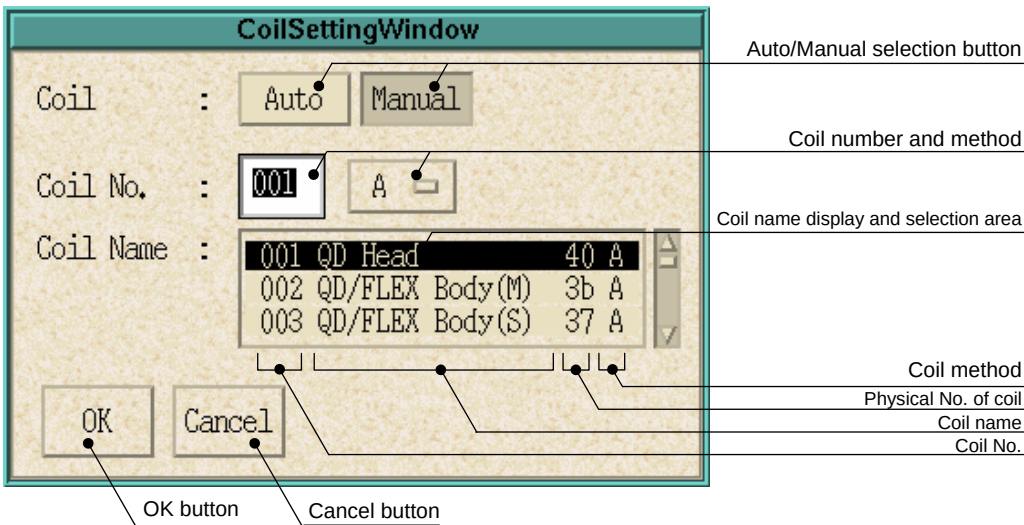


Fig.6.16-16 Coil Setting Window Display Screen (Manual mode)

6.16.4.16.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.16.4.16.5 Selection of coil method

Select the method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.16.4.16.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area.

The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application.

This setting is effective in Manual mode only. The settable range is limited to coil entered numbers only out of [1~128].

6.16.4.16.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.16.4.16.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.16.4.16.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.16.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.16-18 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.



Fig. 6.16-17 Quit Button



Fig. 6.16-18 Exit Confirmation Dialog

6.16.6 Error specifications

Error specifications are shown in Table 6.16-3.

Table 6.16-3 Error Specifications

Message	Status	Solution
Cannot save case to file. Comment not specified.	Data cannot be saved in the file.	Input a comment.
Analysis error.	Analysis error occurred.	Check the protocol for correct setting.
Calculation error.	An error occurred in calculation analysis process.	Check the protocol for correct setting.
Scanning initialization error.	An error occurred in initialization process.	Check the protocol for correct setting.
Scanning error.	An error occurred during scan.	Readjust the parameters and retry a scan.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application

Message	Status	Solution
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.16.7 Other cautions

To obtain correct measurement values, a prescan must be performed at least once before executing this tool.

6.17 Magnetic Field Search

6.17.1 General

This is a tool to measure fluctuation of the magnetic field and display the result graphically. The measured data up to three cases can be stored in the H/D. Maximum number of measurement per one case is 72,000.

6.17.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Mag. Field Search in the displayed items and click [OK] or double-click the selected item.

6.17.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.17-1. In addition, the screen displaying the magnetic field search (Fig. 6.17-2) is displayed.

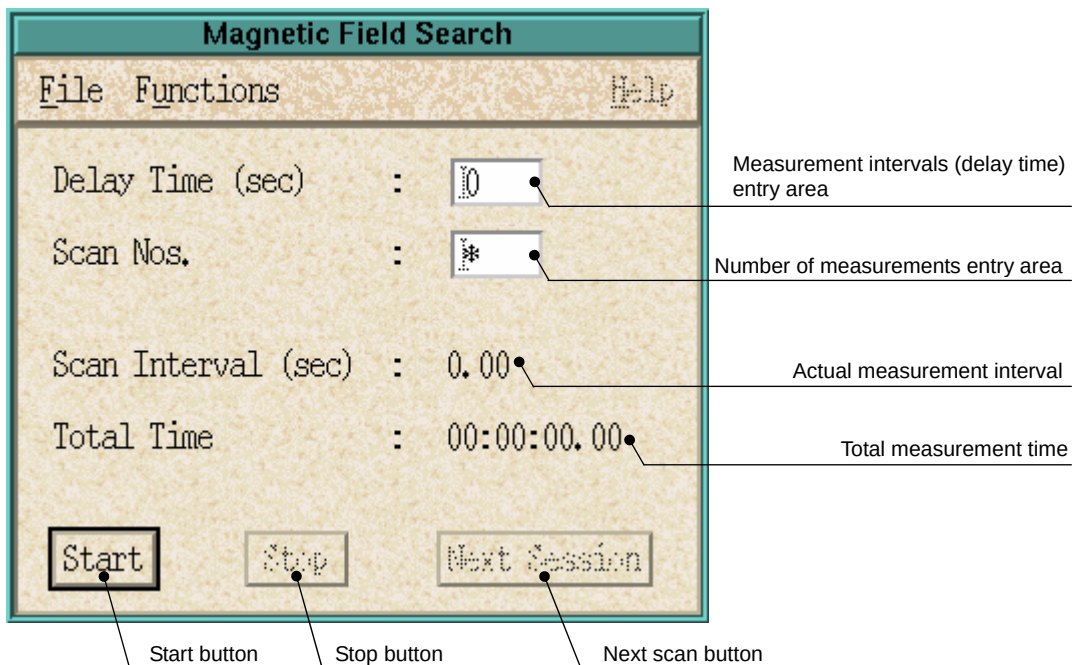


Fig. 6.17-1 Magnetic Field Search Display Screen

6.17.4 Operation procedure

Follow the next steps to execute the process.

6.17.4.1 Setting parameters

Using the keyboard and mouse cursor, change parameters. Place the mouse cursor to the entry area of the parameter to be changed and click the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

Table 6.17-1 Parameter Setting Range

Parameter Name	Available Range for setting	Remarks
Measurement interval	0 ~ 999	
Number of measurements	1 ~ 9999, *	* : Max. number of available scans.

6.17.4.2 Starting measurement

Click [Start] button to start the measurement.

6.17.4.3 Ending measurement

Click [Stop] button to end the measurement.

6.17.4.4 Display of result

In case of the graph display-ON mode, the measurement result is graphically displayed in real time.

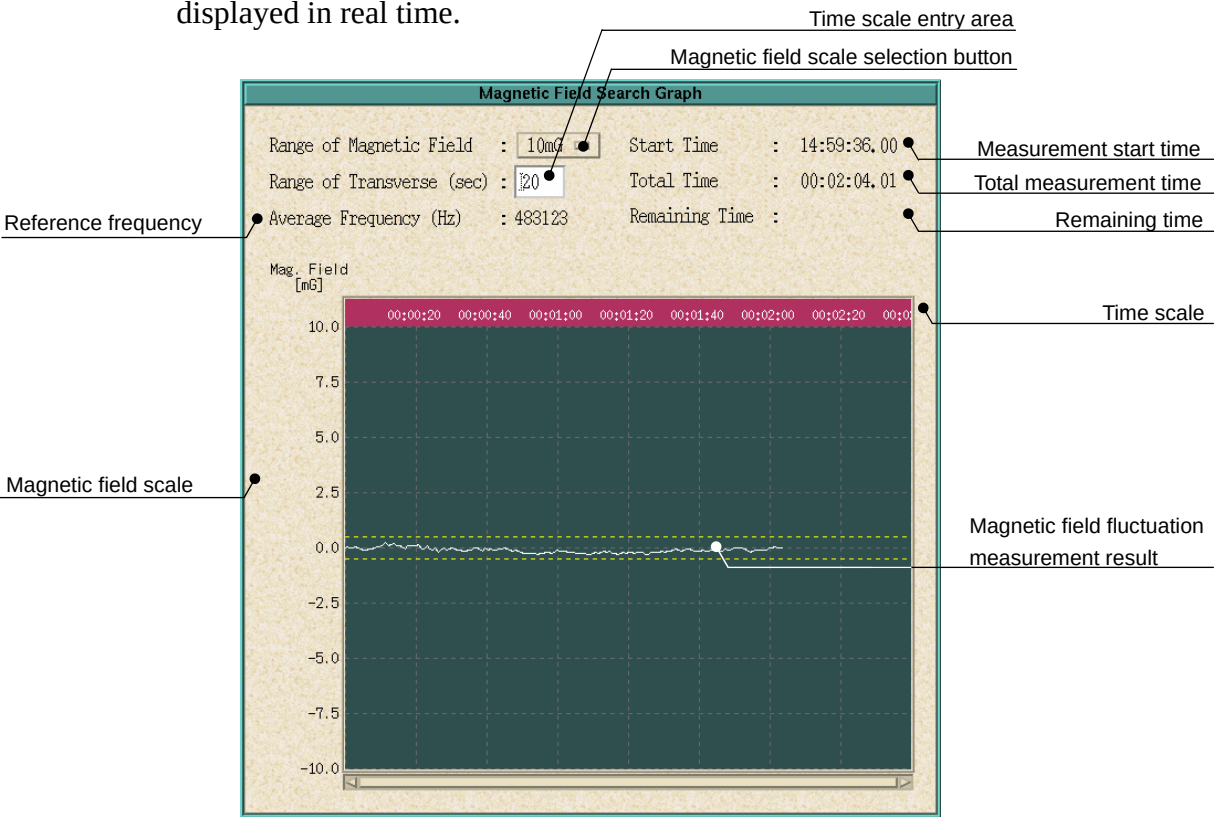


Fig. 6.17-2 An Example of Measurement Result

6.17.4.5 Saving measured data

To save the measured data, select [Save] on "File Menu". The screen shown in Fig. 6.17-4 is displayed. Click the file to be saved from the three files and enter comments in the comment area, then click [OK] to save the measured data. Clicking [Cancel] results in not saving the data. Comments up to 32 characters can be entered.

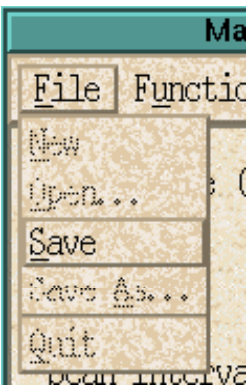


Fig. 6.17-3 Save Button

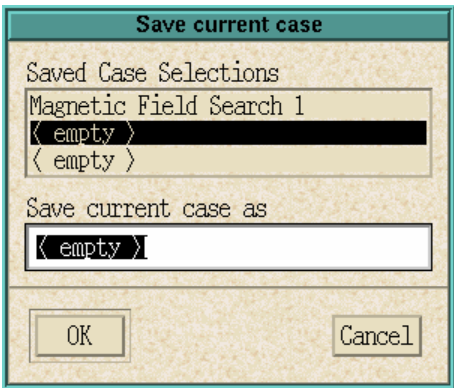


Fig. 6.17-4 Measurement Data Save Screen

6.17.4.6 Next measurement

To perform the next measurement after the first session, in case the measured data have been saved, the [Start] button shown in Fig. 6.17-1 is enabled. So, carry out the same operation procedure from start of the measurement as described in Paragraph 6.17.4.2.

In case the data have not been saved, click [Next Session] button. The measured data clearing confirmation screen as shown in Fig. 6.17-5 is displayed. To clear the measured data, click [Yes], while to save the data, click [No]. In case [No] is selected, follow the procedures described in Paragraph 6.17.4.5. After [Start] button is enabled to click, follow the same operation procedures from start of the measurement as described in Paragraph 6.17.4.2.

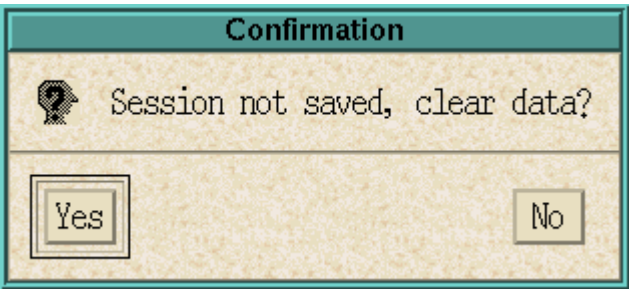


Fig. 6.17-5 Measurement Data Clearing Confirmation Screen

6.17.4.7 Reviewing saved data

The saved measured data can be reviewed. Select [Open] on "File Menu". On the measurement data selection screen as shown in Fig. 6.17-7 is displayed, select the data to be reviewed and click [OK] to display the selected data on the measurement data display screen (Fig. 6.17-2).



Fig. 6.17-6 Open Button

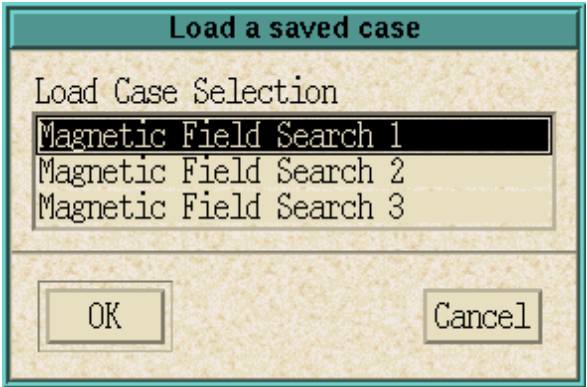


Fig. 6.17-7 Measured Data Selection Screen

6.17.4.8 Operation on graphic screen

6.17.4.8.1 Selection of display/non-display of graphic screen

The graphic screen is displayed by selecting [Graph ON] on "Functions Menu" and is not displayed by [Graph OFF].

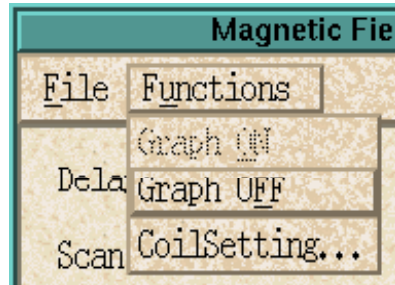


Fig. 6.17-8 Graph Display/Non-display Selection Button

6.17.4.8.2 Setting graph scale

Scales for the ordinate and abscissa axes on the graphical display screen are set. For the ordinate axis, click Magnetic field scale selection button, and select [100mG] or [10mG] or [2mG] on the menu displayed by clicking. The default is 10mG. For the abscissa axis, set seconds per division in the time scale setting field. A value up to 999sec. can be set.

6.17.4.9 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.17.4.9.1 Default

Default at the start time is Auto. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.17.4.9.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

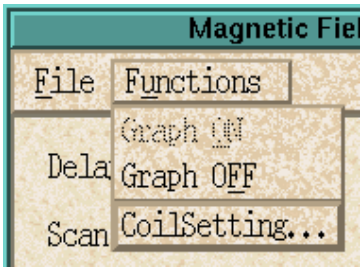


Fig. 6.17-9 Coil setting window display button

6.17.4.9.3 Display screen

The display screen and function names are shown in Fig. 6.17-10 and Fig. 6.17-11.

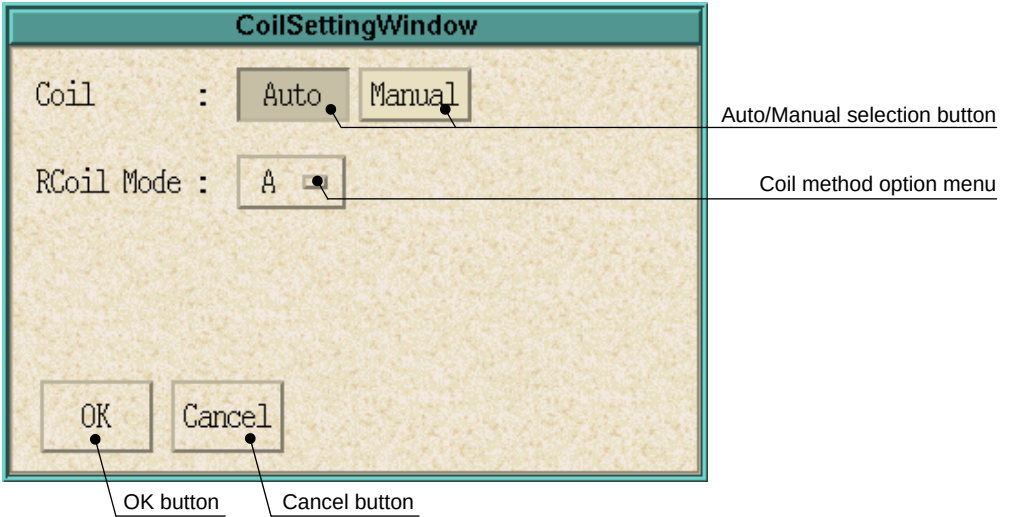


Fig. 6.17-10 Coil Setting Window Display Screen (Auto mode)

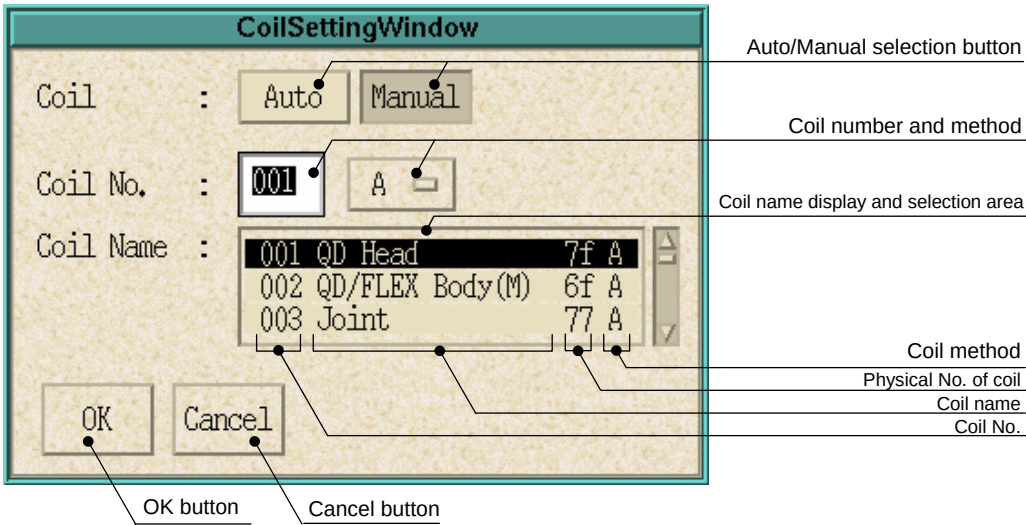


Fig. 6.17-11 Coil Setting Window Display Screen (Manual mode)

6.17.4.9.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.17.4.9.5 Selection of coil method

Select the method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.17.4.9.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area.

The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application.

This setting is effective in Manual mode only. The settable range is limited to coil entered numbers only out of [1~128].

6.17.4.9.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in Manual mode only.

6.17.4.9.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.17.4.9.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.17.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.17-13 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

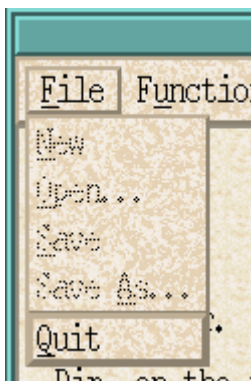


Fig. 6.17-12 Quit Button



Fig. 6.17-13 Exit Confirmation Dialog

6.17.6 Error specifications

Error specifications are shown in Table 6.17-2.

Table 6.17-2 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.17.7 Other cautions

To obtain correct measurement values, a prescan must be performed at least once before executing this tool.

6.18 GC Pulse Generation

6.18.1 General

This is a tool to control the gradient field having the user-defined complicated sequence pattern.

6.18.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click GC Pulse Generation in the displayed items and click [OK] or double-click the selected item.

6.18.3 Display screen

The screen displayed at start-up and function names are shown in Fig. 6.18-1. In parameter data are stored, the Data1 parameters are displayed (refer to Paragraph 6.18.4.6). If no parameter data is stored, the values set in the system are displayed.

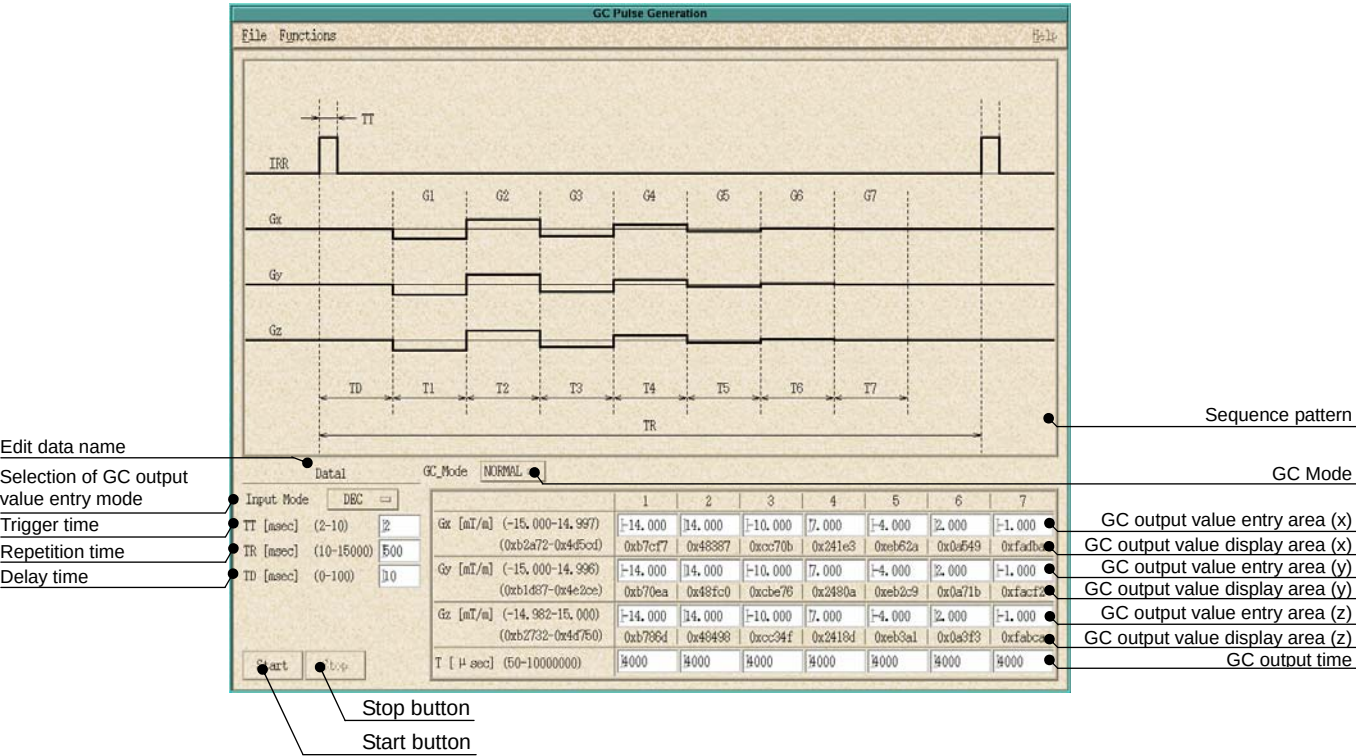


Fig. 6.18-1 GC Pulse Generation Display Screen

6.18.4 Operation procedure

Follow the next steps to execute the process.

6.18.4.1 Updating parameters

Using the keyboard and mouse cursor, update parameters.

Place the mouse cursor to the entry area of the parameter to be changed and click the entry area to select it. From the keyboard, enter figures or character string, and either press **<Enter>** on the keyboard or remove the focus from the entry area to change the parameter.

When a GC output value is input in the GC output value input area in decimal notation mode, a hardware output value is displayed below the input value. This hardware output value is displayed by adding a GC offset value to the input GC output value. In case of a hexadecimal notation mode input, the input value is set as a hardware output value as it is. For the method of switching the GC output value modes, refer to Paragraph 6.18.4.7.

As the GC output value changes, the sequence pattern on the upper part of the window changes correspondingly, so that an outlined sequence pattern waveform can be graphed.

The time parameter must satisfies the input protect formula shown below. If the following formula is not satisfied, the dialog shown in Fig. 6.18-2 is displayed by clicking [Start] button and the sequence is not executed. In such case, reset the time parameter to appropriate value and retry the operation.

Time parameter input protect formula:

$$TD \times 1000 + T1 + T2 + T2 + T3 + T4 + T5 + T6 + T7 + 5000 < TR \times 1000$$

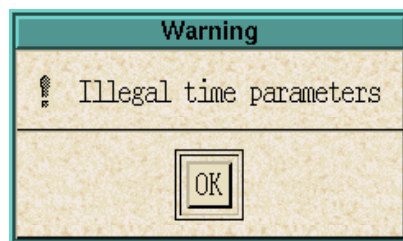


Fig. 6.18-2 Time Parameter Error Dialog

Table 6.18-1 Parameter Setting Range

Parameter Name	Setting range
TT (msec)	2 ~ 10
TD (msec)	0 ~ 100
TR (msec)	10 ~ 15000
T1 ~ T7 (μ sec)	50 ~ 10000000
Gx1 ~ Gz7 (mT/m)	-gm ~ gm (0x80000 ~ 0x7ffff)
GC Mode	NORMAL, Case1 ~ 4

6.18.4.2 Execution of sequence

The sequence is executed by pressing [Start] button. While the sequence being executed, **Running** is displayed on the screen and all the entry operations except [Stop] button are disabled.

6.18.4.3 Stopping sequence

The sequence is stopped by [Stop] button. By stopping the sequence, entry operations are enabled again.

6.18.4.4 Cancellation of change

By opening "Functions Menu" of the Menu Bar and selecting [Reset], change of the parameter can be canceled. Consequently, the parameter is reset to the one used at the last loading, saving or executing the sequence.

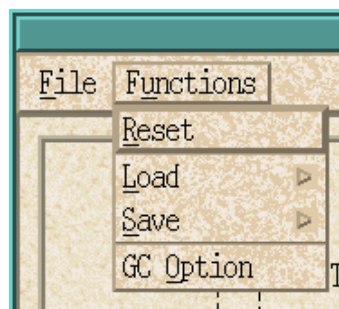


Fig. 6.18-3 Reset Button

6.18.4.5 Loading parameter

Parameter can be loaded by selecting [Load] on "Functions Menu" of the Menu Bar. Clicking [Load] displays the sub-menu of data names. Select either [Data1] or [Data2]. The data name of the currently editing parameter is displayed on the main screen. At start-up of the application, Data1 is loaded.

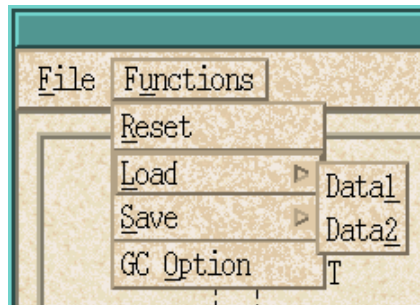


Fig. 6.18-4 Load Button

6.18.4.6 Saving parameter

Parameter can be saved by selecting [Save] on "Functions Menu" of the Menu Bar. Parameter data of up to Data1 and Data2 can be saved.

Clicking [Save] displays the sub-menu of data names, and select either [Data1] or [Data2].

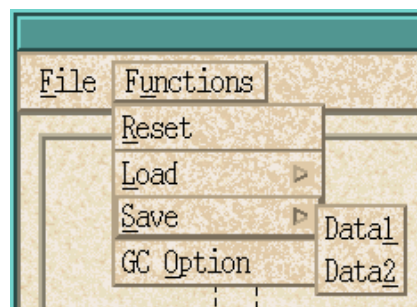


Fig. 6.18-5 Save Button

6.18.4.7 Selection of GC output value mode

Using [Input Mode], a selection whether GC output value is input in a decimal system (mT/m unit) or a hexadecimal system can be made.

To set the decimal input mode, select [DEC] on the menu displayed by clicking [Input Mode], while to set the hexadecimal input mode, select [HEX].

In case of the decimal input mode, hexadecimal figure is displayed under the entry area, while in case of the hexadecimal input mode, decimal figure is displayed. The default is decimal input mode.

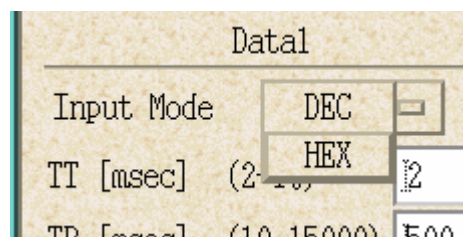


Fig. 6.18-6 Option Menu for Selection of GC Output Value Input Mode

6.18.4.8 GC option

By opening "Functions Menu" of the Menu Bar and selecting [GC Option], the GC Option window is popped up. On the GC option window, the Maximum GC, Slew Rate, Offset and Gain value can be input. By selecting [Reset], the data can be reset to the initial values (values on the system table). Clicking [Close] closes the GC option window, but the changed parameters made on the GC option window are not cleared.



Fig. 6.18-7 GC Option Button

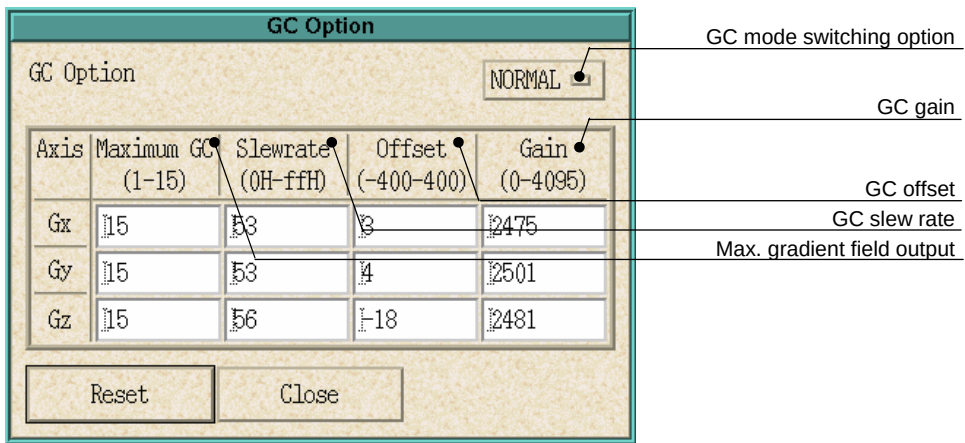


Fig. 6.18-8 GC Option Window

Table 6.18-2 Parameter Setting Range

Parameter Name	Setting range
Max. gradient field output	NORMAL : 1~15, Case 1~4 : 1~25
Slew rate	0H~ffH
Offset	-400~400
Gain	0~4095

6.18.4.9 Selection of GC modes

A changeover of the GC mode to use is enabled with [GC Mode].
NORMAL shows the normal GC mode and Case 1 to Case 4 show the GC modes for High Slew Rate.

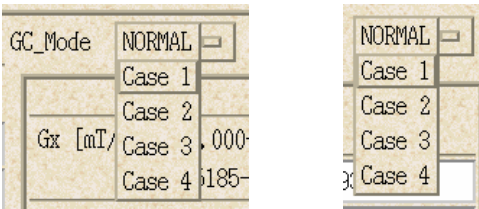


Fig. 6.18-9 GC mode switching option menu
(Left : Main screen Right : GC option setting screen)

6.18.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.18-11 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.



Fig. 6.18-10 Quit Button



Fig. 6.18-11 Exit Confirmation Dialog

6.18.6 Error specifications

Error specifications are shown in Table 6.18-3.

Table 6.18-3 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application

6.19 GC Offset Adjustment

6.19.1 General

This is a tool to adjust the offset of the gradient magnetic field. Setting measurement parameters and number of measurements, and monitoring the measurement results, the optimum value is obtained.

6.19.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click GC Offset Adjustment in the displayed items and click [OK] or double-click the selected item.

6.19.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.19-1.

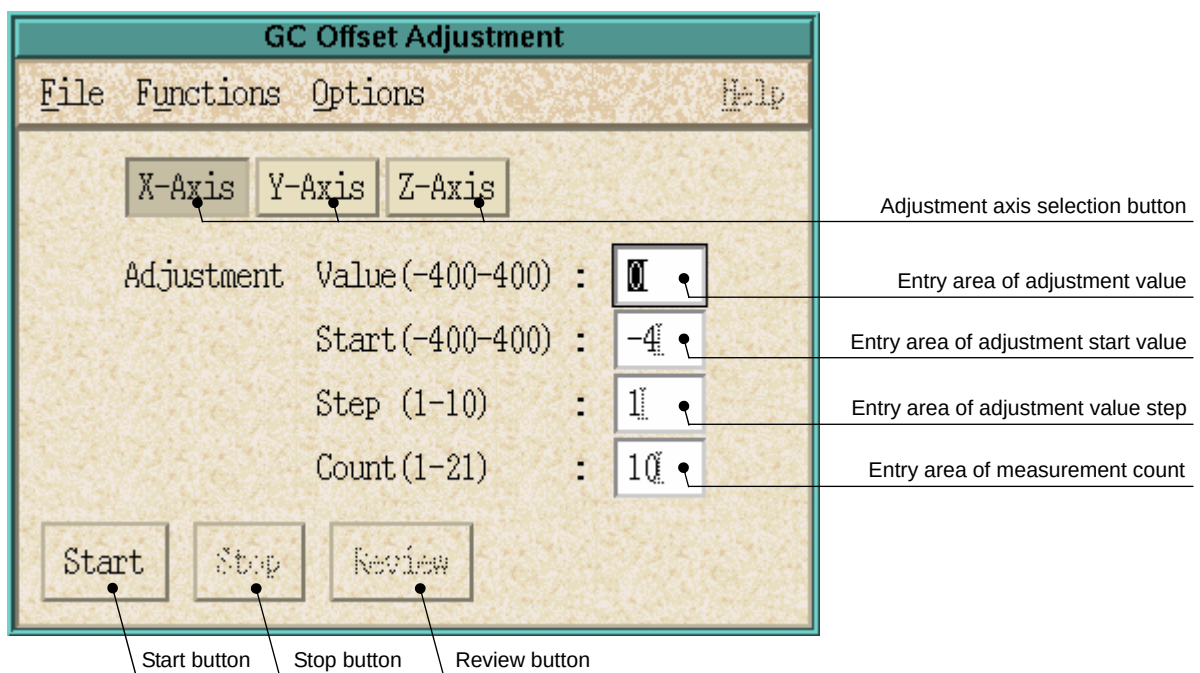


Fig. 6.19-1 GC Offset Adjustment Display Screen

6.19.4 Operation procedure

Follow the next steps to execute the process.

6.19.4.1 Updating parameters

Using the keyboard and mouse cursor, change parameters. The following methods are available to change parameters.

6.19.4.1.1 Selection of parameter by pushbutton

The push-buttons arranged in array are of single-selection type, and by clicking one of them by the mouse cursor, the corresponding parameter is changed.

6.19.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press **<Enter>** on the keyboard or remove the focus from the entry area to change the parameter.

Measurement is started from the Start value and it is executed in number of times set by Count and increments set by Step. The default of Value is the one set in the system table, which is set to be at the center of measurement counts. However, if a value is directly input in Start value, the value is set as Start value.

Table 6.19-1 Parameter Setting Range

Parameter Name	Setting range
Adjustment axis	X, Y, Z
Value	-400 ~ 400
Start	-400 ~ 400
Step	1 ~ 10
Count	1 ~ 21

6.19.4.2 Starting measurement

The measurement is executed by pressing either [Start] button on "Functions Menu" or [Start] on the screen.

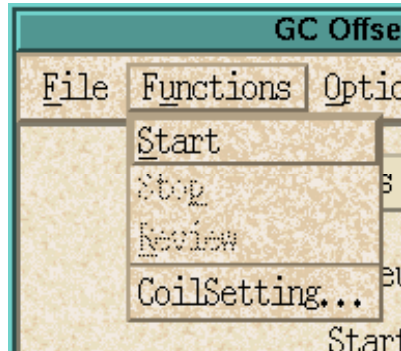


Fig. 6.19-2 Start Button

6.19.4.3 Stopping measurement

The measurement is stopped by either selecting [Stop] button on "Functions Menu" or clicking [Stop] button on the main screen.

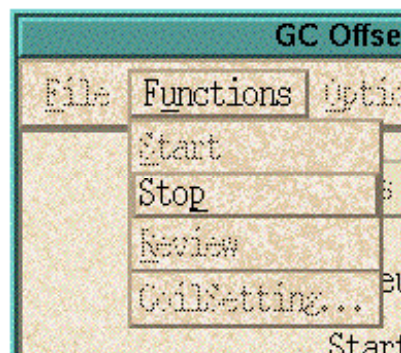


Fig. 6.19-3 Stop Button

6.19.4.4 Result display

The measured data are displayed as shown in Fig. 6.19-4. Of the measured data, the data with the max. peak value is graphically displayed and becomes as the default of the selected value.

To review the graph of other data, click the line corresponding to the data to be reviewed. The value of the clicked line becomes the selected value. Graph is displayed with absolute values.

To obtain phase display, click [Phase] button.

To close this screen, click [Close] button.

To set the adjustment value, click [Set] button and go to Paragraph 6.19.4.6.

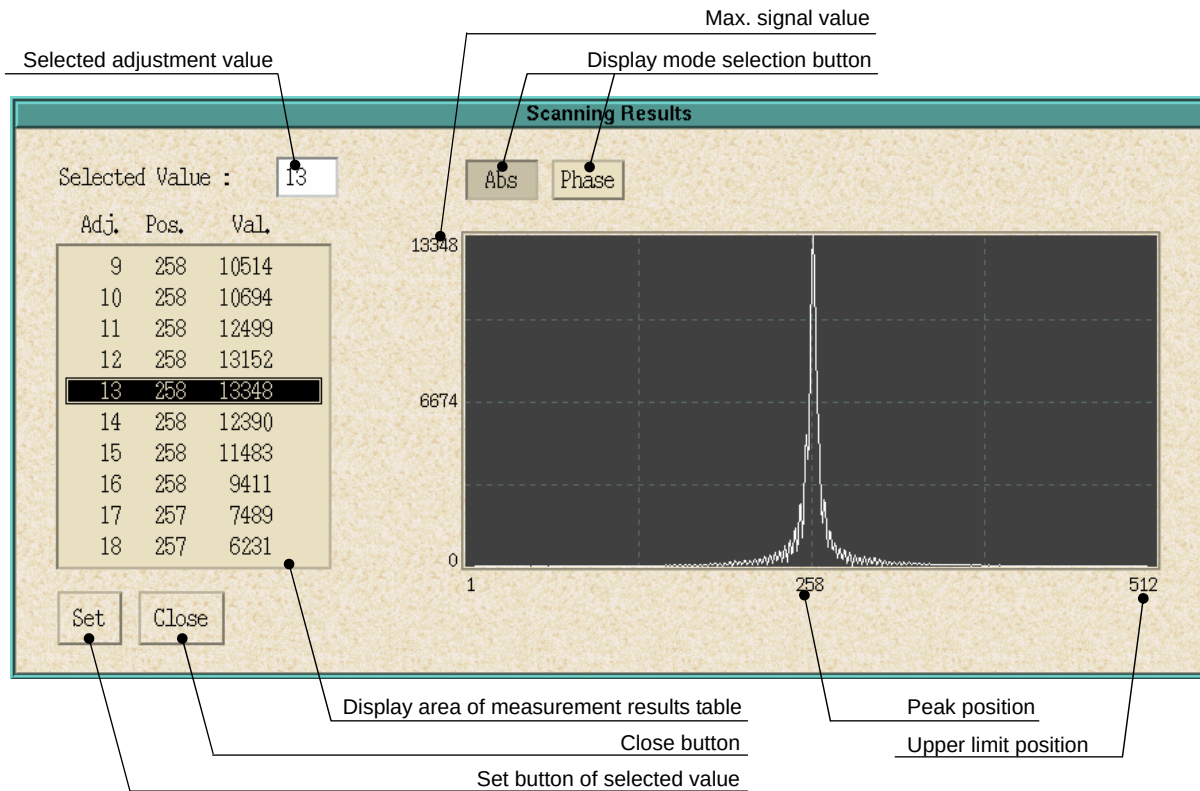


Fig. 6.19-4 Example of Measurement Results
(Absolute value display)

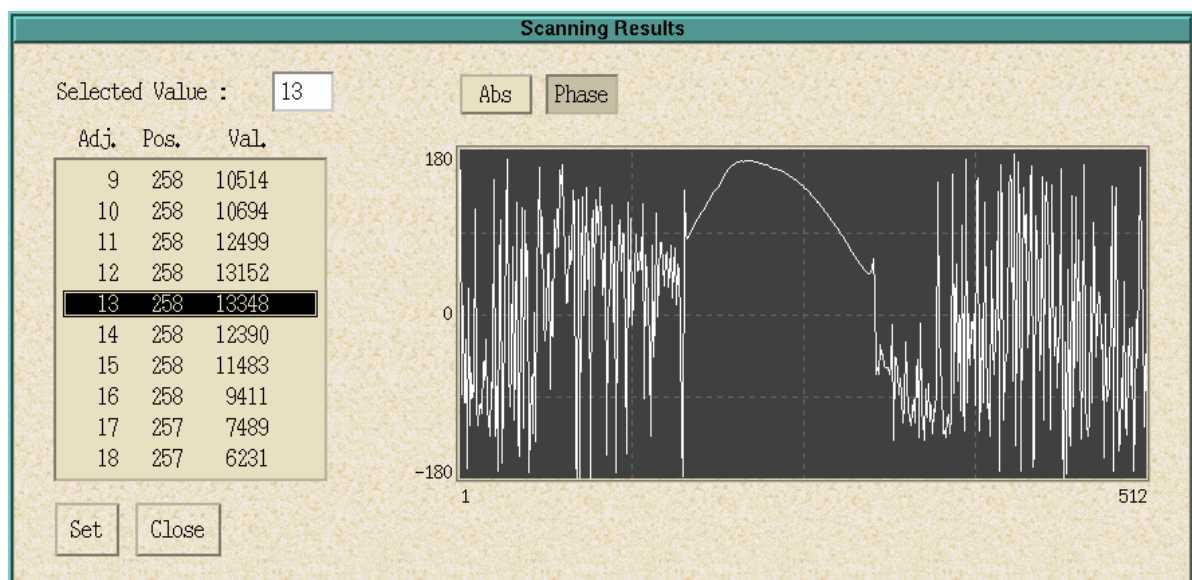


Fig. 6.19-5 Example of Measurement Results
(Phase display)

6.19.4.5 Review of results

By either selecting [Review] on "Functions Menu" or clicking [Review] button on the Main Screen, the results can be reviewed.

6.19.4.6 Setting adjustment values

To set the adjustment values, select [Save] on "File Menu" of the Menu Bar. The Change Confirmation screen as shown in Fig. 6.19-7 is displayed. Clicking [Yes] on this screen registers the adjustment values both in the system table memory and H/D. Clicking [No] does not register them in either of them.

The setting operation can be done either for each axis or for all of the axes together when adjustments for all the axes are completed.



Fig. 6.19-6 Save Button

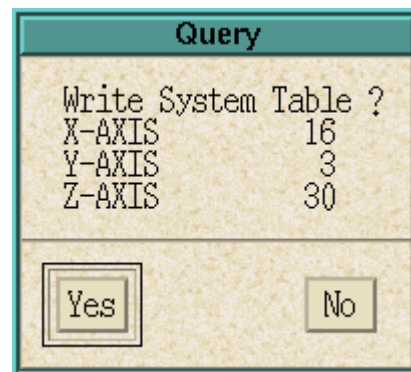


Fig. 6.19-7 Change Confirmation Screen

6.19.4.7 Option

Select [Change Protocol] on "Functions Menu" of the Menu Bar. The protocol change screen is popped up.

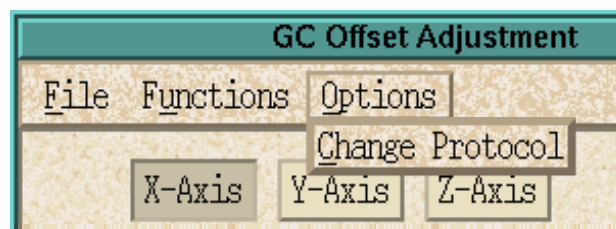


Fig. 6.19-8 Option Menu

6.19.4.7.1 Display screen

The display screen and function names are shown in Fig. 6.19-9.

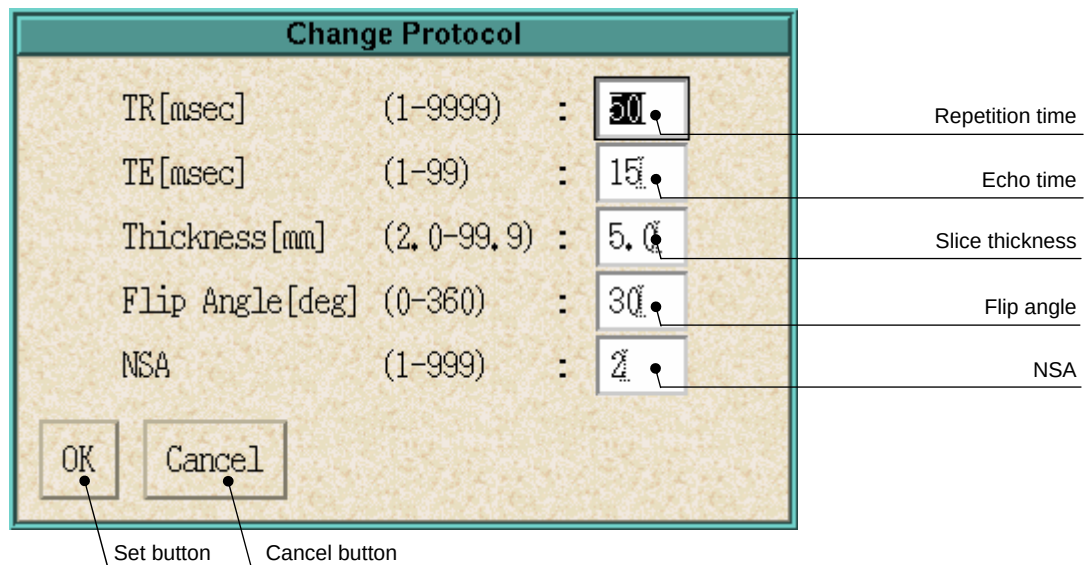


Fig. 6.19-9 Protocol Change Screen

6.19.4.7.2 Changing parameter

For procedure to change parameters on the protocol change screen, refer to Paragraph 6.19.4.1.

Table 6.19-2 Protocol Parameter Setting Range
(Option Screen)

Parameter Name	Setting range
TR	1 ~ 9999
TE	1 ~ 99
Thickness	2.0 ~ 99.9
Flip angle	0 ~ 360
NSA	1 ~ 999

6.19.4.7.3 Determination of parameter change

By clicking [OK] button on the protocol change screen, the changed parameters can be determined and the protocol change screen is closed.

6.19.4.7.4 Cancellation of parameter change

By clicking [Cancel] on the protocol change screen, the change of parameter can be canceled and the protocol change screen is closed.

6.19.4.8 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.19.4.8.1 Default

Default at the start time is Auto. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.19.4.8.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.



Fig. 6.19-10 Coil Setting Window Display Button

6.19.4.8.3 Display screen

The display screen and function names are shown in Fig. 6.19-11 and Fig. 6.19-12.

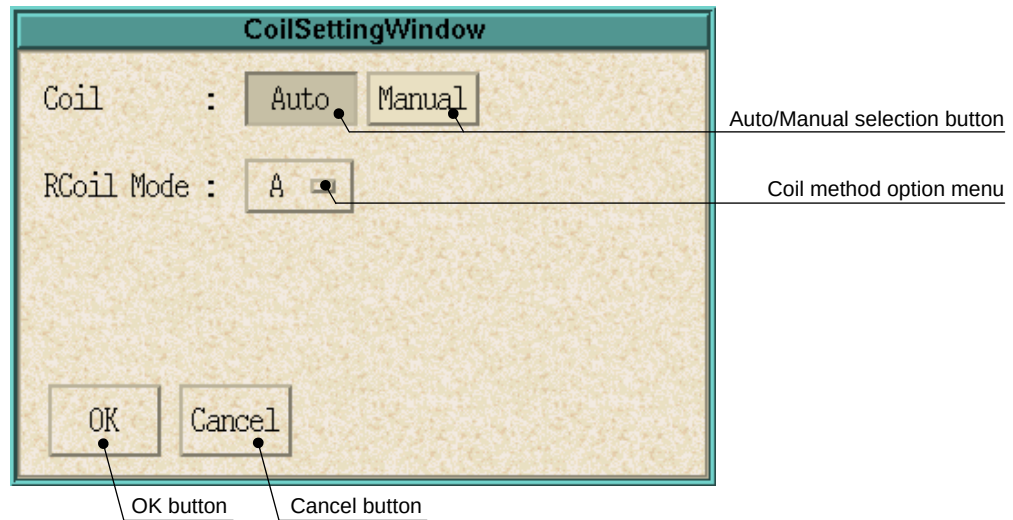


Fig. 6.19-11 Coil Setting Window Display Screen (Auto mode)

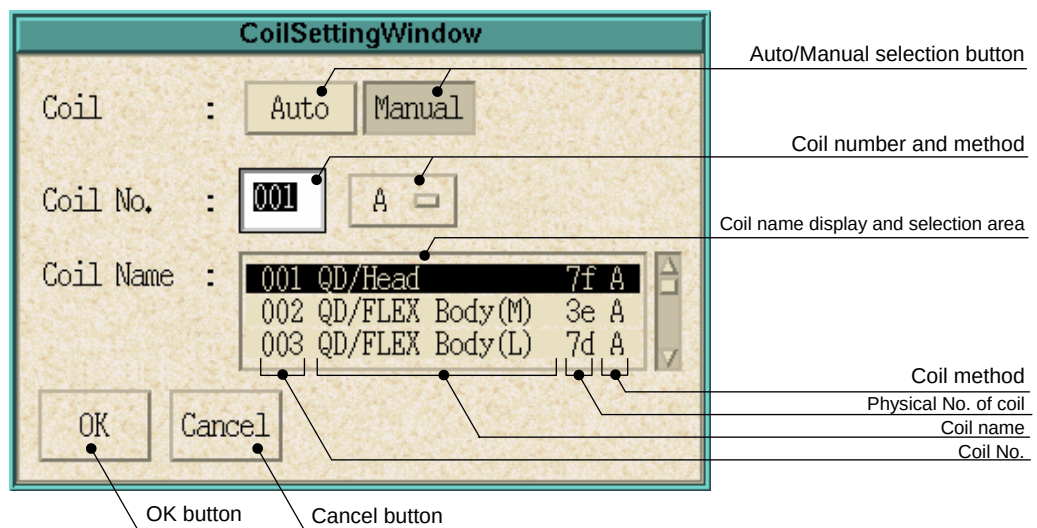


Fig. 6.19-12 Coil Setting Window Display Screen (Manual mode)

6.19.4.8.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the **Manual** screen in case of the **Auto** mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of **Manual** screen is [0], the coil method is [A], and the coil name is not selected.

6.19.4.8.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in **Manual** mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.19.4.8.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area. The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.19.4.8.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in Manual mode only.

6.19.4.8.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.19.4.8.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.19.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.19-14 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application. When the adjustment value has been changed but the system table has not been updated, the change confirmation screen as shown in Fig. 6.19-7 is displayed. Upon responding to the screen, the dialog shown in Fig. 6.19-14 is popped up. Then, follow the operation described above.



Fig. 6.19-13 Quit Button

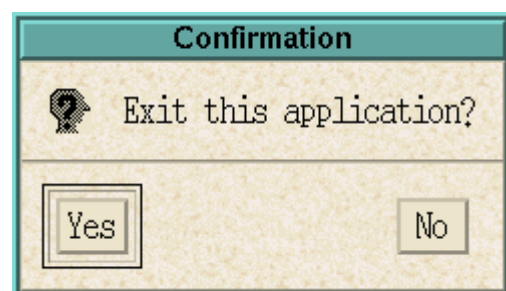


Fig. 6.19-14 Exit Confirmation Dialog

6.19.6 Error specifications

Error specifications are shown in Table 6.19-3.

Table 6.19-3 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.19.7 Other cautions

To obtain correct measurement values, a prescan must be performed at least once before executing this tool.

6.20 GC Slew Rate Adjustment

6.20.1 General

This is a tool to adjust the slew rate of the gradient magnetic field. Setting measurement parameters and number of measurements, and monitoring the measurement results, the optimum value is obtained.

6.20.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click GC Slew Rate Adjustment in the displayed items and click [OK] or double-click the selected item.

6.20.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.20-1.

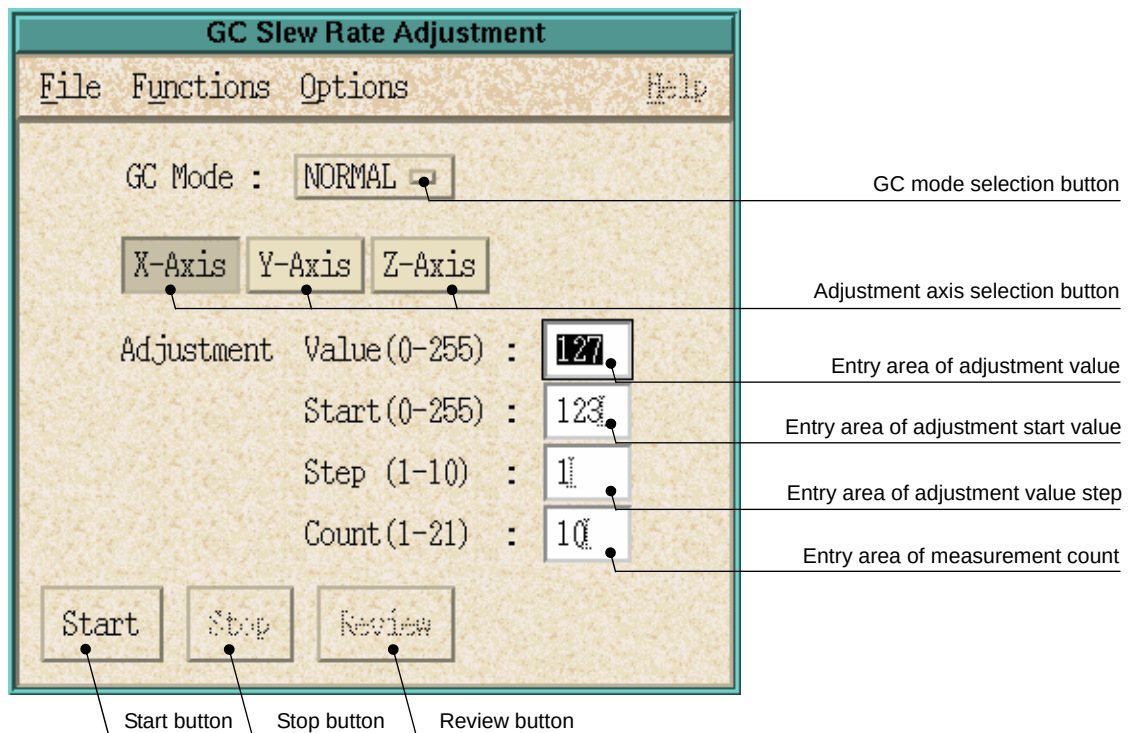


Fig. 6.20-1 GC Slew Rate Adjustment Display Screen

6.20.4 Operation procedure

Follow the next steps to execute the process.

6.20.4.1 Updating parameters

Using the keyboard and mouse cursor, change parameters. The following methods are available to change parameters.

6.20.4.1.1 Selection of parameter by pushbutton

The push-buttons arranged in array are of single-selection type, and by clicking one of them by the mouse cursor, the corresponding parameter is changed.

6.20.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

Measurement is started from the Start value and it is executed in number of times set by Count and increments set by Step. The default of Value is the one set in the system table, which is set to be at the center of measurement counts. However, if a value is directly input in Start value, the value is set as Start value.

6.20.4.1.3 Parameter change by option menu

Displayed items show the set parameters. Click the option menu, and then, click a parameter to be changed from the menu by the mouse cursor. The parameter is updated.

Table 6.20-1 Parameter Setting Range

Parameter Name	Setting range
GC mode	NORMAL, Case1 ~ 4
Adjustment axis	X, Y, Z
Value	0 ~ 255
Start	0 ~ 255
Step	1 ~ 10
Count	1 ~ 21

6.20.4.2 Starting measurement

The measurement is executed by pressing either [Start] button on "Functions Menu" or [Start] on the screen.

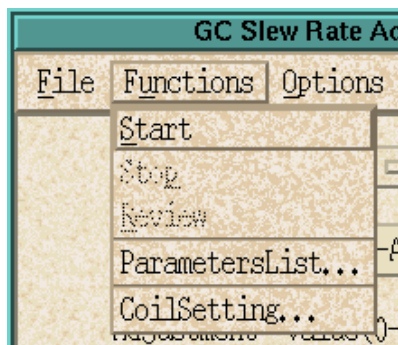


Fig. 6.20-2 Start Button

6.20.4.3 Stopping measurement

The measurement is stopped by either selecting [Stop] button on "Functions Menu" or clicking [Stop] button on the main screen.

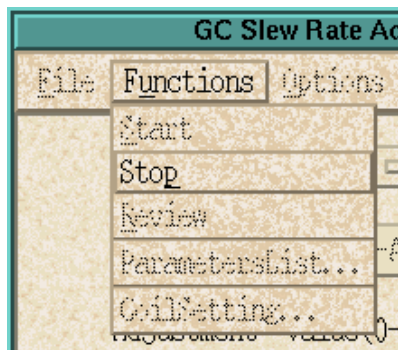


Fig. 6.20-3 Stop Button

6.20.4.4 Result display

The measured data are displayed as shown in Fig. 6.20-4. Of the measured data, the data with the max. peak value is graphically displayed and becomes as the default of the selected value.

To review the graph of other data, click the line corresponding to the data to be reviewed. The value of the clicked line becomes the selected value. Graph is displayed with absolute values.

To obtain phase display, click [Phase] button.

To close this screen, click [Close] button.

To set the adjustment value, click [Set] button and go to Paragraph 6.20.4.7.

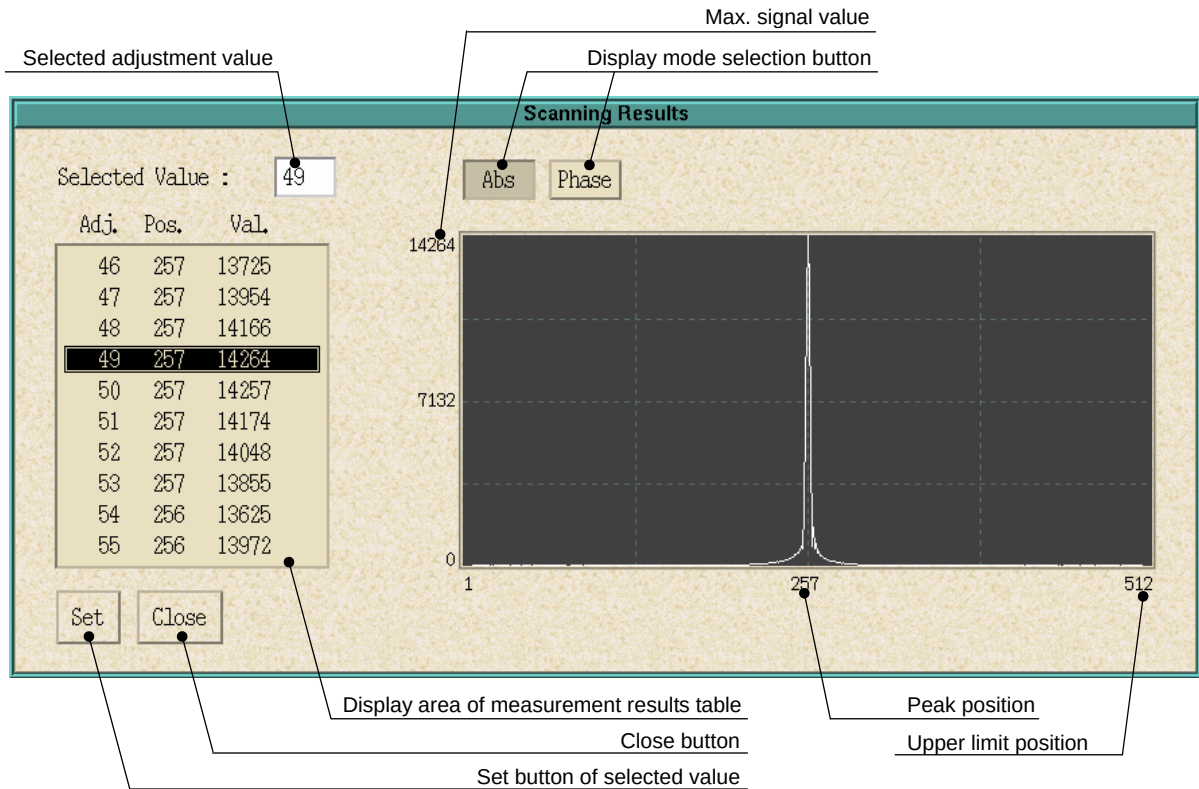


Fig. 6.20-4 Example of Measurement Results
(Absolute value display)

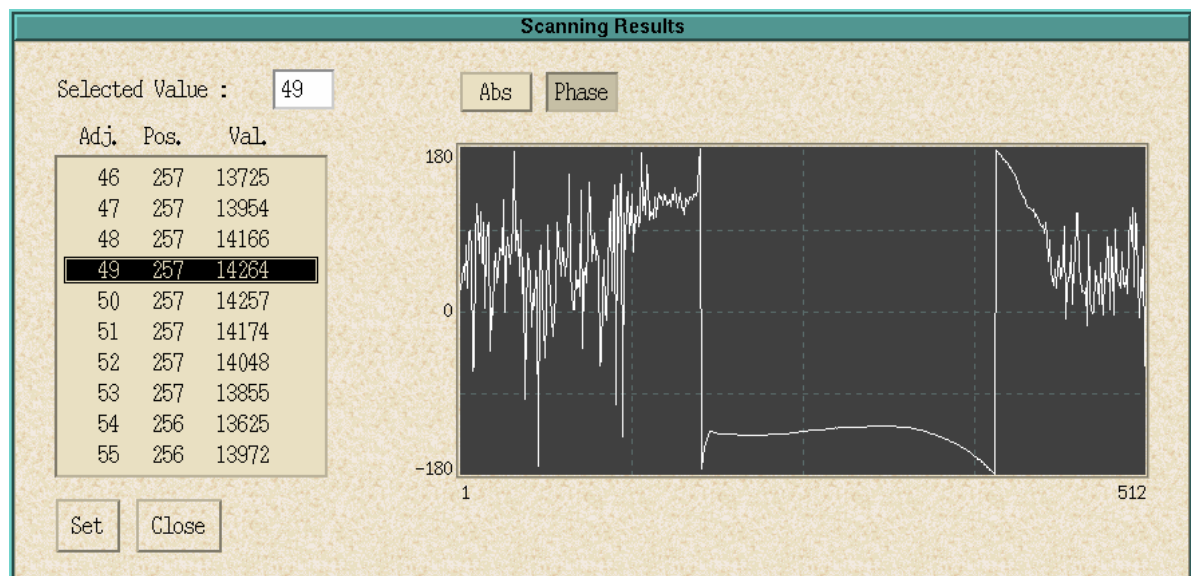


Fig. 6.20-5 Example of Measurement Results
(Phase display)

6.20.4.5 Review of results

By either selecting [Review] on "Functions Menu" or clicking [Review] button on the Main Screen, the results can be reviewed.

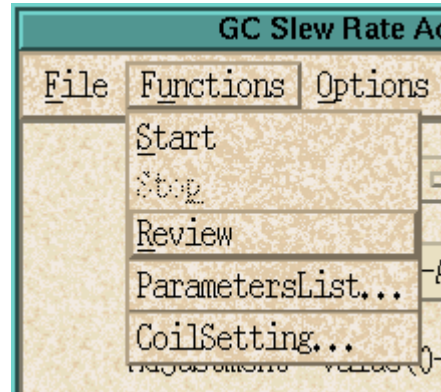


Fig. 6.20-6 Review button

6.20.4.6 Adjusting values check

Select [ParametersList ...] on "Functions Menu" of the Menu Bar, an adjusting values list display screen is displayed as shown in Fig. 6.20-8.

No adjusting value can be changed in this screen. For closing the adjusting values list display screen, click [Close] button.

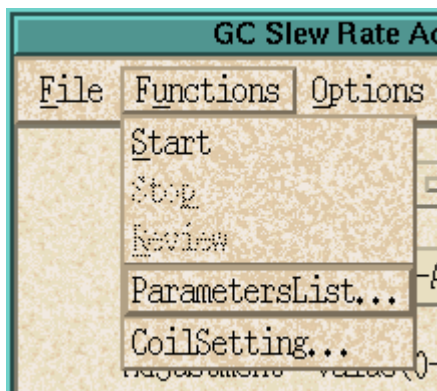


Fig. 6.20-7 Parameters List button

 A screenshot of a window titled "Parameters List". It contains a table with the following data:

	NORMAL	Case 1	Case 2	Case 3	Case 4
X-Axis	127	93	136	102	124
Y-Axis	125	98	134	106	122
Z-Axis	129	94	133	104	119

 Below the table is a "Close" button.

Fig. 6.20-8 Adjusting values list display screen

6.20.4.7 Setting adjustment values

To set the adjustment values, select [Save] on "File Menu" of the Menu Bar.

The setting operation can be done either for each axis or for all of the axes together when adjustments for all the axes are completed.

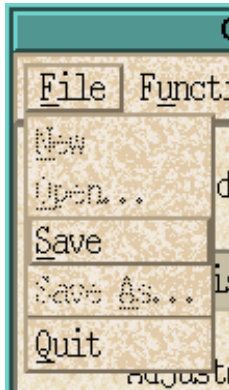


Fig. 6.20-9 Save Button

6.20.4.8 Adjusting protocol change

Select [Change Protocol] on "Functions Menu" of the Menu Bar. The protocol change screen is popped up.

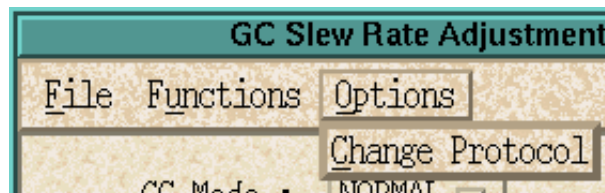


Fig. 6.20-10 Option Menu

6.20.4.8.1 Display screen

The display screen and function names are shown in Fig. 6.20-11.

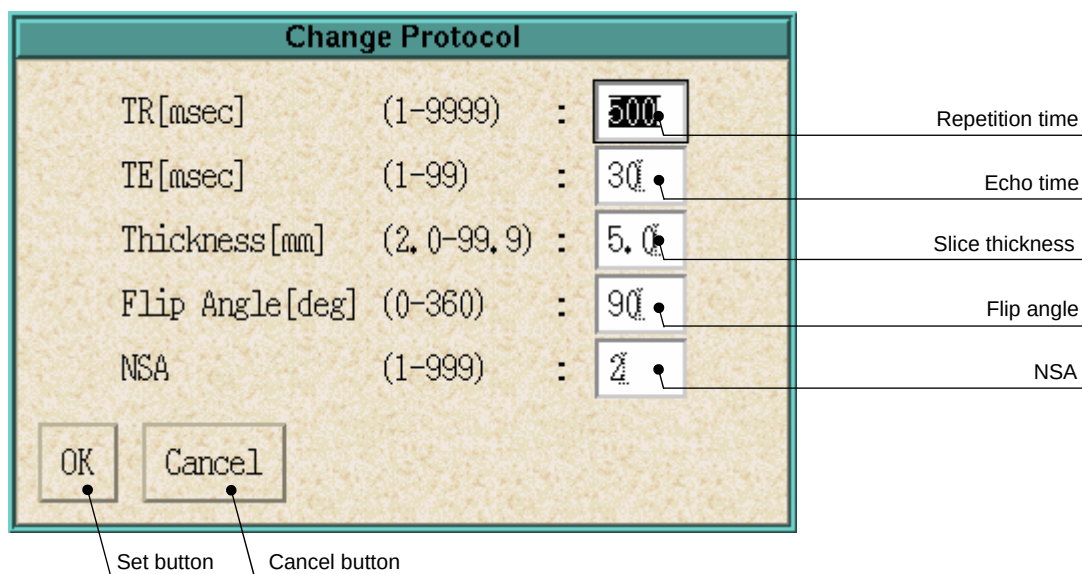


Fig. 6.20-11 Protocol Change Screen

6.20.4.8.2 Changing parameter

For procedure to change parameters on the protocol change screen, refer to Paragraph 6.20.4.1.

Table 6.20-2 Protocol Parameter Setting Range (Option Screen)

Parameter Name	Setting range
TR	1 ~ 9999
TE	1 ~ 99
Thickness	2.0 ~ 99.9
Flip angle	0 ~ 360
NSA	1 ~ 999

6.20.4.8.3 Determination of parameter change

By clicking [OK] button on the protocol change screen, the changed parameters can be determined and the protocol change screen is closed.

6.20.4.8.4 Cancellation of parameter change

By clicking [Cancel] on the protocol change screen, the change of parameter can be canceled and the protocol change screen is closed.

6.20.4.9 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.20.4.9.1 Default

Default at the start time is Auto. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.20.4.9.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

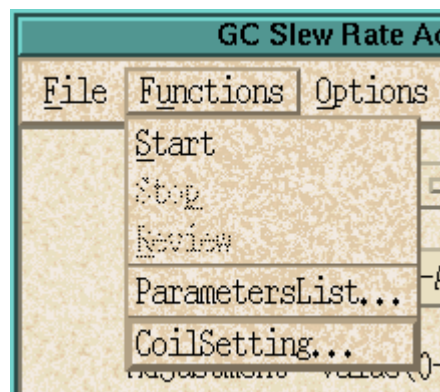


Fig. 6.20-12 Coil Setting Window Display Button

6.20.4.9.3 Display screen

The display screen and function names are shown in Fig. 6.20-13 and Fig. 6.20-14.

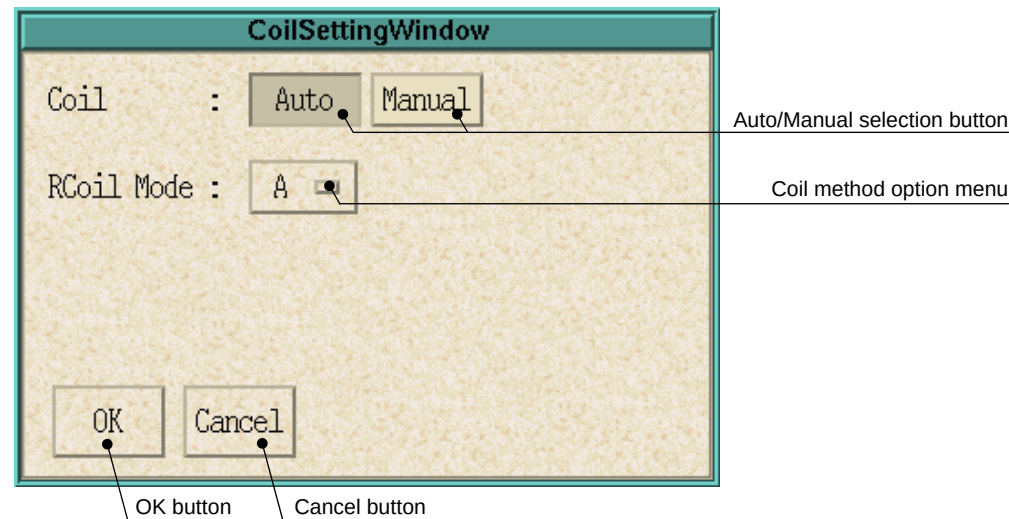


Fig. 6.20-13 Coil Setting Window Display Screen (Auto mode)

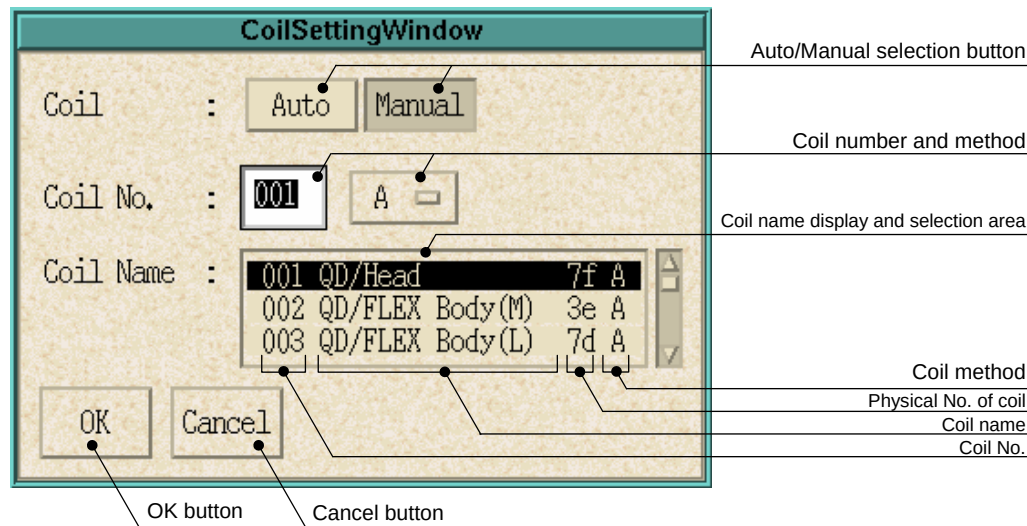


Fig. 6.20-14 Coil Setting Window Display Screen (Manual mode)

6.20.4.9.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.20.4.9.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.20.4.9.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area. The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in Manual mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.20.4.9.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in Manual mode only.

6.20.4.9.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.20.4.9.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.20.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.20-16 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

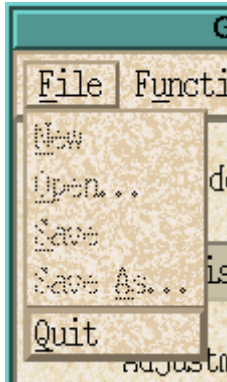


Fig. 6.20-15 Quit Button

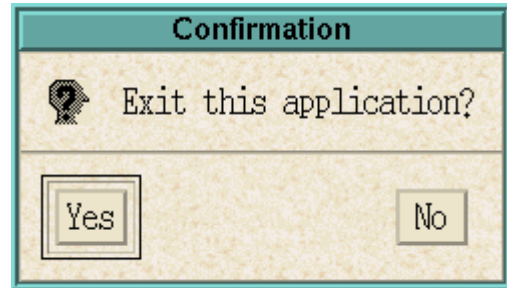


Fig. 6.20-16 Exit Confirmation Dialog

When an adjusting value is changed, the dialogue in Fig. 6.20-17 is popped up. When clicking [Yes], the changed system table adjusting value is written into H/D and the application terminates. When clicking [No], the application terminates without writing any adjusting value. If [Cancel] is clicked, the screen returns to the original screen without terminating the application.

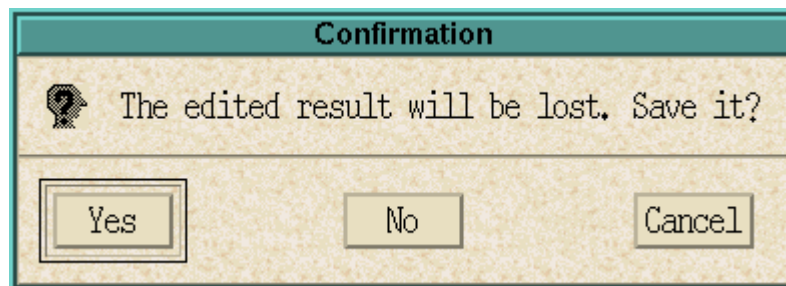


Fig. 6.20-17 System table save and exit confirmation dialog

6.20.6 Error specifications

Error specifications are shown in Table 6.20-3.

Table 6.20-3 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.20.7 Other cautions

To obtain correct measurement values, a prescan must be performed at least once before executing this tool.

6.21 GC Gain Parameter Setting

6.21.1 General

This is a tool to calculate a GC gain value from the entered measurement value and target value as a result of FOV measurement and make change of adjustment value easy.

6.21.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click GC Gain Parameter Set in the displayed items and click [OK] or double-click the selected item.

6.21.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.21-1.

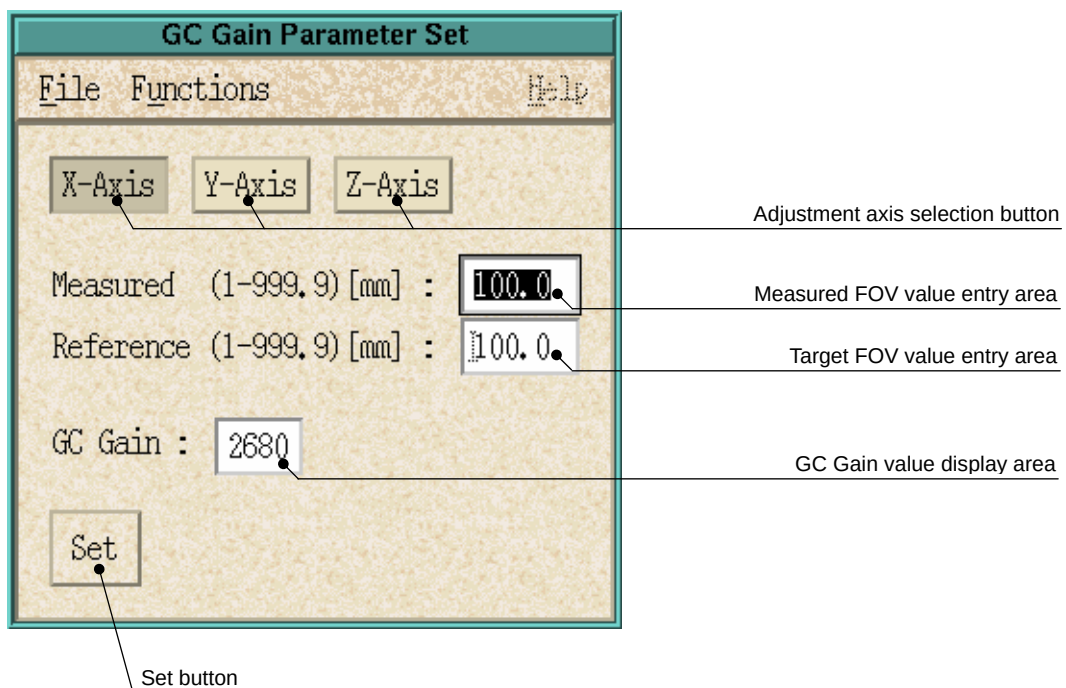


Fig. 6.21-1 GC Gain Parameter Setting Display Screen

6.21.4 Operation procedure

Follow the next steps to execute the process.

6.21.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following methods are available for changing parameters.

6.21.4.1.1 Selection of parameter by pushbutton

Each of the arrayed pushbuttons is for selection of a single parameter, and clicking one of them by the mouse cursor changes the corresponding parameter.

6.21.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

Table 6.21-1 Parameter Setting Range

Parameter Name	Setting Range
Adjustment axis	X, Y, Z
Measured FOV value	1.0 ~ 999.9
Target FOV value	1.0 ~ 999.9

6.21.4.2 Calculation and storage of GC correction value

Calculation and storage of the GC Gain value for the changed parameter can be executed by clicking [Set] button. GC Gain value is calculated by the following formula and the result is stored in the H/D or Memory of the system table.

$$\text{New Parameter} = \text{Current Parameter} * \text{Reference} / \text{Measured}$$

The value of each GC mode is obtained from the maximum inclined magnetic field strength by proportional calculation using these results, and updated automatically at the same time. Updated values can be confirmed on the GC gain value table display screen shown in the next paragraph.

6.21.4.3 GC gain values check

Select [ParametersList ...] on "Functions Menu" of the Menu Bar, a GC gain values list display screen is displayed as shown in Fig. 6.21-3. No GC gain value can be changed in this screen.

For closing the GC gain values list display screen, click [Close] button.

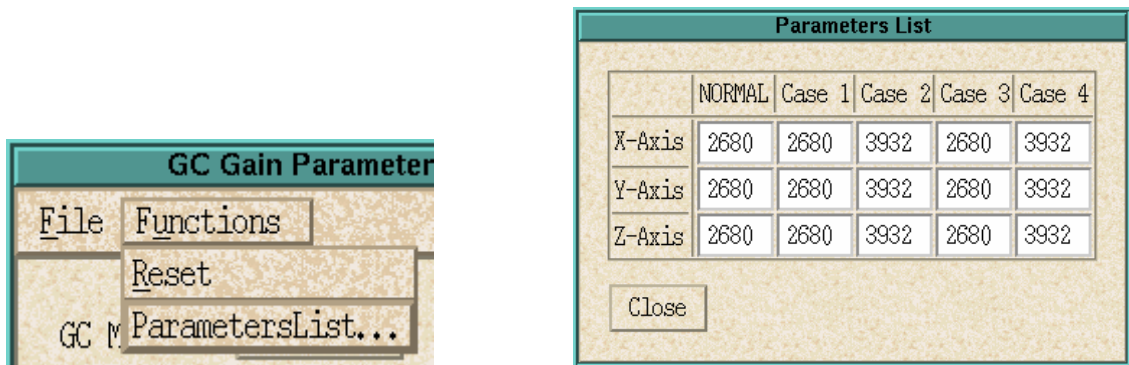


Fig. 6.21-2 Parameters List button

Fig. 6.21-3 GC gain values list display screen

6.21.4.4 Recovery of GC Gain value

The changed GC Gain value can be recovered by selecting [Reset] of "Functions Menu" to the one after start-up.

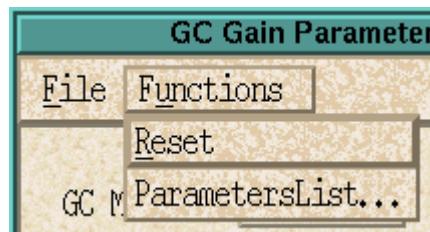


Fig. 6.21-4 Reset Button

6.21.5 End

Selecting [Quit] button from "File Menu" in the menu bar pops up the dialog shown in Fig. 6.21-6. Clicking [Yes] button on this dialog ends the process, while clicking [No] button returns to the previous screen without ending the process.

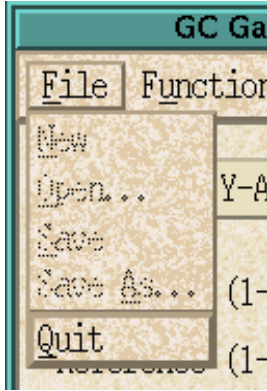


Fig. 6.21-5 Quit Button



Fig. 6.21-6 Exit Confirmation Dialog

6.21.6 Error specifications

Error specifications are shown in Table 6.21-2.

Table 6.21-2 Error Specifications

Message	Status	Solution
Calculated value was out of permissible range 0 ~ 4095.	The calculated GC Gain value from the parameter exceeds the permissible setting range.	Reset the parameter.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Checking the system status, retry the same operation procedure.

6.22 FOV Adjustment

6.22.1 General

This is a tool to calculate the GC correction value from the entered measurement value as a result of FOV measurement and target value and to make change of adjustment value easy.

6.22.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click FOV Adjustment in the displayed items and click [OK] or double-click the selected item.

6.22.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.22-1.

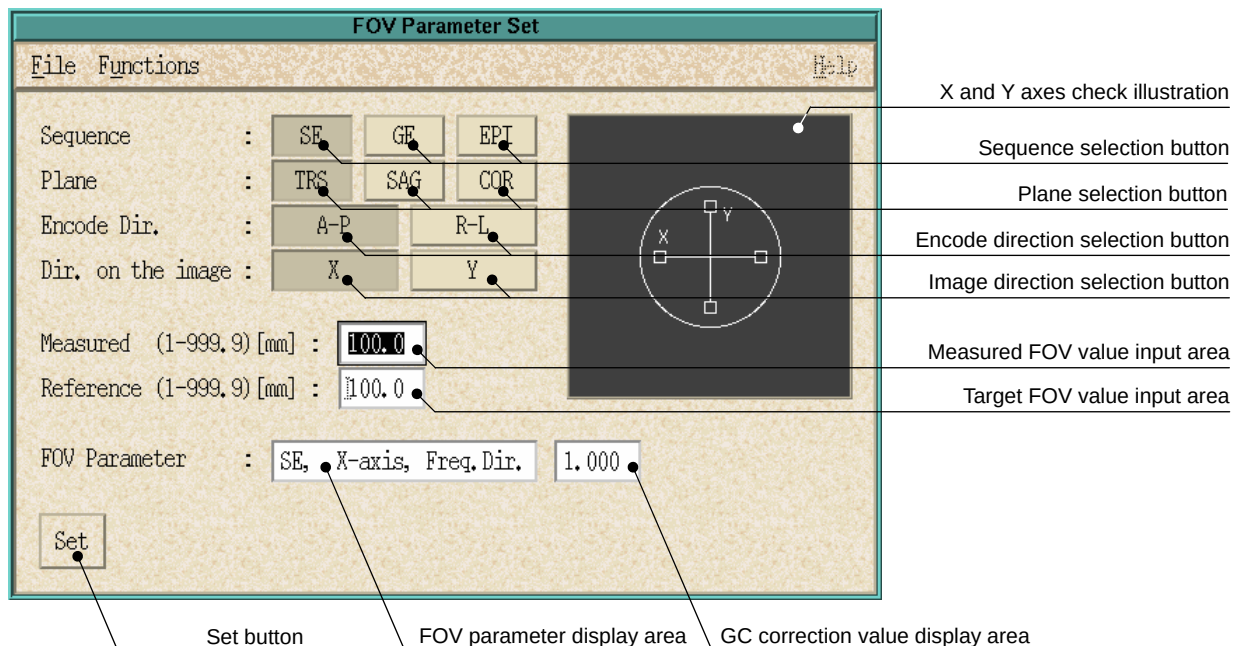


Fig. 6.22-1 FOV Adjustment Display Screen

6.22.4 Operation procedure

Follow the next steps to execute the process.

6.22.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following methods are available for changing parameters.

6.22.4.1.1 Selection of parameter by pushbutton

Each of the arrayed pushbuttons is for selection of a single parameter, and clicking one of them by the mouse cursor changes the corresponding parameter.

6.22.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

Table 6.22-1 Parameter Setting Range

Parameter Name	Setting Range
Sequence	SE, GE, EPI
Plane	TRS, SAG, COR
Encode direction	A-P, R-L, H-F
Image direction	X, Y
Measured FOV value	1.0 ~ 999.9
Target FOV value	1.0 ~ 999.9

6.22.4.2 Calculation and storage of GC correction value

Calculation and storage of the GC correction value for the changed parameter can be executed by clicking [Set] button.

GC correction value is calculated by the following formula and the corresponding parameter (always displayed in the FOV parameter display area) is stored in the H/D or Memory of the system table.

$$\text{New Parameter} = \text{Current Parameter} * \text{Reference} / \text{Measured}$$

6.22.4.3 Recovery of GC correction value

The changed GC correction value can be recovered by selecting [Reset] of "Functions Menu" to the one after start-up.

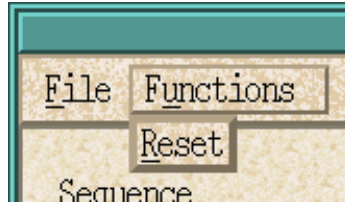


Fig. 6.22-2 Reset Button

6.22.5 End

Selecting [Quit] button from "File Menu" in the menu bar pops up the dialog shown in Fig. 6.22-4. Clicking [Yes] button on this dialog ends the process, while clicking [No] button returns to the previous screen without ending the process.



Fig. 6.22-3 Quit Button



Fig. 6.22-4 Exit Confirmation Dialog

6.22.6 Error specifications

Error specifications are shown in Table 6.22-2.

Table 6.22-2 Error Specifications

Message	Status	Solution
Calculated value was out of permissible range 0.900 ~ 1.100.	The calculated GC correction value from the parameter exceeds the permissible setting range.	Reset the parameter.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Checking the system status, retry the same operation procedure.

6.23 FSE Auto Adjustment

6.23.1 General

This is a tool to automatically adjust the gradient field pulse intensity for correction in the slice, phase and frequency axes in Fast Spin Echo (FSE) sequence. Carrying out measurements by changing correction value according to the given parameters, the optimum value among the obtained values is used as the adjustment value.

6.23.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click FSE Auto Adjustment in the displayed items and click [OK] or double-click the selected item.

6.23.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.23-1.

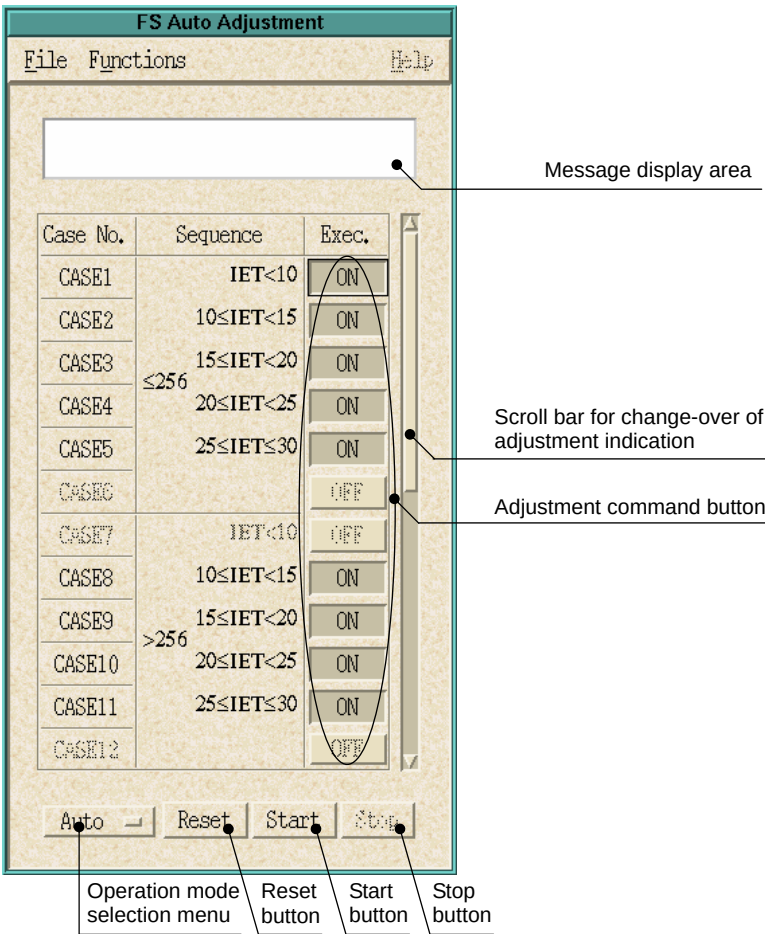


Fig. 6.23-1 FSE Auto Adjustment Display Screen

6.23.4 Operation procedure

Follow the next steps to execute the process.

6.23.4.1 Setting adjustment parameters

On the FSE auto adjustment main screen (hereafter called Main screen) shown in Fig. 6.23-1, set the adjustment command button to ON for the case to be adjusted, while set it to OFF for the case not be adjusted.

This ON/OFF setting can be done by moving the mouse cursor on the desired button and clicking the left mouse button. For change-over of Frequency direction rephase (CASE 1~12)/Slice direction rephase (CASE 13~24), operate the scroll bar for change-over adjustment indication button by using the mouse cursor (CASE 6,12,18,24 is not used).

For the case set to ON, the default adjustment item is set. To change the adjustment item, move the mouse cursor to the adjustment command button set to ON and click the right mouse button. The detailed condition setting screen as shown in Fig. 6.23-2 is displayed. Then, click the left mouse button on the item to be adjusted to set ON/OFF.

To close the setting, click [OK] button.

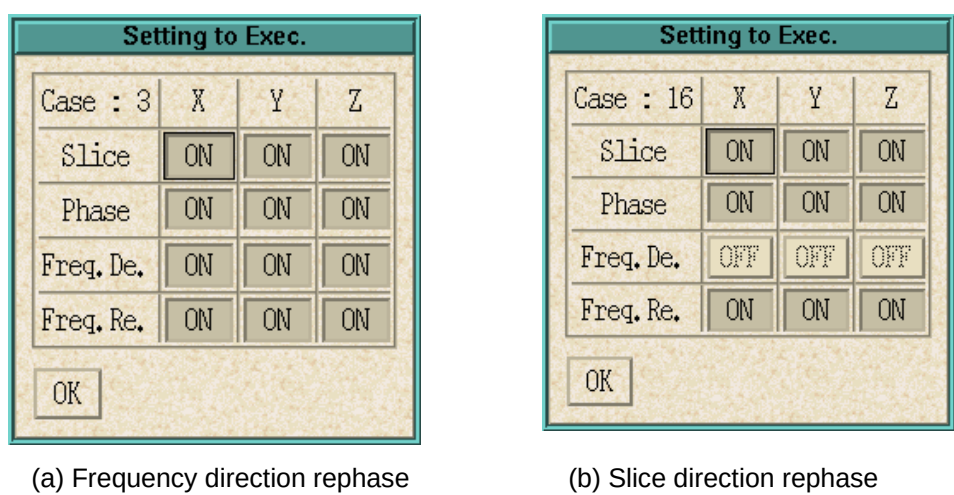


Fig. 6.23-2 Detailed Condition setting Screen

6.23.4.2 Selection of Auto/Manual mode

Select **Auto** or **Manual** on the mode selection menu of the Main screen. In **Auto** mode, a result of each measurement is not displayed, but in **Manual** mode, a measurement result after each measurement is displayed (Fig. 6.23-4 and 6.23-5).

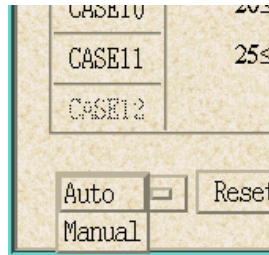


Fig. 6.23-3 Mode Selection Menu

6.23.4.3 Starting measurement

Click [**Start**] button to start the measurement.

In the **Manual** mode measurement, the measurement result for each adjustment is displayed on the screen. The measurement result for **Slice** is displayed in two upper and lower areas as shown in Fig. 6.23-4. The measurement result for **Dephase**, **Rephase** and **Phase** is displayed only in the upper area respectively. To go to the measurement of the next adjustment item, click [**OK**] button. After the measurement of the last adjustment item, the adjustment result display screen as shown in Fig. 6.23-5 is displayed. To suspend the measurement before completing all the specified adjustment items, click [**Cancel**] button. In this case, the parameter entry screen (Fig. 6.23-12) is displayed, so follow the operation procedure described in Paragraph 6.23.4.9.

In the **Auto** mode measurement, the result of each measurement is not displayed. After all the measurements are completed, the adjustment result screen as shown in Fig. 6.23-5 is displayed.

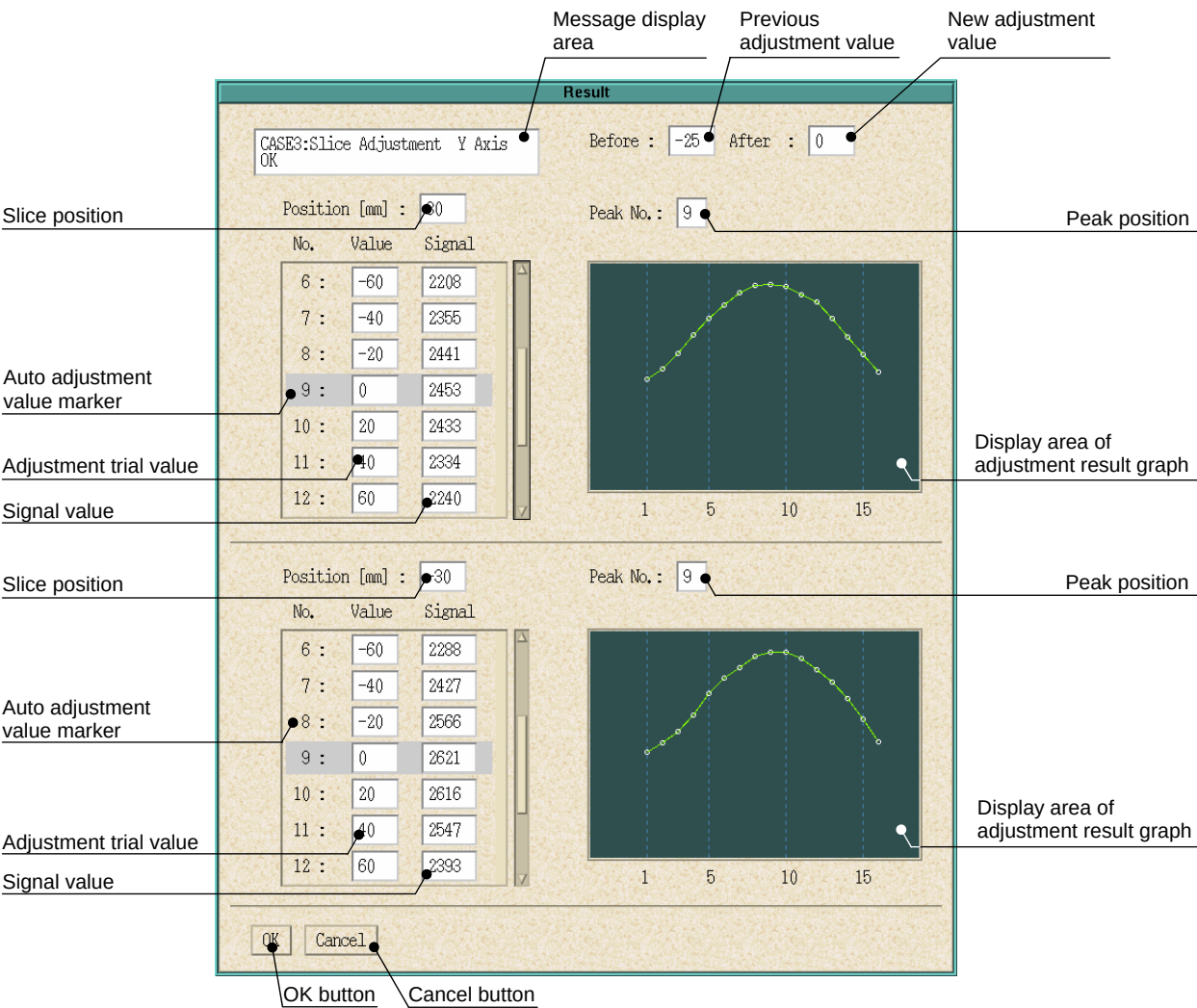


Fig. 6.23-4 Example of Measurement Result (Slice)

In the message display area are displayed the case No. being adjusted, adjustment item, axis and status in the following display format.

Case No. + Adjustment item $\left\{ \begin{array}{l} \text{Slice} \\ \text{Phase} \\ \text{Dephase} \\ \text{Rephase} \end{array} \right\} + \text{Adjustment axis} \left\{ \begin{array}{l} \text{X} \\ \text{Y} \\ \text{Z} \end{array} \right\} + \text{Status} \left\{ \begin{array}{l} \text{Execution} \\ \text{OK} \\ \text{NG} \\ \text{Abort} \\ \text{Error} \end{array} \right\}$

For example, it is displayed as follows:

Case 10 : Slice Adjustment X Axis OK

The previous adjustment value is the one set in the system table before starting this adjustment, and the new adjustment value is the one obtained by the latest measurement if the result is OK, but is the previous one if the result is NG. This value is not written in the H/D but is registered in the system table memory.

The position of the peak value of the measurement result graph is displayed in the peak position, and the position is automatically marked as the optimum value among the adjustment trial values. When any peak value is not found, no marker is put on.

6.23.4.4 Stopping measurement

Click [Stop] button. In any of Auto and Manual mode, the measurement is stopped and the adjustment results by that time are displayed as shown in Fig. 6.23-5.

6.23.4.5 Checking adjustment results

Upon completion of adjustments of the specified items, the adjustment result screen as shown in Fig. 6.23-5 is displayed. ○ mark indicates the item been adjusted, while × mark indicates the item not been adjusted. The adjustment command button for the adjusted item is automatically turned OFF. But the button turns to the initial status by restarting the application. Change-over of Frequency direction rephase (CASE 1~12)/Slice direction rephase (CASE 13~24) can be carried out with Rephase Type change-over menu (CASE 6,12,18,24 is not used).

This screen is also displayed when any measurement error occurs during measurement. Reset the measurement parameter (refer to Paragraph 6.23.4.9) and retry the adjustment.

In case items can't be adjusted exist, Rephase Type change-over menu becomes yellow.

Clicking [Close] button closes this screen and the step goes to Paragraph 6.23.4.6 Checking adjustment values.

Case No. Sequence

Case No.	Sequence	Result											
		①			②			③			④		
Case No.	Sequence	s-X	s-Y	s-Z	p-X	p-Y	p-Z	f-X	f-Y	f-Z	fr-X	fr-Y	fr-Z
CASE1	IET<10	•	•	•	×	×	•	•	•	•			
CASE2	10≤IET<15	•	•	•	×	×	•	•	•	•			
CASE3	15≤IET<20	•	•	•	•	×	•	•	•	•	•	•	•
CASE4	20≤IET<25	•	•	•	•	×	•	•	•	•	•	•	•
CASE5	25≤IET≤30	•	•	•	•	×	•	•	•	•	•	•	•
CASE6													
CASE7	IET<10												
CASE8	10≤IET<15	•	•	•	×	×	•	•	•	•			
CASE9	15≤IET<20	•	•	•	•	•	•	•	•	•	•	•	•
CASE10	20≤IET<25	•	•	•	•	×	•	•	•	•	•	•	•
CASE11	25≤IET≤30	•	•	•	•	×	•	•	•	•	•	•	•
CASE12													

Close Clear

Rephase Type : Freq.

Close button Clear button Rephase Type change-over menu

- ① Adjustment items in slice direction (X,Y,Z)
- ② Adjustment items in phase direction (X,Y,Z)
- ③ Adjustment items in frequency direction (X,Y,Z)
- ④ Adjustment items in frequency direction (Rephase)(X,Y,Z)

Fig. 6.23-5 Adjustment Result Screen

Clicking [Clear] button displays the dialog shown in Fig. 6.23-6. Clicking [Yes] on this dialog clears the resultant data and returns the screen to the initial status (but not to clear the adjustment value). Clicking [No] does not clear the data.

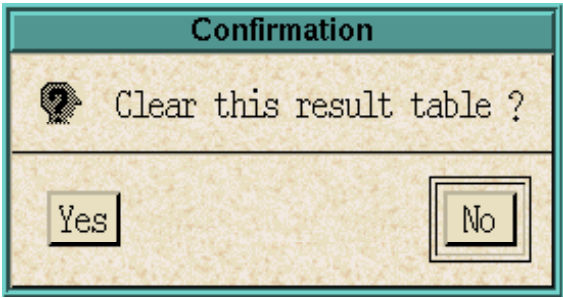


Fig. 6.23-6 Confirmation Screen to clear Adjustment Result

It is also possible to display this adjustment result screen by selecting [Result] on "Functions Menu".

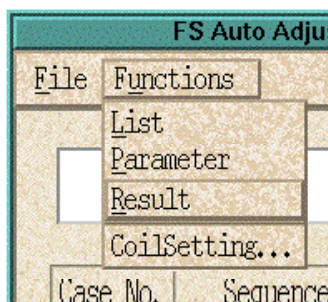


Fig. 6.23-7 Button to display Adjustment Result Screen

6.23.4.6 Checking adjustment value

After completion of the adjustment, select [List] on "Functions Menu" and check the adjustment value on the adjustment value screen shown in Fig. 6.23-9.

The automatically adjusted item is registered in the system table memory.

Change-over of Frequency direction rephase (CASE 1~12)/Slice direction rephase (CASE 13~24) can be carried out with Rephase Type change-over menu (CASE 6,12,18,24 is not used).

The adjustment value list display screen can be closed by clicking [Close] button on the screen.

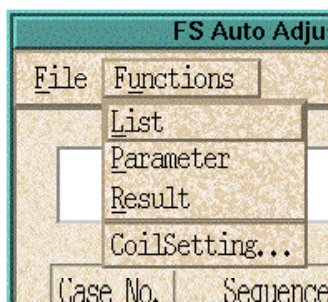
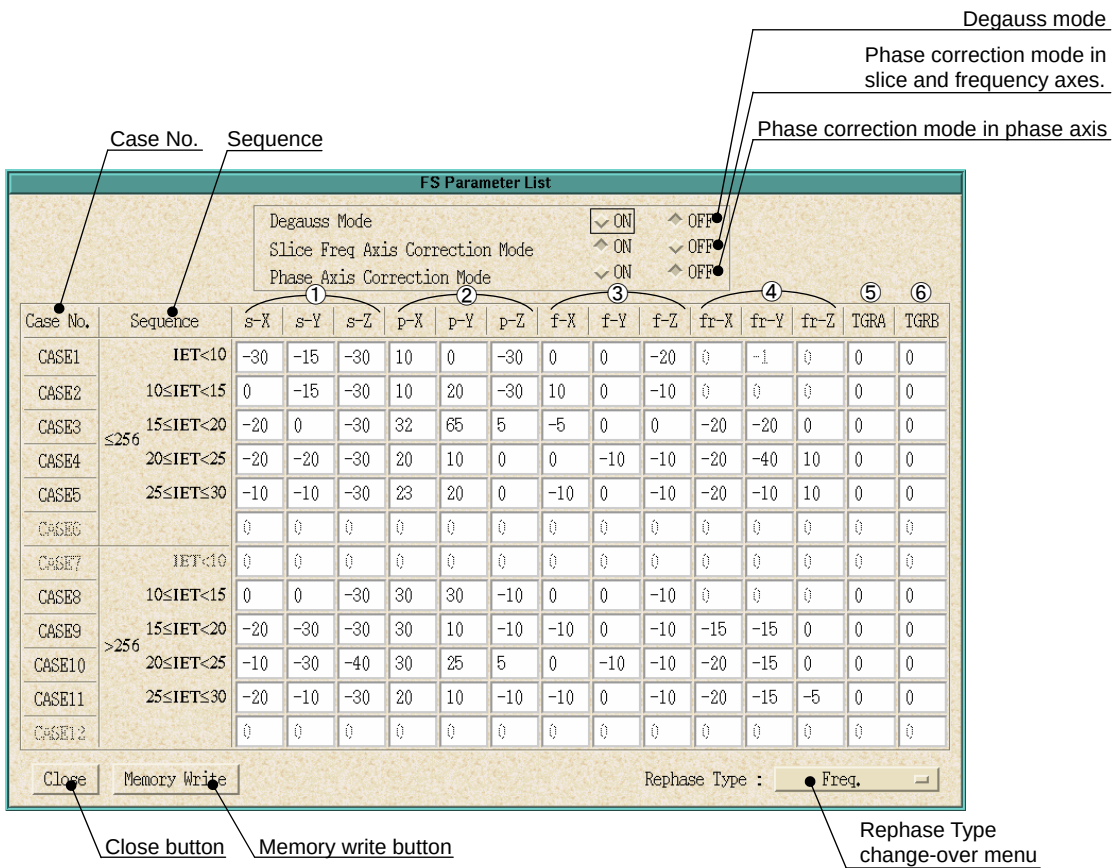


Fig. 6.23-8 Button to display Adjustment Value



- ① Adjustment items in slice direction (X,Y,Z)
- ② Adjustment items in phase direction (X,Y,Z)
- ③ Adjustment items in frequency direction (X,Y,Z)
- ④ Adjustment items in frequency direction (Rephase)(X,Y,Z)
- ⑤ 1-echo correction value in frequency direction
- ⑥ 2-echo correction value in frequency direction

Fig. 6.23-9 Screen to display Adjustment Values List

6.23.4.7 Direct entry of adjustment value

Adjustment values can be changed by direct input without using measurement values. Directly input adjustment values on the screen shown in Fig. 6.23-9. To set numerical parameter, place the mouse cursor on the entry area of the parameter to be changed and select the entry area. Input value from the keyboard and change the parameter value by either pressing <Enter> key or removing the focus from the entry area.

Change-over of Frequency direction rephase (CASE 1~12)/Slice direction rephase (CASE 13~24) can be carried out with Rephase Type change-over menu (CASE 6,12,18,24 is not used).

Table 6.23-1 Parameter Setting Range (Adjustment value)

Parameter Name	Setting range
Adjustment items in slice direction (X, Y, Z)	-999 ~ 999
Adjustment items in phase direction (X, Y, Z)	-999 ~ 999
Adjustment items in frequency direction (X,Y,Z)	-999 ~ 999
Adjustment items in frequency direction (Rephase) (X,Y,Z)	-999 ~ 999
1-echo correction in frequency direction	-999 ~ 999
2-echo correction in frequency direction	-999 ~ 999
Degauss mode	ON, OFF
Phase correction mode in slice and frequency axes	ON, OFF
Phase correction mode in phase axis	ON, OFF

To register the changed adjustment value in the system memory, click [Memory Write] button.

By clicking [Close] button on the screen, the screen to display adjustment values list can be closed. When trying to close the screen without registering the values in the system table memory, the screen shown in Fig. 6.23-10 is displayed. Clicking [Yes] determines the change, while clicking [NO] can cancel the change.

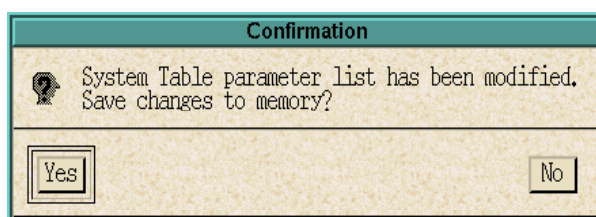


Fig. 6.23-10 Confirmation Screen for System Table Memory Change

6.23.4.8 Resetting adjustment item

The adjustment command button on the Main Menu are reset to the default at start-up by clicking [Reset] button on the screen.

6.23.4.9 Setting parameter

Select [Parameter] on "Functions Menu" of the Menu Bar and the parameter entry screen shown in Fig. 6.23-12 is displayed. Input measurement parameters as required. If no change is required, this operation can be omitted.

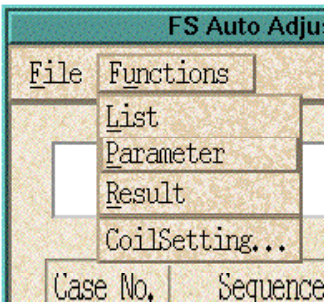


Fig. 6.23-11 Parameter Button

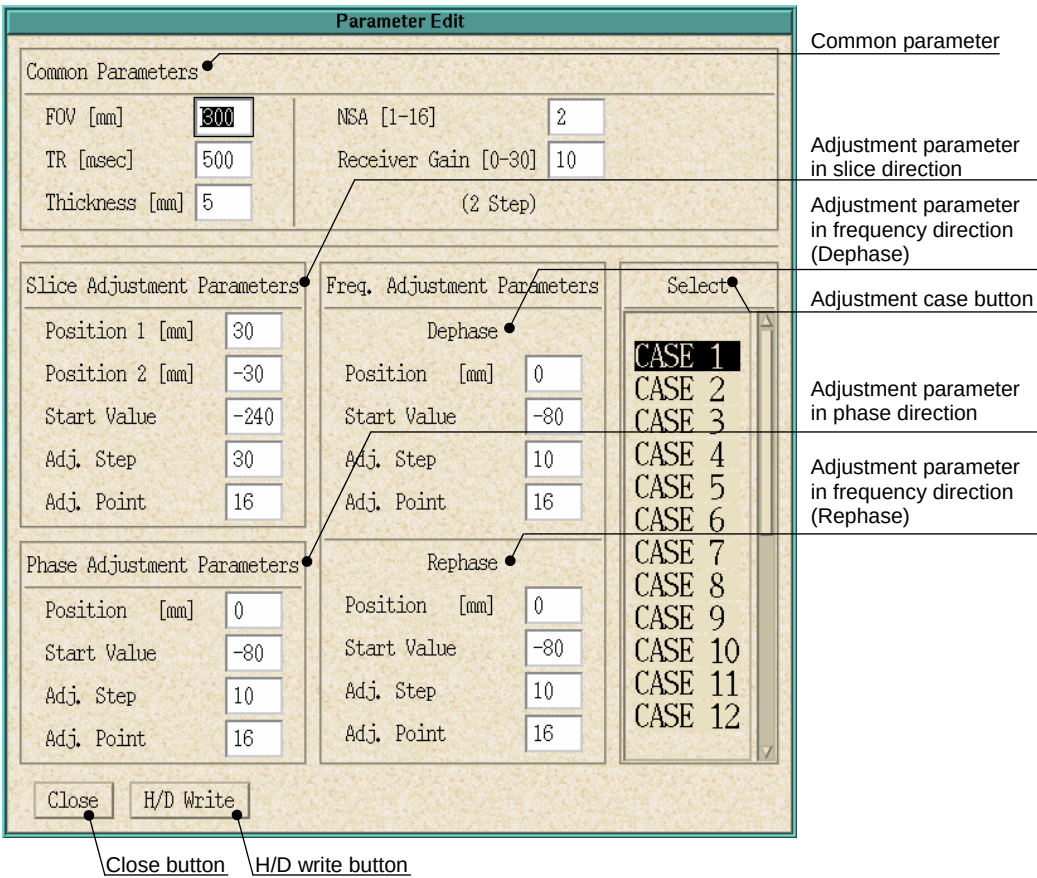


Fig. 6.23-12 Parameter Entry Screen

The changed parameter is stored in the H/D by clicking [H/D Write] button and is used as default parameter for the next measurement.

Clicking [Close] button closes this screen. If the screen is tried to close without storing the changed parameter, the H/D Write Confirmation screen as shown in Fig. 6.23-13 is displayed. To store it in the H/D, click [Yes], while not to store it, click [No].

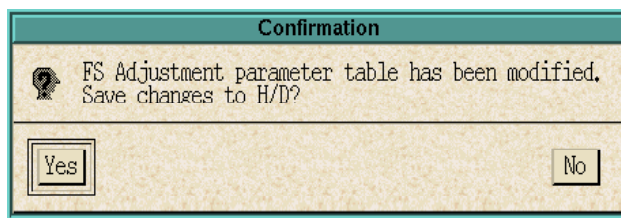


Fig. 6.23-13 H/D Write Confirmation Screen

Common Parameters on the parameter entry screen is common to all the cases. All of other parameters must be set for each case. To set numerical parameter, place the mouse cursor on the entry area of the parameter to be changed and select the entry area. Input a numerical value from the keyboard and change the parameter by either pressing <Enter> key on the keyboard or removing the focus from the entry area. Each parameter has a setting range as shown in Table 6.23-2 ~ 6.23-6.

Table 6.23-2 Parameter Setting Range (Common Parameter)

Parameter Name	Setting range	Remarks
FOV	50 ~ 500	
TR	200 ~ 9999	
Thickness	2 ~ 100	
NSA	1 ~ 16	
Receiver Gain	0 ~ 30	Only even figures can be input.

Table 6.23-3 Parameter Setting Range
(Adjustment Parameter in Slice Direction)

Parameter Name	Parameter description	Setting range
Position 1	Adjustment position	-999 ~ 999
Position 2	Adjustment position	-999 ~ 999
Start Value	Initial value of adjustment value	-999 ~ 999
Adj. Step	Incremental value of adjustment value	1 ~ 100
Adj. Point	Number of measurement points for addition of incremental adjustment value	1 ~ 16

Table 6.23-4 Parameter Setting Range
(Adjustment Parameter in Phase Direction)

Parameter Name	Parameter description	Setting range
Position	Adjustment position	-999 ~ 999
Start Value	Initial value of adjustment value	-999 ~ 999
Adj. Step	Incremental value of adjustment value	1 ~ 100
Adj. Point	Number of measurement points for addition of incremental adjustment value	1 ~ 16

Table 6.23-5 Parameter Setting Range
(Adjustment Parameter in Frequency (Dephase) Direction)

Parameter Name	Parameter description	Setting range
Position	Adjustment position	-999 ~ 999
Start Value	Initial value of adjustment value	-999 ~ 999
Adj. Step	Incremental value of adjustment value	1 ~ 100
Adj. Point	Number of measurement points for addition of incremental adjustment value	1 ~ 16

Table 6.23-6 Parameter Setting Range
(Adjustment Parameter in Frequency (Rephase) Direction)

Parameter Name	Parameter description	Setting range
Position	Adjustment position	-999 ~ 999
Start Value	Initial value of adjustment value	-999 ~ 999
Adj. Step	Incremental value of adjustment value	1 ~ 100
Adj. Point	Number of measurement points for addition of incremental adjustment value	1 ~ 16

6.23.4.10 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.23.4.10.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.23.4.10.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

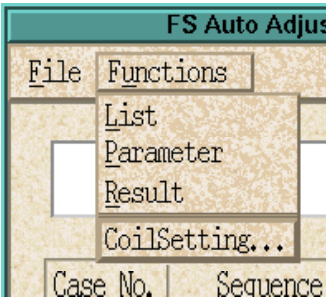


Fig. 6.23-14 Coil Setting Window Display Button

6.23.4.10.3 Display screen

The display screen and function names are shown in Fig. 6.23-15 and Fig. 6.23-16.

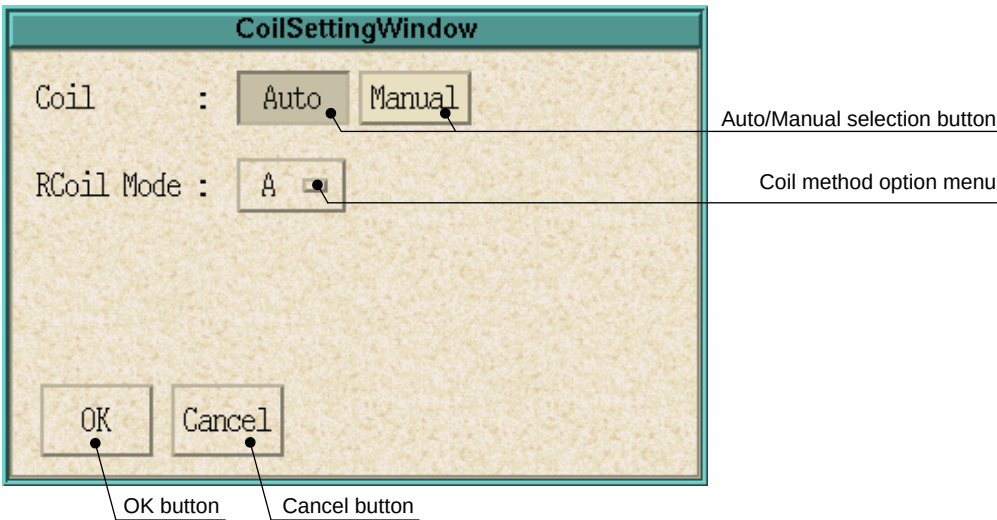


Fig. 6.23-15 Coil Setting Window Display Screen (Auto mode)

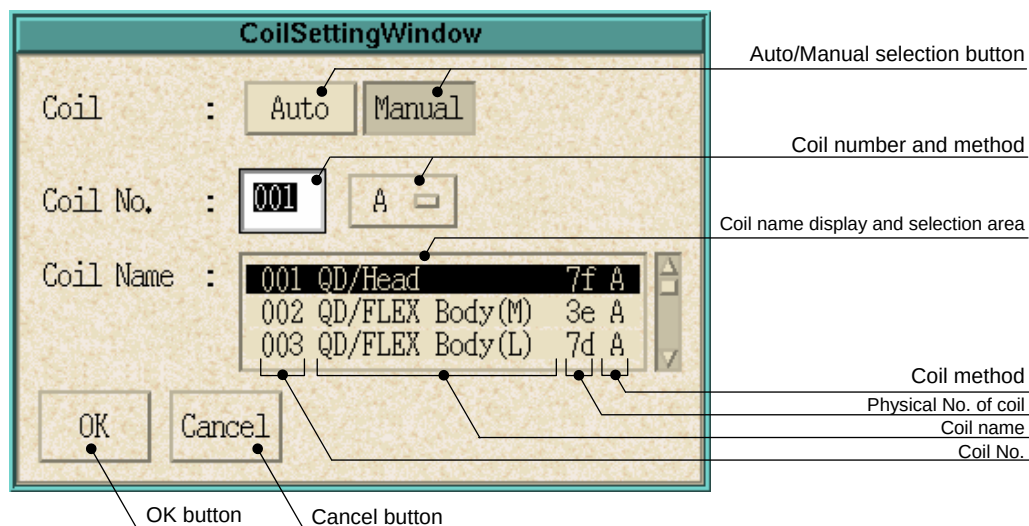


Fig. 6.23-16 Coil Setting Window Display Screen (Manual mode)

6.23.4.10.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.23.4.10.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.23.4.10.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area

The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.23.4.10.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.23.4.10.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.23.4.10.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.23.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.23-18 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.



Fig. 6.23-17 Quit Button



Fig. 6.23-18 Exit Confirmation Dialog

When the adjustment value on the system table memory is changed, the dialog box as shown in Fig. 6.23-19 is popped up. Clicking [Yes] on this box writes the changed system table adjustment value in the H/D and closes the application, while clicking [No] closes the application without writing the value. Clicking [Cancel] returns to the previous screen without closing the application.

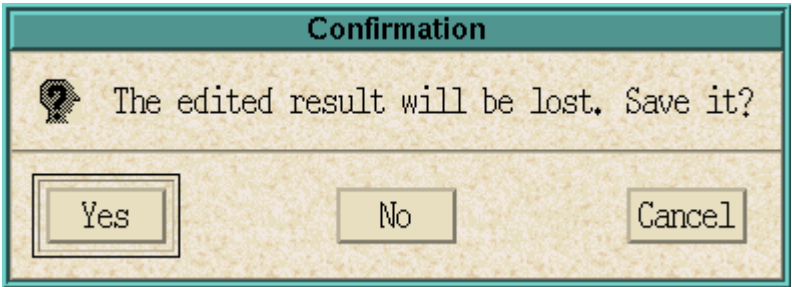


Fig. 6.23-19 System Table Storage and Exit Confirmation Dialog

6.23.6 Error specifications

Error specifications are shown in Table 6.23-7.

Table 6.23-7 Error Specifications

Message	Status	Solution
Signal has underflowed	Signal intensity is too low.	Adjust Gain.
Signal has overflowed	Signal intensity is too high.	Adjust Gain.
Parameter Error	Measurement error	Adjust FOV, TR, Thickness, etc.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.23.7 Other cautions

To obtain correct measurement values, a prescan must be performed at least once before executing this tool.

6.24 Easy Raw Data Analysis

6.24.1 General

This is a tool to analyze raw data and image data after reconstruction. Raw data are read from the reconstruction memory or H/D and the signals are imaged and displayed. By applying analysis functions such as W/L adjustment, profile display, image zooming, etc. to this image, the signal values can be visually analyzed. Simultaneously, quantitative analysis is also possible by displaying digital values of the signals on the profile and detecting the maximum and minimum values.

Up to 10 pieces of the read raw data can be stored in the H/D.

Furthermore, two or more analysis functions can be simultaneously run, thus enabling to perform comparative analysis.

6.24.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Raw Data Analysis II in the displayed items and click [OK] or double-click the selected item.

6.24.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.24-1.

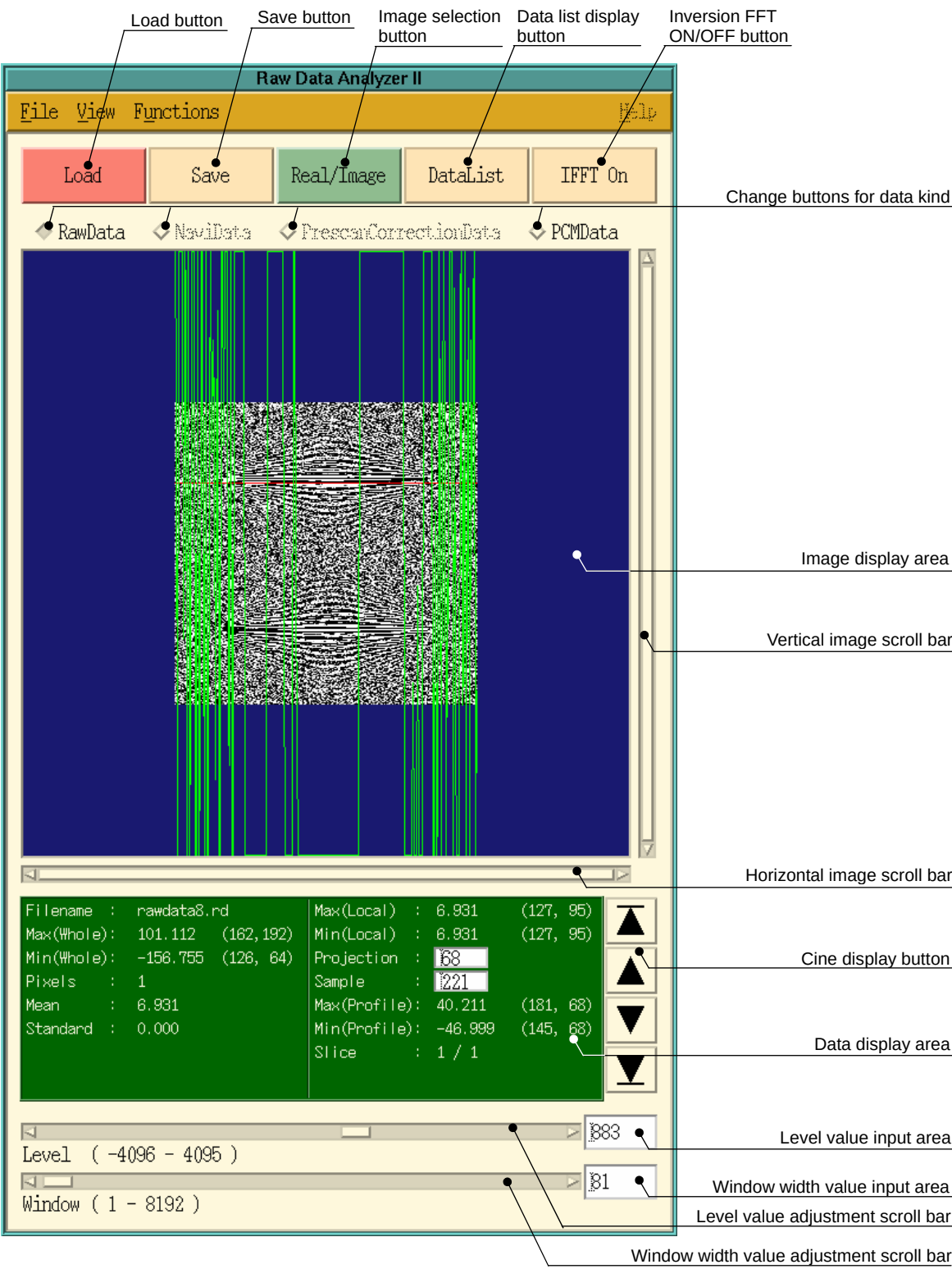


Fig. 6.24-1 Easy Raw Data Analysis Display Screen

6.24.4 Operation procedure

Follow the next steps to execute the process.

6.24.4.1 Loading raw data

Either select [Load] from "Functions Menu" in the Menu Bar or click [Load] button on the screen to pop up the screen shown in Fig. 6.24-3.

Click [Memory] to load raw data from the reconstruction memory, while click [H/D] to load it from the H/D (default at pop-up is always loading from the H/D).

To cancel loading of raw data, click [Cancel] button. The loading screen is closed and the main screen is displayed.

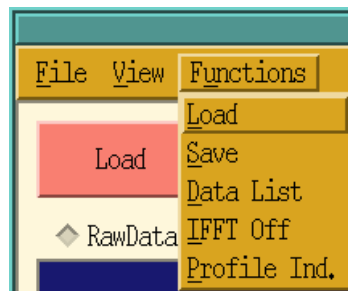


Fig. 6.24-2 Load Button

6.24.4.1.1 Loading from H/D

Clicking [H/D] button displays the raw data file list. With this application, only the file which is in the directory hierarchy lower than the directory shown below and of which extension is [.rd] or [.img] can be loaded.

Select the desired data file from the list and, if necessary, change IFFT ON/OFF selection, Data type, Matrix and Slice No. Either clicking [Load] button or double-click the selected item executes the load process of the raw data.

If the data file is the one saved by the raw data saving function of this application, appropriate data type, matrix and slice No. are automatically set when the data file is selected from the list, therefore the data file can be immediately loaded unless any parameter change is required.

Retrieval : /opt/MR/h0

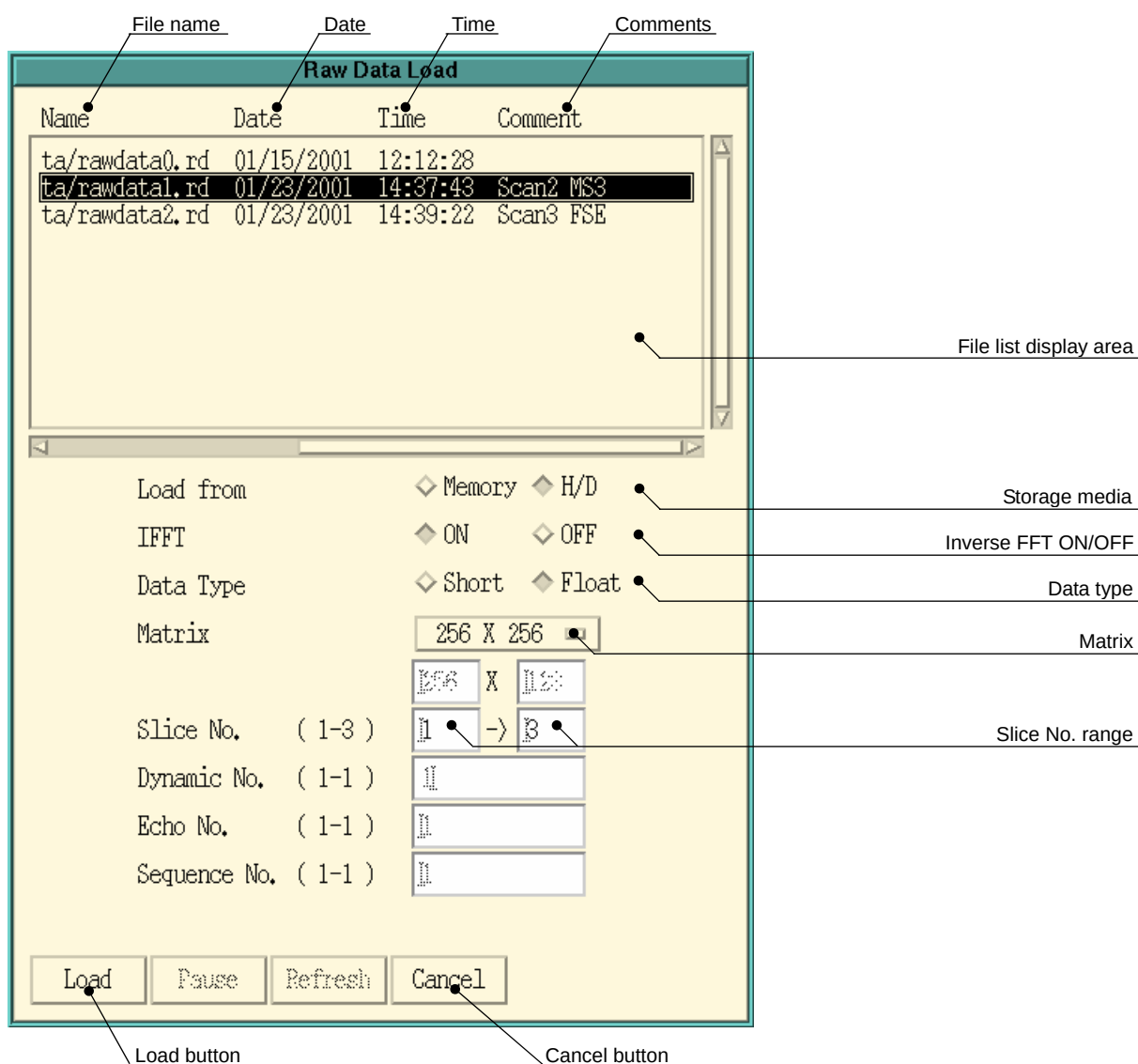


Fig. 6.24-3 Load Screen (H/D)

6.24.4.1.2 Loading from reconstruction memory

Clicking [Memory] displays the channel name list. Select the desired channel by the mouse. If change IFFT ON/OFF selection, Slice No., Dynamic No., Echo No. and Sequence No.

Since Matrix is automatically set, it needs not be changed. Clicking [Load] executes the loading process of the raw data.

Without taking any scan, data loading from the memory cannot be done. In case this application is started before taking a scan, click [Refresh] button after the scan, then load the data. In case more than one of this application are running, more than one data cannot be simultaneously loaded from the memory.

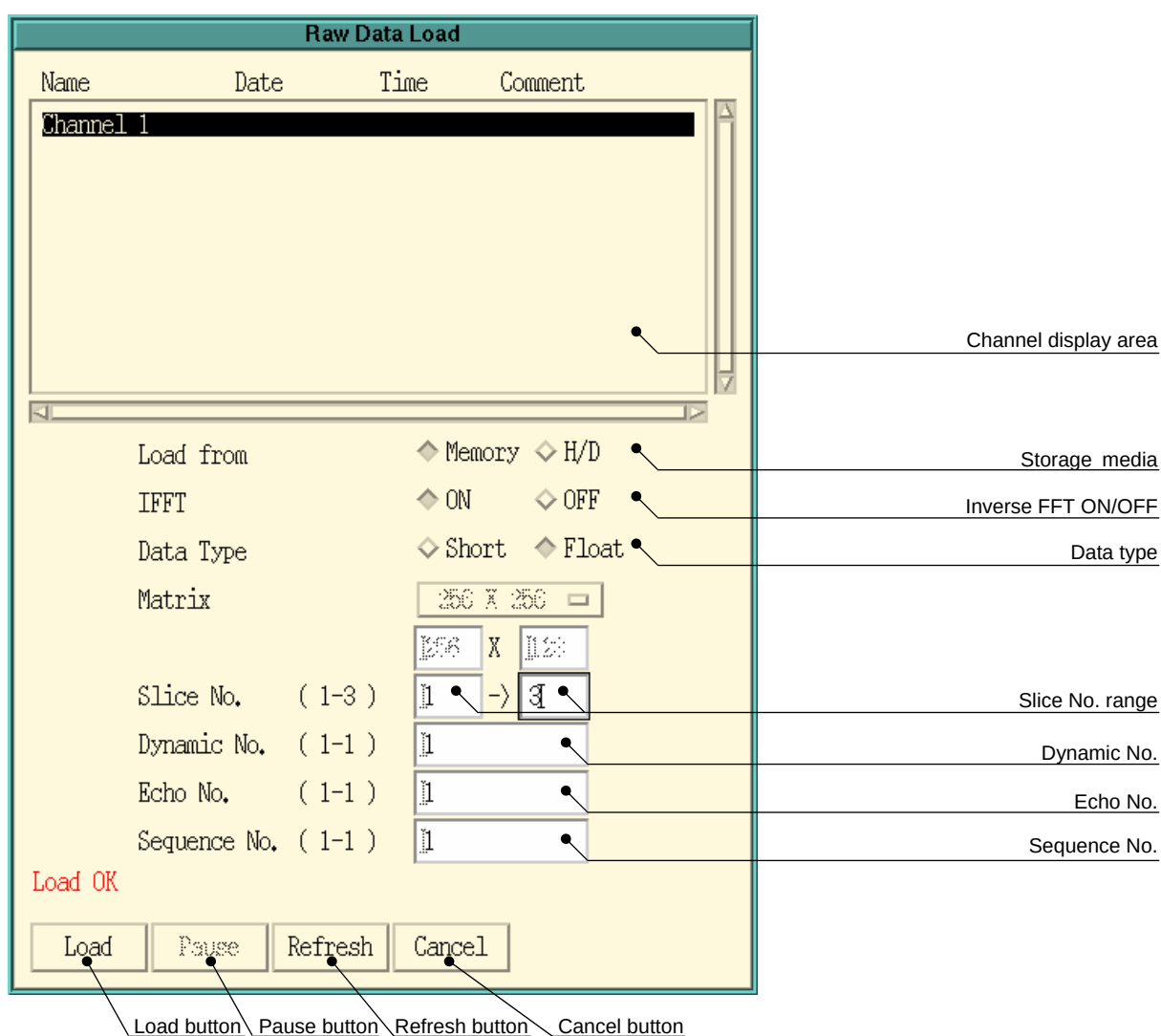


Fig.6.24-4 Load Screen (Reconfiguration memory)

6.24.4.2 Suspension of raw data loading process

To suspend the raw data saving process being executed, click [Pause] button. The suspension dialog shown in Fig. 6.24-5 is popped up. Clicking [Stop] button here terminates the saving process, while clicking [Continue] button restart the saving process.

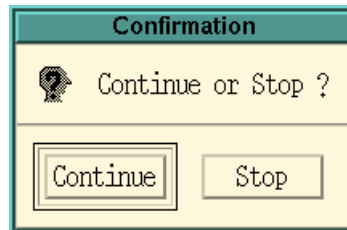


Fig. 6.24-5 Suspension Dialog

6.24.4.3 Change of data kind

When changing the kind of loaded data, the selection can be done by changing the selection button shown in Fig. 6.24-6 to the kind of intended kind.

There are four kinds of data, Raw data, Navi data and Pre-scan correction data, PCM data. And after completion of loading data, the default becomes [RawData].

The Navi data and Pre-scan correction data become non-active except at EPI scan.

Even at the EPI scan, it is impossible to select the button when there is no data.

The PCM data become non-active except at the scan which is not using the PCM mode of RAPID.

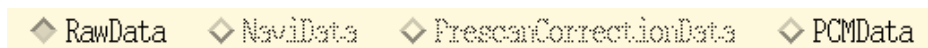


Fig. 6.24-6 Change button for data kind

6.24.4.4 Profile display

The signals on one line in an image are displayed in graph.

6.24.4.4.1 Projection display

By putting the mouse cursor in an image and pressing the left mouse button, the projection at the position where the mouse cursor is placed is displayed with a green line.

The coordinates showing the projection position are displayed in the projection input area of the data display area. By inputting coordinate values directly in this area, the projection can also be displayed. In Max(Profile) and Min(Profile) items are displayed the max. and min. values on the profile.

By using < ↑ > and < ↓ > keys, the projection position can be shifted in a unit of pixel.

6.24.4.4.2 Sample display

By placing the mouse cursor on the screen and pressing the middle mouse button, the sample at the position under the mouse cursor is displayed with green line. The coordinates representing the sample position are displayed in the sample input area of the data display field. By directly inputting coordinate values there, the sample can also be displayed.

The maximum value and minimum value on the profile and their positions are displayed in Max(Profile) and Min(Profile) items of the data display area.

Using $\leftarrow$ and $\rightarrow$ keys, the position can be moved in 1 pixel steps.

6.24.4.4.3 Clearing profile

Select [Profile Clear] from "View Menu" of the Menu Bar. The profile line being displayed is cleared.

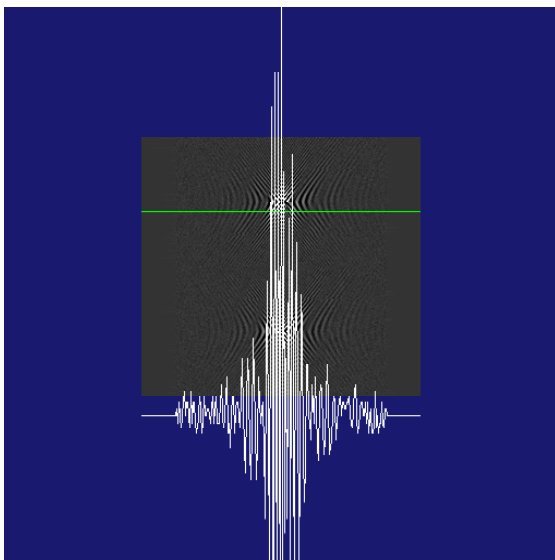


Fig. 6.24-7 Example of Projection Display

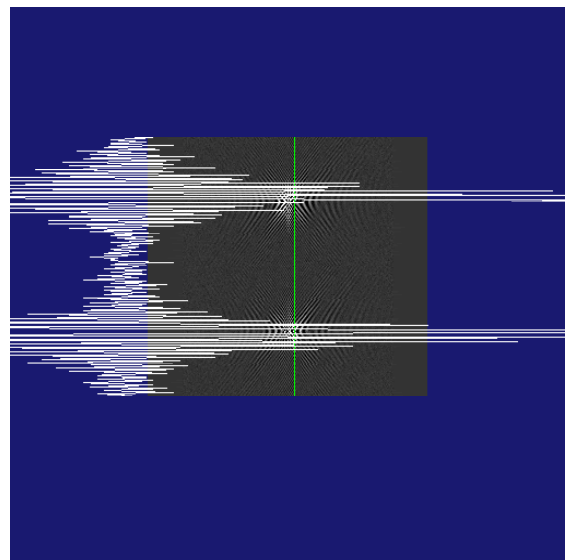


Fig. 6.24-8 Example of Sample Display

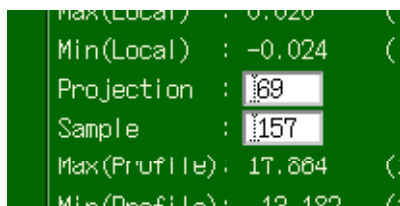


Fig. 6.24-9 Profile Position display and Input Area

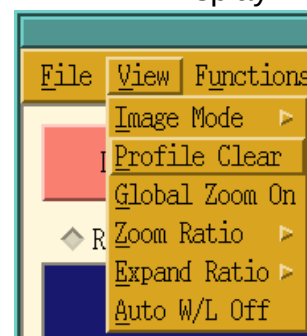


Fig. 6.24-10 Profile Clear Button

6.24.4.5 Regional data analysis

Placing the mouse cursor on the screen and dragging the mouse with its right button pressed displays a violet frame. By releasing the right button at the desired position, the region is defined, and number of pixels (Pixels), mean value (Mean), standard deviation (Standard), maximum value (Max(Local)) and minimum value (Min(Local)) are displayed in the data display area. At the same time, the positions of the signals showing the maximum value and minimum value on the image are marked with [○] and [×] respectively.

To clear the frame and marks displayed on the image, select [Profile Clear] from "View Menu" of the Menu Bar.

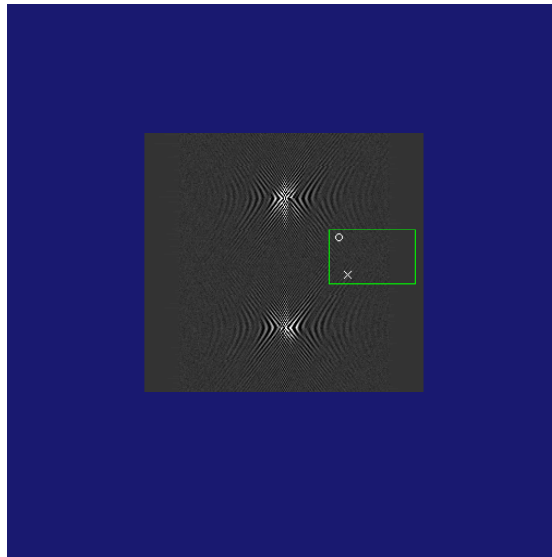


Fig. 6.24-11 Regional Data Analysis

6.24.4.6 Adjustment of displayed image W/L

Place the mouse cursor in the image display area, and with its right button pressed together with **Shift** button pressed move the mouse up-down and right-left to adjust W/L. The relation between movement of the mouse and change in W/L is shown in Fig. 6.24-12.

W/L can also be adjusted by the scroll bar and value entry area at the bottom of the screen. In case of using the scroll bar, as shown in Fig. 6.24-13, W/L values change depending on position to be clicked and dragged. Each of the current values is displayed in the corresponding input area.

To set W/L directly with values, change the figures in the input areas and press **<Enter>** key.

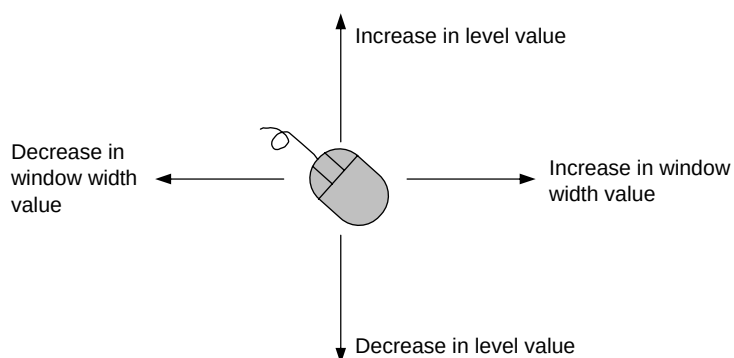


Fig. 6.24-12 Relation between Mouse Move Direction and W/L

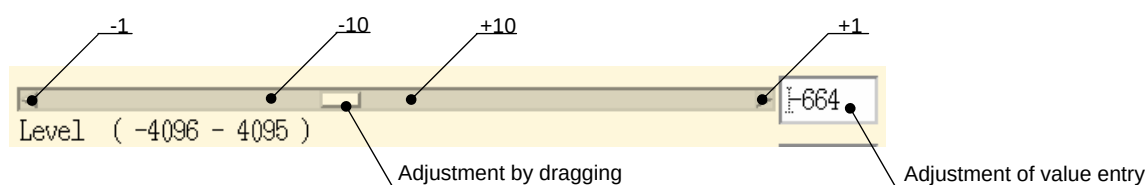


Fig. 6.24-13 Adjustment Scroll Bar and Input area

6.24.4.7 Numerical display of signal values on profile

Either selecting [Data List] from "Functions Menu" of the Menu Bar or clicking [Data List] button on the main screen displays the signal values on the profile as the numerical value list. On the screen are displayed 28 pieces of data at the same time, and by dragging the scroll bar the whole data of the profile can be displayed.

To search a specific data, three search methods shown below can be used.

To search data around the desired coordinate position, click [Coordinate] and input the value of an adjacent input area and press <Enter> key.

To search the maximum value, click [Maximum] button, while to search the minimum value, click [Minimum] button.

With the data list screen displayed, it is also possible to observe changing signal values by moving the profile position.

To close the data list display screen, click [Close] button.

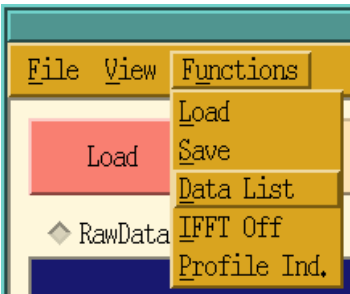


Fig. 6.24-14 Data List Display Button

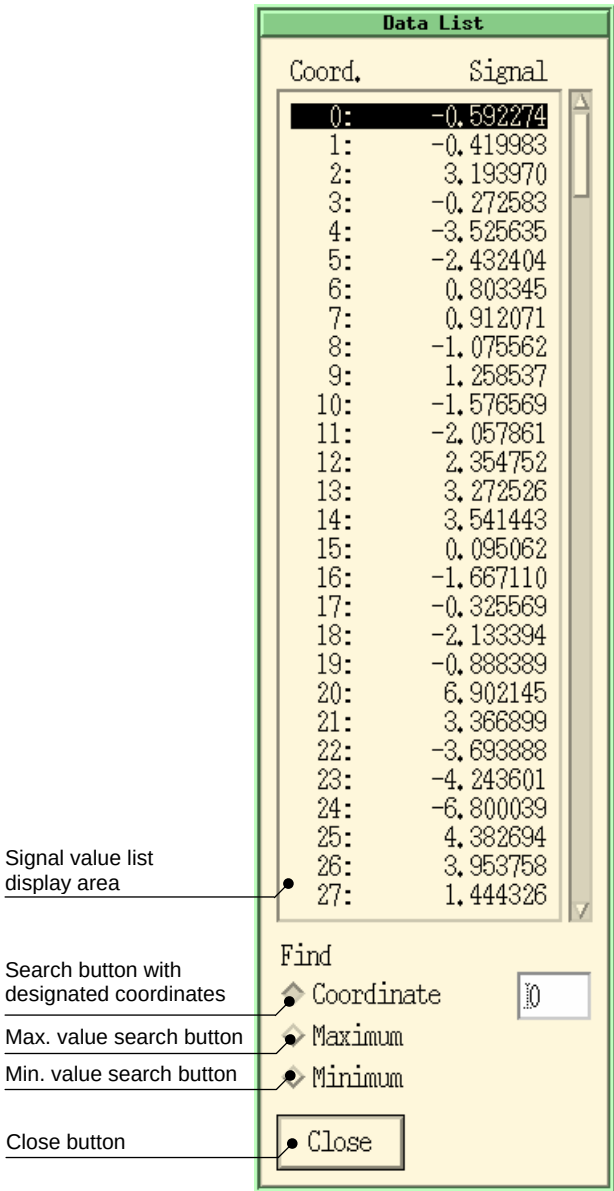


Fig. 6.24-15 Data List Display Screen

6.24.4.8 Setting profile independency and profile W/L adjustment

Select [Profile Ind.] from "Functions Menu" of the Menu Bar.

The screen shown in Fig. 6.24-17 is popped up.

To change Profile Independency to Dependency, click [Dependency] button, while to change Dependency to Independency click [Independency] button. W/L of the profile can be adjusted in the same procedure described in Paragraph 6.24.4.6 by using the scroll bar and input area in the window.

Profile independency means that the profile W/L and displayed image W/L are independent each other. If its relation is dependent, as changing the displayed image W/L the profile W/L also changes accordingly. In case of independency on the other hand, even if the display image W/L is changed the profile W/L will not change and vice versa.

With the profile independency setting screen displayed, other application can be executed. To close the profile independency setting screen, click [Close] button.

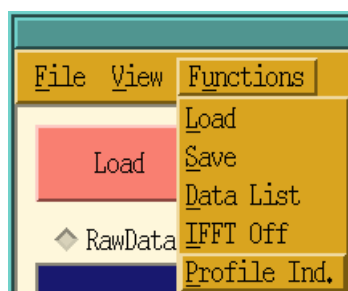


Fig. 6.24-16 Profile Independency Button

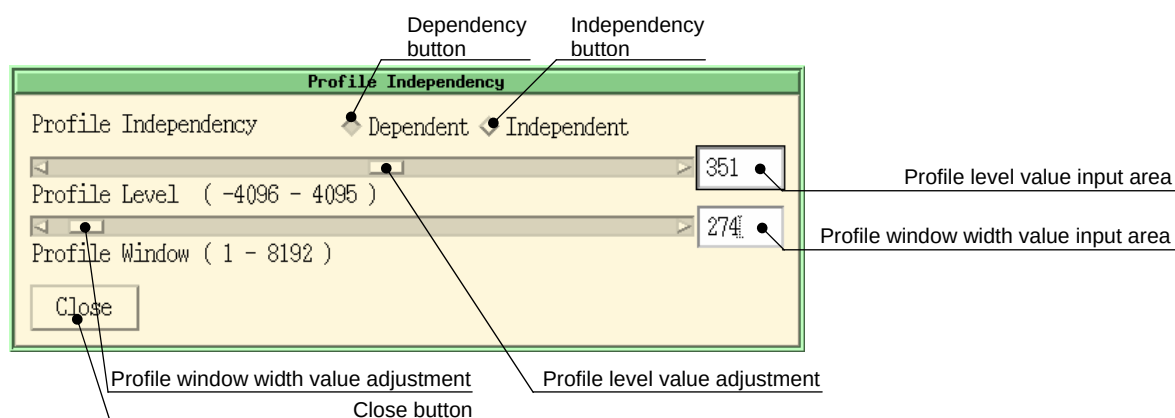


Fig. 6.24-17 Profile Independency Setting Screen

6.24.4.9 Zoom function

This is a tool to zoom an image.

6.24.4.9.1 Regional Zooming

Place the mouse cursor on the screen, and with Ctrl key pressed click the mouse button. Then, the square area around the mouse cursor is zoomed up.

Depending on the button of the mouse, the zooming area differs. Clicking the left mouse button, middle button or right button zooms up a small region, middle region or large region respectively.

To change a magnification factor of the regional zooming, select a desired magnification factor in the menu displayed by clicking [Zoom Ratio] from "View Menu" of the Menu Bar. The default magnification factor is [2].

The profile display and regional analysis functions cannot be applied to the zoomed image. To apply analysis functions to zoomed image, use the Global Zoom function (refer to Paragraph 6.24.4.9.2).

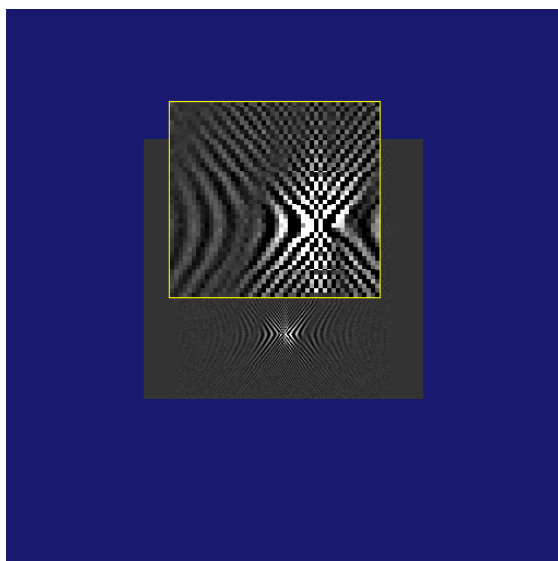


Fig. 6.24-18 Regional Zooming

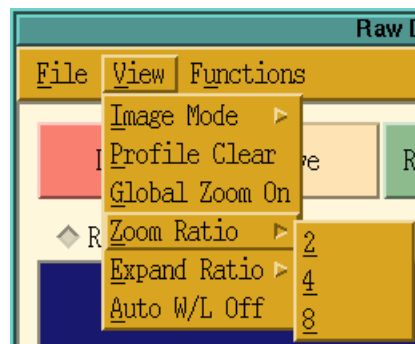


Fig. 6.24-19 Zoom Ratio Button

6.24.4.9.2 Global Zooming

Selecting [Global Zoom On] from "View Menu" of the Menu Bar zooms up the whole image 2 times.

Profile display and regional analysis functions can be applied to the image zoomed up here in the same procedures. However, W/L adjustment cannot be applied to the zoomed image, therefore adjust W/L of the image before global zooming.

To close the global zoom function, select [Global Zoom Off] from "View Menu" of the Menu Bar.

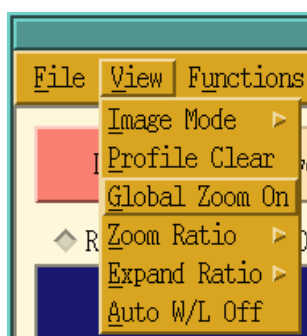


Fig. 6.24-20 Global Zoom
On/Off Button

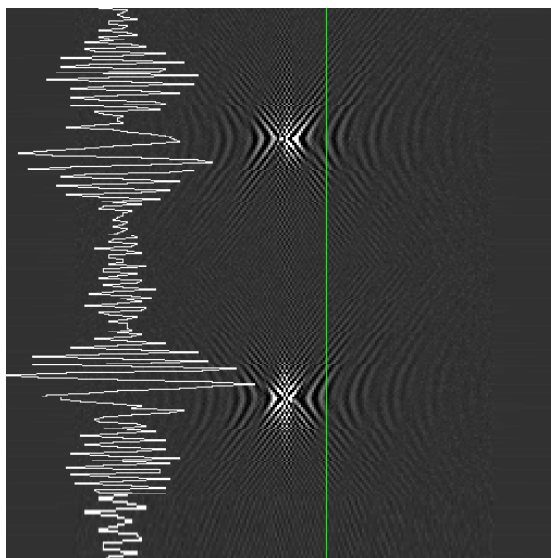


Fig. 6.24-21 Global Zooming

6.24.4.10 Selection of image mode

By selecting [Image Mode] from "View Menu" of the Menu Bar, the sub-menu shown in Fig. 6.24-22 is displayed. To display the real part and imaginary part image, absolute image or phase image, select [Real/Image], [Absolute] and [Phase] button respectively. Image mode can also be selected by the green button on the main screen. In this case, every time the button is clicked, the displayed image is sequentially changed as shown in Fig. 6.24-23.

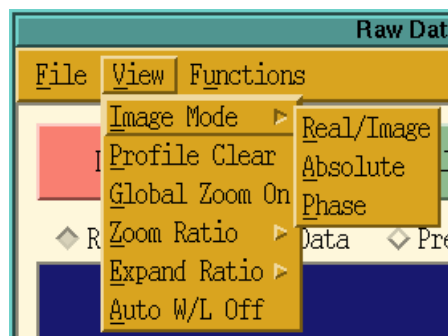


Fig. 6.24-22 Selection of Image Mode

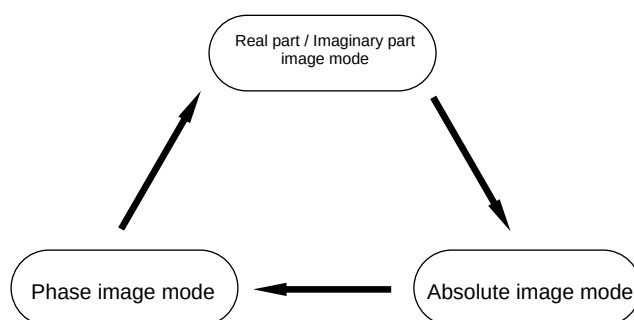


Fig. 6.24-23 Sequential Selection of Image Mode

6.24.4.11 Inverse Fourier Transformation On/Off

Selecting [IFFT On] or [IFFT Off] from "Functions Menu" of the Menu Bar sets the IFFT function to ON or OFF. The same operation can be done by using the button on the main screen.

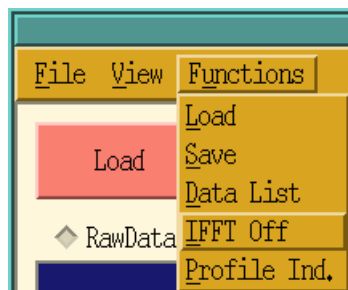


Fig. 6.24-24 IFFT On/Off

6.24.4.12 Auto W/L ON/OFF

Selecting [Auto W/L On] or [Auto W/L Off] from "View Menu" of the Menu Bar sets the Auto W/L function to On or Off.

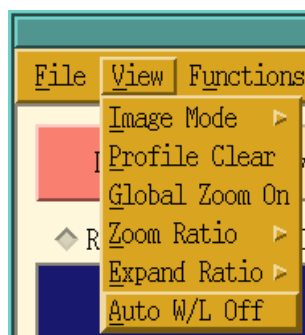


Fig. 6.24-25 Auto W/L On/Off

6.24.4.13 Expansion of signal value

Select [Expand Ratio] from "View Menu" of the Menu Bar and select the desired expansion ratio from the opened sub-menu. The whole signal data is expanded by the selection expansion ratio and is displayed so that 0 signal value becomes 0 level value. However, the large parts of the signal amplitude are cut off. The default is normalized so that the whole signals are within the maximum value of the window width. To return to the default, select [Default].

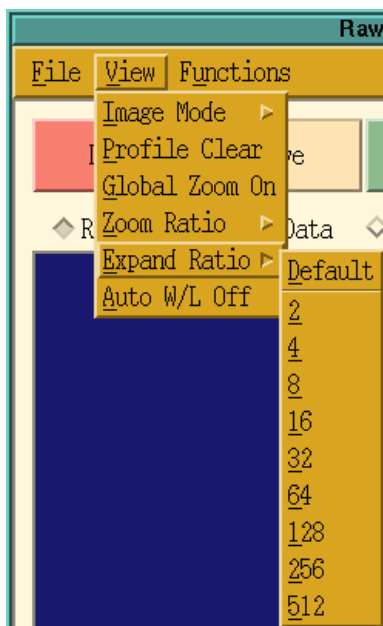


Fig. 6.24-26 Expansion of Signal Value

6.24.4.14 Saving raw data

Either select [Save] from "Functions Menu" of the Menu Bar or click [Save] button on the main screen. The saving screen shown in Fig. 6.24-28 is popped up. Select the file name to be saved on this screen and, if necessary, input slice No. (number of slices of the image being loaded is set as default) and comments. Then, clicking [Save] button executes raw data saving. To close the process without saving raw data, click [Cancel] button.

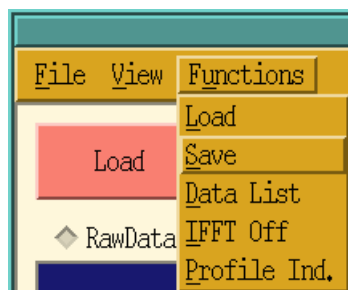


Fig. 6.24-27 Save Button

6.24.4.15 Deletion of raw data file

Either select [Save] from "Functions Menu" of the Main Bar or click [Save] button on the main screen. The saving screen shown in Fig. 6.24-28 is popped up. Select the file name to be deleted from the file list displayed on this screen and click [Delete] button to delete the file. To close the process, click [Cancel] button.

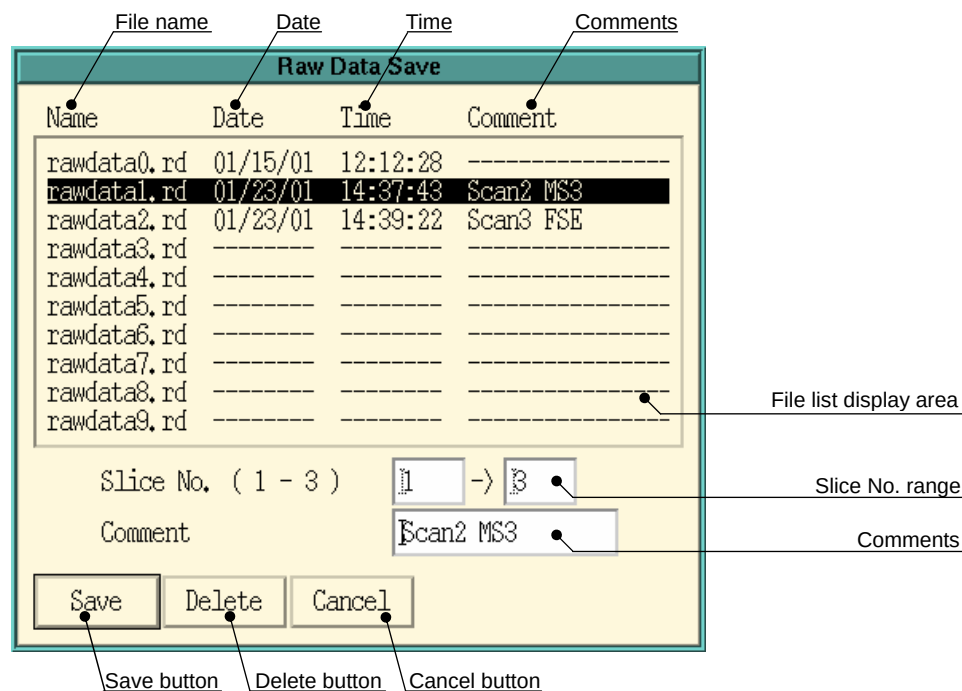


Fig. 6.24-28 Save Screen

6.24.4.16 Cine display

In case more than 2 slices of the loaded raw data are available, the cine display button is used to select a display slice. The following four buttons have the meanings shown in Fig. 6.24-29.

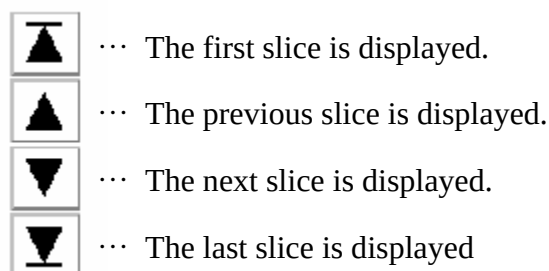


Fig. 6.24-29 Cine Display Button

6.24.4.17 Data displayed in data display area

The following is a list of data shown in the data display area.

Table 6.24-1 Data Display Area

File Name	File name
Max(Whole)	Max. value and its position in whole image
Min(Whole)	Min. value and its position in whole image
Pixels	Number of pixels in the selected region
Mean	Mean value in the selected region
Standard	Standard deviation in the selected region
Max(Local)	Max. value and its position in the selected region
Min(Local)	Min. value and its position in the selected region
Projection	Projection position
Sample	Position of sample
Max(profile)	Max. value and its position on profile
Min(Profile)	Min. value and its position on profile
Slice	Display slice No./Number of all slices

Filename : fantom1.rd	Max(Local) : 0.020 (18, 119)
Max(Whole): 421.124 (128, 192)	Min(Local) : -0.024 (18, 125)
Min(Whole): -442.714 (129, 192)	Projection : 169
Pixels : 684	Sample : 157
Mean : 0.001	Max(Profile): 17.804 (120, 60)
Standard : 0.009	Min(Profile): -13.182 (128, 69)
	Slice : 1 / 1

Fig. 6.24-30 Data Display Area

6.24.4.18 Short-cut key

To proceed operations smoothly, several short-cut keys are available. In case any of the short-cut keys is not effective, display the profile once, then press the key again.

Table 6.24-2 Short-cut Key

<c> key	Cancellation of profile and region selection frame
<p> key	Pop-up of profile independency setting screen
<d> key	Pop-up of data list screen
<l> key	Pop-up of load screen
<s> key	Pop-up of save screen
<z> key	Global zoom ON/OFF
<i> key	Selection of image mode
<a> key	Auto W/L: ON
<f> key	Inverse FFT ON/OFF
<r> key	Selection of regional expansion ratio
<q> key	Closing application
<↑> and <↓> keys	Movement of projection position (± 1)
<←> and <→> keys	Movement of sampling position (± 1)
Shift + <↑> and shift + <↓>	Movement of projection position (± 10)
Shift + <←> and shift + <→>	Movement of sampling position (± 10)
Ctrl + <↑> and Ctrl + <↓>	Movement of projection position (to end)
Ctrl + <←> and Ctrl + <→>	Movement of sampling position (to end)

6.24.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.24-32. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.



Fig. 6.24-31 Quit Button

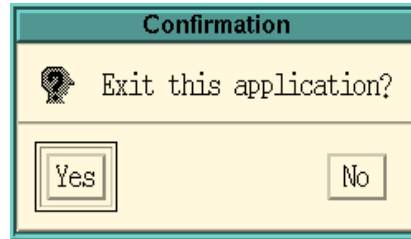


Fig. 6.24-32 Exit Confirmation Dialog

6.24.6 Error specifications

Error specifications are shown in Table 6.24-3.

Table 6.24-3 Error Specifications

Message	Status	Solution
The file of specification cannot be opened.	The designated file could not be opened.	Check for the existence of the file and permission.
Can't start the application, more than 32.	This applications can be only started up to 32 at the same time.	Terminate some applications or use the application which has already started.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't start task.	The application started unsuccessfully.	Check the system status and retry the same operation procedure.
Can't end task.	The application ended unsuccessfully.	Check the system status and retry the same operation procedure.
The translation of parameter failed.	Parameter transfer failed.	Check the system status and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Checking the system status, retry the same operation procedure.

6.25 Prescan Data Display

6.25.1 General

This is a tool to display and analyze the prescan result and to use for adjustment of the system. The prescan data display application includes 9 tools, and patient data up to for 3 patients can be handled for each tool.

6.25.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click PreScan Data Display in the displayed items and click [OK] or double-click the selected item.

6.25.3 Display screen

The screen displayed after start-up is shown in Fig. 6.25-1.

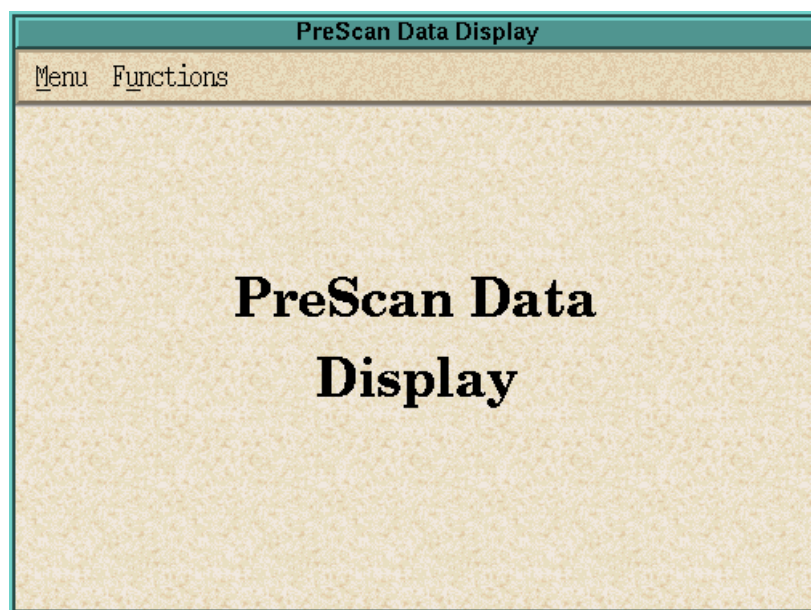


Fig. 6.25-1 PreScan Data Start-up Screen

6.25.4 Operation procedure

Follow the next steps to execute the process.

6.25.4.1 Start-up

Any tool can be started by selecting the tool to be started on "Menu" of the Menu Bar and further selecting the patient in its Submenu (Fig. 6.25-2).

The list of the tools is shown in Table 6.25-1.

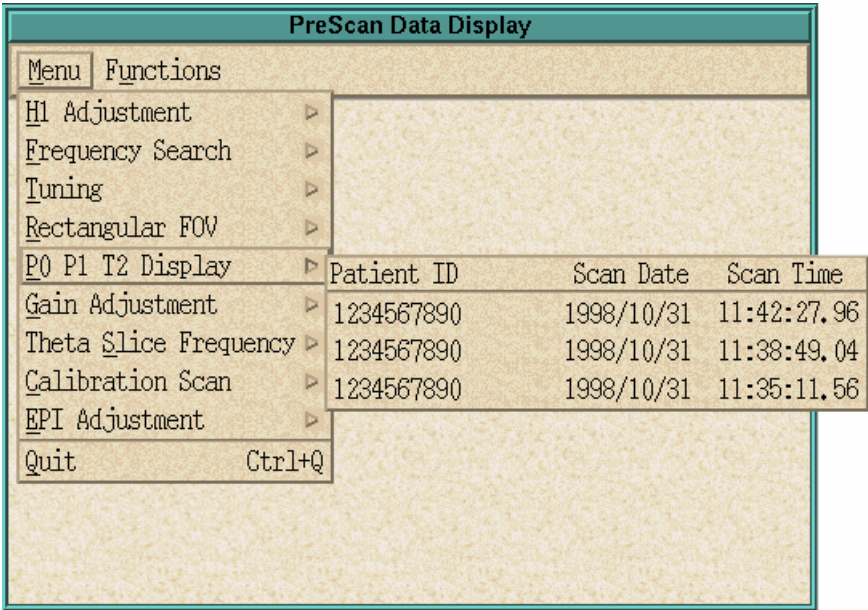


Fig. 6.25-2 Example of Start-up Operation of Tool

Table 6.25-1 List of Tools

Reference Section	Title
6.25.4.3	H1 Adjustment
6.25.4.4	Frequency Search
6.25.4.4.1	Frequency Search
6.25.4.4.2	Frequency Lock
6.25.4.5	Tuning
6.25.4.6	Rectangular Field of View
6.25.4.7	P0 P1 T2 Display
6.25.4.8	Gain Adjustment
6.25.4.8.1	Multi-Slab
6.25.4.8.2	Phase-Encode
6.25.4.8.3	Slice-Encode
6.25.4.8.4	Receiver-Gain
6.25.4.8.5	Gain-Search
6.25.4.9	Theta Slice Frequency
6.25.4.9.1	Theta Slice Frequency
6.25.4.9.2	Theta Phase
6.25.4.10	Calibration Scan
6.25.4.10.1	X-Axis
6.25.4.10.2	Y-Axis
6.25.4.10.3	Z-Axis
6.25.4.11	EPI Adjustment

6.25.4.2 Refresh of logging data

In case log data is changed after start-up, select [Refresh] on "Functions Menu".



Fig. 6.25-3 Refresh Button

6.25.4.3 H1 Adjustment

The ordinate axis and abscissa axis represent echo signal value and H1 value respectively. Color of the line on the graph has the meanings shown in Table 6.25-2.

Signal values between measurement points are obtained by quadratic curve approximation, but they are displayed by linear interpolation. Because the approximation is made by quadratic curve, the resultant H1 values are not always on the measurement points.

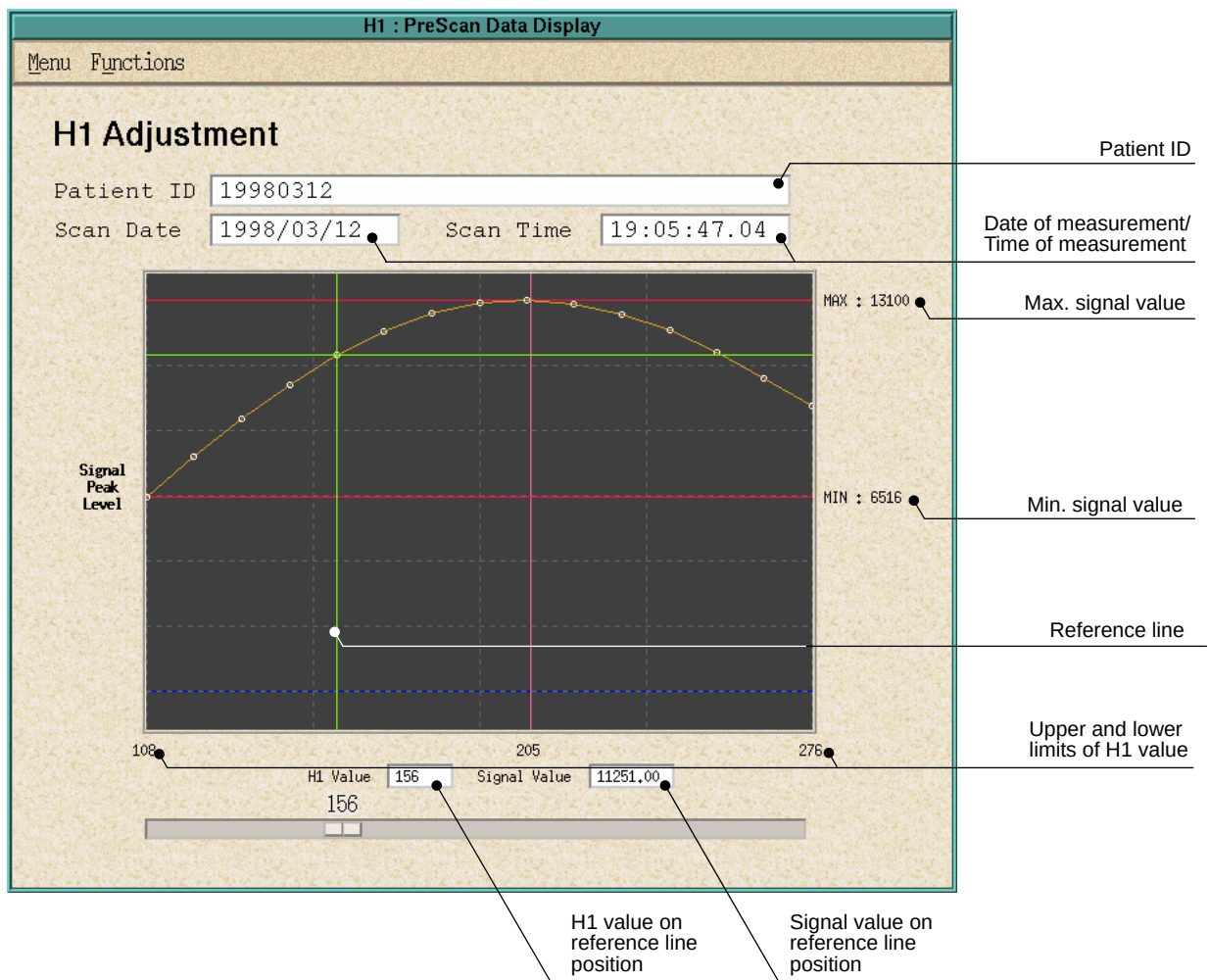


Fig. 6.25-4 H1 Adjustment Display Screen

Table 6.25-2 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to H1 value on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-3.

Table 6.25-3 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.4 Frequency Search

Either Frequency Search or Frequency Lock is displayed. Frequency Search and Frequency Lock can be switched over on the option menu displayed on the screen. When the desired menu is displayed in light color, it indicates that no measurement data is available.

6.25.4.4.1 Frequency Search

The ordinate axis and abscissa axis represent echo signal value and frequency (Hz) respectively. Color of the line on the graph has the meanings shown in Table 6.25-4.

The data with ○ is the actually measured value.

Data are measured in 1 kHz steps but data between measurement points are linear-interpolated and displayed.

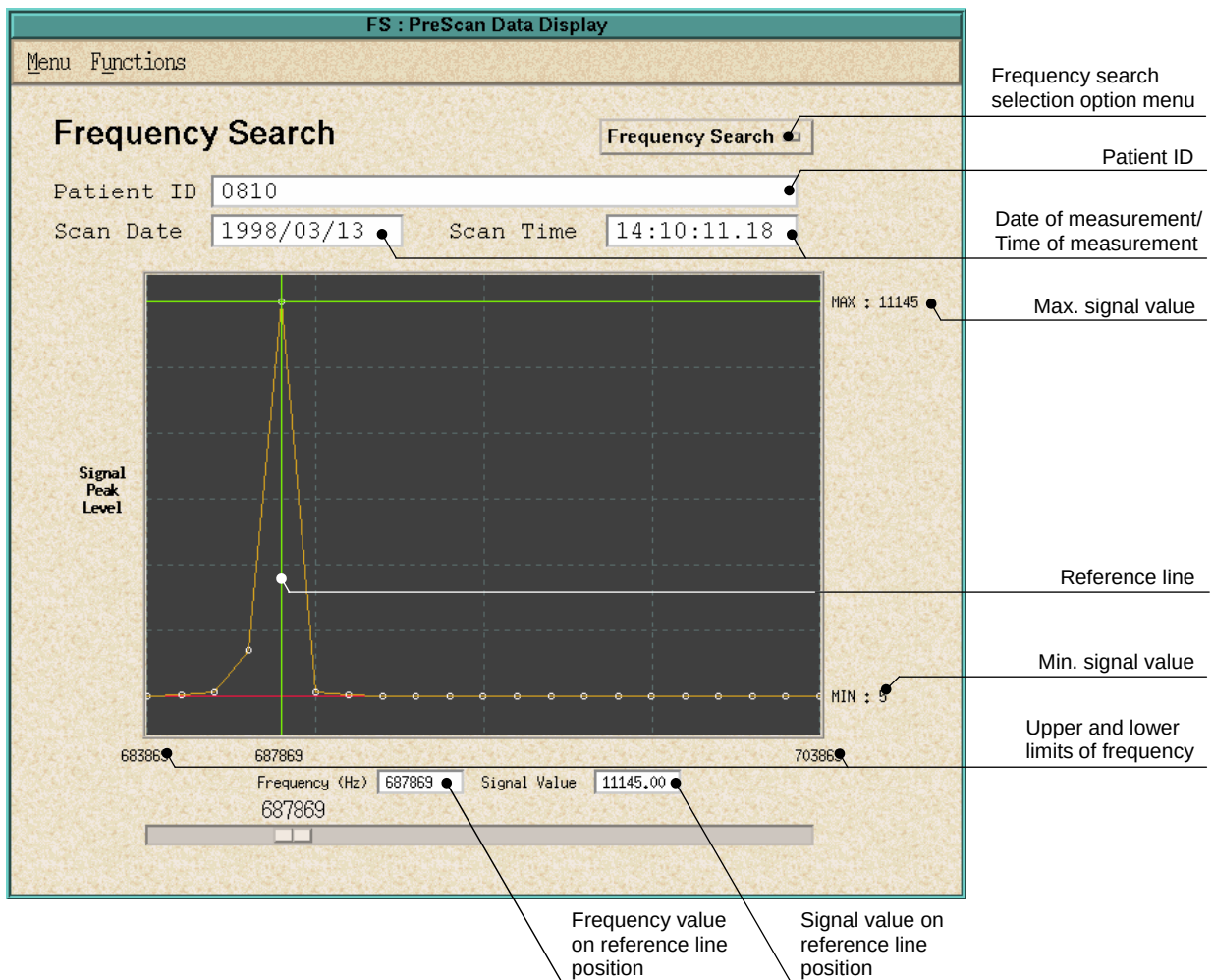


Fig. 6.25-5 Frequency Search Display Screen

Table 6.25-4 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to frequency on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-5.

Table 6.25-5 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.4.2 Frequency Lock

The ordinate axis and abscissa axis represent FFT signal value and frequency (Hz) respectively. Color of the line on the graph has the meanings shown in Table 6.25-6.

Because data are approximated by gravity center method, the maximum signal value and resultant frequency are not always agreeable.

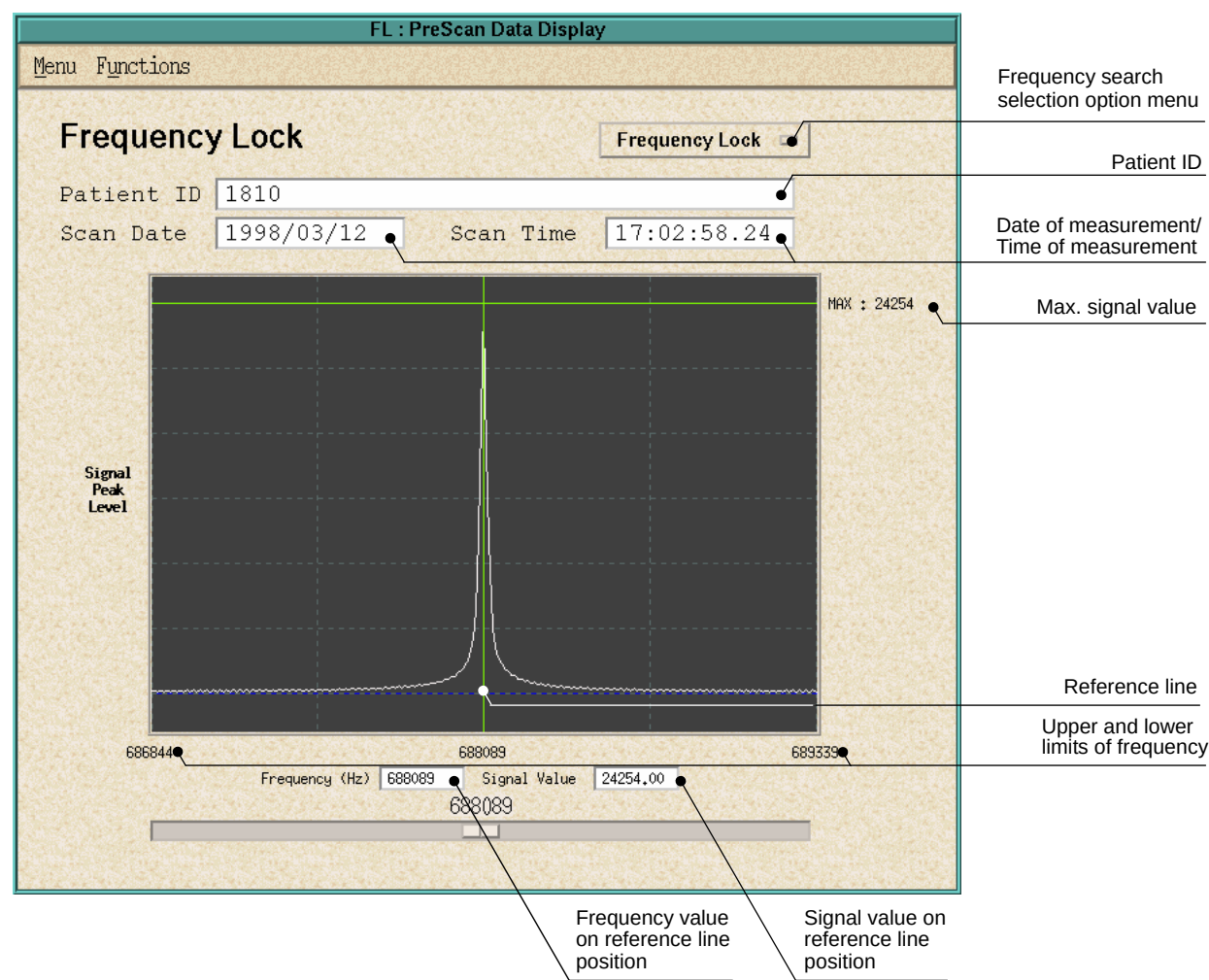


Fig. 6.25-6 Frequency Lock Display Screen

Table 6.25-6 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to frequency on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-7.

Table 6.25-7 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 250.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.5 Tuning

The ordinate axis and abscissa axis represent echo signal value and tuning value of the reception coil respectively. Color of the line on the graph has the meanings shown in Table 6.25-8.

The data with ○ is the actually measured value.

Display between measurement points are linear-interpolated.

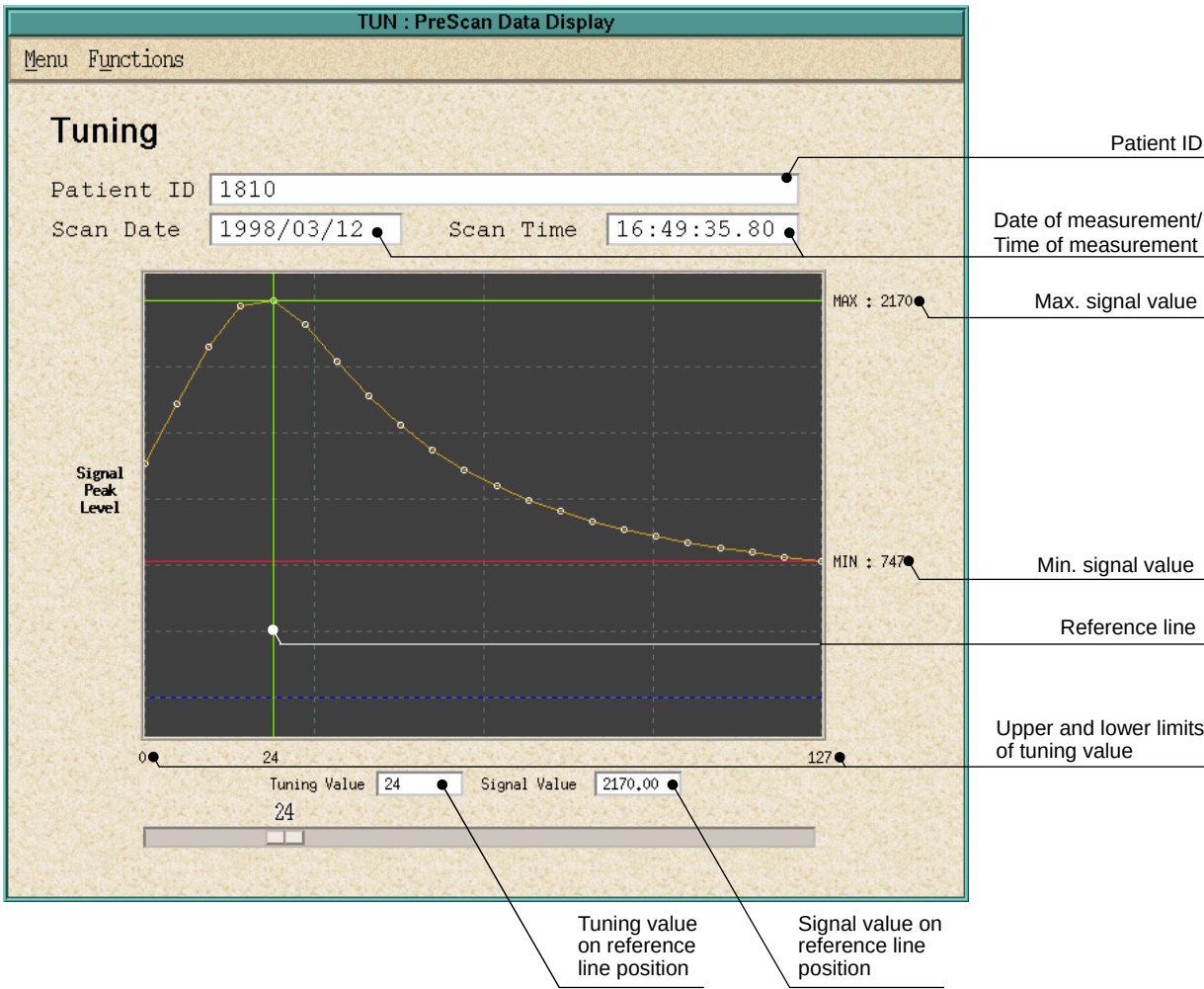


Fig. 6.25-7 Tuning Display Screen

Table 6.25-8 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to tuning value on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-9.

Table 6.25-9 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.6 Rectangular Field of View

The ordinate axis and abscissa axis represent FFT signal value and position from the FOV center respectively. Color of the line on the graph has the meanings shown in Table 6.25-10.

In 2D multi-slice mode and 3D multi-slab mode, the data are displayed as graph obtained by projection of the maximum values of FFT data of all slices.

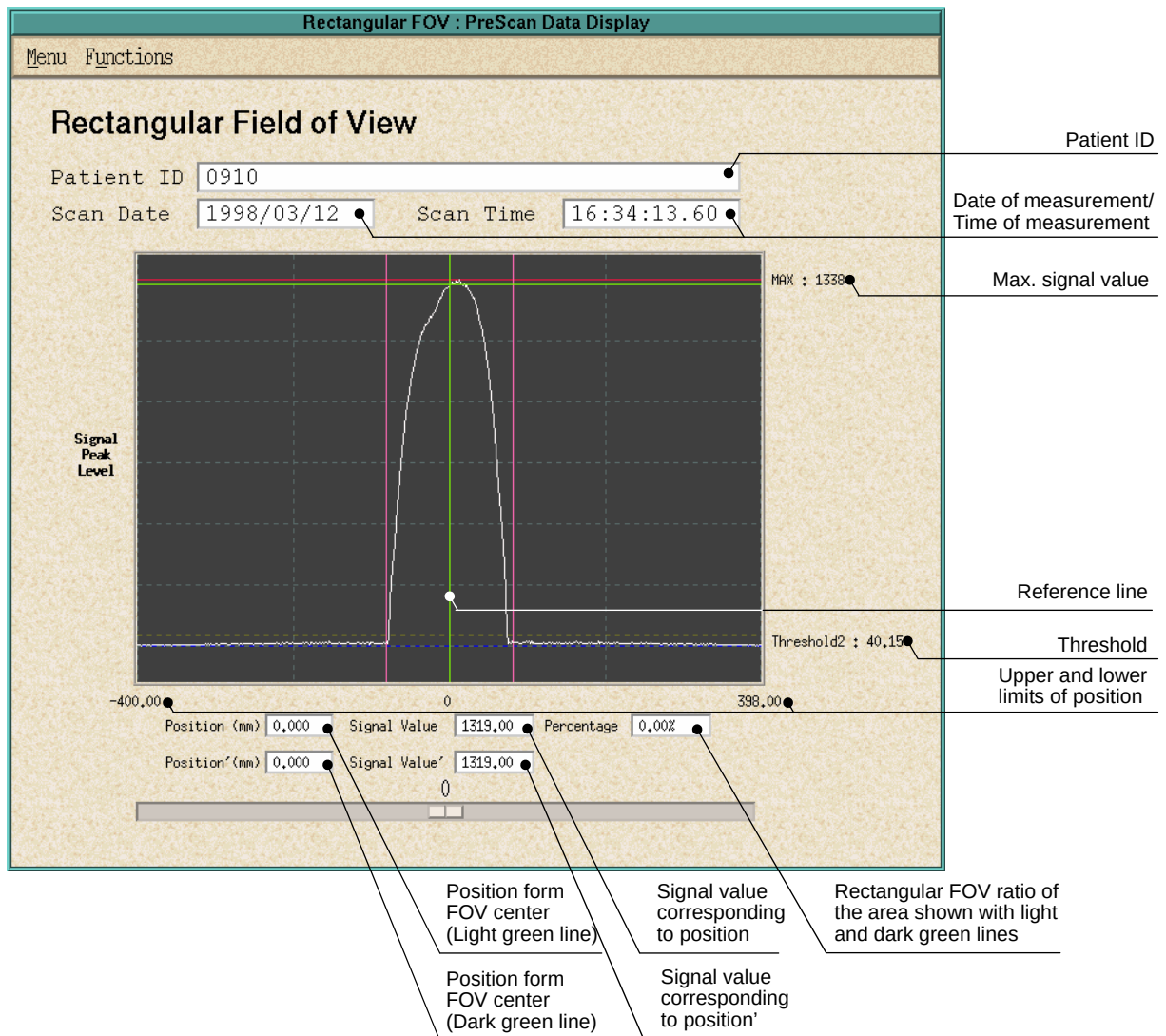


Fig. 6.25-8 Rectangular Field of View Display Screen

Table 6.25-10 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to the position of the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)
Orange	Threshold value

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-11.

Table 6.25-11 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.7 P0 P1 T2 Display

The ordinate axis and abscissa axis represent signal value and data No. respectively.
Color of the line on the graph has the meanings shown in Table 6.25-13.

The data with ○ is the actually measured value.

For detailed explanation of various values refer to Table 6.25-12.

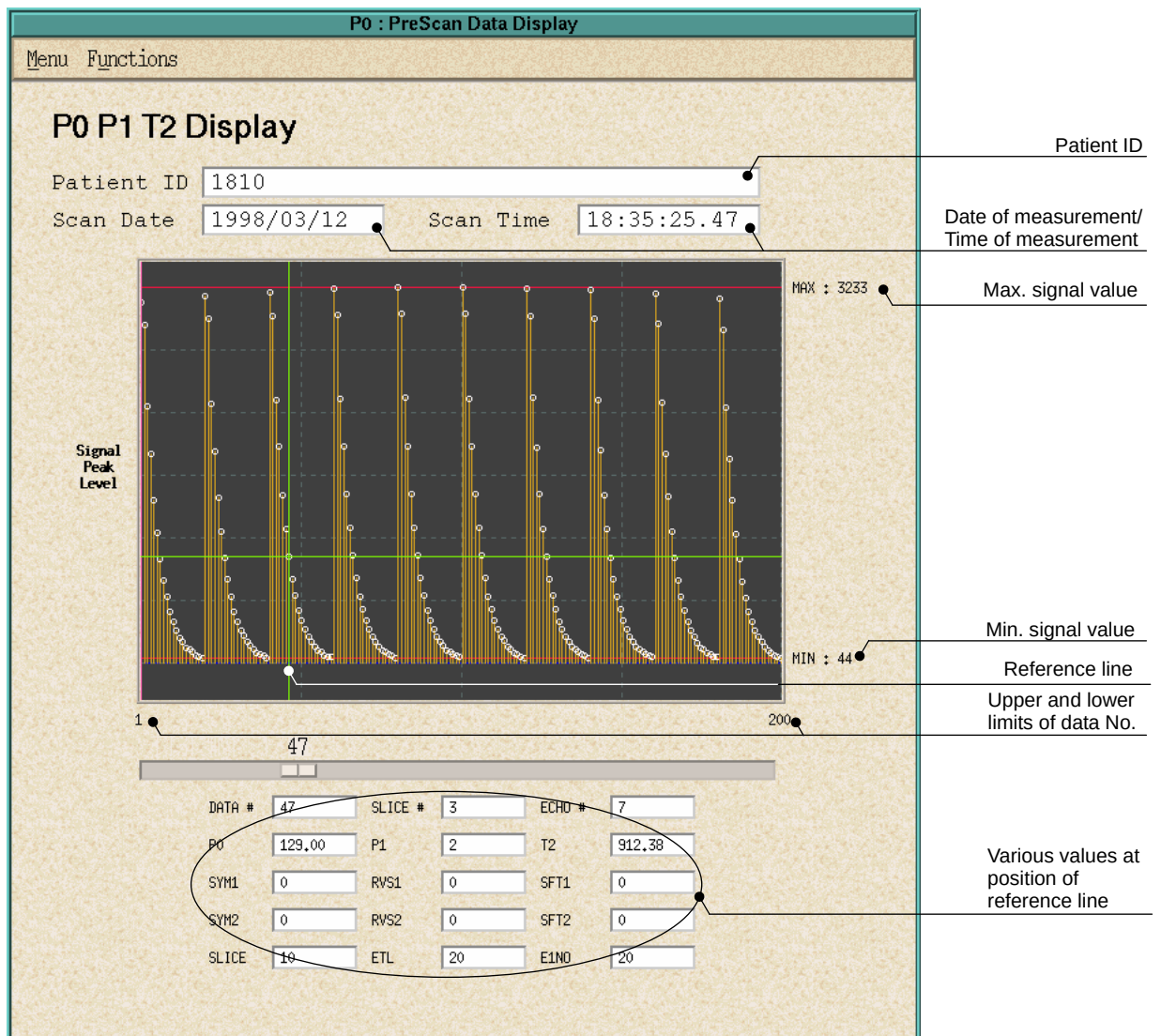


Fig. 6.25-9 P0 P1 T2 Display Screen

Table 6.25-12 Detailed Explanation of Various Values

Item	Description	Item	Description
DATA#	Data No. from the top	RVS2	Flag reversely placed to correction data for 2nd echo.
P0	P0 value at reference line position	ETL	Number of all echoes
SYM1	Flag symmetrically placed with correction data for 1st echo	ECHO#	Echo No. at reference line position
SYM2	Flag symmetrically placed with correction data for 2nd echo	T2	T2 value of reference line position
SLICE	Number of all slices in scan	SFT1	Number of correction data shift for 1st echo
SLICE#	Slice No. at reference line position	SFT2	Number of correction data shift for 2nd echo
P1	P1 value at the reference line position	E1NO	Number of echoes used for 1st echo image
RVS1	Flag reversely placed to correction data for 1st echo.		

Table 6.25-13 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to data No. on the reference line (horizontal line)
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-14.

Table 6.25-14 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.8 Gain Adjustment

One of multi-slab, phase-encode, slice-encode, reception gain and gain search is displayed. One of

Multi-Slab/Phase-Encode/Slice-Encode/Receiver-Gain/Gain-Search is selected on the option menu displayed on the screen. When the selected menu is displayed in light color, it indicates that no measurement data is available.

6.25.4.8.1 Gain Adjustment (Multi-Slab)

The ordinate axis and abscissa axis represent echo signal value and GB value respectively.

Color of the line on the graph has the meanings shown in Table 6.25-15.

The data with ○ is the actually measured value.

Gain adjustment is carried out for multi-slice in 2D scan and for multi-slab in 3D scan.

The range of values is number of multi-slices and number of multi-slab. GB value is sequence No. of measurements which is different from sequence No. of reconstruction.

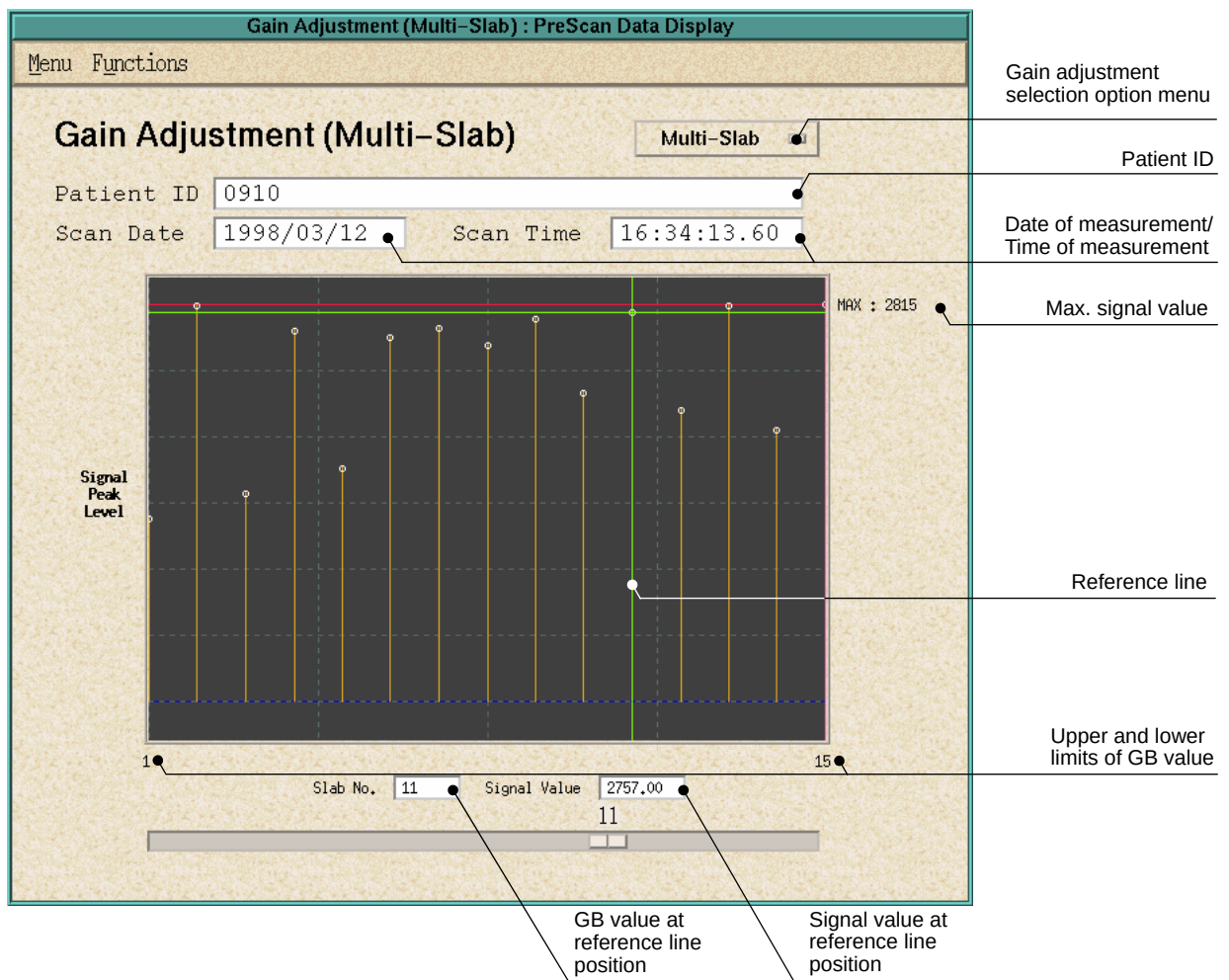


Fig. 6.25-10 Gain Adjustment (Multi-Slab) Display Screen

Table 6.25-15 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to GB value on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-16.

Table 6.25-16 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.8.2 Gain Adjustment (Phase-Encode)

The ordinate axis and abscissa axis represent echo signal value and phase-encode No. respectively. Color of the line on the graph has the meanings shown in Table 6.25-17.

The data with ○ is the actually measured value.

The range of values of phase-encode No. depends on phase of the BASIC page in main scan, and is 5 encodes around the center in SE sequence (SE, IR, FSE, FIR) and 7 encodes around the center in GFE sequence (GE, TOF, PC).

Display values between measurement points are obtained by linear-interpolation.

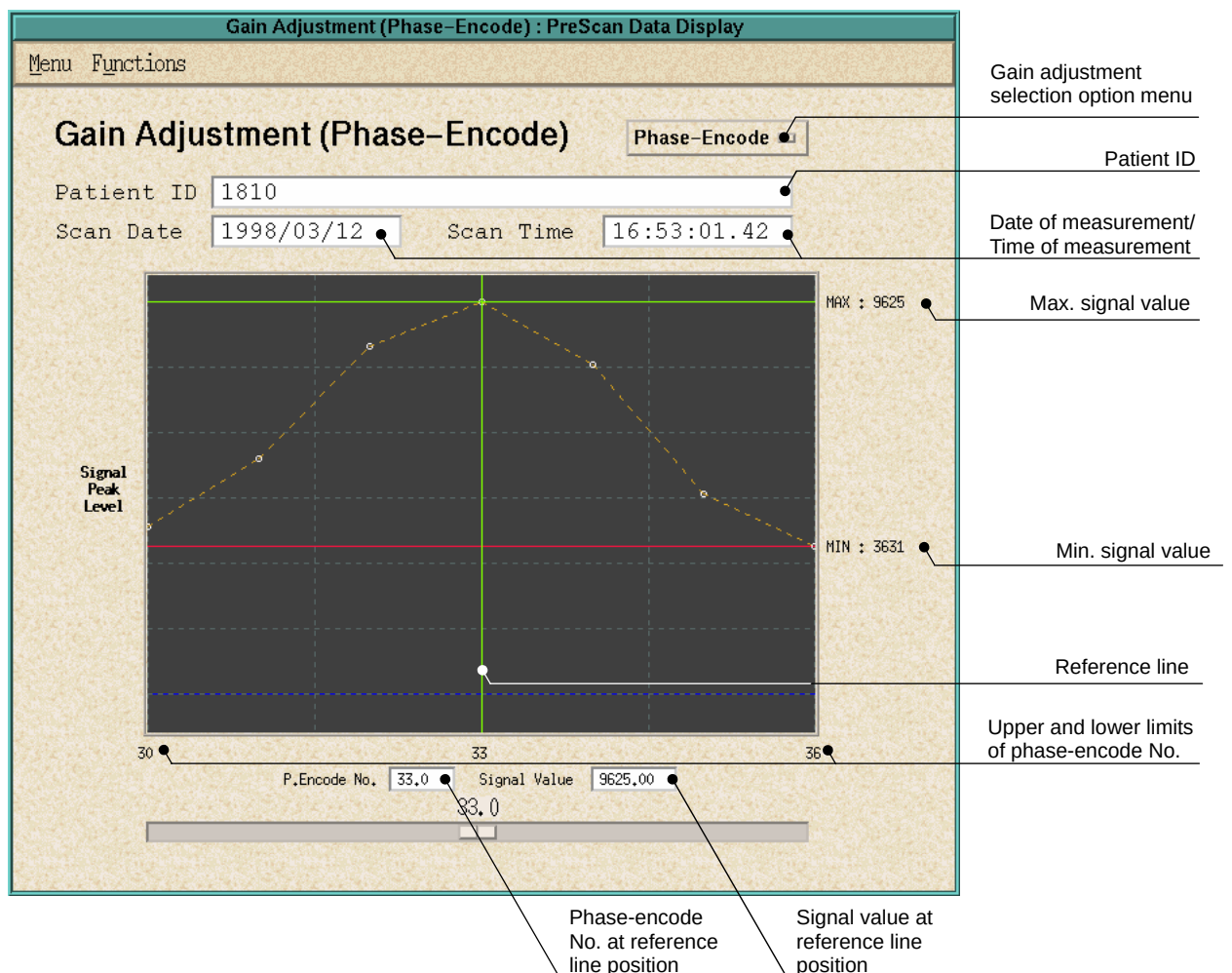


Fig. 6.25-11 Gain Adjustment (Phase-Encode) Display Screen

Table 6.25-17 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to Phase-encode No. on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-18.

Table 6.25-18 Key Operation

<←> and <→>	Reference line moves horizontally by 0.1 encode value.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.8.3 Gain Adjustment (Slice-Encode)

The ordinate axis and abscissa axis represent echo signal value and Slice-encode No. respectively. Color of the line on the graph has the meanings shown in Table 6.25-19.

The data with ○ is the actually measured value.

The range of values of Slice-encode No. in the abscissa axis depends on Slice-encode of the BASIC page in main scan, and is 5 encodes around the center.

Display values between measurement points are obtained by linear-interpolation.

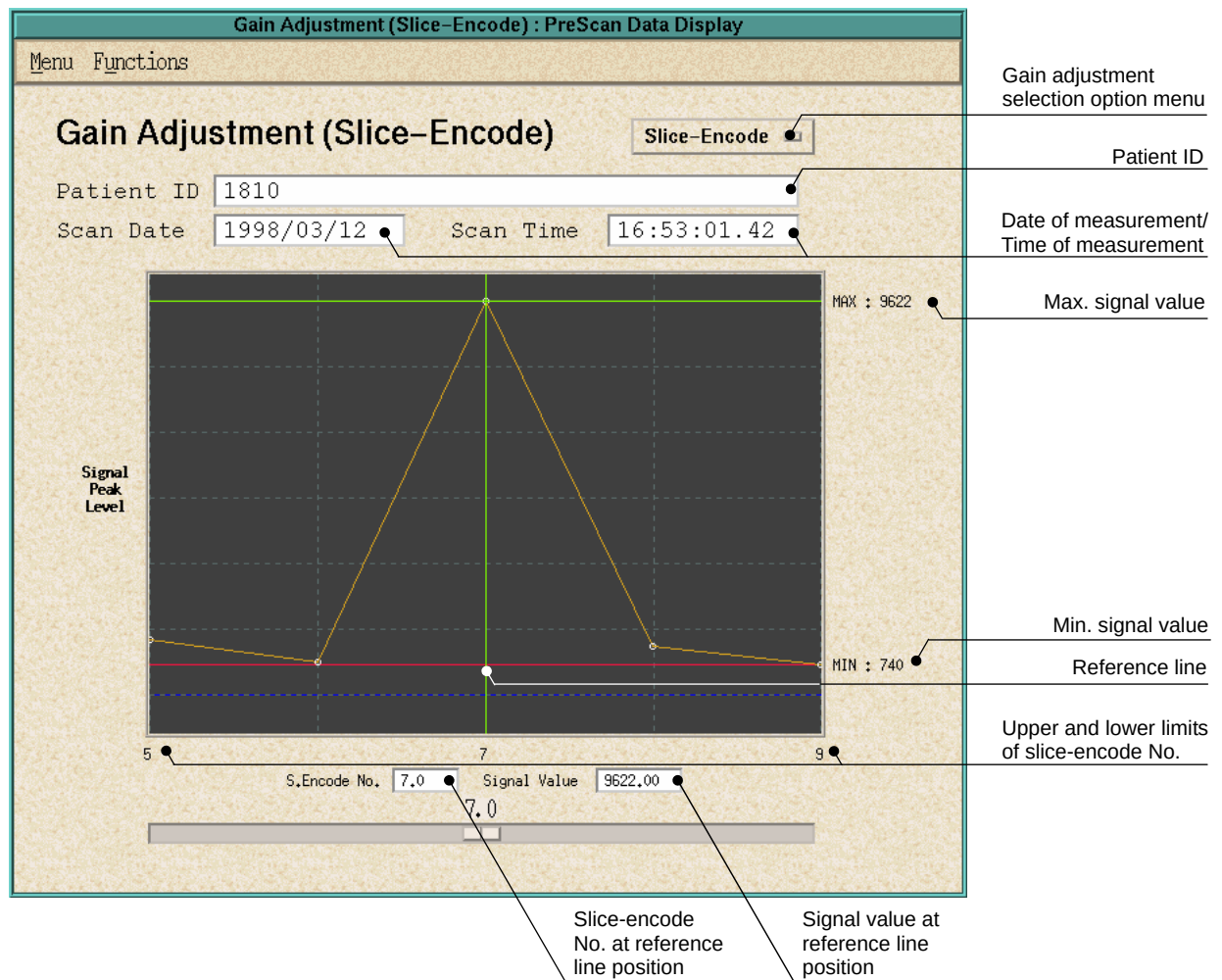


Fig. 6.25-12 Gain Adjustment (Slice-Encode) Display Screen

Table 6.25-19 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to Slice-encode No. on the reference line (horizontal line)
Pink	Resultant signal value
Red	Maximum and minimum value of signal value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-20.

Table 6.25-20 Key Operation

<←> and <→>	Reference line moves horizontally by 0.1 step.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.8.4 Gain Adjustment (Receiver-Gain)

The ordinate axis and abscissa axis represent echo signal value and gain value respectively. Color of the line on the graph has the meanings shown in Table 6.25-21.

The mark ○ represents the intersection point of the reception gain and signal value for scan, while the mark × represents the intersection point of the reception gain and estimated value of the resultant signal value for the main scan.

Since the value displayed in LED of the RF unit is HEX value, its value range is 0 ~ 1EH.

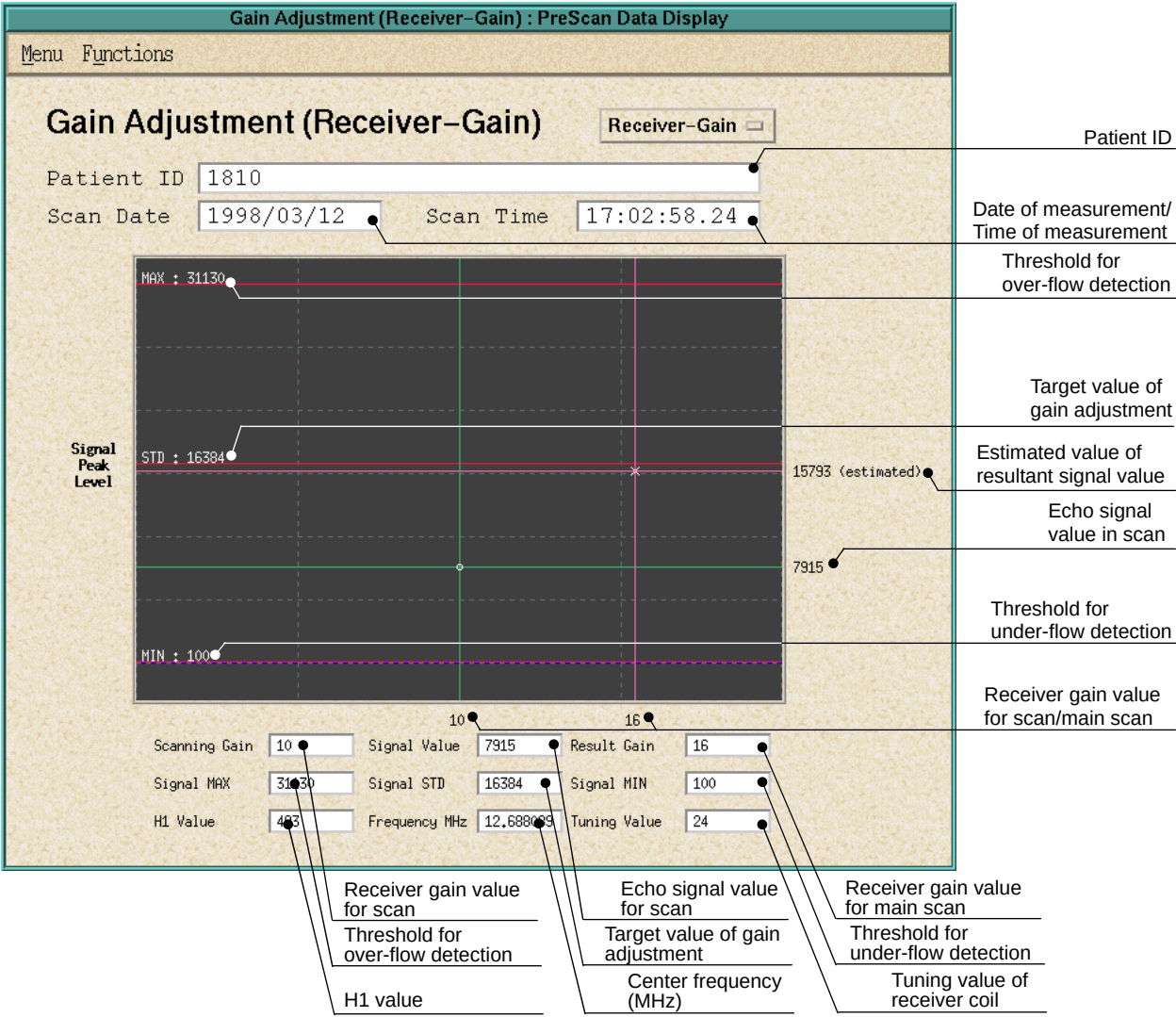


Fig. 6.25-13 Gain Adjustment (Receiver-Gain) Display Screen

Table 6.25-21 Meanings of Colors

Color	Meanings
Green	Reception gain value for scan (vertical line) and signal value (horizontal line)
Pink	Reception gain value for main scan (vertical line) and estimated value of resultant signal value (horizontal line)
Red	Over-flow detection level, under-flow detection level and target level of gain adjustment
Blue	Coordinate axes (signal value: 0)

6.25.4.8.5 Gain Adjustment (Gain-Search)

The ordinate axis and abscissa axis represent the maximum value of AD conversion data and reception gain value respectively. Color of the line on the graph has the meanings shown in Table 6.25-22.

The data with ○ is the actually measured value.

Display values between measurement points are obtained by linear-interpolation. The maximum value of AD conversion data is 32767. If reception signal is too high, it saturates at this value.

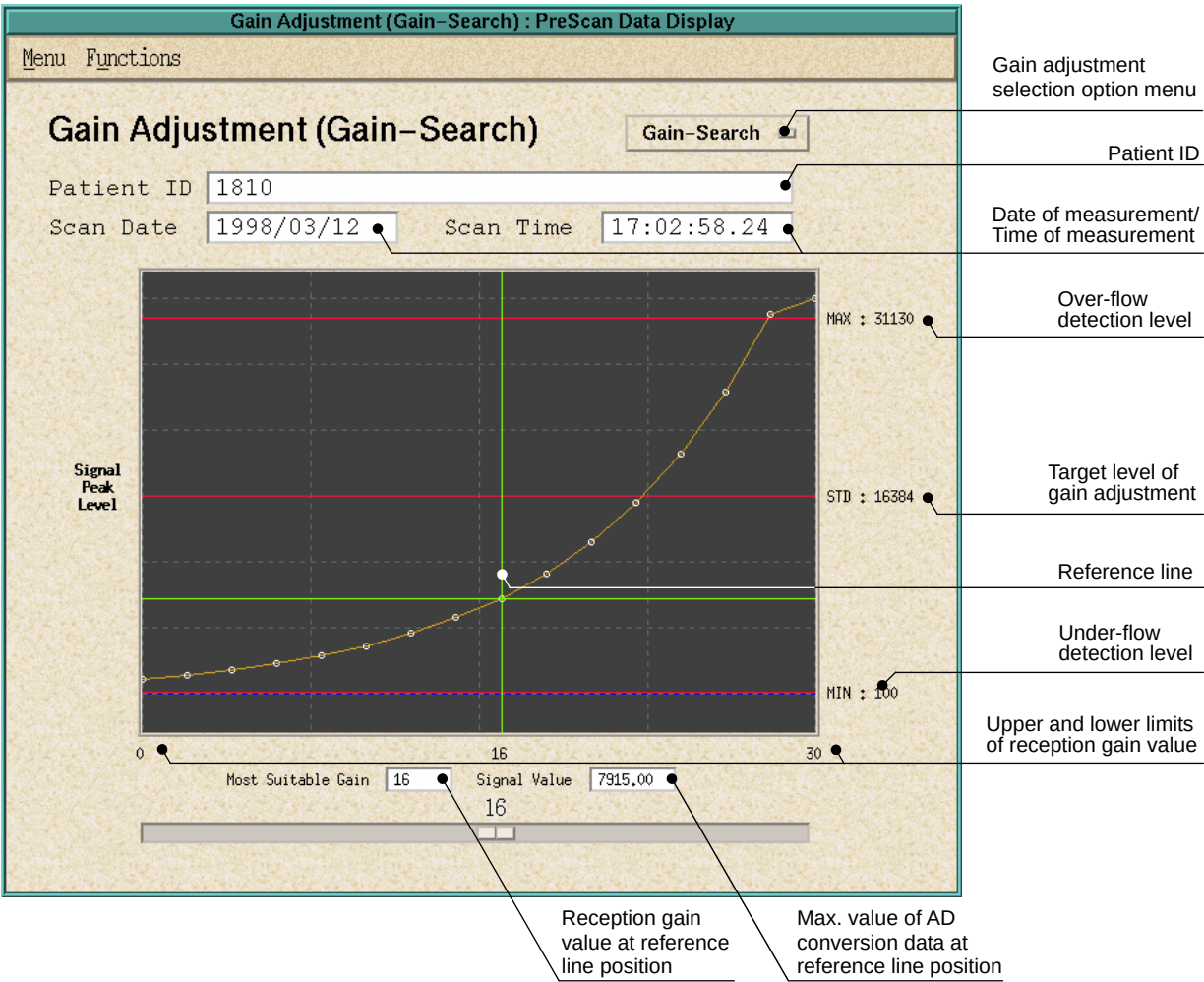


Fig. 6.25-14 Gain Adjustment (Gain-Search) Display Screen

Table 6.25-22 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and the maximum value of AD conversion data corresponding to the reception gain value on the reference line (horizontal line)
Pink	Resultant signal value
Red	Over-flow detection level, under-flow detection level and target level of gain adjustment
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-23.

Table 6.25-23 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
Ctrl + <←> and Ctrl <→> keys	Reference line moves horizontally on measured points by 1 step.
<r> key	Reference line moves on resultant signal value (pink color) upward.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.9 Theta Slice Frequency

Either Theta Slice Frequency or Theta Phase screen is displayed. Slice Frequency or Phase can be selected on the option menu displayed on the screen. If the selected menu is displayed in light color, it means that no scan data is available.

6.25.4.9.1 Theta Slice Frequency

The ordinate axis and abscissa axis represent P0 value and data No. respectively. Color of the line on the graph has the meanings shown in Table 6.25-24.

The data with ○ is the actually measured value.

Only in case of FSE sequence and its execution is designated on the system table, the log is kept.

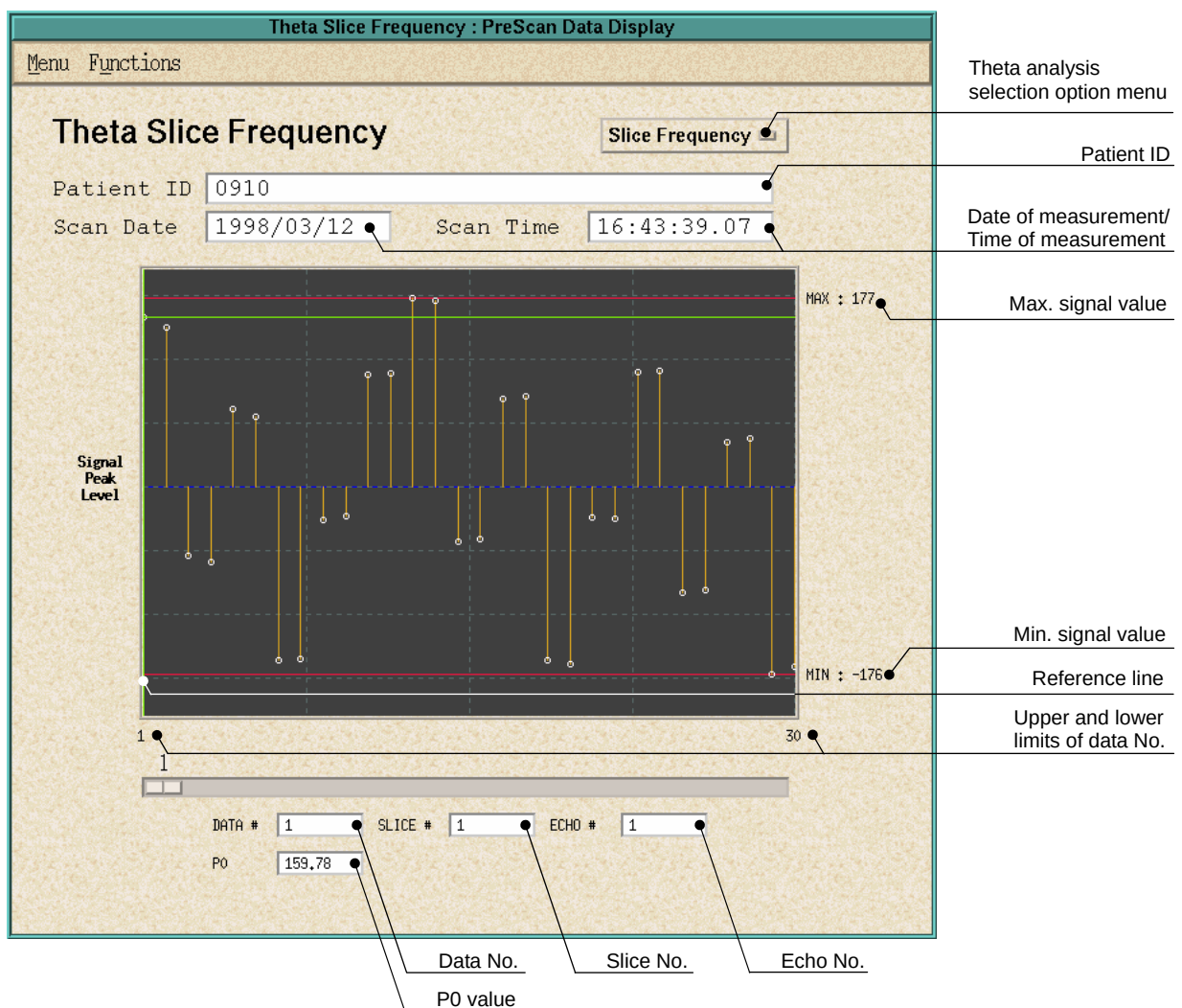


Fig. 6.25-15 Theta Slice Frequency Display Screen

Table 6.25-24 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and P0 value corresponding to the data No. on the reference line (horizontal line)
Pink	Resultant signal value
Red	Max. and Min. values of P0 value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-25.

Table 6.25-25 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.9.2 Theta Phase

The ordinate axis and abscissa axis represent P0 value and data No. respectively.
Color of the line on the graph has the meanings shown in Table 6.25-26.

The data with ○ is the actually measured value.

Only in case of FSE sequence and its execution is designated on the system table, the log is kept.

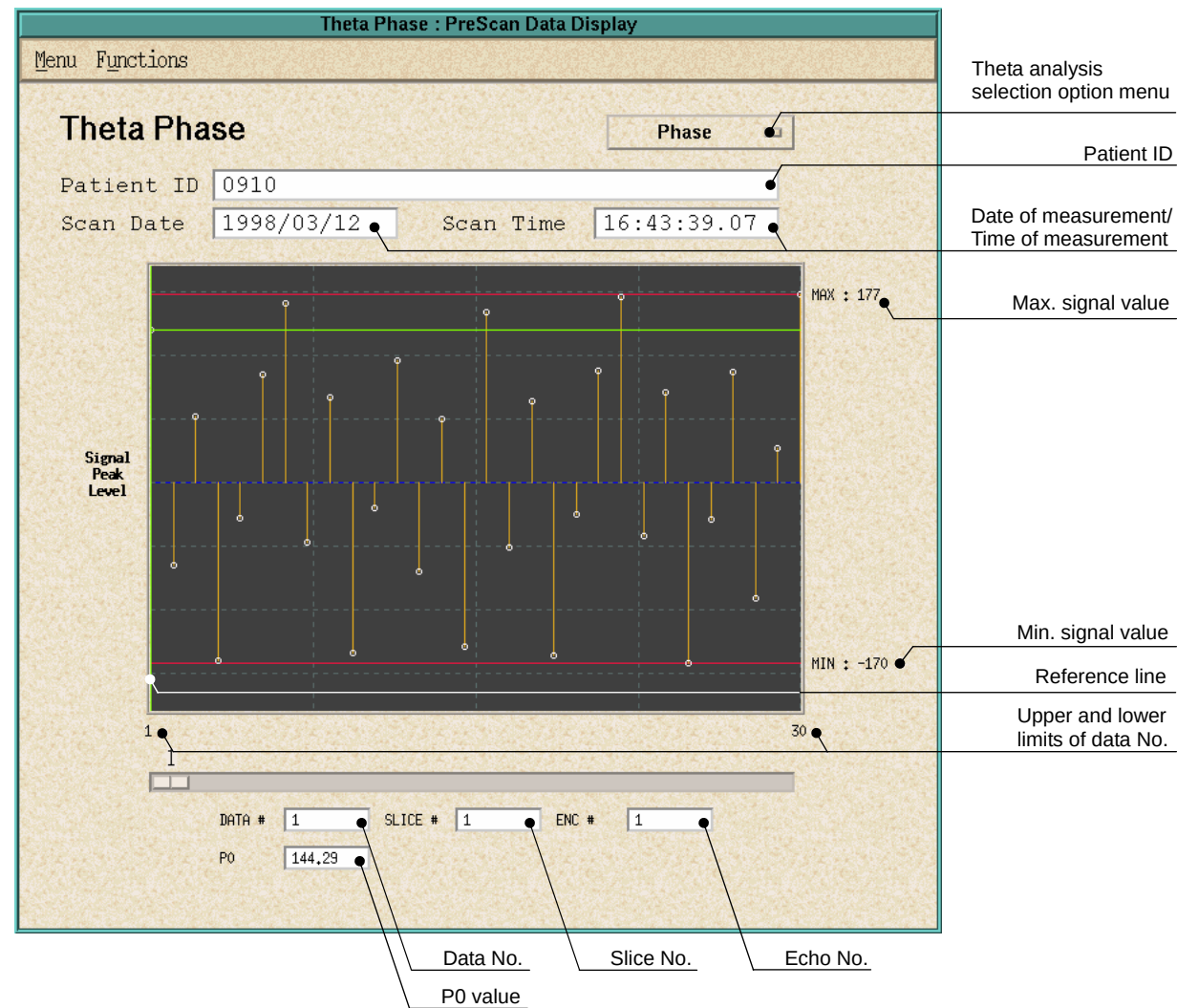


Fig. 6.25-16 Theta Phase Display Screen

Table 6.25-26 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and P0 value corresponding to the data No. on the reference line (horizontal line)
Red	Max. and Min. values of P0 value
Blue	Coordinate axes (signal value: 0)

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-27.

Table 6.25-27 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.10 Calibration Scan

One of X, Y and Z-Axis is displayed. Any of X, Y and Z can be selected on the Option Menu displayed on the screen. If the selected menu is displayed in light color, it means that no scan data is available.

6.25.4.10.1 Calibration Scan (X-Axis)

The ordinate axis and abscissa axis represent signal value and encode No. respectively. Color of the line on the graph has the meanings shown in Table 6.25-28.

The data with ○ is the actually measured value.

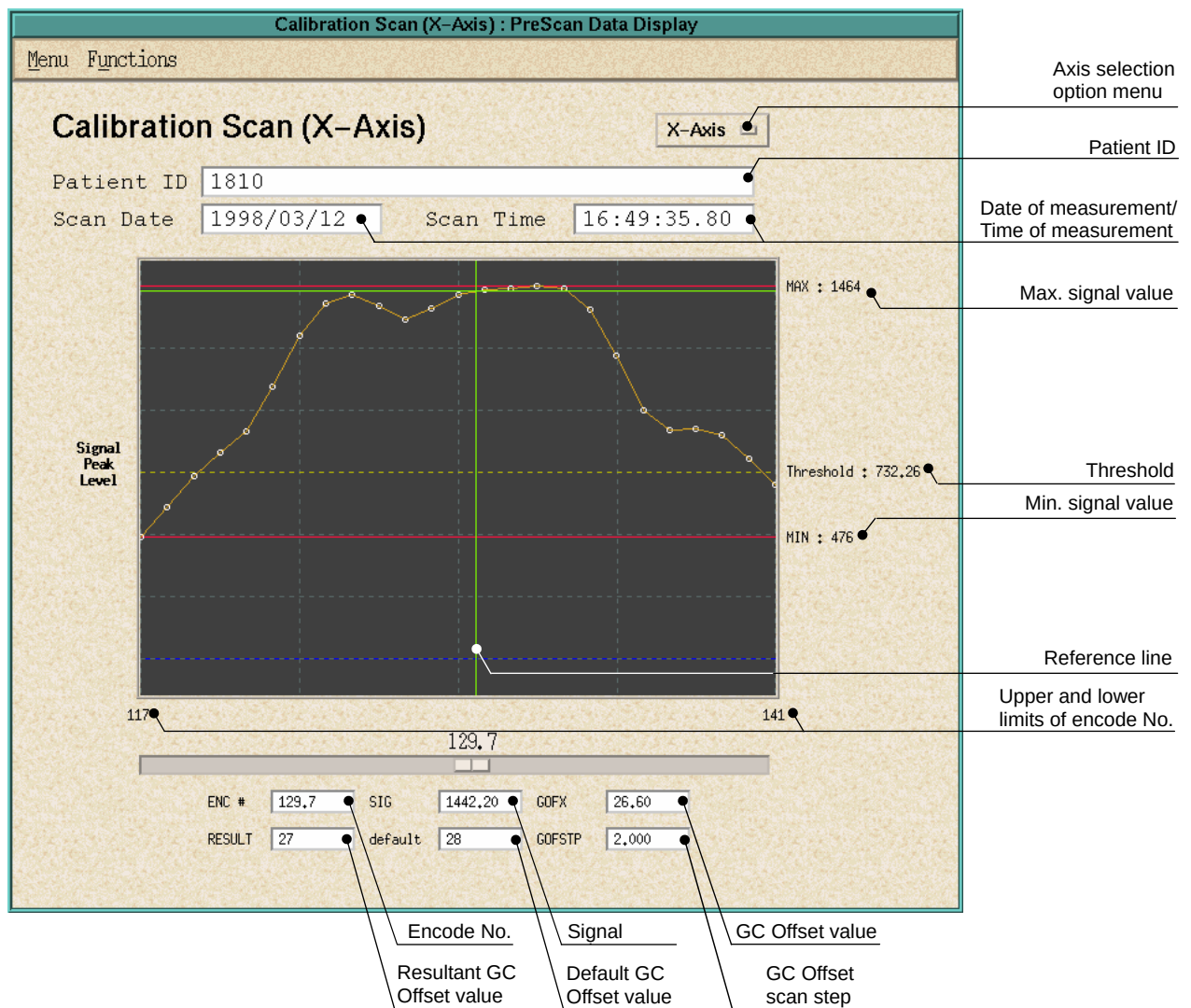


Fig. 6.25-17 Calibration Scan (X-Axis) Display Screen

Table 6.25-28 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to the encode No. on the reference line (horizontal line)
Pink	Resultant signal value
Red	Max. and Min. values of signal value
Blue	Coordinate axes (signal value: 0)
Orange	Threshold value

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-29.

Table 6.25-29 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.10.2 Calibration Scan (Y-Axis)

The ordinate axis and abscissa axis represent signal value and encode No. respectively. Color of the line on the graph has the meanings shown in Table 6.25-30.

The data with ○ is the actually measured value.

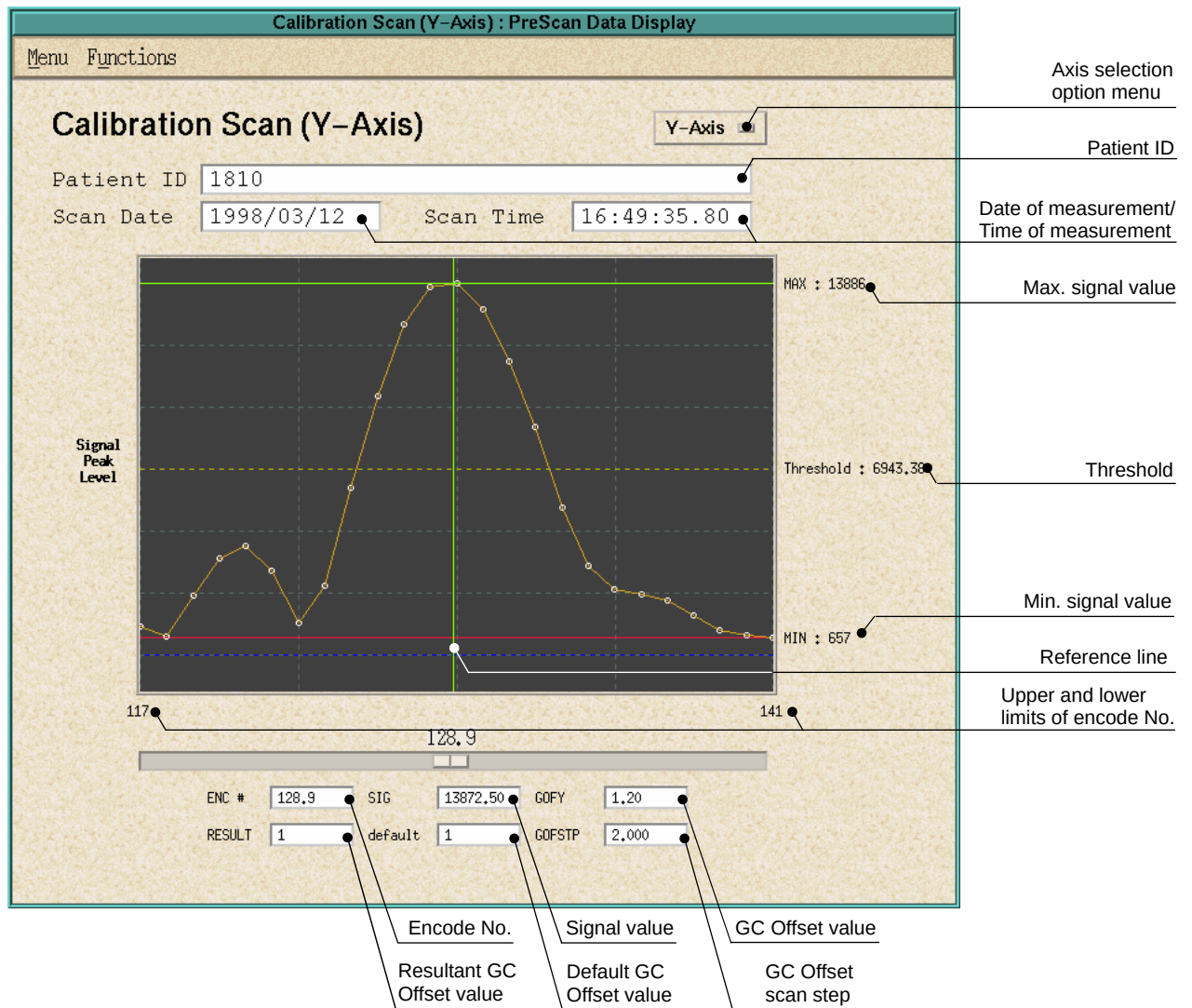


Fig. 6.25-18 Calibration Scan (Y-Axis) Display Screen

Table 6.25-30 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to the encode No. on the reference line (horizontal line)
Pink	Resultant signal value
Red	Max. and Min. values of signal value
Blue	Coordinate axes (signal value: 0)
Orange	Threshold value

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-31.

Table 6.25-31 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.10.3 Calibration Scan (Z-Axis)

The ordinate axis and abscissa axis represent signal value and encode No. respectively. Color of the line on the graph has the meanings shown in Table 6.25-32.

The data with ○ is the actually measured value.

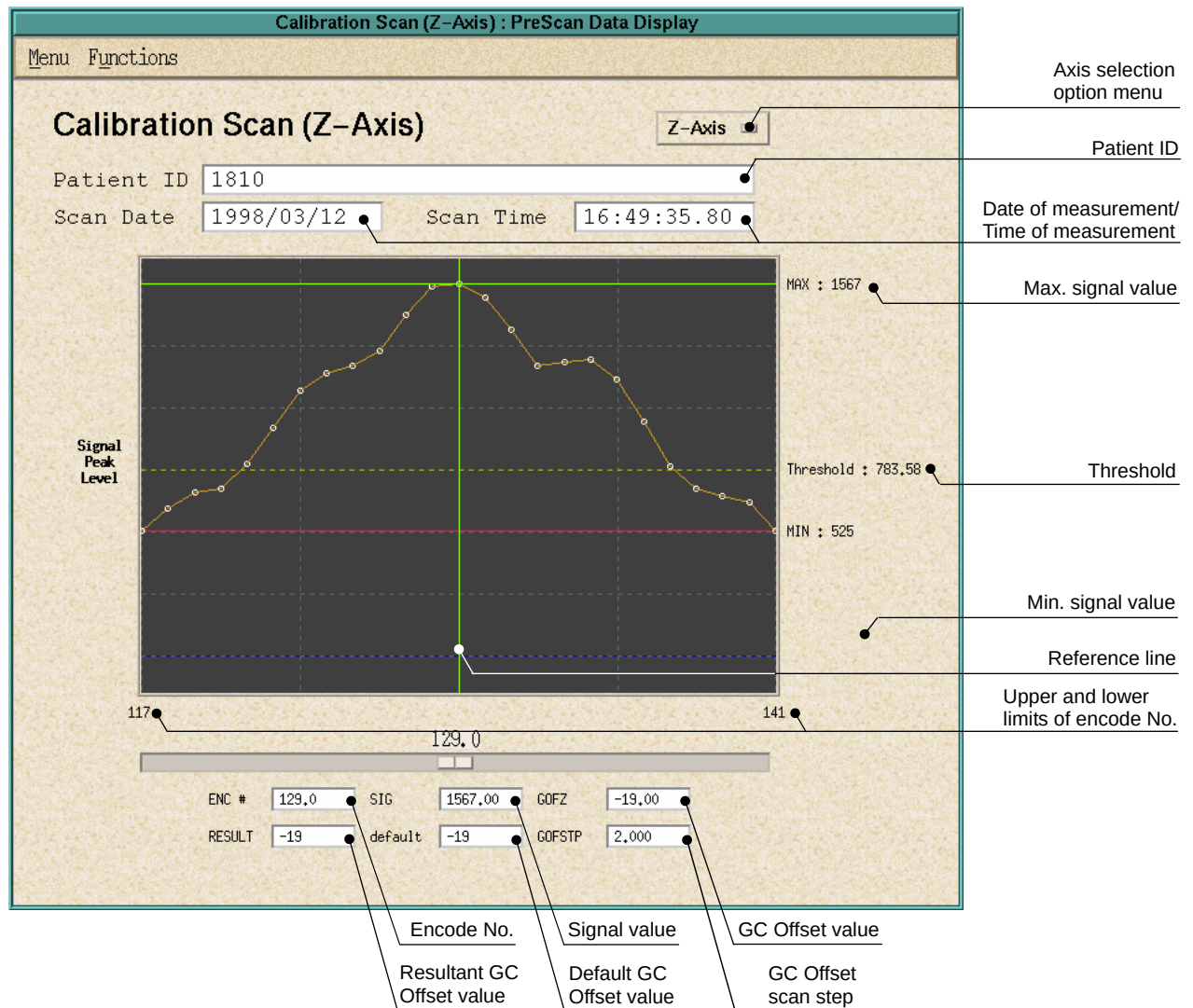


Fig. 6.25-19 Calibration Scan (Z-Axis) Display Screen

Table 6.25-32 Meanings of Colors

Color	Meanings
Green	Reference line (vertical line) and signal value corresponding to the encode No. on the reference line (horizontal line)
Pink	Resultant signal value
Red	Max. and Min. values of signal value
Blue	Coordinate axes (signal value: 0)
Orange	Threshold value

By clicking and dragging the slider by the mouse, the reference line (green vertical line) can be moved. It can also be moved from the keyboard. Key operation is shown in Table 6.25-33.

Table 6.25-33 Key Operation

<←> and <→>	Reference line moves horizontally by 1 step.
<Home> key	Reference line moves to the leftmost end.
<End> key	Reference line moves to the rightmost end.

6.25.4.11 EPI Adjustment

The ordinate axis and abscissa axis represent echo signal value and gain value respectively. Color of the line on the graph has the meanings shown in Table 6.25-35.

The mark ○ represents the intersection point of the receiver gain and signal value for scan, while the mark × represents the intersection point of the receiver gain and estimated value of the resultant signal value for the main scan.

For detailed explanation of various values refer to Table 6.25-34.

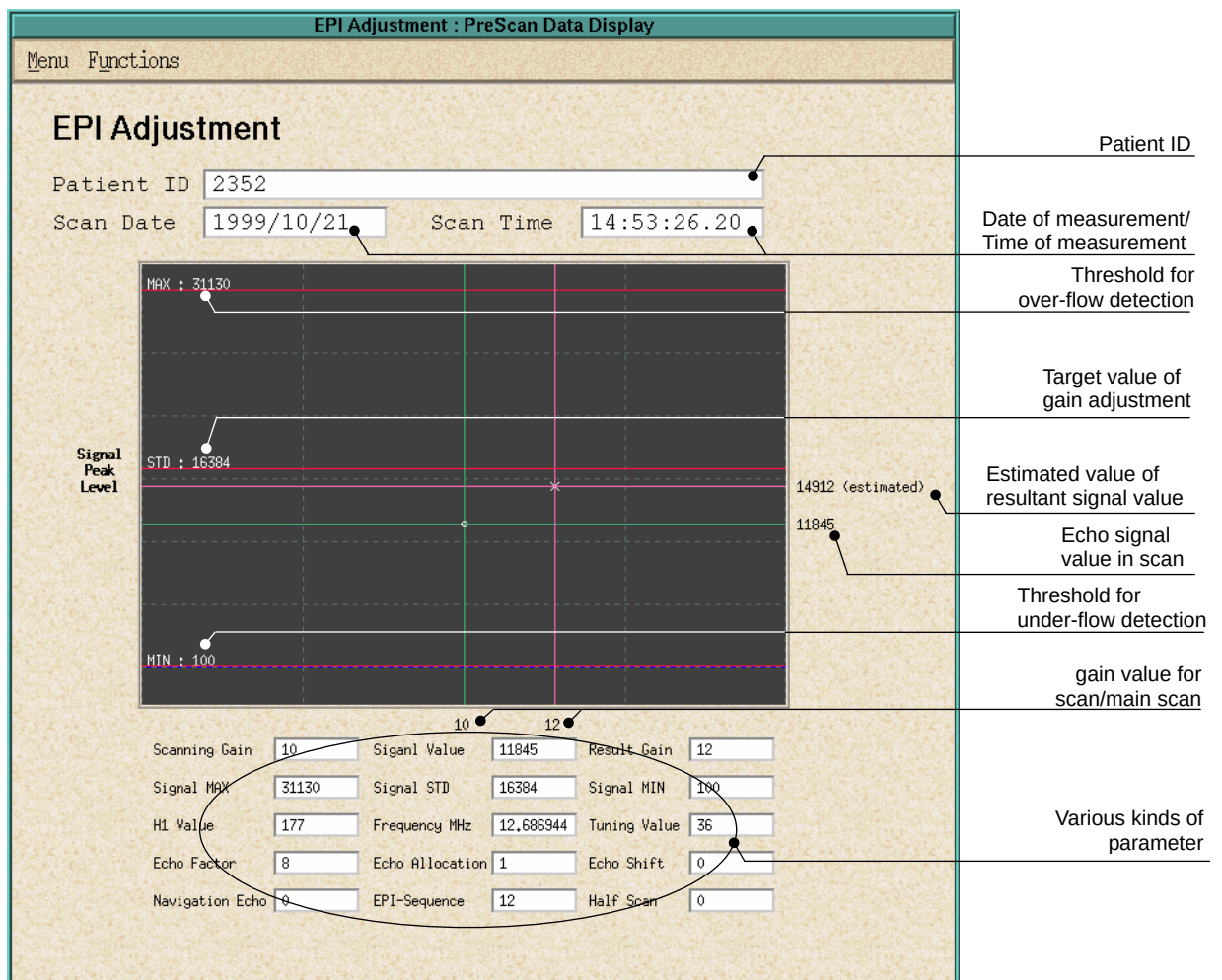


Fig. 6.25-20 EPI Adjustment Display Screen

Table 6.25-34 Detailed Explanation of Various Values

Item	Description	Item	Description
Scanning Gain	Gain value in scan	Echo Allocation	Echo allocation method
Signal MAX	Upper limit value of signal	EPI-Sequence	Scan sequence name
H1 Value	Default H1 value	Result Gain	Result gain value
Echo Factor	Number of echo factor	Signal MIN	Lower limit value of signal
Navigation Echo	Number of navigation echo	Tuning Value	Turning value at scan time
Signal Value	Max. value of AD conversion	Echo Shift	Number of echo shift
Signal STD	Signal target value	Half Scan	Half scan flag
Frequency [MHz]	Frequency value at scan time		

Table 6.25-35 Meanings of Colors

Color	Meanings
Green	Reception gain value for scan (vertical line) and signal value (horizontal line)
Pink	Reception gain value for main scan (vertical line) and estimated value of resultant signal value (horizontal line)
Red	Over-flow detection level, under-flow detection level and target level of gain adjustment
Blue	Coordinate axes (signal value: 0)

6.25.5 End

By selecting [Quit] on "Menu" of the Menu bar, the dialog box shown in Fig. 6.25-22 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

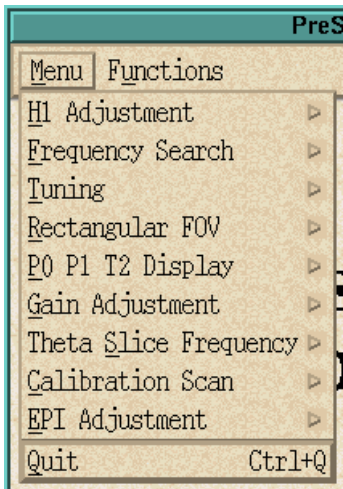


Fig. 6.25-21 Quit Button



Fig. 6.25-22 Exit Confirmation Dialog

6.25.6 Error specification

Error specification is shown in Table 6.25-36.

Table 6.25-36 Error Specification

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.

6.26 Frequency Log Display

6.26.1 General

The frequency log display tool is used to store the measured frequency data obtained in a pre-scan with 1024 data at maximum up to the latest data and displays those data in graph. Frequency, H1, tuning and gain values at each measuring point are also displayed at the same time.

6.26.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Frequency Log Display in the displayed items and click [OK] or double-click the selected item.

6.26.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.26-1.

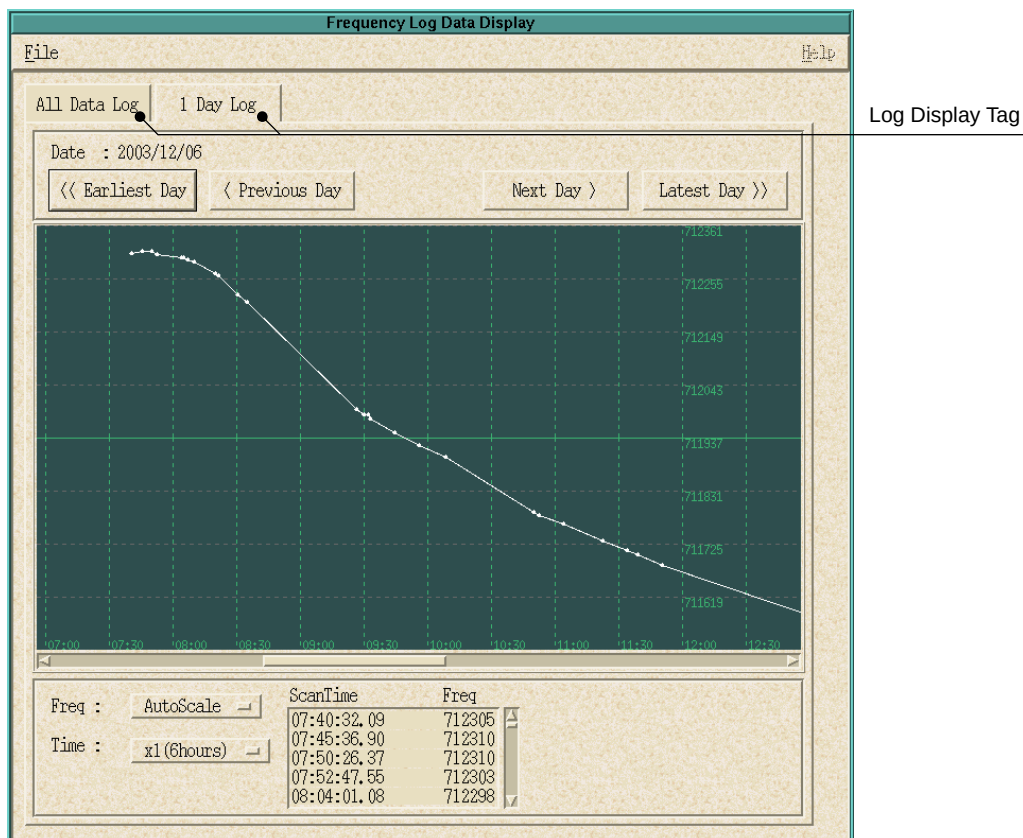


Fig. 6.26-1 Frequency Log Display Screen

6.26.4 Operation procedure

Follow the next steps to execute the process.

6.26.4.1 All Data Log

If All Data Log is selected with a log display tag, the graph that displayed all the samples of a log can be seen. Displayed frequency log data are the latest at the left end and they become older as they are displayed rightward. At the lower part of the screen, the frequency, H1, tuning, and gain values are displayed for the data at the reference line.

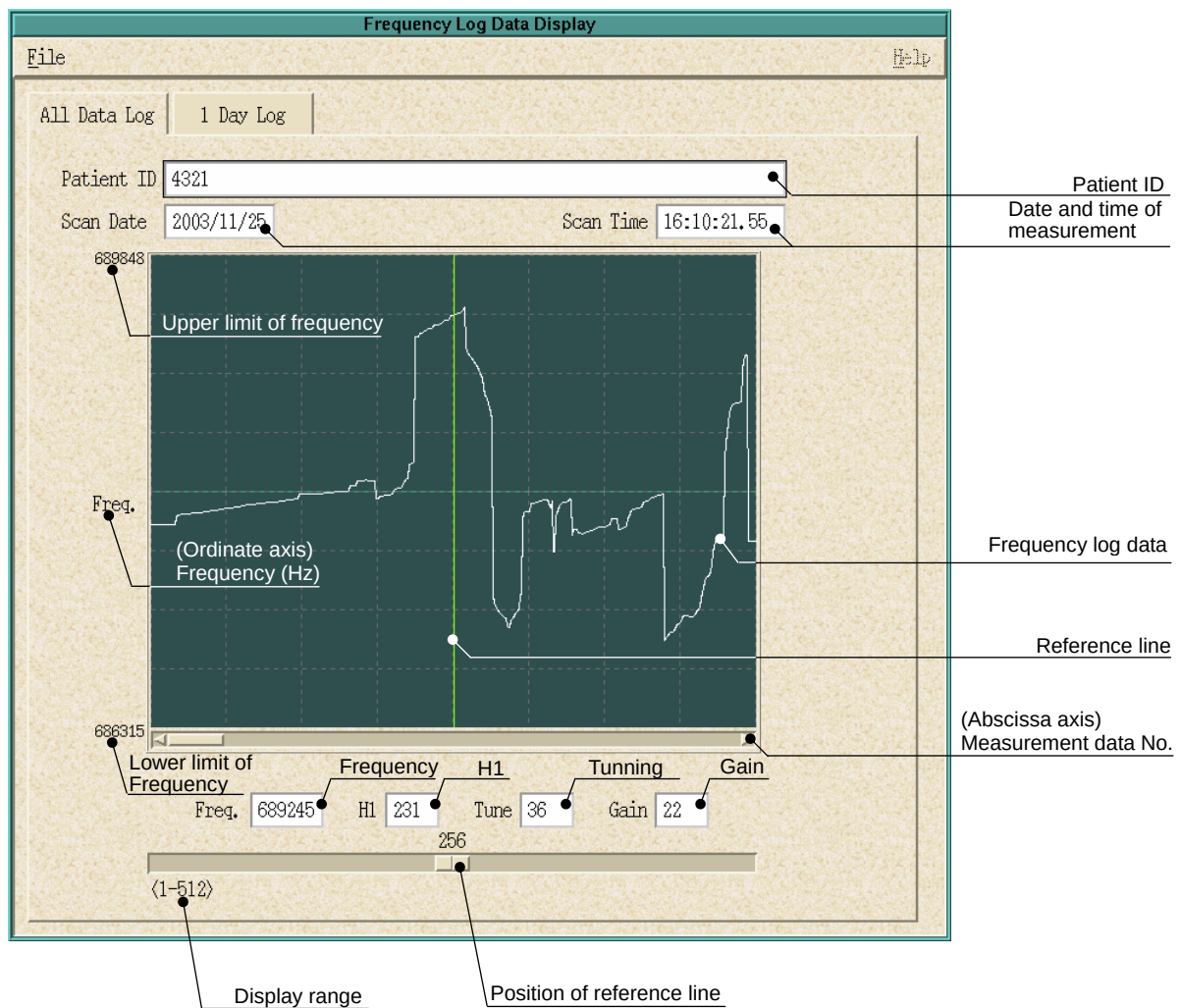


Fig. 6.26-2 All Data Log Display Screen

6.26.4.1.1 Movement of display position

Only 512 data of the total 1024 data are displayed on the screen. After start-up, measurement data of No. from 1 to 512 are displayed. Dragging the scroll bar to scroll the screen, all the 1024 data can be displayed.

The display position can be scrolled not only by the mouse cursor but also by the keyboard.

Table 6.26-1 Keyboard Operation for scrolling Display Position

Shift + <←>	Scroll to left by [1]
Shift + <→>	Scroll to right by [1]
<PageDown>	Scroll to left by [64]
<PageUp>	Scroll to right by [64]
Shift + <Home>	Scroll to the left end
Shift + <End>	Scroll to the right end

6.26.4.1.2 Movement of reference line

By dragging the scale bar, the reference line can be moved within the range being displayed.

The reference line can be moved not only by the mouse cursor but also by the keyboard.

Table 6.26-2 Keyboard Operation for moving Reference Line

<←>	Move to left by [1]
<→>	Move to right by [1]
Ctrl + <←>	Move to left by [64]
Ctrl + <→>	Move to right by [64]
<Home>	Move to the left end
<End>	Move to the right end

6.26.4.2 Progress for one day

If 1Day Log is selected with a log display tag, the graph that displayed by the time progress for one day can be seen.

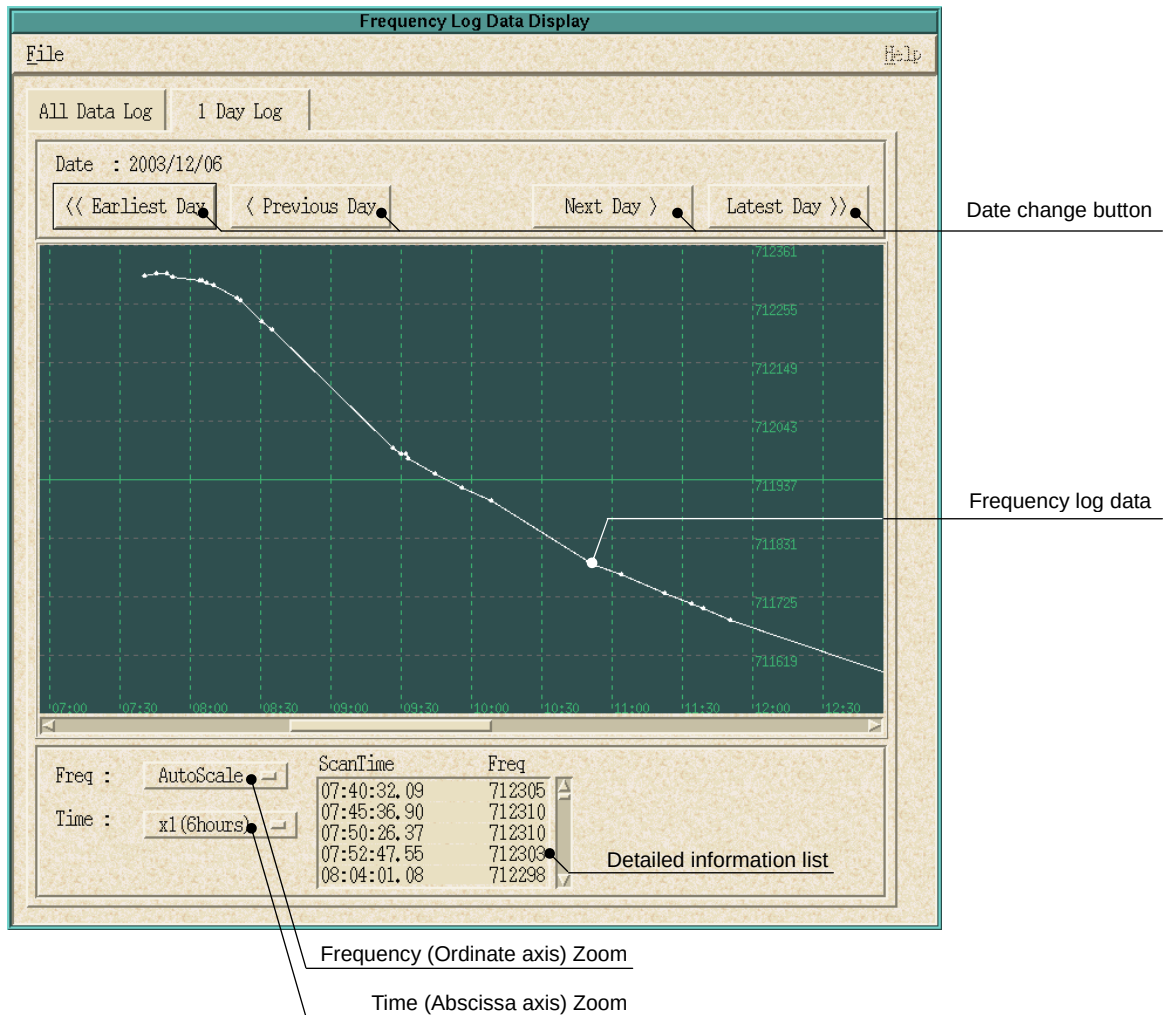


Fig. 6.26-3 All Data Log Display Screen

6.26.4.2.1 Change of the date to display

A click of the date change button (Earliest Day, Previous Day, Next Day, Latest Day) displays the log of each date.

6.26.4.2.2 Expansion and reduction of frequency graph

It can change by changing expansion / reduction menu. The frequency that is a ordinate axis can be chosen from **Auto Scale**, **x2**, and **x4**. The time that is a abscissa axis can be chosen from **x1(6hours)**, **x2(3hours)**, and **x6(1hour)**.

6.26.4.2.3 Detailed information on a sample

The frequency value and time that each sample of a date on display is actual on a detailed information list are displayed.

If a sample is chosen from a list, a line will be displayed on the time when it corresponds on graph. Moreover, if a graph is clicked, the nearest sample will be chosen. These states of choosing are canceled by choosing again.

6.26.5 End

Selecting [Quit] button from "File Menu" in the menu bar pops up the dialog shown in Fig. 6.26-5. Clicking [Yes] button on this dialog closes the application, while clicking [No] button returns to the previous screen without closing the application.

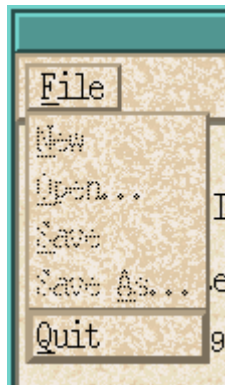


Fig. 6.26-4 Quit Button



Fig. 6.26-5 Exit Confirmation Dialog

6.26.6 Error specification

Error specification are shown in Table 6.26-3.

Table 6.26-3 Error Specification

Message	Status	Solution
File not found.	Because the logging process has not been executed at all, any logging file was not found.	Because no data to be displayed is available, close the application.

6.27 Check Sum Display

6.27.1 General

This is a tool to calculate the sum of files and save it. As a result of the sum, whether any file is changed or destroyed is checked from the difference from the previously calculated value. Number of registerable directories is 20.

6.27.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Check Sum in the displayed items and click [OK] or double-click the selected item.

6.27.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.27-1.

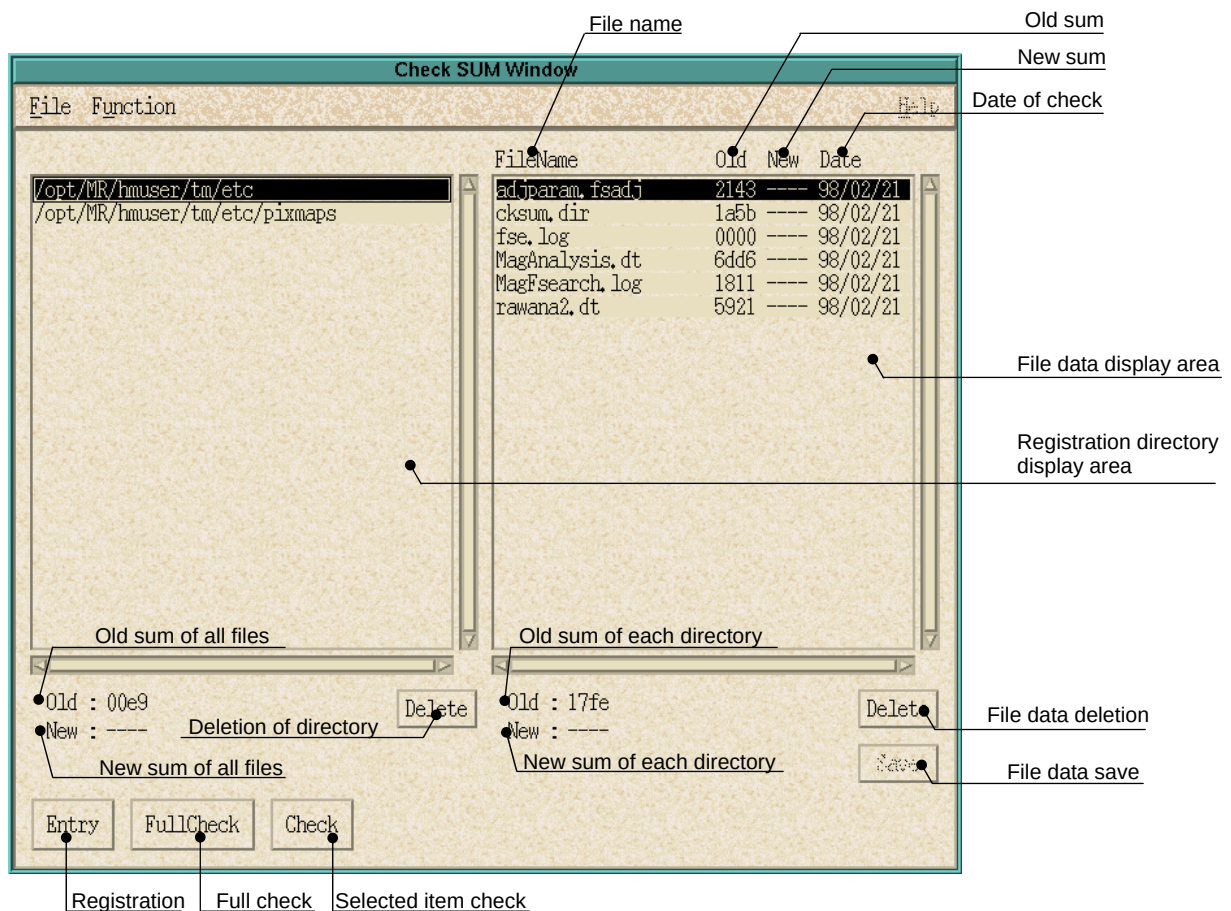


Fig. 6.27-1 Check Sum Display Screen

6.27.4 Operation procedure

Follow the next steps to execute the process.

6.27.4.1 Registration of check items

Either select [Entry...] from "Functions Menu" in the Menu Bar or click [Entry] button on the screen to pop up the File Selection Box shown in Fig. 6.27-3.

Select the desired directory. [...] shows one upper hierarchy. The selected directory is highlighted to black. The selected directory is defined by double-clicking on the selected item or clicking on [Filter].

Select the desired file. Only one file can be selected. The selected file is highlighted to black. The selected file is defined by double-clicking on the selected item or clicking on [OK].

In this case, if the selected file is the already registered one, an alarm sounds and the file is highlighted to black on the main screen.

The file selection box is closed by clicking [Cancel].

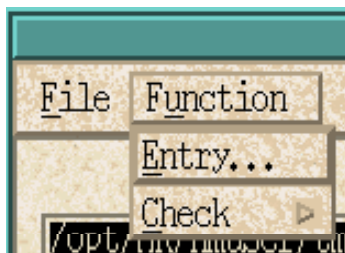


Fig. 6.27-2 Registration Button

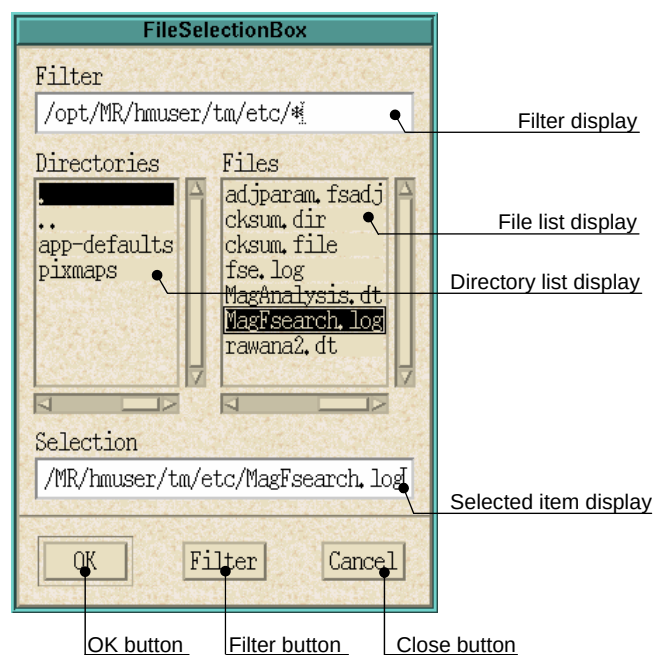


Fig. 6.27-3 File Selection Box

6.27.4.2 Selection of file to be operated

The following methods are available for selection of the file data list display area.

Table 6.27-1 Selection Method of File to be operated

Single selection	Move the mouse cursor to the item to be selected and click on it. Carry out the same operation to select another item.
Selection of more than one item (change)	Move the mouse cursor to the top of the items. Then, with the mouse pressed either drag the mouse upward or downward or click the top item and then click on the last item to be selected with the Shift key pressed.
Selection of more than one item (addition)	With the Ctrl key pressed, click the mouse button on the item to be selected. The newly selected item is added to the list of the already selected items. The already selected items are deselected.

6.27.4.3 Sum check

Select the directory to which the sum check is applied.

To carry out the sum check for all items, either select [All] on [Check] sub-menu from "Functions Menu" in the Menu Bar or click [Full Check] button.

In this case, even if some items have been selected, they are neglected.

To check the selected item, either select [Selected Items] on [Check] sub-menu from "Functions Menu" in the Menu Bar or click [Check] button.



Fig. 6.27-4 Check Sub-menu

6.27.4.4 Result display

For the file for which the check was previously carried out, the old result is displayed in the Old Sum display area. The date shown there is the date of the checking. [----] indicates none of old data.

After executing the check the result is displayed in the new sum display area. In case the old sum and new sum are same, [----] is displayed, but the date is updated.

After the check process is completed, the sum values of all the registered directories are displayed in the area under the registration directory list display area, and the sum value of each registered directory is displayed in the lower part of the file data list display area.

6.27.4.5 Suspension of sum check

To suspend the sum check during the process being executed, click [Cancel] button in the working dialog.

6.27.4.6 Saving file data

Click [Save] button to save the sum check results.

With the sum value changed, if the directory is tried to be changed or the check is retried, the dialog shown in Fig. 6.27-5 is popped up. To save the sum value click [Yes] button, while to abandon it click [No] button.

Once the sum value is saved, the new sum value is displayed as old sum value and the date is also saved. If the data is abandoned, the screen returns to the one before the sum check was made.

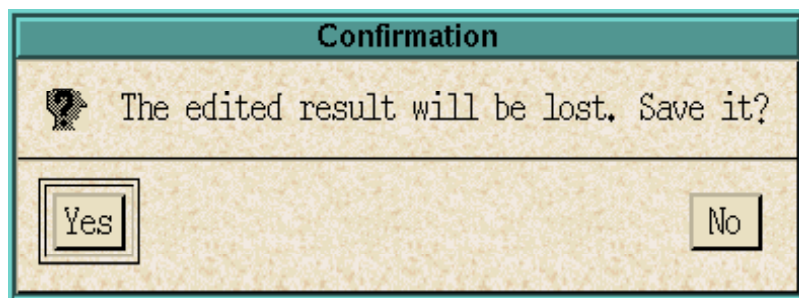


Fig. 6.27-5 Save Confirmation Dialog

6.27.4.7 Deletion of registered directory

Click [Delete] button located at the lower part of the registered directory list display area. The directory is deleted from the registration list, but because the file data in the directory remains, it is possible to confirm the file data for registration.

6.27.4.8 Deletion of file data

Click [Delete] button located at the lower part of the file data list display area.

More than one file can be selected. Note that even if the deleted files is reregistered the previous sum value is not displayed.

6.27.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.27-7. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.

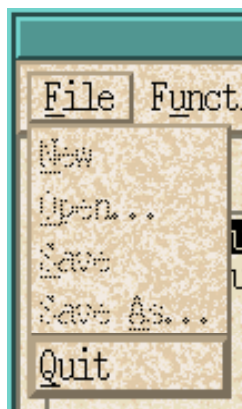


Fig. 6.27-6 Quit Button



Fig. 6.27-7 Exit Confirmation Dialog

In case the sum value is updated, the dialog shown in Fig. 6.27-8 is popped up. To save the sum value click [Yes] button, while to abandon it click [No] button. The application is closed. Clicking [Cancel] button returns to the previous screen without exiting from the application.

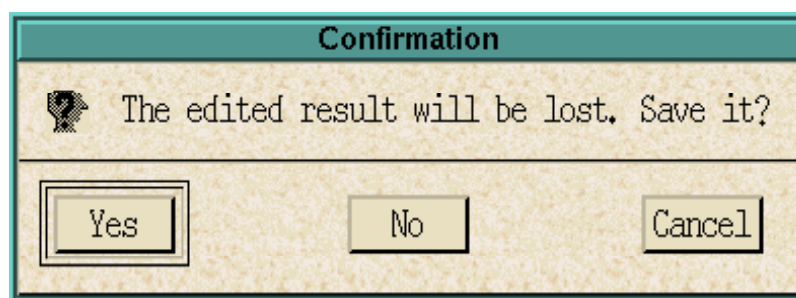


Fig. 6.27-8 Save Confirmation Dialog

6.27.6 Error specification

Error specification is shown in Table 6.27-2.

Table 6.27-2 Error Specification

Message	Status	Solution
The file of specification cannot be opened.	The designated file could not be opened.	Please report the error to the KASHIWA factory to need to confirm existence of the file and the permission etc.

6.28 Frequency Search

6.28.1 General

This is a tool to carry out frequency search and frequency lock and display the measurement results in order to obtain the optimum center frequency for the system.

6.28.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and display "Others Group" (refer to Paragraph 1.3.5). Then, select F-Search from "Maint. Menu" on the scan parameter setting area.

6.28.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.28-1.

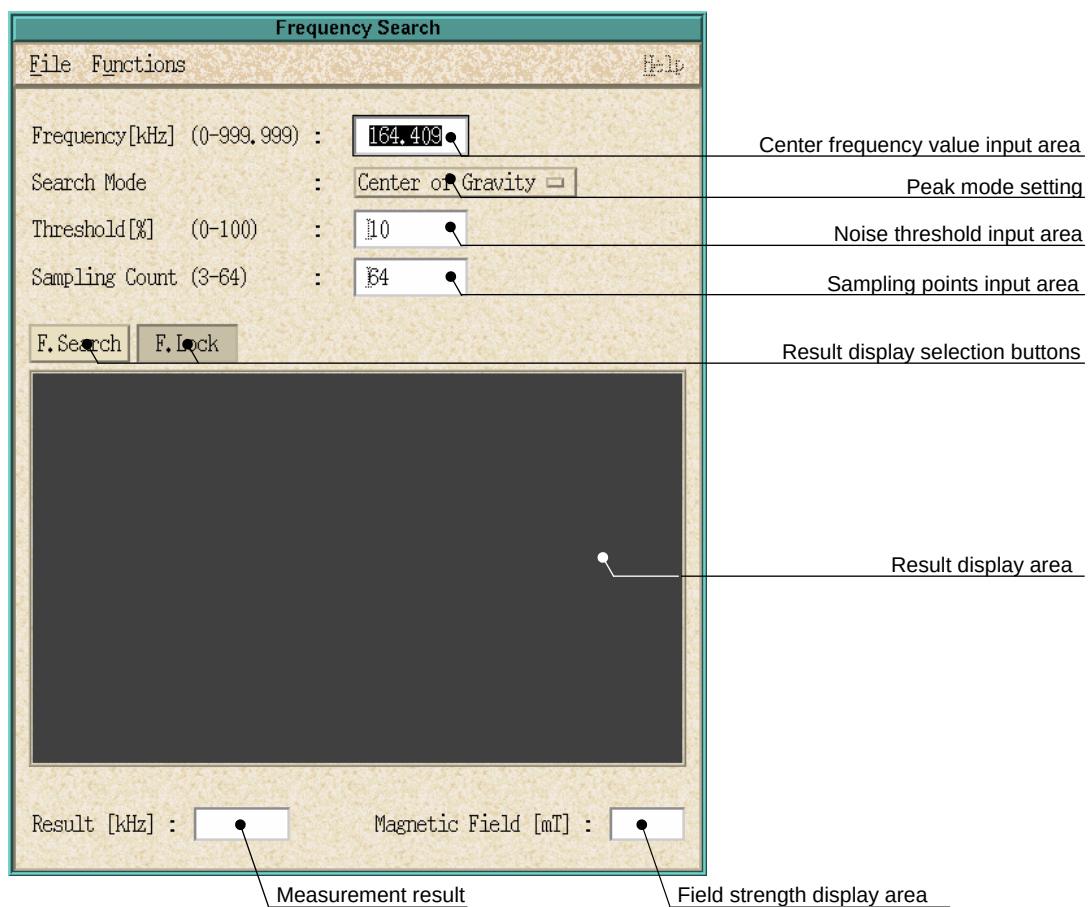


Fig. 6.28-1 Frequency Search Display Screen

6.28.4 Operation procedure

Follow the next steps to execute the process.

6.28.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following procedures are available to change parameters.

6.28.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. Using the keyboard, input figures or character string, then either press <Enter> key on the keyboard or remove the focus from the input area to change the parameter.

6.28.4.1.2 Parameter change by option menu

The displayed items are the set parameters. Click the option menu and click the mouse cursor on the parameter to be changed on the opened menu. The parameter is changed.

Table 6.28-1 Parameter Setting Range

Parameter Name	Setting Range
Center frequency	0.000 ~ 999.999
Search mode	Peak/Center of Gravity
Noise threshold	0 ~ 100
Sampling points	3 ~ 64

6.28.4.2 Measurement start

Click [Start] button on the scan card to start the measurement.

6.28.4.3 Measurement stop

Click [Stop] button on the scan card. The measurement is stopped.

6.28.4.4 Result display

According to the selected mode button, either frequency search or frequency lock result is displayed in the measurement result display area. The abscissa axis represents frequency value and ordinate axis represents signal value.

The maximum and minimum values in the measurement range are displayed at the both bottom ends of the graph.

Sampling points for frequency search are the ones designated in the measurement points input area, while they are 512 points for frequency lock, and they are connected with a straight line on graph.

For frequency lock, the resultant line is displayed on the points of the resultant frequency values. Each of the resultant frequency values is also updated and displayed according to the selected graph.

After measurement, the peak frequency obtained by frequency lock is automatically set in the frequency value input area, and the magnetic field strength calculated from the peak frequency is displayed.

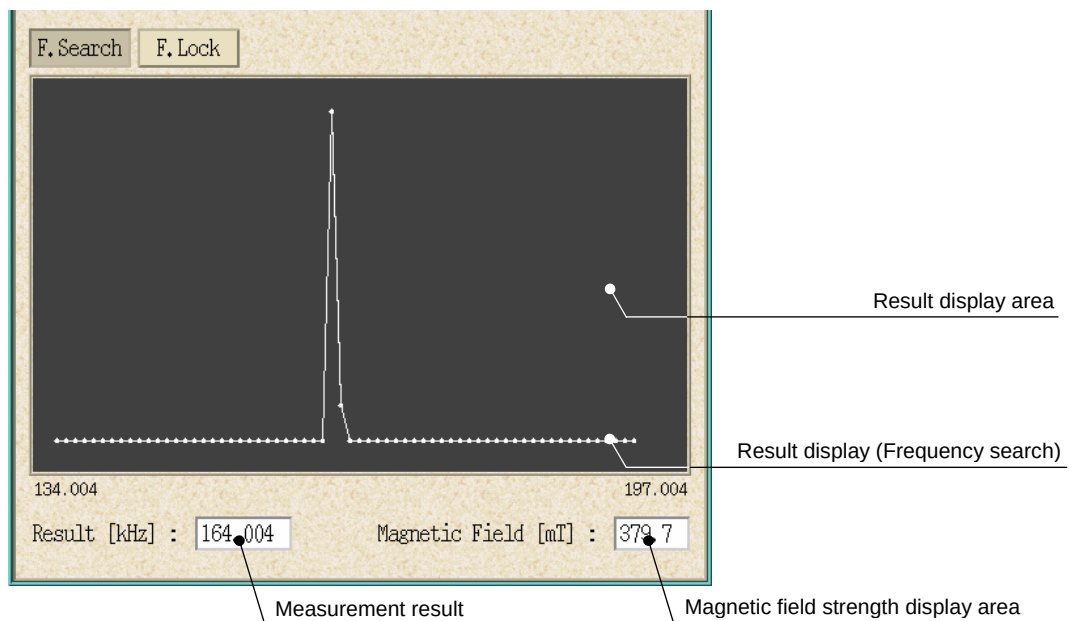


Fig. 6.28-2 Example of Measurement Result (Frequency search)

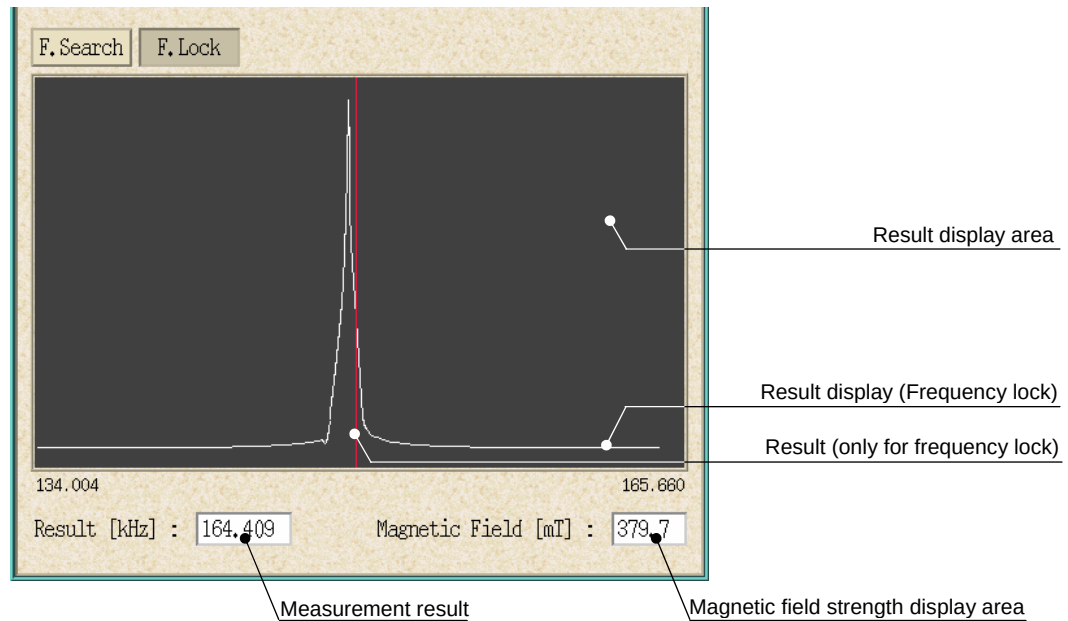


Fig. 6.28-3 Example of Measurement Result (Frequency lock)

6.28.4.5 Selection of measurement result display area

Click either [F. Search] or [F. Lock] on the upper part of the measurement result display area, or select it from [Graph] on "Functions Menu" of the Menu Bar.

The result of either Frequency search or Frequency lock is instantly selected.

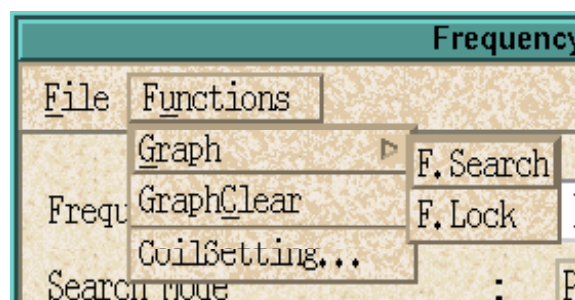


Fig. 6.28-4 Result Selection Buttons

6.28.4.6 Saving result

Select [Save] from "File Menu" of the Menu Bar. The value displayed in the frequency value input area is saved into the system table (H/D and Memory).

This button is selectable in case the frequency value set in the frequency value input area and scan resultant frequency value are different each other.

6.28.4.7 Clearing result display area

Select [Clear] button from "Functions Menu" of the Menu Bar. The data in the result display area is cleared.

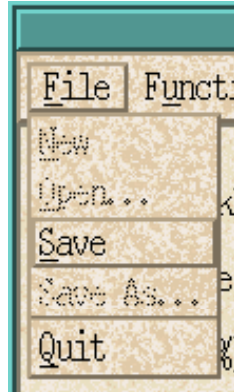


Fig. 6.28-5 Save Button

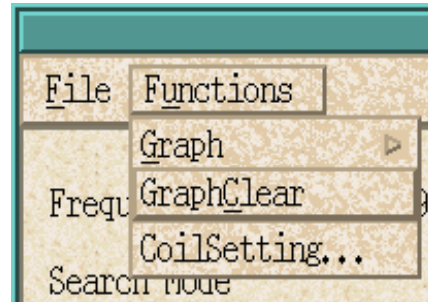


Fig. 6.28-6 Clear Button

6.28.4.8 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.28.4.8.1 Default

Default at the start time is Auto. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.28.4.8.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

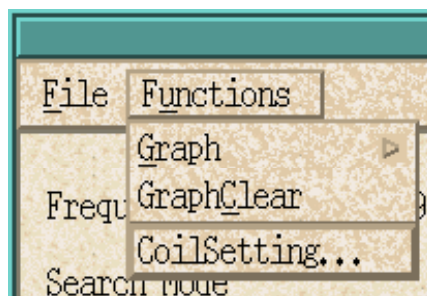


Fig. 6.28-7 Coil Setting Window Display Button

6.28.4.8.3 Display screen

The display screen and function names are shown in Fig. 6.28-8 and Fig. 6.28-9.

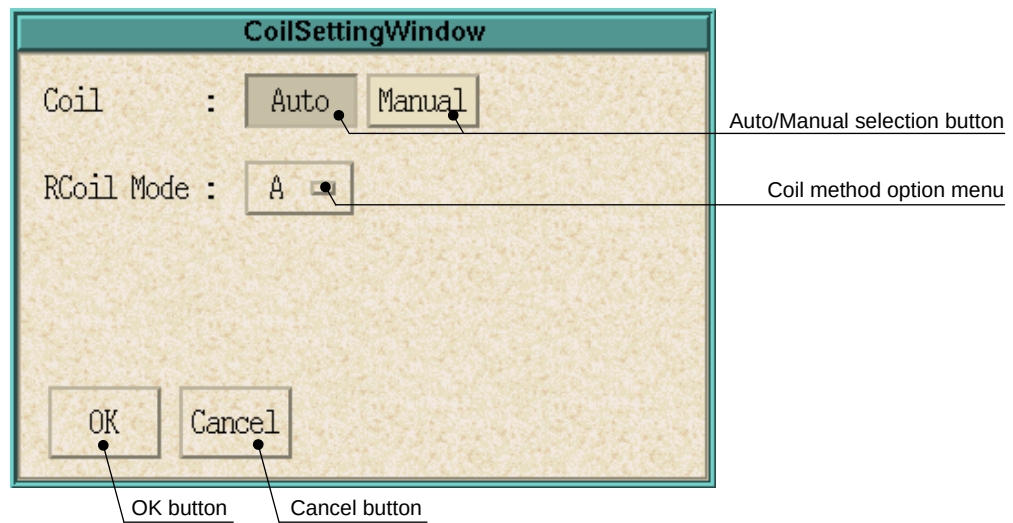


Fig. 6.28-8 Coil Setting Window Display Screen (Auto mode)

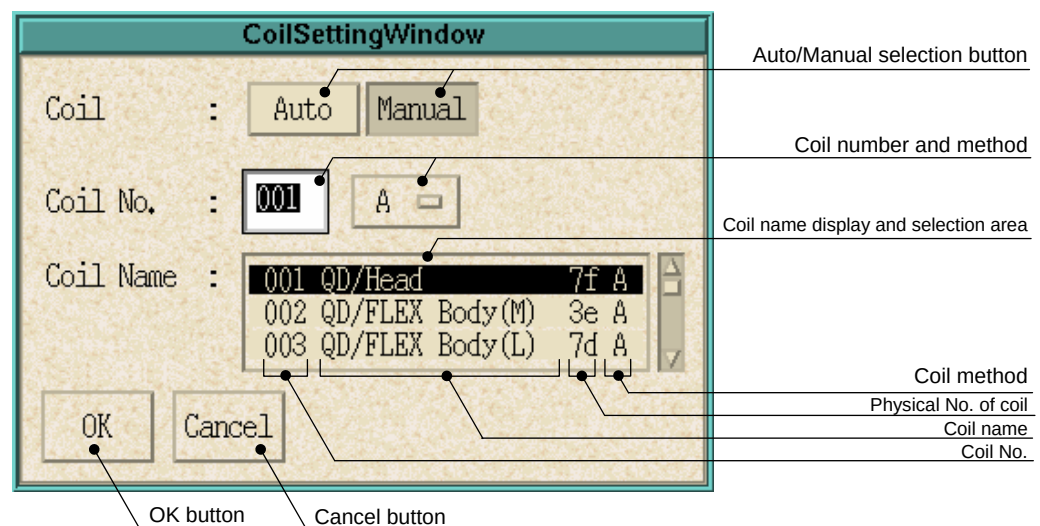


Fig. 6.28-9 Coil Setting Window Display Screen (Manual mode)

6.28.4.8.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the **Manual** screen in case of the **Auto** mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of **Manual** screen is [0], the coil method is [A], and the coil name is not selected.

6.28.4.8.5 Selection of coil application method

Select the method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in **Manual** mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.28.4.8.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area. The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.28.4.8.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.28.4.8.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.28.4.8.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.28.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.28-11. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.

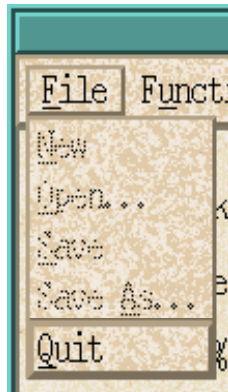


Fig. 6.28-10 Quit Button



Fig. 6.28-11 Exit Confirmation Dialog

In case the frequency value has been changed as a result of measurement, the dialog shown in Fig. 6.28-12 is popped up. Clicking [Yes] button save the result into the system table (H/D and Memory) and closes the application. To close the application without saving the result into the system table, click [No] button. Clicking [Cancel] button returns to the previous screen without exiting from the application.

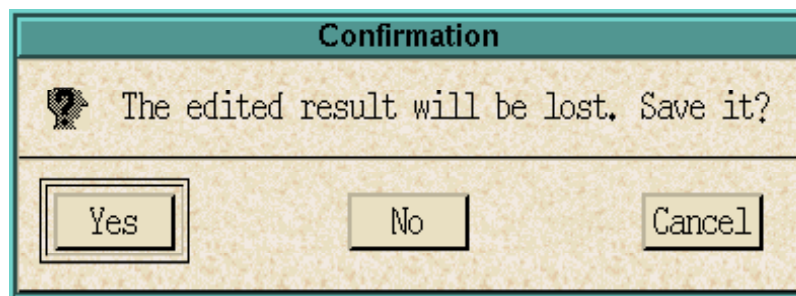


Fig. 6.28-12 Save Confirmation Dialog

6.28.6 Error specifications

Error specifications are shown in Table 6.28-2.

Table 6.28-2 Error Specifications

Message	Status	Solution
Couldn't draw result line (Flock).	The resultant line could not be drawn.	The center frequency is deviated. With the changed value, retry the measurement.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system status and retry the same operation procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the registered coils list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the registered coils list.
No coil is connected to the system.	The coil is not connected yet to the system.	Check the coil connection conditions.
Error occurred in the patient table.	Communication with the patient table failed.	Check the patient table condition.

6.28.7 Other cautions

Because this application is operated on the scan card, the buttons on the card are used to start and stop a scan.

Accordingly normal scan cannot be performed while this application is running, therefore close this application to start a normal scan.

Do not change the card while this application is running, otherwise the connection between this application and scan card may be broken. Before changing the card, be sure to close this application.

Should the card be changed without closing this application, immediately close this application. In this case, the data can be saved into the system table.

6.29 H1 Adjustment

6.29.1 General

This is a tool to acquire signals as changing H1 value and display the measurement result of the process to be retrieved in order to obtain the optimum H1 value for the system.

6.29.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and display "Others Group" (refer to Paragraph 1.3.5). Then, select H1 Adjust from "Maint. Menu" on the scan parameter setting area.

6.29.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.29-1.

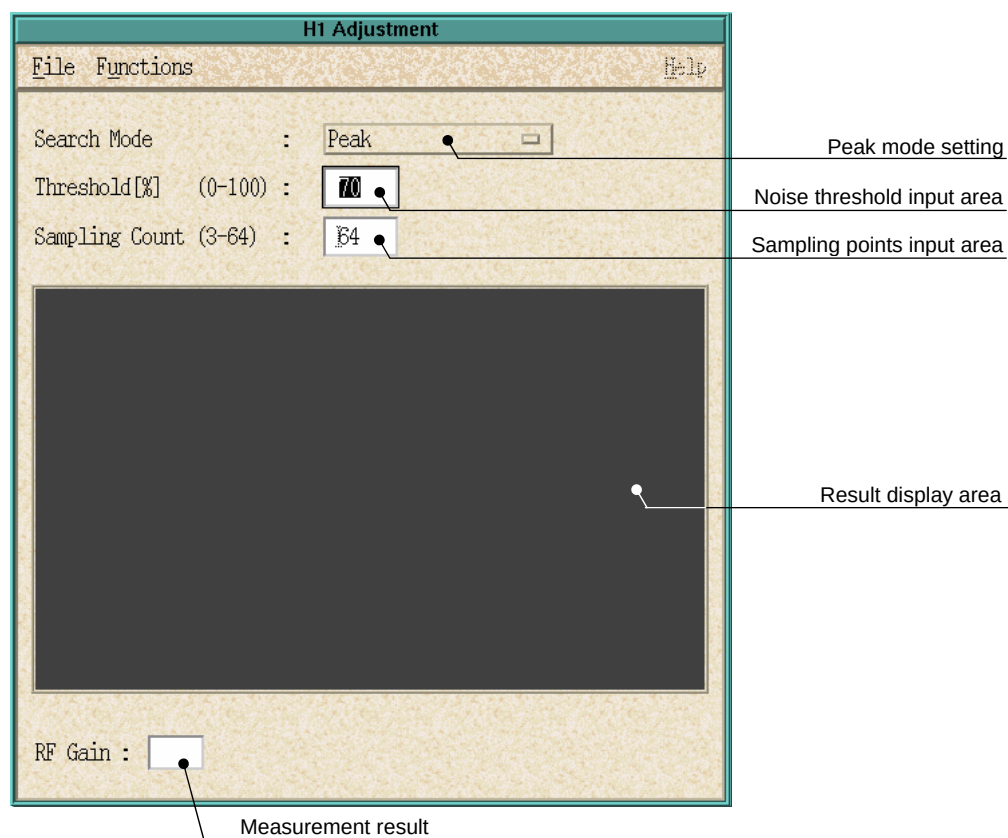


Fig. 6.29-1 H1 Adjustment Display Screen

6.29.4 Operation procedure

Follow the next steps to execute the process.

6.29.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following procedures are available to change parameters.

6.29.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area.

Using the keyboard, input figures or character string, then either press **<Enter>** key on the keyboard or remove the focus from the input area to change the parameter.

6.29.4.1.2 Parameter change by option menu

The displayed items are the set parameters. Click the option menu and click the mouse cursor on the parameter to be changed on the opened menu. The parameter is changed.

Table 6.29-1 Parameter Setting Range

Parameter Name	Setting Range
Search mode	Peak/Center of Gravity/Approximation
Noise threshold	0 ~ 100
Sampling points	3 ~ 64

6.29.4.2 Measurement start

Click **[Start]** button on the scan card to start the measurement.

6.29.4.3 Measurement stop

Click **[Stop]** button on the scan card. The measurement is stopped.

6.29.4.4 Result display

The H1 adjustment results are displayed in graph.

Sampling points are the ones designated in the measurement points input area, and they are connected with a straight line on graph. The resultant line is displayed along the resultant values. After the measurements are completed, H1 value is displayed in the result display area.

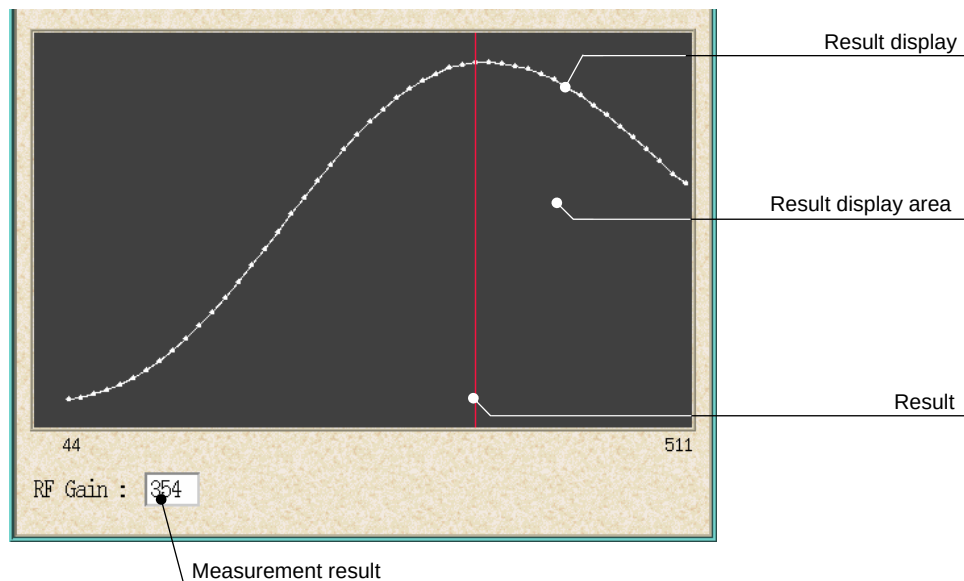


Fig. 6.29-2 Example of Result Display

6.29.4.5 Saving result

Select [Save] from "File Menu" of the Menu Bar. The results are saved into the system table (H/D and Memory). This button is not selectable when the measurement result is not to be displayed.

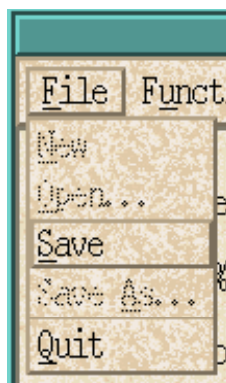


Fig. 6.29-3 Save Button

6.29.4.6 Clearing result display area

Select [Clear] button from "Functions Menu" of the Menu Bar. The data in the result display area is cleared. At this time, because the data for the resultant values are also cleared, the dialog shown in Fig. 6.29-5 is popped up. If loss of the data is allowed, click [OK] button, while it is not allowed click [Cancel] and save the data.

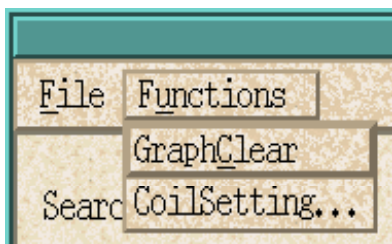


Fig. 6.29-4 Clear Button

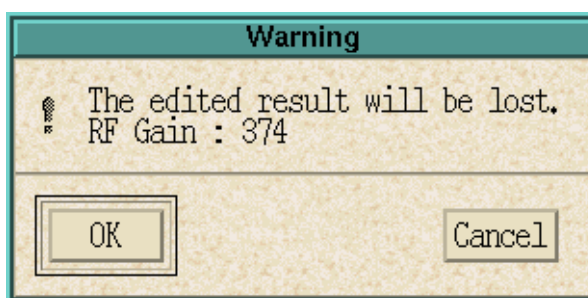


Fig. 6.29-5 Cancel Confirmation Dialog

6.29.4.7 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.29.4.7.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.29.4.7.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

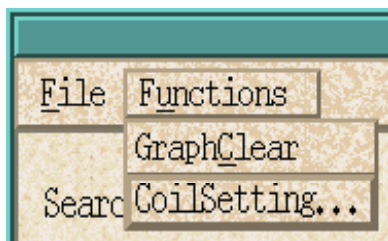


Fig. 6.29-6 Coil Setting Window Display Button

6.29.4.7.3 Display screen

The display screen and function names are shown in Fig. 6.29-7 and Fig. 6.29-8.

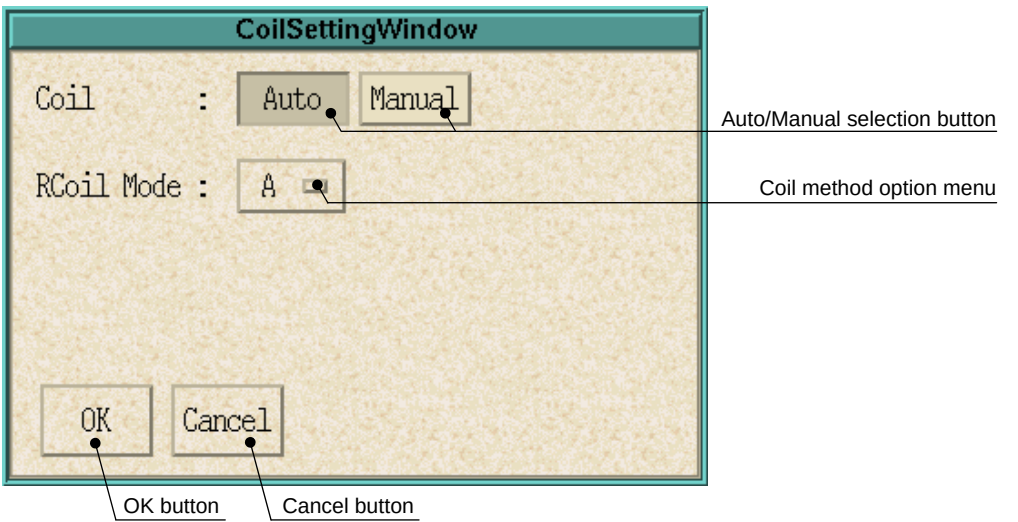


Fig. 6.29-7 Coil Setting Window Display Screen (Auto mode)

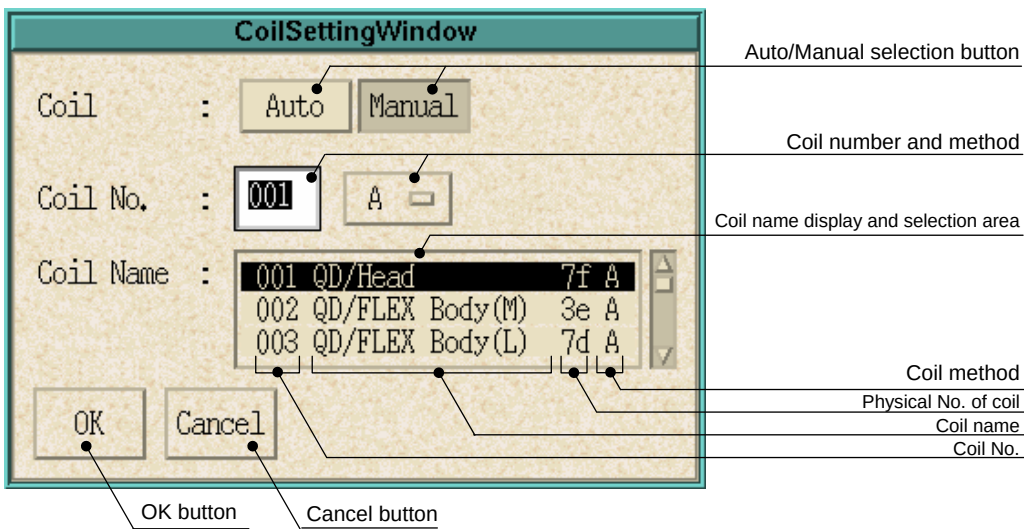


Fig. 6.29-8 Coil Setting Window Display Screen (Manual mode)

6.29.4.7.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the **Manual** screen in case of the **Auto** mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of **Manual** screen is [0], the coil method is [A], and the coil name is not selected.

6.29.4.7.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in **Manual** mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.29.4.7.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area. The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.29.4.7.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.29.4.7.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.29.4.7.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.29.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.29-4. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.



Fig. 6.29-9 Quit Button

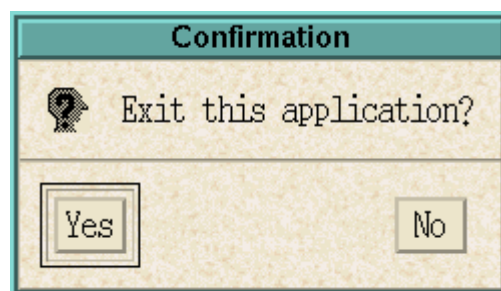


Fig. 6.29-10 Exit Confirmation Dialog

In case the frequency value has been changed as a result of measurement, the dialog shown in Fig. 6.29-11 is popped up. Clicking [Yes] button save the result into the system table (H/D and Memory) and closes the application. To close the application without saving the result into the system table, click [No] button. Clicking [Cancel] button returns to the previous screen without exiting from the application.

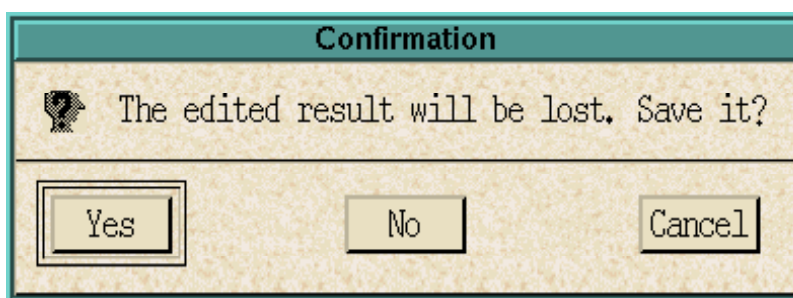


Fig. 6.29-11 Save Confirmation Dialog

6.29.6 Error specifications

Error specifications are shown in Table 6.29-2.

Table 6.29-2 Error Specifications

Message	Status	Solution
Couldn't draw result line.	The resultant line could not be drawn.	The H1 value is deviated. With the changed value, retry the measurement.
Gravity Error.	Error in measurement. (for Search mode: Gravity method)	Change the value and retry the measurement.
MSE Error.	Error in measurement. (for Search mode: Method of Least Squares)	Change the value and retry the measurement.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Can't write the SYSTEM_TABLE.	Writing the system table has failed.	Check the system status and retry the same operation procedure.
Event Invalid.	Request is not accepted.	Check the system status and retry the same operation procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the registered coils list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the registered coils list.
No coil is connected to the system.	The coil is not connected yet to the system.	Check the coil connection conditions.
Error occurred in the patient table.	Communication with the patient table failed.	Check the patient table condition.

6.29.7 Other cautions

Because this application is operated on the scan card, the buttons on the card are used to start and stop a scan.

Accordingly normal scan cannot be performed while this application is running, therefore close this application to start a normal scan.

Do not change the card while this application is running, otherwise the connection between this application and scan card may be broken. Before changing the card, be sure to close this application.

Should the card be changed without closing this application, immediately close this application. In this case, the data can be saved into the system table.

6.30 Manual Scan

6.30.1 General

This is a tool to make a scan and reconstruction with the set conditions and to simplify the check operation of the hardware. Setting center frequency, RF gain value, receiver gain value and tuning value, take a scan. Even if T2 amplitude correction filter is set, it does not function.

The default parameter values shown after start-up are the ones set for the previous scan.

6.30.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and display "Others Group" (refer to Paragraph 1.3.5). Then, select M-Scan from "Maint. Menu" on the scan parameter setting area.

By the way, to prohibit the action of pre-scan, make sure that the following flags have been set at off status.

- P0/P1 flag (refer to item 6.15.)
- Slice and frequency axes phase correction mode (refer to item 6.23.)
- Phase axis Phase correction mode (refer to item 6.23.)

6.30.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.30-1.

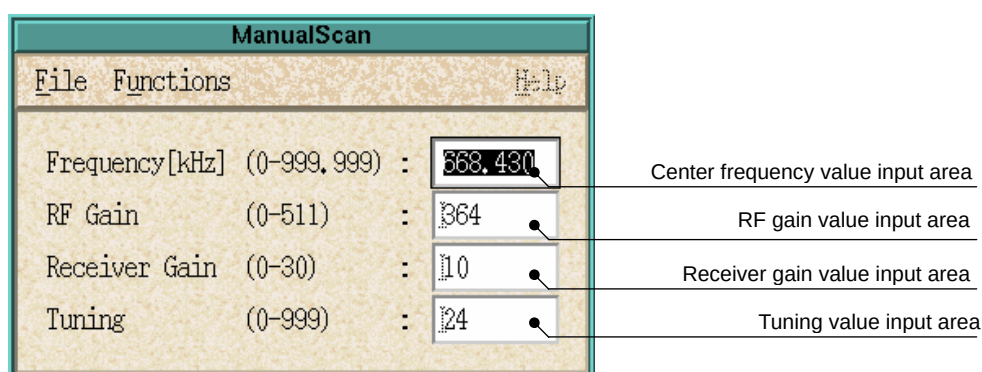


Fig. 6.30-1 Manual Scan Display Screen

6.30.4 Operation procedure

Follow the next steps to execute the process.

6.30.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following procedures are available to change parameters.

6.30.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. Using the keyboard, input figures or character string, then either press <Enter> key on the keyboard or remove the focus from the input area to change the parameter.

Table 6.30-1 Parameter Setting Range

Parameter Name	Setting Range	Remarks
Center frequency	0.000 ~ 999.999	
RF gain	0 ~ 511	
Receiver gain	0 ~ 30	Only even numbers
Tuning	0 ~ 999	

6.30.4.2 Measurement start

Click [Start] button on the scan card to start the measurement.

6.30.4.3 Measurement stop

Click [Stop] button on the scan card. The measurement is stopped.

6.30.4.4 Recovery of parameter

The changed parameter can be recovered by selecting [Reset] of "Functions Menu" of the Menu Bar to the one after start-up.

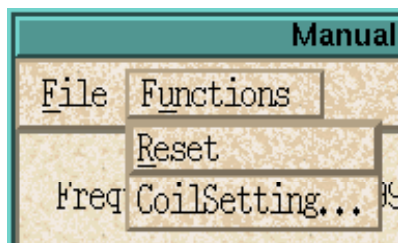


Fig. 6.30-2 Reset Button

6.30.4.5 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.30.4.5.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.30.4.5.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

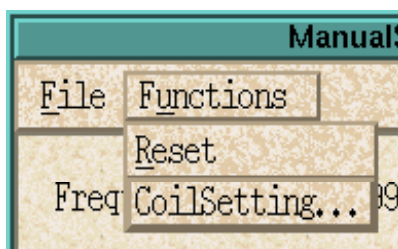


Fig. 6.30-3 Coil Setting Window Display Button

6.30.4.5.3 Display screen

The display screen and function names are shown in Fig. 6.30-4 and Fig. 6.30-5.

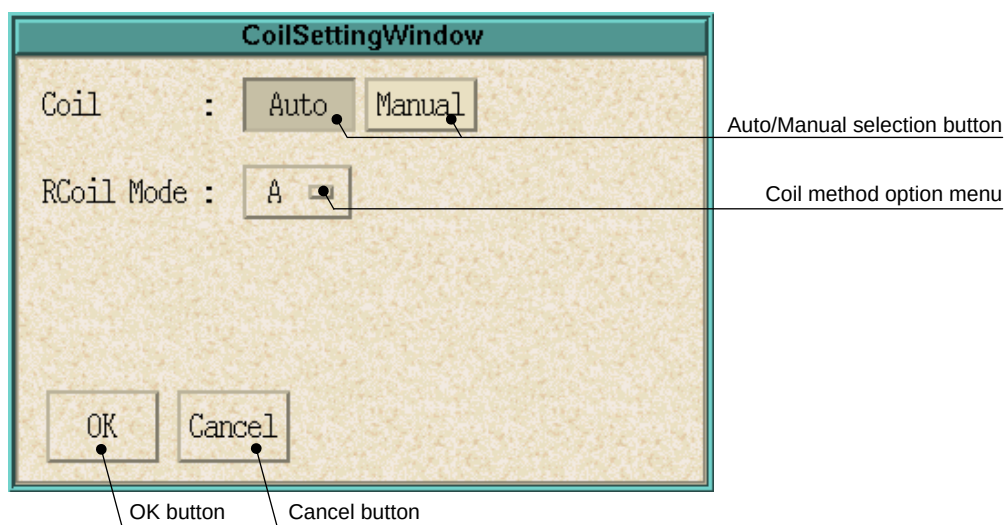


Fig. 6.30-4 Coil Setting Window Display Screen (Auto mode)

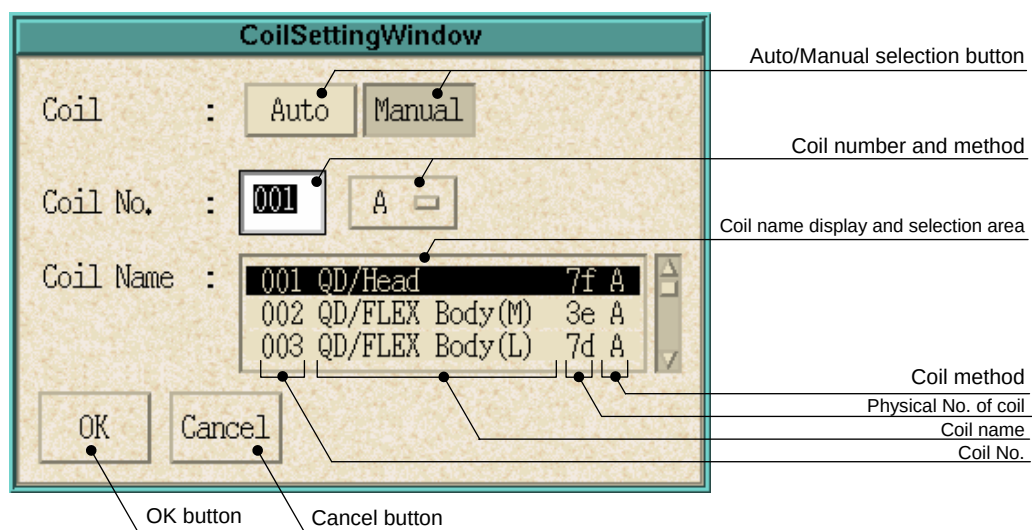


Fig. 6.30-5 Coil Setting Window Display Screen (Manual mode)

6.30.4.5.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.30.4.5.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.30.4.5.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area. The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in Manual mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.30.4.5.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in Manual mode only.

6.30.4.5.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.30.4.5.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.30.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.30-7. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.

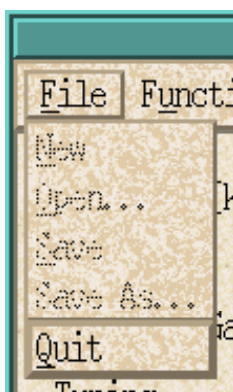


Fig. 6.30-6 Quit Button

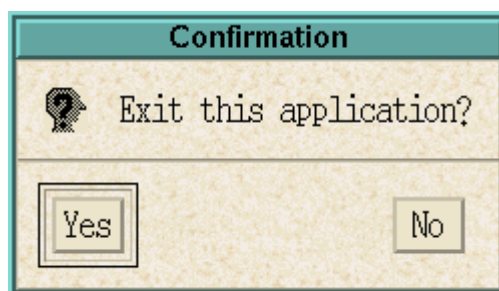


Fig. 6.30-7 Exit Confirmation Dialog

6.30.6 Error specifications

Error specifications are shown in Table 6.30-2.

Table 6.30-2 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system status and retry the same operation procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the registered coils list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the registered coils list.
No coil is connected to the system.	The coil is not connected yet to the system.	Check the coil connection conditions.
Error occurred in the patient table.	Communication with the patient table failed.	Check the patient table condition.

6.30.7 Other cautions

Because this application is operated on the scan card, the buttons on the card are used to start and stop a scan.

Accordingly normal scan cannot be performed while this application is running, therefore close this application to start a normal scan.

Do not change the card while this application is running, otherwise the connection between this application and scan card may be broken. Before changing the card, be sure to close this application.

Should the card be changed without closing this application, immediately close this application.

6.31 Signal Generation Check

6.31.1 General

This is a tool to generate the measurement signal for adjustment of the hardware. This measurement signal can be used even when a hard error occurs. Generation of the measurement signal continues until the Stop button is pressed. Setting the center frequency, RF gain value, receiver gain value and tuning value, a scan is made.

The default parameter values shown after start-up are the ones set for the previous scan.

6.31.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and display "Others Group" (refer to Paragraph 1.3.5). Then, select Sgnl.Gen. from "Maint. Menu" on the scan parameter setting area.

By the way, to prohibit the action of pre-scan, make sure that the following flags have been set at off status.

- P0/P1 flag (refer to item 6.15.)
- Slice and frequency axes phase correction mode (refer to item 6.23.)
- Phase axis Phase correction mode (refer to item 6.23.)

6.31.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.31-1.

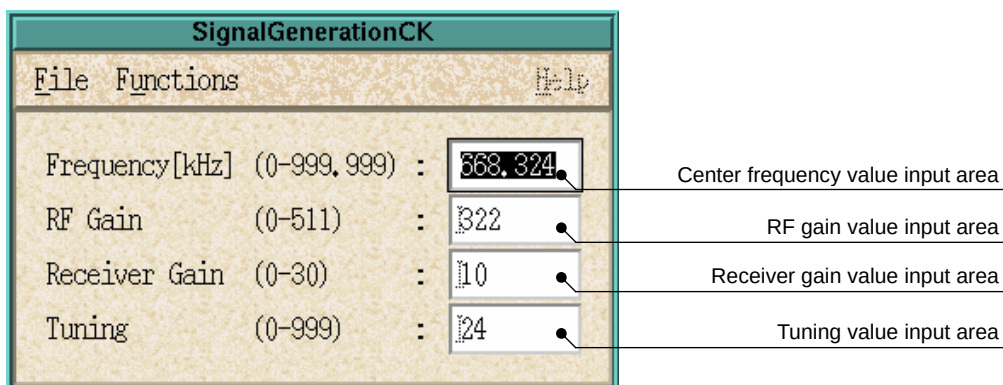


Fig. 6.31-1 Signal Generation Check Display Screen

6.31.4 Operation procedure

Follow the next steps to execute the process.

6.31.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following procedures are available to change parameters.

6.31.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. Using the keyboard, input figures or character string, then either press <Enter> key on the keyboard or remove the focus from the input area to change the parameter.

Table 6.31-1 Parameter Setting Range

Parameter Name	Setting Range	Remarks
Center frequency	0.000 ~ 999.999	
RF gain	0 ~ 511	
Receiver gain	0 ~ 30	Only even numbers
Tuning	0 ~ 999	

6.31.4.2 Measurement start

Click [Start] button on the scan card to start generation of the signal.

6.31.4.3 Measurement stop

Click [Stop] button on the scan card. Generation of the signal is stopped.

6.31.4.4 Recovery of parameter

The changed parameter can be recovered by selecting [Reset] of "Functions Menu" of the Menu Bar to the one after start-up.

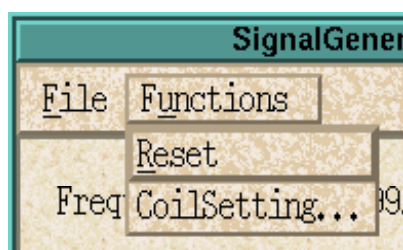


Fig. 6.31-2 Reset Button

6.31.4.5 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.31.4.5.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.31.4.5.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

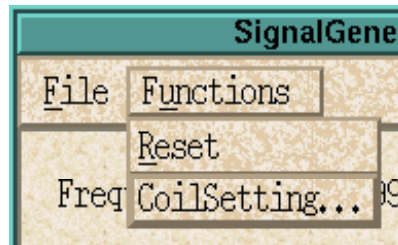


Fig. 6.31-3 Coil Setting Window Display Button

6.31.4.5.3 Display screen

The display screen and function names are shown in Fig. 6.31-4 and Fig. 6.31-5.

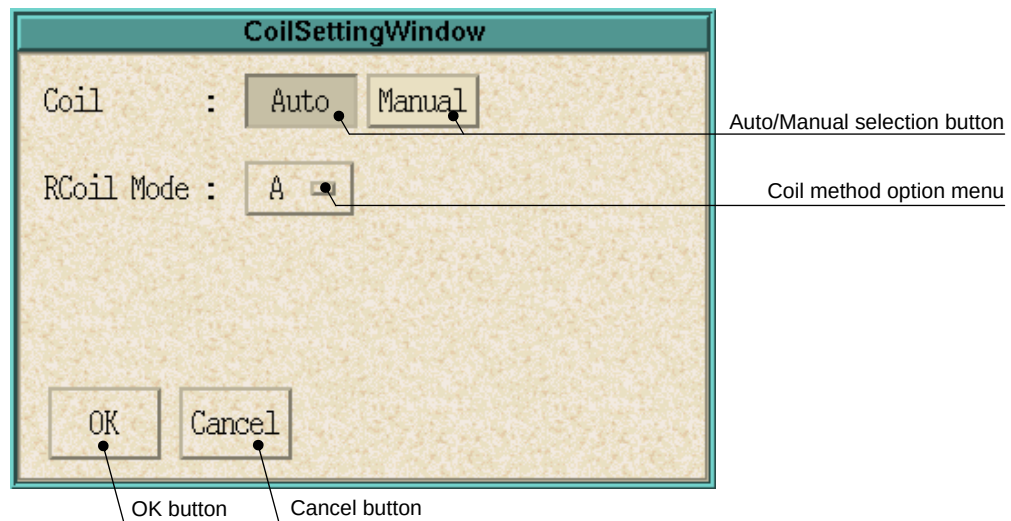


Fig. 6.31-4 Coil Setting Window Display Screen (Auto mode)

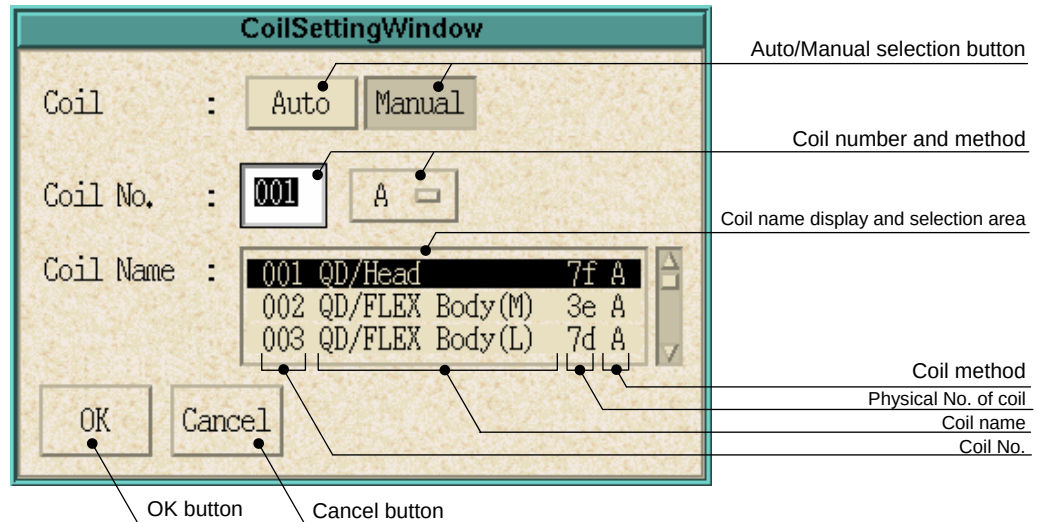


Fig. 6.31-5 Coil Setting Window Display Screen (Manual mode)

6.31.4.5.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.31.4.5.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.31.4.5.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area

The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.31.4.5.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.31.4.5.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.31.4.5.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.31.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.31-7. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.

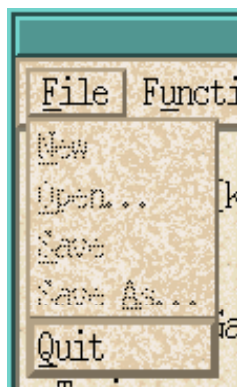


Fig. 6.31-6 Quit Button

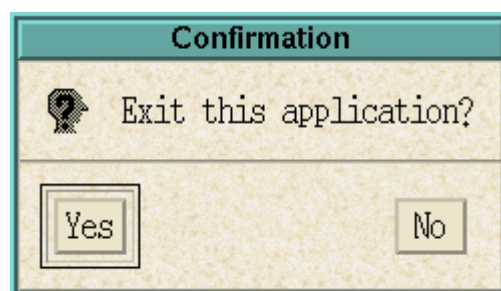


Fig. 6.31-7 Exit Confirmation Dialog

6.31.6 Error specifications

Error specifications are shown in Table 6.31-2.

Table 6.31-2 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system status and retry the same operation procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the registered coils list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the registered coils list.
No coil is connected to the system.	The coil is not connected yet to the system.	Check the coil connection conditions.
Error occurred in the patient table.	Communication with the patient table failed.	Check the patient table condition.

6.31.7 Other cautions

Because this application is operated on the scan card, the buttons on the card are used to start and stop a scan. Accordingly normal scan cannot be performed while this application is running, therefore close this application to start a normal scan.

Do not change the card while this application is running, otherwise the connection between this application and scan card may be broken. Before changing the card, be sure to close this application.

Should the card be changed without closing this application, immediately close this application.

6.32 Coil Adjustment

6.32.1 General

This is a tool to set the decoupling current for transmission coil tuning and also to set the tuning voltage necessary for the tuning of receiving coil.

6.32.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click Coil Adjustment in the displayed items and click [OK] or double-click the selecting item.

6.32.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.32-1.

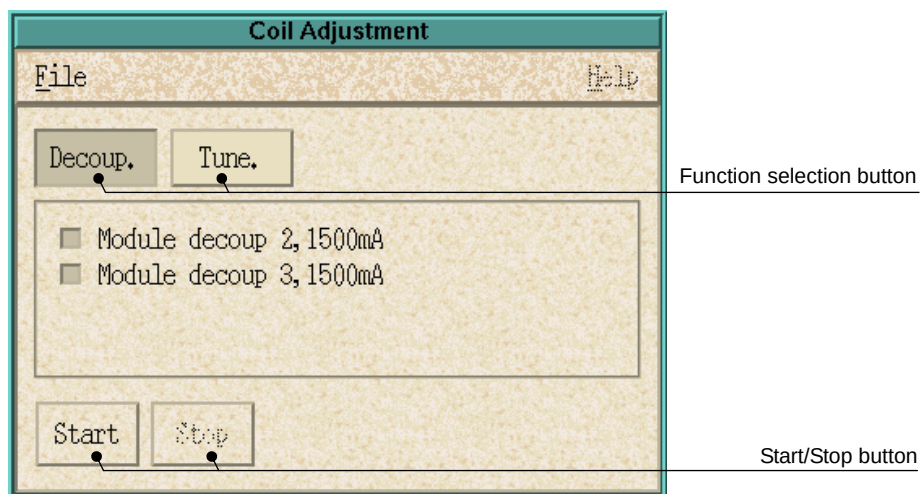


Fig. 6.32-1 Coil Adjustment Tool Display Screen
(When setting decoupling current)

6.32.4 Operation procedure

Follow the next steps to execute the process.

6.32.4.1 Parameter change

Using the keyboard or mouse cursor, change parameters. The following methods are available for changing parameters.

6.32.4.1.1 Selection of item by push button

A group of push buttons arranged on a lateral line are buttons for each single selection, and parameters can be changed by clicking with the mouse cursor.

6.32.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. Using the keyboard, input figures or character string, then either press <Enter> key on the keyboard or remove the focus from the input area to change the parameter.

Table 6.32-1 Parameter Setting Range

Parameter Name	Setting Range	Remarks
Tuning voltage value	00 ~ 7F	Tuning voltage Setting

6.32.4.1.3 Selection of item by toggle button

Item can be selected by clicking the button part (□, ■) with the mouse cursor.

6.32.4.2 Selection of function

This tool includes two functions shown in Table below. Each function is described in the following.

Table 6.32-2 List of Functions

Item No.	Function Name	Content
6.32.4.3	Decoupling current setting	Sets decoupling current for transmission coil tuning.
6.32.4.4	Tuning voltage setting	Sets tuning voltage necessary for receiving coil tuning.

6.32.4.3 Setting decoupling current

Follow the next steps to set decoupling current.

- (1) Select [Decup.] button with function selection buttons to display decoupling current setting screen. Decoupling current setting screen is displayed when the function is started.
- (2) According to the requirement, set On/Off of the decoupling current.
- (3) Click [Start] button. Decoupling current value (fixed value) shown in the screen is set.
- (4) To release the setting, click [Stop] button.

By the way, when [Stop] button is not clicked even elapsing a designated time (fixed at 1200 seconds), the setting is automatically released.

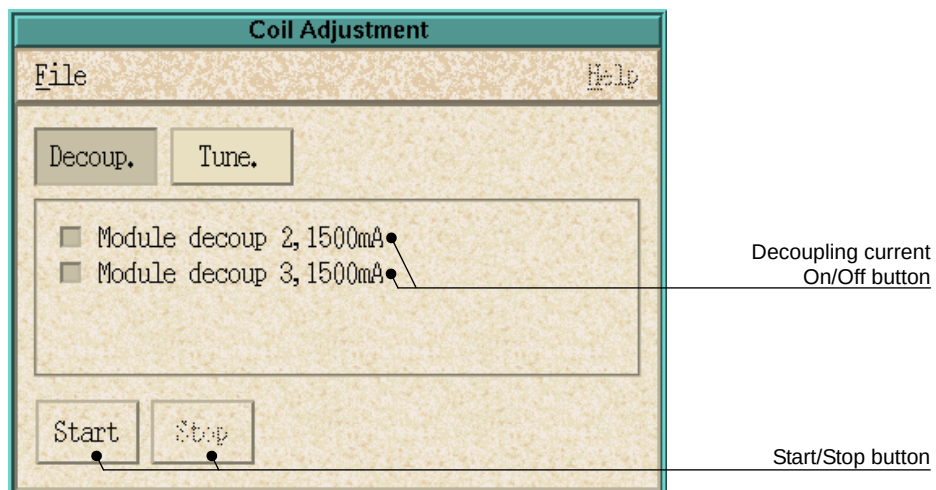


Fig. 6.32-2 Decoupling current Setting Screen

6.32.4.4 Setting tuning voltage

Follow the next steps to set tuning voltage.

- (1) Select [Tune.] button with function selection buttons to display tuning voltage setting screen.
- (2) Input voltage value (hexadecimal) into the entry area. The default is the tuning value of the previous measurement.
- (3) Click [Start] button to set tuning voltage.
- (4) Click [Stop] button to released the setting.

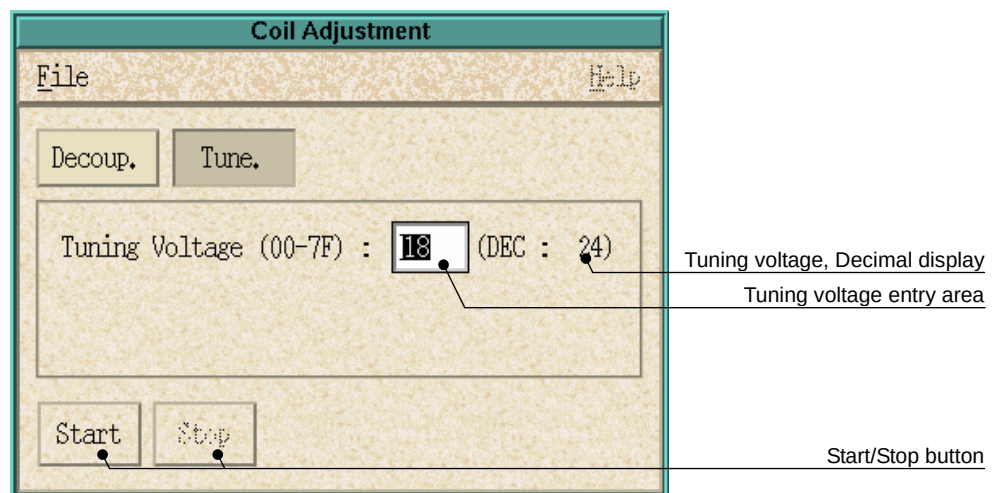


Fig. 6.32-3 Tuning voltage Setting Screen

6.32.5 End

Selecting [Quit] from "File Menu" in the menu bar pops up the dialog shown in Fig. 6.32-5. Clicking [Yes] button on this dialog ends the process, while clicking [No] button returns to the previous screen without ending the process.

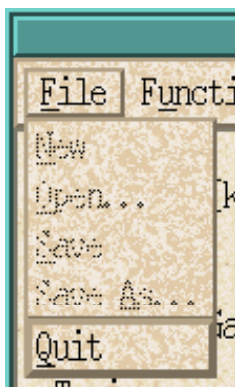


Fig. 6.32-4 Quit Button

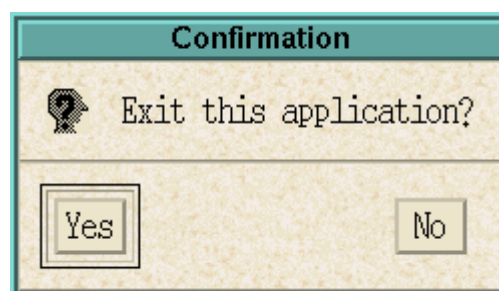


Fig. 6.32-5 Exit Confirmation Dialog

6.32.6 Error specification

Error specifications are shown in Table 6.32-3.

Table 6.32-3 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Checking the system status, retry the same operation procedure.

6.33 UPS Setup

6.33.1 General

This is a tool to set the type of UPS being connected.

6.33.2 Start-up

Register the T/M license with the key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click **UPS Setup** in the displayed items and click [OK] or double-click the selecting item.

6.33.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.33-1.

When starting up, if the kind of UPS has been already set, the UPS type setting has been selected state.

(With the default, Power RiteProII is selected.)

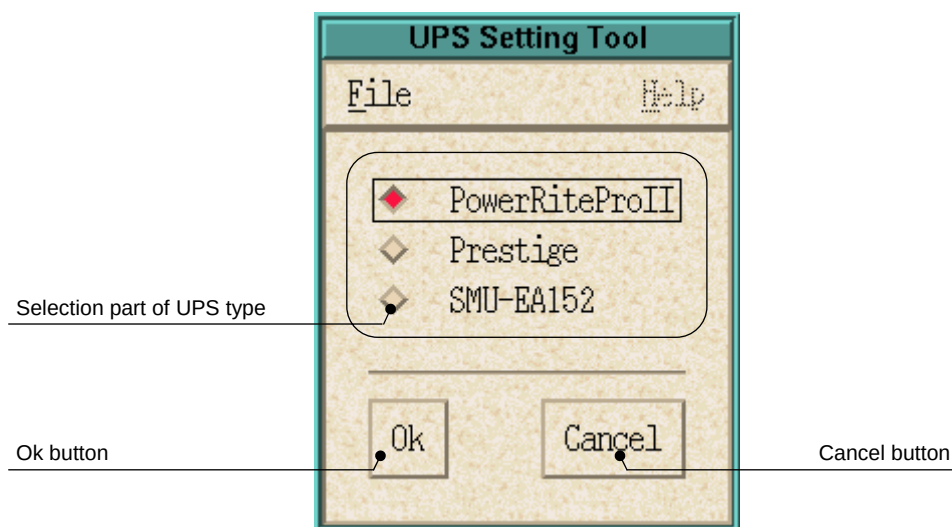


Fig. 6.33-1 UPS Setting Tool Display Screen

6.33.4 Setting type of UPS

Select UPS type to be set with Selection part of UPS type. For the type of UPS, refer to Table 6.33-1.

After selecting the type, click [OK] button. The UPS type setting is executed and the UPS type setting tool ends.

NOTE: The UPS type setting executed in this procedure becomes effective from the restart of the system. After starting the system again, check the function in accordance with Item 6.34.

Table 6.33-1 Type of UPS

System type	Type of UPS
MR-DP-7A, MR-DP-7B	Prestige
MR-RGP-1, MR-IP-1	PowerRitePro II
MR-IP-1,MR-OP-3	SMU-EA152

6.33.5 Cancel of UPS Setting Tool

For terminating UPS Setting Tool process without executing UPS type setting, click [Cancel] button to pop up the dialog shown in Fig. 6.33-2. Clicking [Yes] button terminates UPS Setting Tool process without setting UPS type, while clicking [No] button returns to the previous screen without exiting from the process.

[Quit] button on "File Menu" is the same function as [Cancel] button.



Fig. 6.33-2 Exit Confirming Dialog



Fig. 6.33-3 Quit Menu

6.34 UPS Check

6.34.1 General

This is a tool to check the function of UPS connected.

6.34.2 Start-up

Register T/M license with the key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click UPS View in the displayed items and click [OK] or double-click the selected item.

6.34.3 Display screen

After starting-up, the dialog shown in Fig. 6.34-1 is popped up. Click [OK] with inputting nothing. The screen shown in Fig. 6.34-2 is displayed.

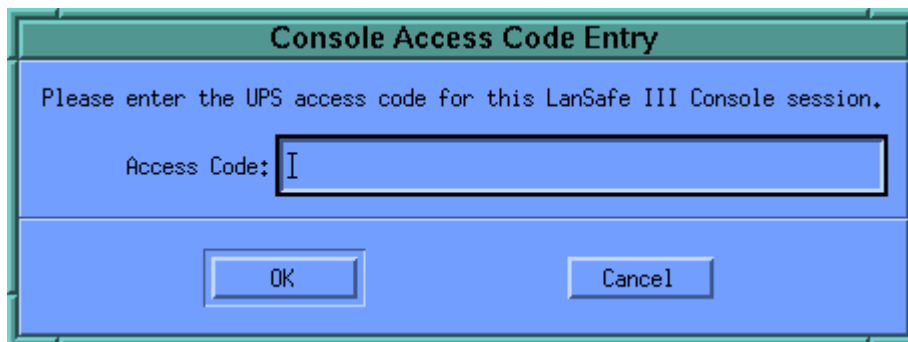
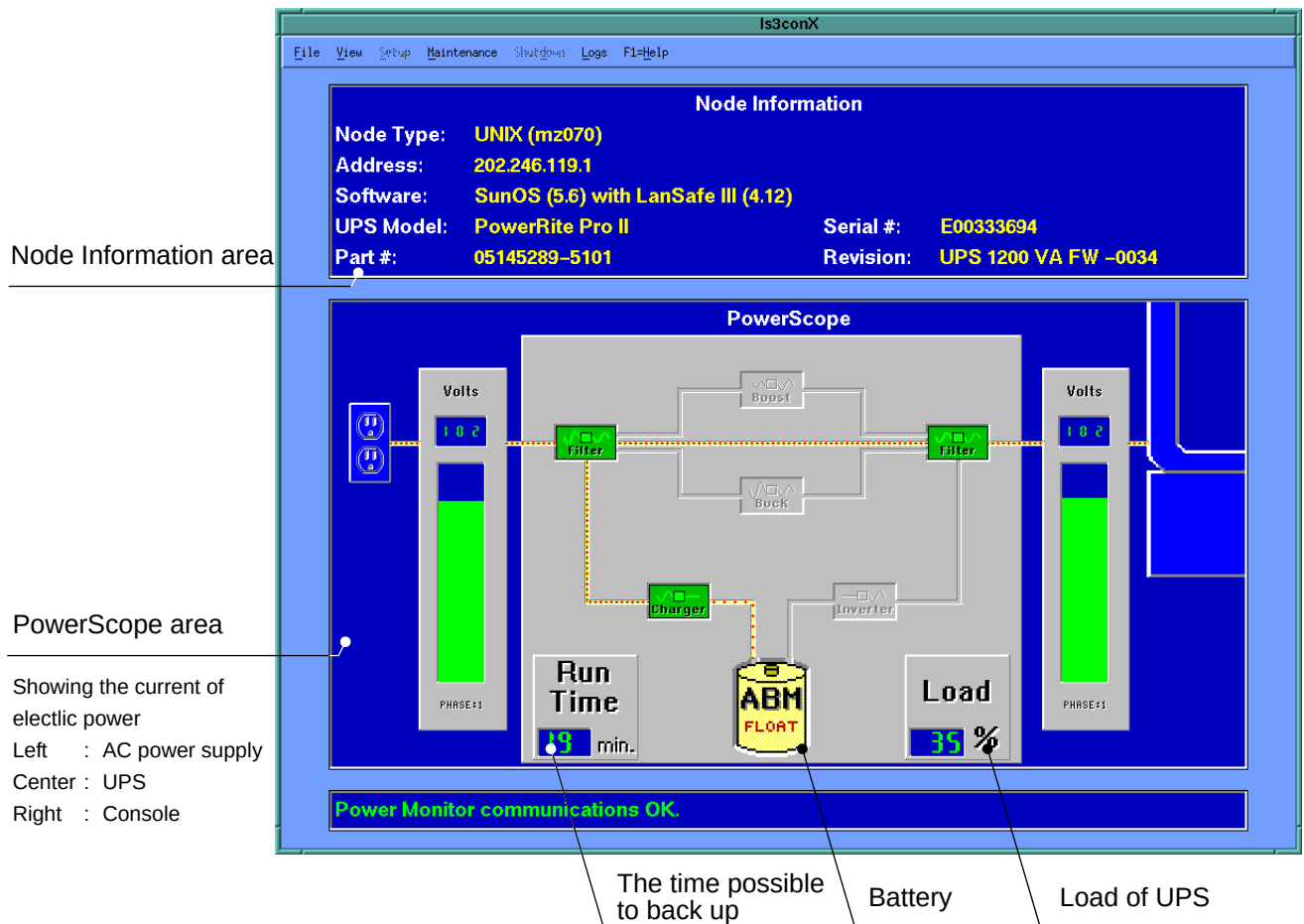


Fig. 6.34-1 Access Code Input Screen



NOTE: "The time possible to back up" shown in the screen is only the approximate value during which UPS can back up to supply the power. The actual back up time may vary in accordance with the load applied to UPS and there may be such case that the indicated value is not same as that of actual time possible to back up.

Fig. 6.34-2 UPS View Screen

6.34.4 Function check

Check the following functions check items with Fig. 6.34-2 View Screen. If all items mentioned below are confirmed, UPS functions correctly.

[Function Check Items]

- 1) Node Type, Address and Software are displayed in the Node Information Area.
- 2) The name of UPS type set in the procedure of UPS type setting is displayed in UPS Model in the Node Information Area.
- 3) Serial #, Part # and Revision of UPS are displayed in the Node Information Area.
- 4) Electric power flows in direction of (AC Power supply) → (UPS) → (Console) in PowerScope Area. However, while the UPS is under charging, electric power flows also in direction of (AC Power supply) → (UPS Battery).

6.34.5 End

Select [Exit] button from "File Menu" in the Menu Bar to end UPS View process.

6.34.6 Error Specification

When an abnormality in function check procedure occurs, check the following items.

- Did you take system re-start action after setting the type of UPS with UPS Setting Tool?
(If the system re-start action is not taken, the setting of UPS type is ineffective.)
- Was the type of UPS correctly set with UPS Setting Tool?
If incorrect type was set, reset the correct type.
- Check if UPS and WS board are correctly connected with a serial cable.

6.35 ECC Tool

6.35.1 General

This is a tool to adjust GC/B0 correction value by operating the GC controller (DGBC) and outputting GC waveform.

6.35.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click ECC Tool in the displayed items and click [OK] or double-click the selected item.

6.35.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.35-1.

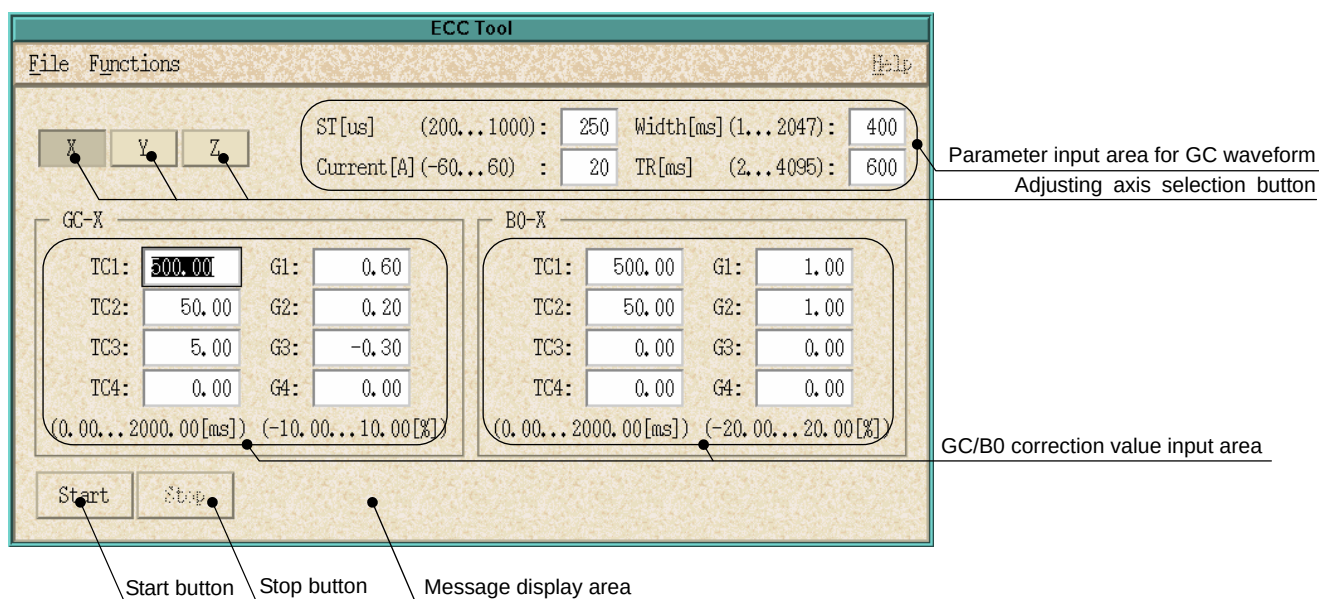


Fig. 6.35-1 ECC Tool Display Screen

6.35.4 Operation procedure

Follow the next steps to execute the process.

6.35.4.1 Updating parameters

Using the keyboard and mouse cursor, change parameters. The following methods are available to change parameters.

6.35.4.1.1 Selection of parameter by pushbutton

The push-buttons arranged in array are of single-selection type, and by clicking one of them by the mouse cursor, the corresponding parameter is changed.

6.35.4.1.2 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard or remove the focus from the entry area to change the parameter.

NOTE: It takes some time until the window will be operable after pressing <Enter>. This is because the input parameter is set in the GC controller (DGBC) from time to time.

Table 6.35-1 Parameter Setting Range

Parameter Name			Available Range for setting	Remarks
Adjustment axis			X, Y, Z	
GC Pulse	ST[μs]	Slew Time	200 ~ 1000	
	Current[A]	Current Value	-60 ~ 60	
	Width	Pulse Width	1 ~ 2047※	
	TR	Repetition Time	2 ~ 4095※	
Correction value	GC	TC[ms]	0.00 ~ 2000.00	
		G	-10.00 ~ 10.00	
	B0	TC[ms]	0.00 ~ 2000.00	
		G	-20.00 ~ 20.00	× 0.03[%]

※ Note that because Duty ratio of 80% is set as the limit for input the setting is no possible unless Width (pulse width) and TR (repetition time) satisfy the following relation. If any erroneous input is made, the input area of Width and TR turns to yellow, and under this condition GC waveform cannot be output.

$$(\text{Duty ratio } 80\%) \geq (\text{Width} \div \text{TR}) \times 100$$

6.35.4.2 Output of GC waveform

Click on [Start] button. The GC waveform of the currently displayed axis is output and the flashing axis name of the GC being output is displayed in the message display area.

The GC waveform parameter is the value common to all the axes. For input of GC waveform parameter, refer to Paragraph 6.35.4.1.

While GC waveform being output, any parameter for GC waveform and GC/B0 correction value other than GC waveform output axis cannot be changed. To change any of these parameters, stop output of GC waveform.

6.35.4.3 Stopping GC waveform output

Click on [Stop] button. Output of GC waveform is stopped.

6.35.4.4 Reset

Selecting [Reset] from “Functions Menu” displays the following dialog. Clicking on [Yes] returns to the default value, while clicking on [NO] returns to the previous screen.

Depending on amount of parameters to be set, time to perform this process varies.

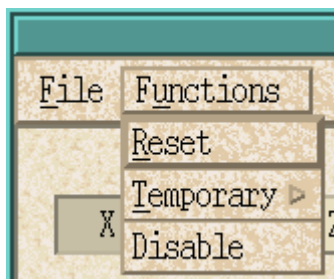


Fig. 6.35-2 Reset Button

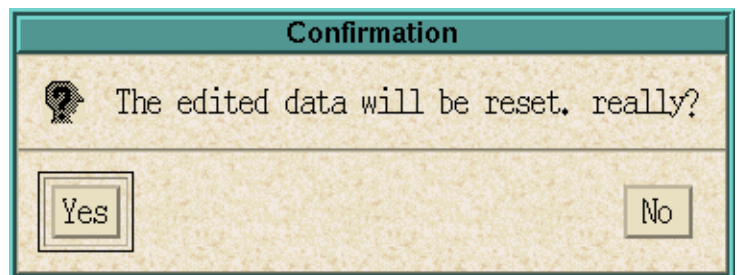


Fig. 6.35-3 Message Dialog for Reset

6.35.4.5 Temporary save of parameter

To save parameter temporarily, select [Save...] from [Temporary] on “Functions Menu”. On the dialog to be displayed as shown in Fig. 6.35-5, click on the area to be saved from the three parameters, input a comment in the comment box and then click on [Save]. The parameter can be temporarily saved. By clicking on [Cancel], temporary save can be canceled. Comment can be input with up to 16 characters and date is automatically input.

If any erroneous input is made for Width (pulse width) and TR (repetition time), temporary save cannot be performed. For details refer to Paragraph 6.35.4.1.

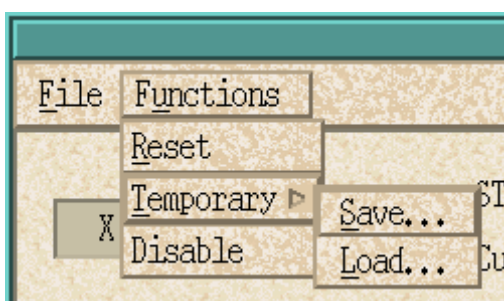


Fig. 6.35-4 Save Button

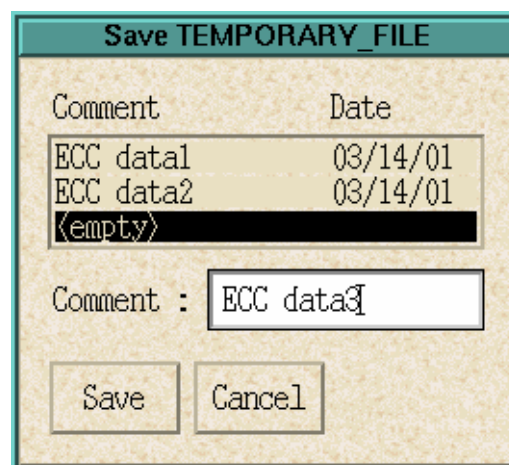


Fig. 6.35-5 Dialog for Temporary Parameter Save

6.35.4.6 Setting temporarily saved parameters

Setting temporarily saved parameters, select [Load...] from [Temporary] on “Functions Menu”. The temporarily saved parameter selection dialog as shown in Fig. 6.35-7 is displayed. Select the data to be set and click on [Load]. The selected data parameter is set.

Depending on amount of parameters to be set, time to perform this process varies.

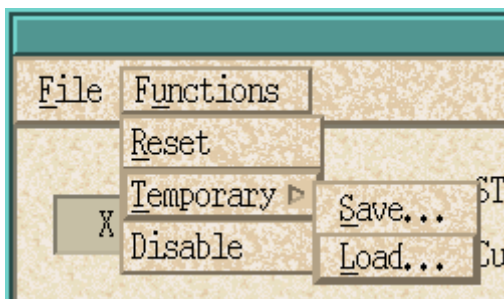


Fig. 6.35-6 Load Button

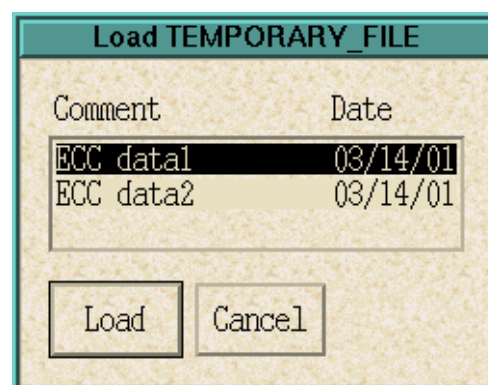


Fig. 6.35-7 Temporarily Saved Data Selection Dialog

6.35.4.7 Setting operation Enable/Disable

To use this function together with Easy Raw Data Analysis (refer to Paragraph 6.24) for adjustment of image quality, operation of this tool must be temporarily stopped.

To load raw data from the memory for Easy Raw Data Analysis, select [Disable] from “Functions Menu” to disable the operation. While the operation being disabled, any operation other than checking the set parameters cannot be operated with this tool.

During this status, the following is displayed in the message display area.

Can't operate this application.

To enable the operation with this tool, select [Enable] from “Functions Menu”. The previous operation-enable status is restored.

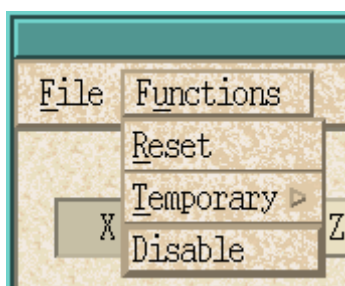


Fig. 6.35-8 Operation-disable Setting Fig. 6.35-9 Operation-enable Setting

6.35.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.35-11 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

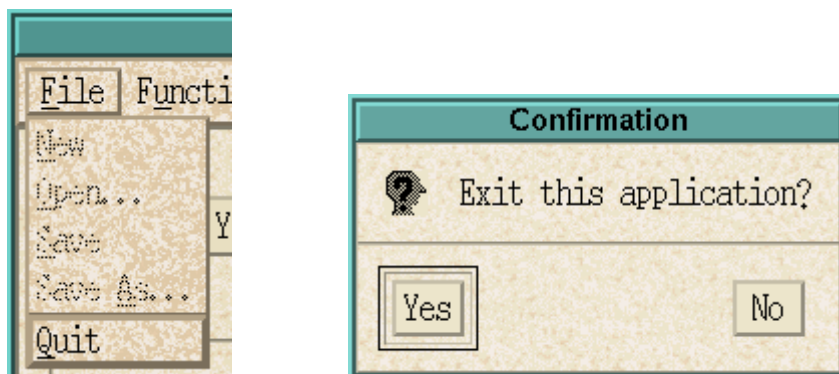


Fig. 6.35-10 Quit Button Fig. 6.35-11 Exit Confirmation Dialog

If any correction is made for GC/B0 correction value, the dialog shown in Fig. 6.35-12 is popped up. Clicking on [Yes] on this dialog saves the corrected GC/B0 correction value in the system file and closes the application. Clicking on [No] closes the application without saving the corrected value in the system file and time to perform this process varies depending on amount of parameters to be set. Clicking on [Cancel] returns to the previous screen without closing the application.

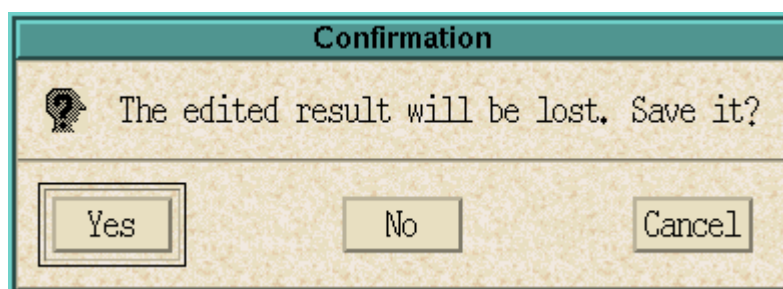


Fig. 6.35-12 System file save and exit confirmation dialog

6.35.6 Error specifications

Error specifications are shown in Table 6.35-2.

Table 6.35-2 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Can't write the SYSTEM_TABLE.	Writing SYSTEM_TABLE has failed.	Check the system condition and retry the operation in the same procedure.
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.

6.35.7 Other cautions

With the TC (time constant) set at 0, if the set value of G (Gain) is any value other than 0, there may be cases where a malfunction occurs on the hardware. Therefore, if TC (time constant) is set at 0, set 0 for G (Gain).

6.36 Language Setup

6.36.1 General

Setting of display languages (English/Japanese) is switched.

As a result of setting change, the message language at a part of dialogues, etc. is switched.

6.36.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click **Language Setup** in the displayed items and click [OK] or double-click the selected item.

6.36.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.36-1. If the language is preset at the start time, the preset language is selected.

(English is selected in default)

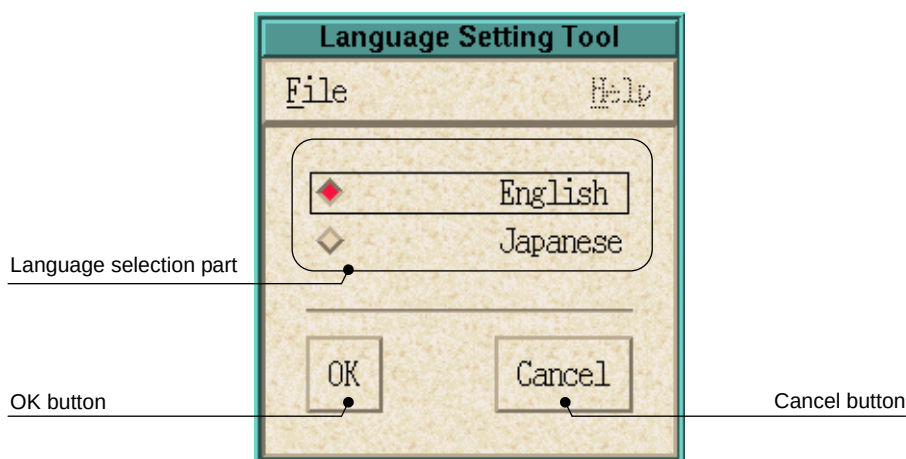


Fig. 6.36-1 Language Setting Tool Display Screen

6.36.4 Language setting

Select the language to be set in the language selection part, and then, click [OK].

Language setting is executed and exiting from the language setting tool is completed.

Note: The language setting executed here becomes effective after the system has been started again.

6.36.5 Cancellation of language setting tool

Clicking [Cancel] exits from the language setting tool without executing the language setting. Fig. 6.36-2 shows the pop up dialog. Clicking [Yes] exits from the language setting tool without executing any language setting. Clicking [No] returns the screen to the last screen.

[Quit] of “File Menu” provides the same function as in [Cancel].

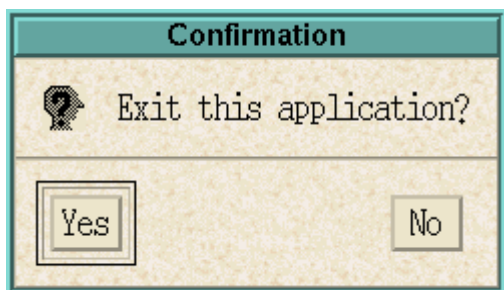


Fig. 6.36-2 Exit Confirmation Dialog



Fig. 6.36-3 Quit Button

6.37 ECC Adjustment

6.37.1 General

This is a tool to measure the magnetic field intensity varying in accordance with the time elapsed caused by eddy current and, by analyzing its result, to adjust GC/B0 correction values.

6.37.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and display "Others Group" (refer to Paragraph 1.3.5). Then, select ECC Adjust from "Maint. Menu" on the scan parameter setting area.

For obtaining the correct adjustment value, starting-up from the "SysScanCard " of Scan Card Library is required.

6.37.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.37-1.

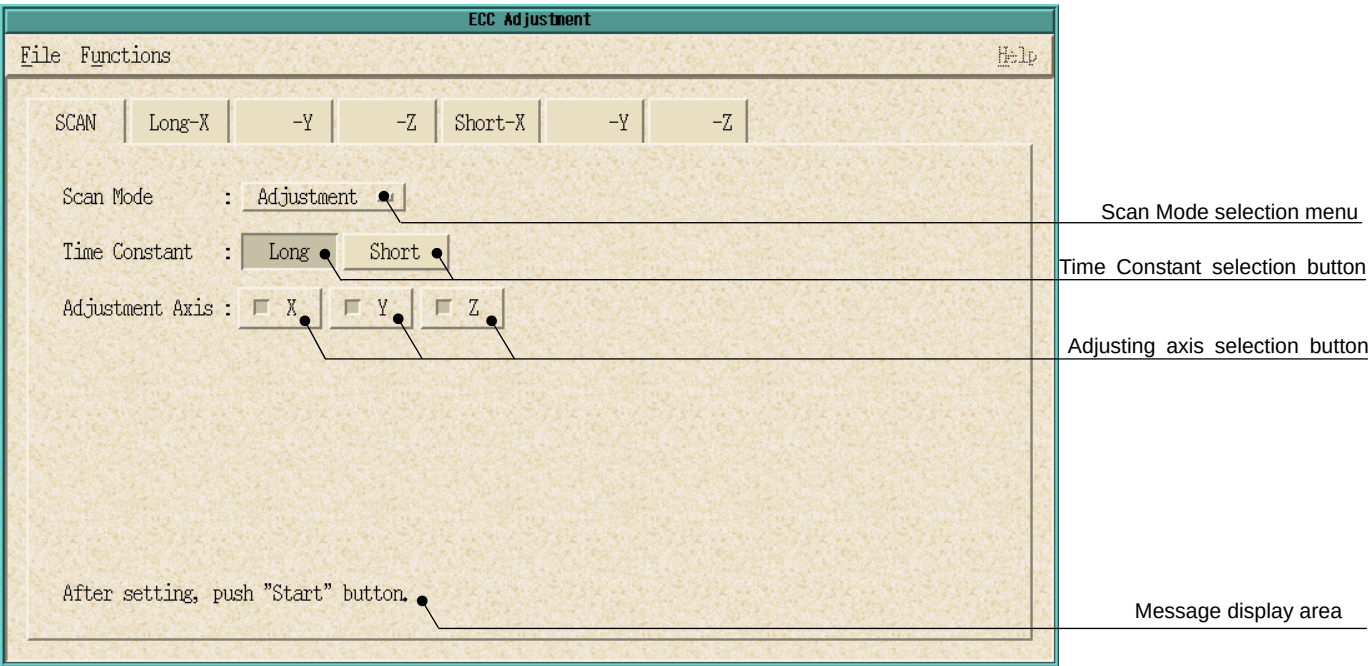


Fig. 6.37-1 ECC Adjustment Display Screen(SCAN)

6.37.4 Operation procedure

Follow the next steps to execute the process.

6.37.4.1 Updating parameters

Using the keyboard and mouse cursor, change parameters. The following methods are available to change parameters.

6.37.4.1.1 Selection of parameter by pushbutton

The push-buttons arranged in array are of single-selection type, and by clicking one of them by the mouse cursor, the corresponding parameter is changed.

6.37.4.1.2 Selection of single selection button

Parameter displayed by a single selection button can be changed by clicking the deselected button.

6.37.4.1.3 Changing the screen by tag

The screen is changed by the mouse cursor clicking on the tag.

6.37.4.1.4 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. From the keyboard, enter figures or character string, and either press <Enter> on the keyboard to change the parameter.

6.37.4.1.5 Parameter change by option menu

Displayed items show the set parameters. Click the option menu, and then, click a parameter to be changed from the menu by the mouse cursor. The parameter is updated.

Table 6.37-1 Parameter Setting Range

Parameter Name			Setting Range	Remarks
SCAN	Scan Mode		Adjustment, Evaluation	
	Time Constant		Long, Short	
	Adjustment Axis		X, Y, Z	Plural selection
Long-X ~ Z	Phase Variation	Point1	ON, OFF	
		Point2	ON, OFF	
	GC Component	TC1, TC2	0.00, 30.00 ~ 2000.00	
		G1, G2	-10.00 ~ 10.00	
		Scan	ON, OFF	
		Fitted	ON, OFF	
		Diff.	ON, OFF	
		Eval.	ON, OFF	
	B0 Component	TC1, TC2	0.00, 30.00 ~ 2000.00	
		G1, G2	-20.00 ~ 20.00	
		Scan	ON, OFF	
		Fitted	ON, OFF	
		Diff.	ON, OFF	
		Eval.	ON, OFF	
Short-X ~ Z	Phase Variation	Point1	ON, OFF	
		Point2	ON, OFF	
	GC Component	TC3, TC4	0.00, 0.40 ~ 30.00	
		G3, G4	-10.00 ~ 10.00	
		Scan	ON, OFF	
		Fitted	ON, OFF	
		Diff.	ON, OFF	
		Eval.	ON, OFF	
	B0 Component	TC3, TC4	0.00, 0.40 ~ 30.00	
		G3, G4	-20.00 ~ 20.00	
		Scan	ON, OFF	
		Fitted	ON, OFF	
		Diff.	ON, OFF	
		Eval.	ON, OFF	

6.37.4.2 Selection of Scan Condition

Designate the time constant range and adjustment axis by the [SCAN] tag.

Because evaluation measurement cannot be made for the combination of time constant and axis for which adjustment has not been made, [Evaluation] button in Scan Mode Selection menu cannot be selected.

Note: It takes some time until the window will be operable after selecting.

This is because the input condition is set in the Scan Card and the GC controller (DGBC) from time to time.

6.37.4.3 Starting measurement

Click [Start] button on the scan card to start the measurement.

6.37.4.4 Stopping measurement

Click [Stop] button on the scan card. The measurement is stopped.

6.37.4.5 Result display

The measurement result and its analyzed result are displayed on the [Long-X~Z] and [Short-X~Z] tags respectively. After the measurement has finished, the analysis starts automatically and the contents on the adjusting axis in the designated time constant range is updated.

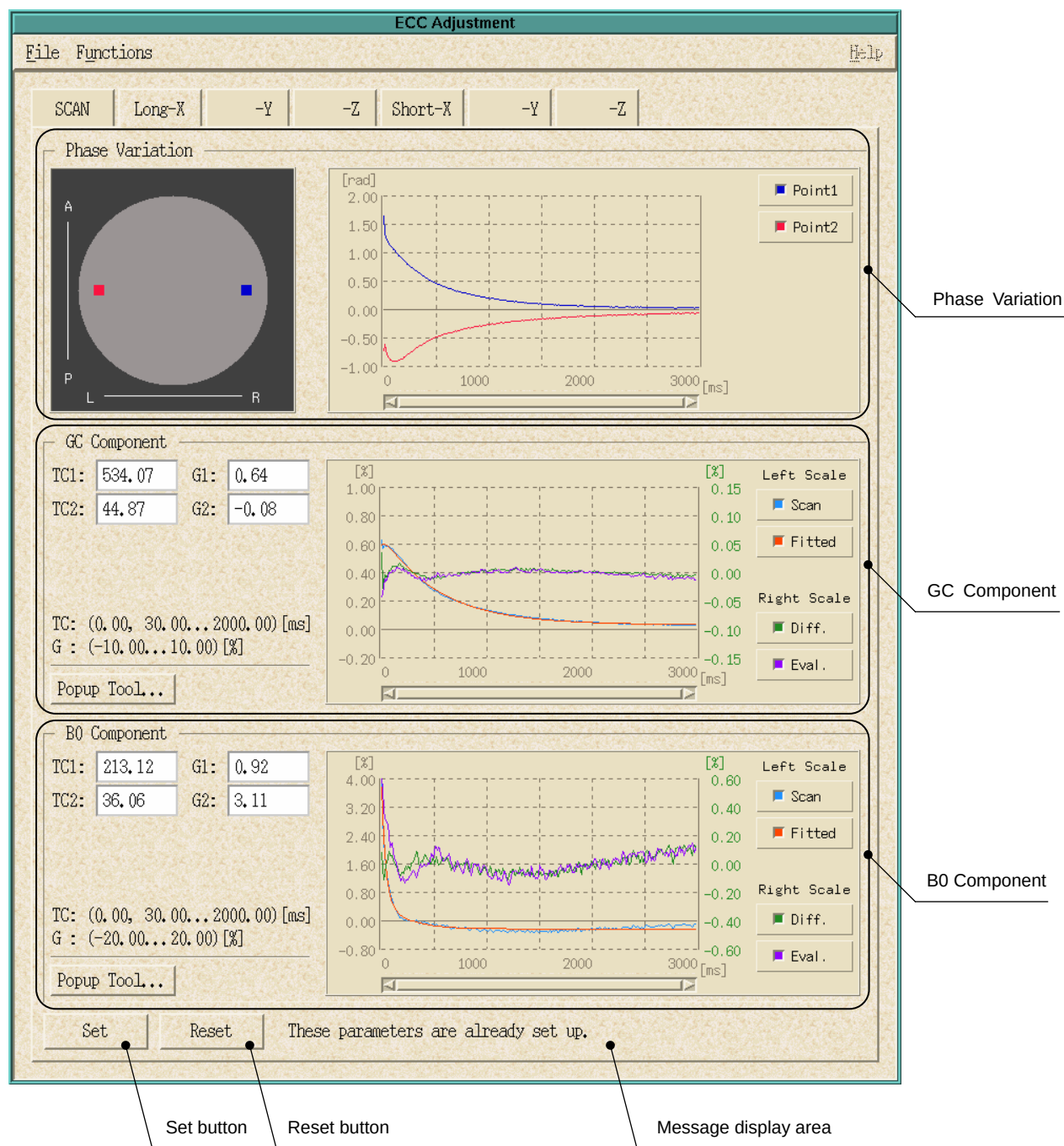


Fig. 6.37-2 ECC Adjustment Display Screen(Analyze)

6.37.4.5.1 Phase Variation

Displays the phase variation at 2 points shown on the simulation image on the graph. The colors of tow points on the simulation image, lines of graph and displayed On/Off toggle switch buttons are all unified.

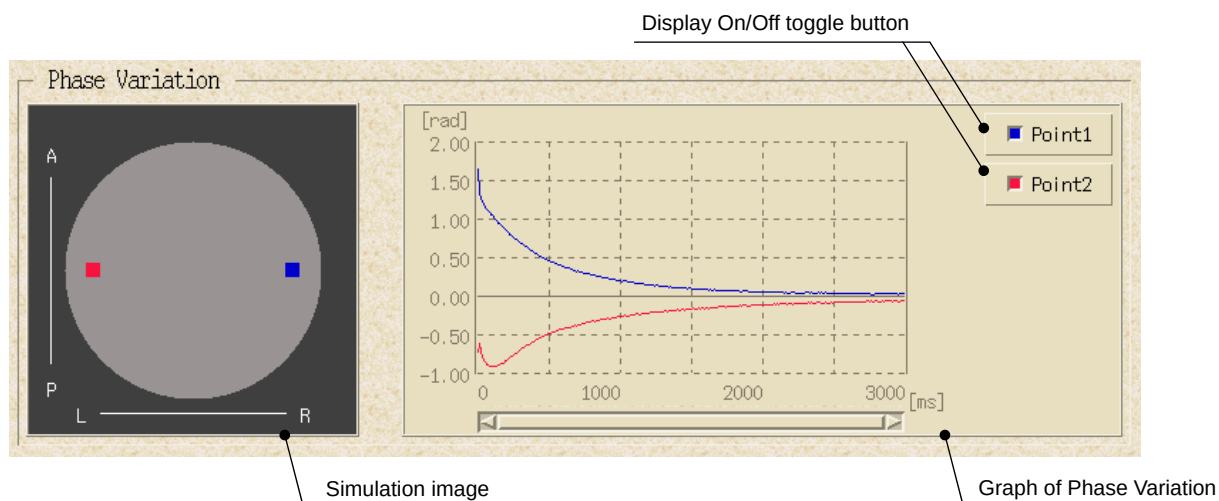


Fig. 6.37-3 Phase Variation

6.37.4.5.2 GC Component / B0 Component

Displays the measurement values of GC component and B0 component which are calculated from the phase variation described in the previous item on the Scan line, the approximation result of its Scan line with the pair of the time constant and the gain on the Fitted line, and the differential value between the Scan line and the Fitted line on the Diff. line.

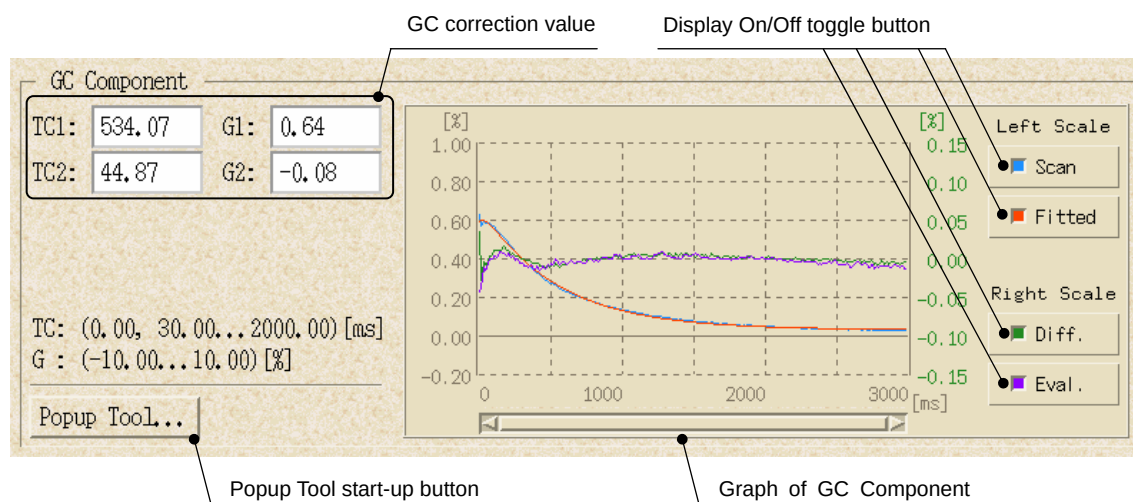


Fig. 6.37-4 GC Component

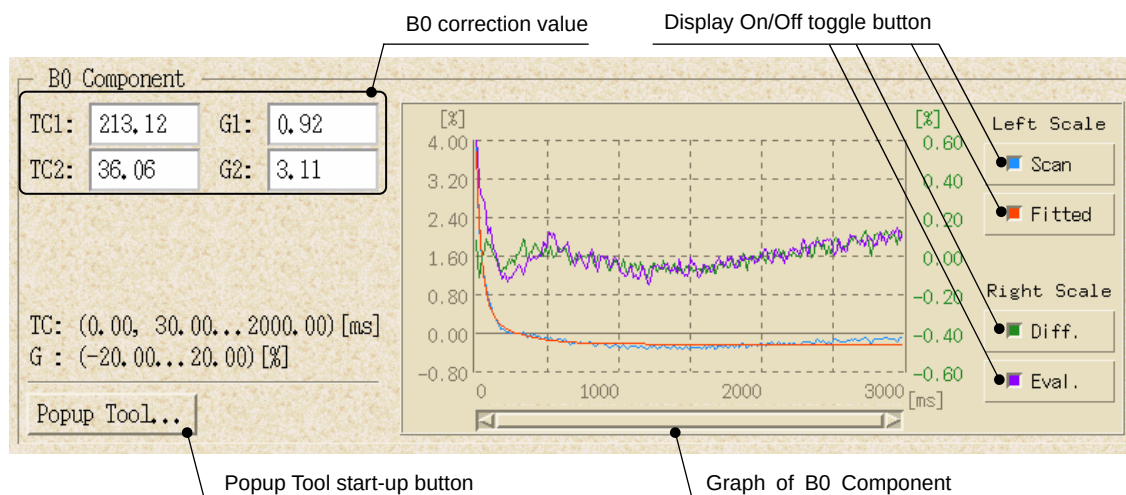


Fig. 6.37-5 B0 Component

6.37.4.6 Changing Result of Analyze

The result of automatic analysis executed when the analysis has ended can be edited in the following method.

6.37.4.6.1 Direct Entry

The entry of analysis result is possible to change by direct entry. Enter the correction values of GC/B0 on the screen of Figs. 6.37-4 and 6.37-5.

The graph is updated according to the entry.

6.37.4.6.2 Manual Analyze

With [Manual Fit] of Popup Tool, the condition for non-linear approximation is possible to change by designating it optionally. The designation items for manual approximation are as shown in Table 6.37-2. It is also possible to check with the possible entry range display by moving the focus to the text area of each item on the screen. For executing the manual approximation, click on the [Fit] button. By executing the manual analysis, the graph is updated according to the result. For ending the manual analysis, click on the [Close] button. By clicking on the [Default] button, the automatic analysis condition is restored.

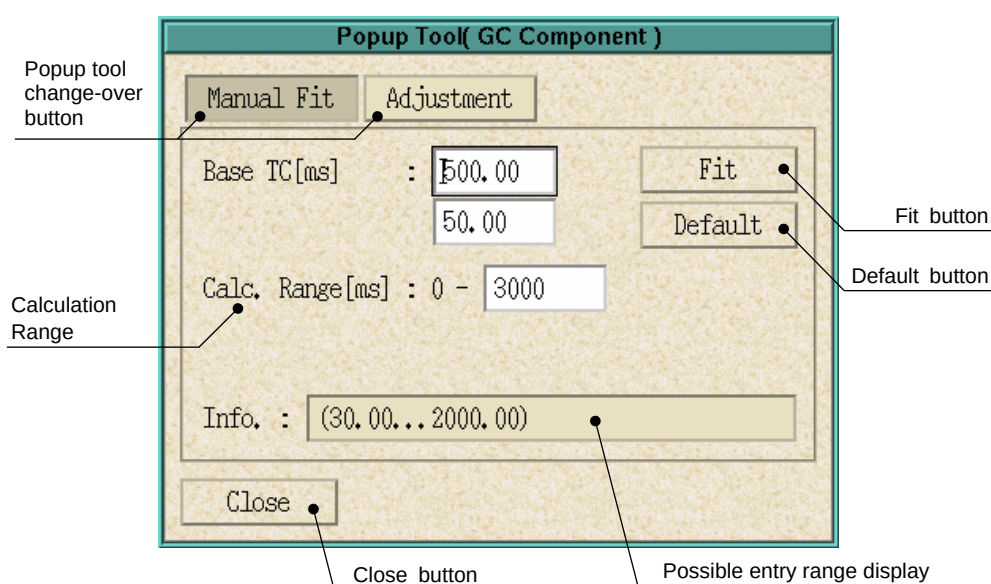


Fig. 6.37-6 Popup Tool Screen Display(Manual Fit)

Table 6.37-2 Parameter Setting Range(Manual Fit)

Parameter Name		Setting Range	Meaning	Remarks
Base TC		Same as TC in Table 6.37-1.	Time constant which becomes the initial value of approximation process.	
Calc. Range	Long-X~Z	300~3000	Calculation Range	
	Short-X~Z	9~96		

6.37.4.7 Adjustment of Fitted Line

Both of GC Component and B0 Component theoretically ought to converge to 0 (zero) according to the time elapsed. Some offset, however, may occur causing the error at the time of measurement. For adjusting the offset, enter the offset with [Adjustment] of Popup Tool and click on the [Set] button. According to this, the graph is updated.

Note: Offset of both of automatic and manual approximation can be adjusted automatically.

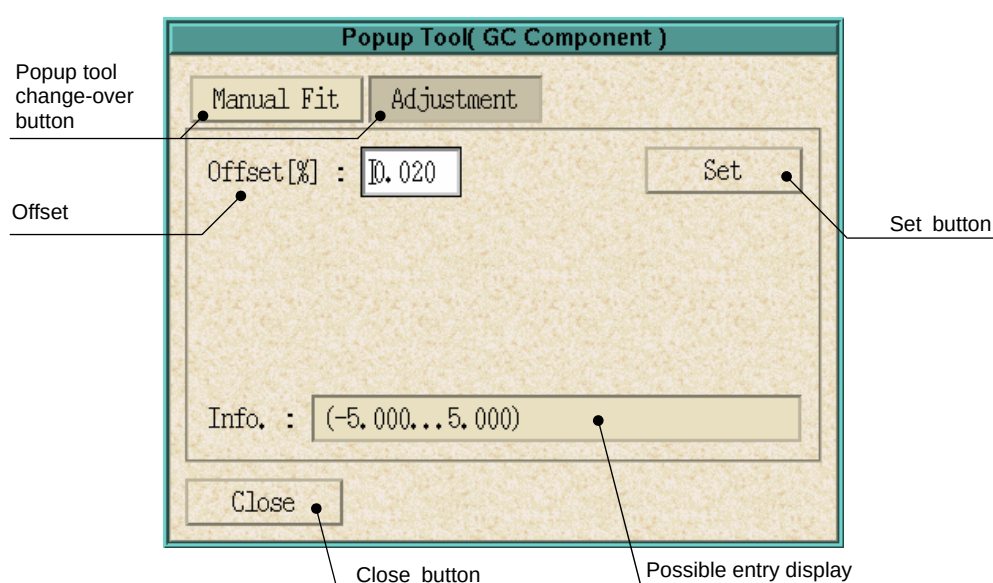


Fig. 6.37-7 Popup Tool Screen Display(Adjustment)

Table 6.37-3 Parameter Setting Range(Adjustment)

Parameter Name		Setting Range	Meaning	Remarks
Offset	GC Component	-5.000~5.000	Offset	
	B0 Component	-10.000~10.000		

6.37.4.8 Deciding Edition Result and Canceling

For deciding the edition result for GC/B0 correction values by manual analysis and direct entry, click on the [Set] button. For canceling the current edition result and returning to the data at last decided state, click on the [Reset] button. By these operation, analysis result becomes the decided state. It is possible to check by the message whether the state is decided or under editing. The color of tags under editing state become yellow other than the tags currently displayed.

Decided state : These parameters are already set up.

Editing state : These parameters NOT set up yet.

6.37.4.9 Reverse

Selecting [Reverse] from "Functions Menu" displays the following dialog. Clicking on [Yes] returns to the default value, while clicking on [NO] returns to the previous screen.

Depending on amount of parameters to be set, time to perform this process varies.

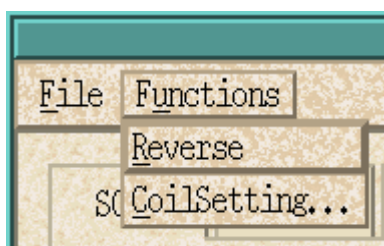


Fig. 6.37-8 Reverse Button

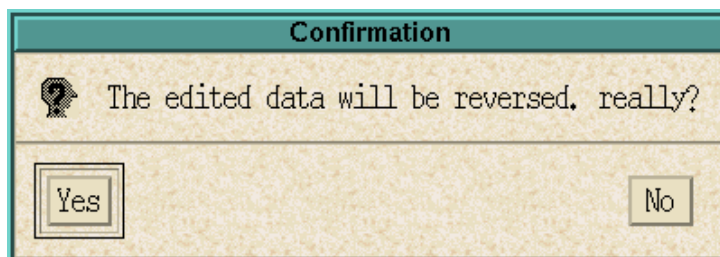


Fig. 6.37-9 Message Dialog for Reverse

6.37.4.10 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.37.4.10.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.37.4.10.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

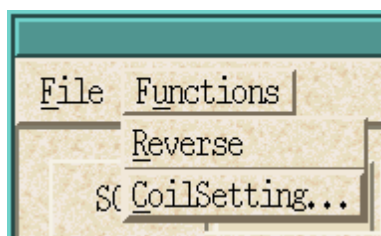


Fig. 6.37-10 Coil setting window display button

6.37.4.10.3 Display screen

The display screen and function names are shown in Fig. 6.37-10 and Fig. 6.37-11.

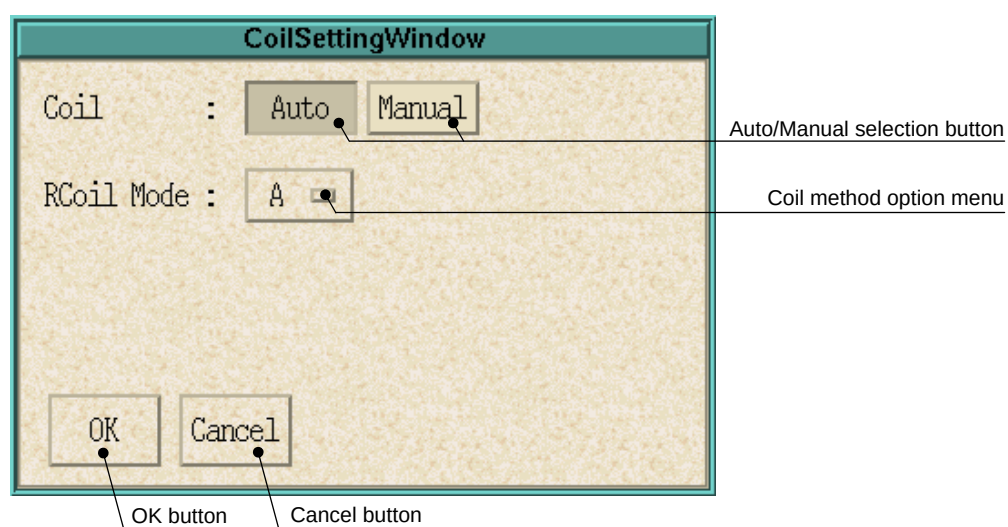


Fig. 6.37-11 Coil Setting Window Display Screen (Auto mode)

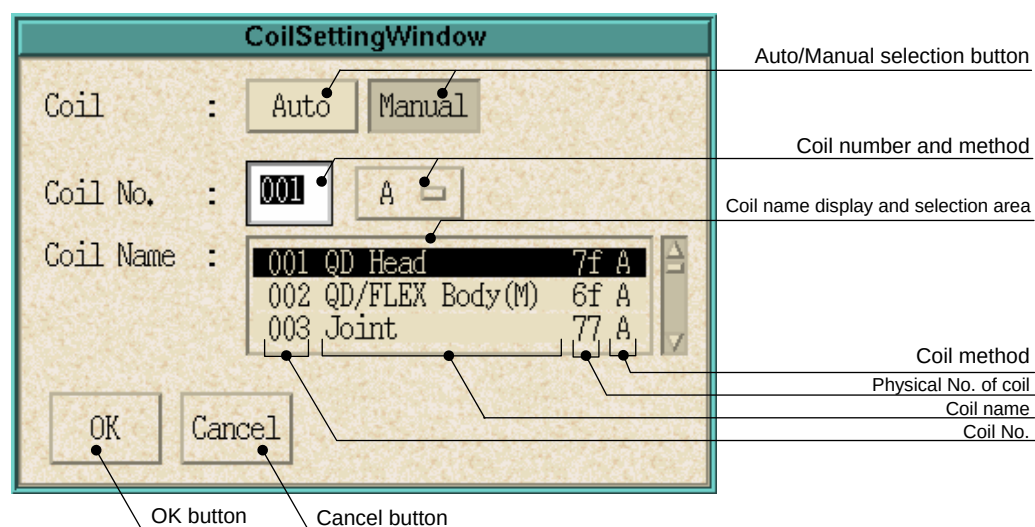


Fig. 6.37-12 Coil Setting Window Display Screen (Manual mode)

6.37.4.10.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.37.4.10.5 Selection of coil method

Select the method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.37.4.10.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area.

The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application.

This setting is effective in Manual mode only. The settable range is limited to coil entered numbers only out of [1~128].

6.37.4.10.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in Manual mode only.

6.37.4.10.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.37.4.10.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.37.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.37-14 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

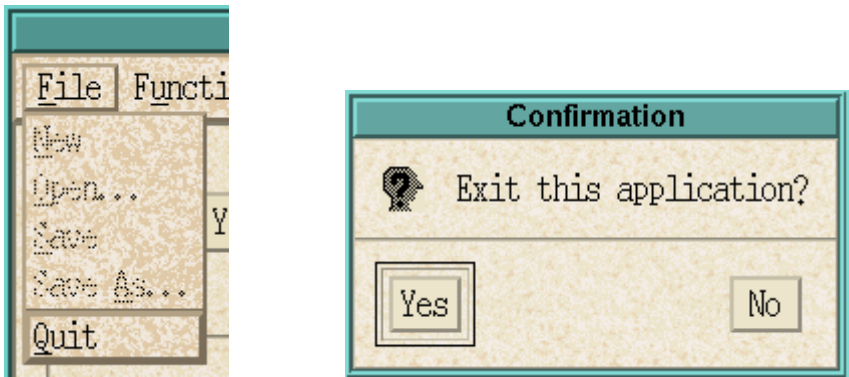


Fig. 6.37-13 Quit Button Fig. 6.37-14 Exit Confirmation Dialog

If any correction is made for GC/B0 correction value, the dialog shown in Fig. 6.37-15 is popped up. Clicking on [Yes] on this dialog saves the corrected GC/B0 correction value in the system file and closes the application. Clicking on [No] closes the application without saving the corrected value in the system file and time to perform this process varies depending on amount of parameters to be set. Clicking on [Cancel] returns to the previous screen without closing the application.

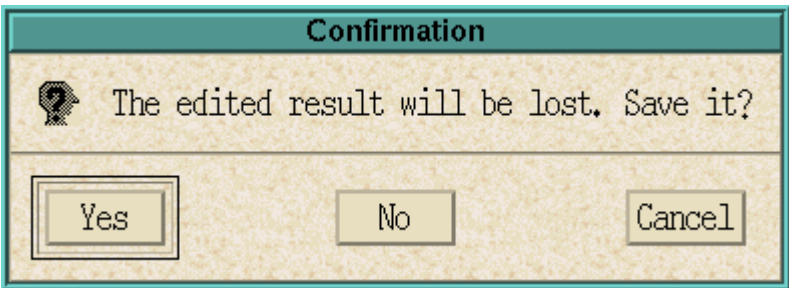


Fig. 6.37-15 System file save and exit confirmation dialog

6.37.6 Error specifications

Error specifications are shown in Table 6.37-4.

Table 6.37-4 Error Specifications

Message	Status	Solution
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Can't write the SYSTEM_TABLE.	Writing SYSTEM_TABLE has failed.	Check the system condition and retry the operation in the same procedure.
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.37.7 Other cautions

- Because this application is operated on the scan card, the buttons on the card are used to start and stop a scan. Accordingly normal scan cannot be performed while this application is running, therefore close this application to start a normal scan.
- Do not change the card while this application is running, otherwise the connection between this application and scan card may be broken. Before changing the card, be sure to close this application. Should the card be changed without closing this application, immediately close this application. In this case, the data can be saved into the system table.
- With the TC (time constant) set at 0, if the set value of G (Gain) is any value other than 0, there may be cases where a malfunction occurs on the hardware. Therefore, if TC (time constant) is set at 0, set 0 for G (Gain).
- In each axis, adjustment of Short is using the adjustment result of Long. Be sure to adjust Long first.
- Evaluation measurement is made to evaluate propriety of the adjustment result. Therefore, because the evaluation data will be lost if the adjustment is made again after evaluation measurement is made, do not carry out the evaluation measurement again.

6.38 B0 Search

6.38.1 General

This is a tool which acquires frequency spectrum and display the analysis results in order to evaluate magnetic field uniformity of the system.

6.38.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and display "Others Group" (refer to Paragraph 1.3.5). Then, select B0 Search from "Maint. Menu" on the scan parameter setting area.

6.38.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.38-1.

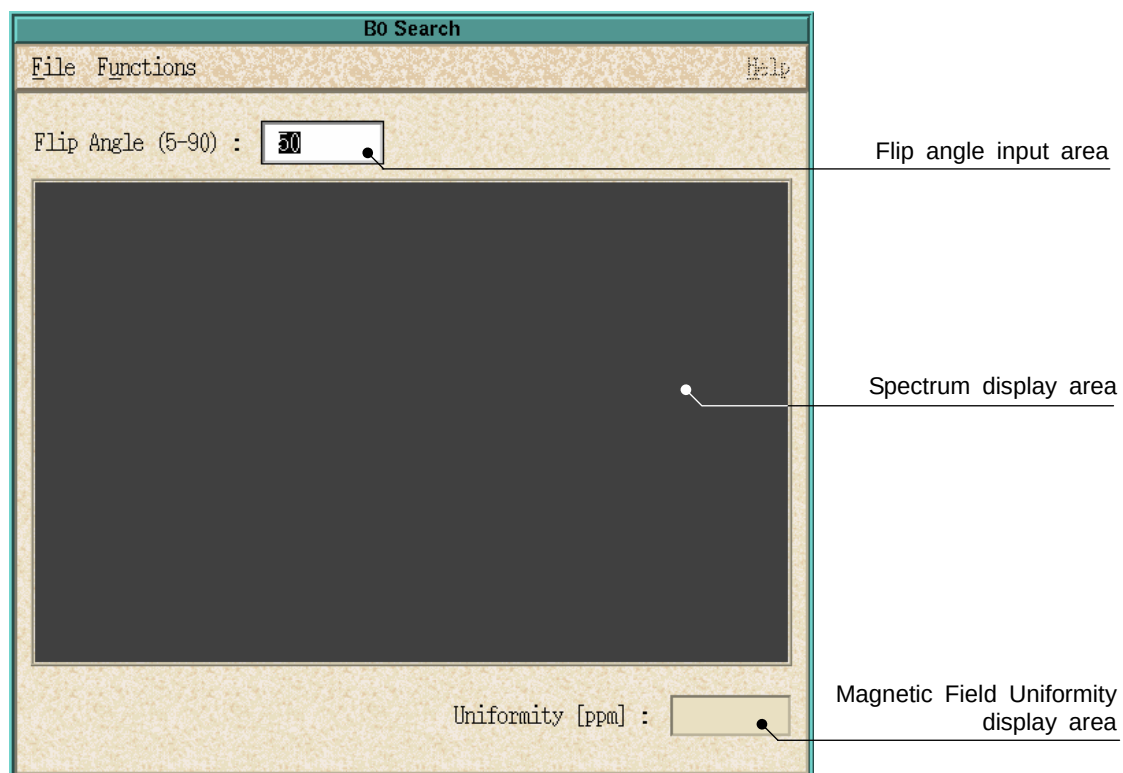


Fig. 6.38-1 B0 Search Display Screen

6.38.4 Operation procedure

Follow the next steps to execute the process.

6.38.4.1 Parameter change

Using the keyboard and mouse cursor, change parameters. The following procedures are available to change parameters.

6.38.4.1.1 Input of character string in entry area

Move the mouse cursor on the entry area of the parameter to be changed and click on the entry area. Using the keyboard, input figures or character string, then either press <Enter> key on the keyboard or remove the focus from the input area to change the parameter.

Table 6.38-1 Parameter Setting Range

Parameter Name	Setting Range
Flip Angle	5 ~ 90

6.38.4.2 Measurement start

Click [Start] button on the scan card to start the measurement.

6.38.4.3 Measurement stop

Click [Stop] button on the scan card. The measurement is stopped.

6.38.4.4 Result display

The result of the acquired frequency spectrum is displayed by graph in the spectrum display area. The abscissa axis represents frequency value and ordinate axis represents signal value.

The evaluation value calculated from the Full Width at Half Maximum of the spectrum is displayed in the magnetic field uniformity display area per ppm.

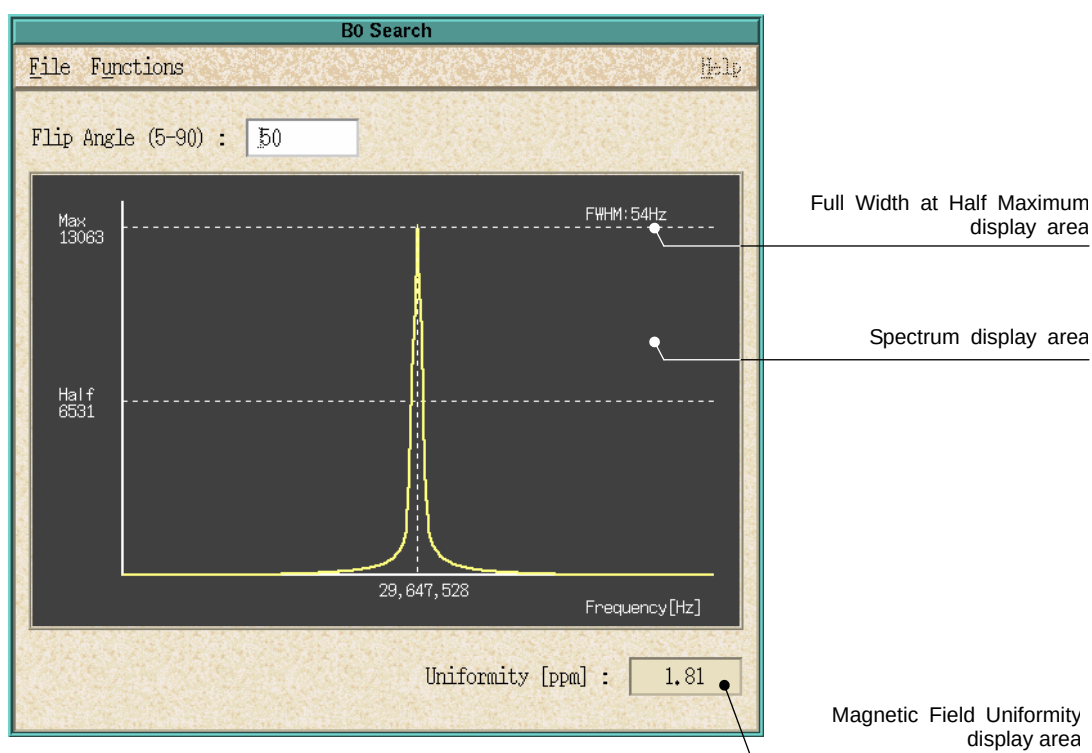


Fig. 6.38-2 Example of Measurement Result

6.38.4.5 Clearing result display area

Select [GraphClear] button from "Functions Menu" of the Menu Bar. The data in the spectrum display area and magnetic field uniformity display area is cleared.

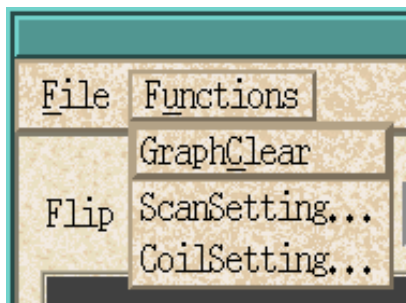


Fig. 6.38-3 Graph Clear Button

6.38.4.6 Scan Setting

Select [ScanSetting...] button from "Functions Menu" of the Menu Bar. The scan setting window is popped up.

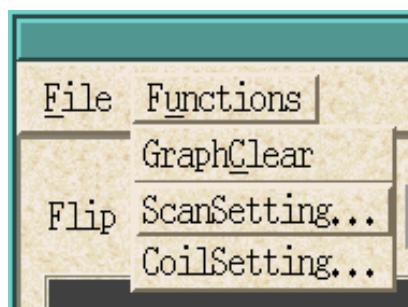


Fig. 6.38-4 Scan Setting Window Display Button

6.38.4.6.1 Display screen

The display screen and function names are shown in Fig. 6.38-5.

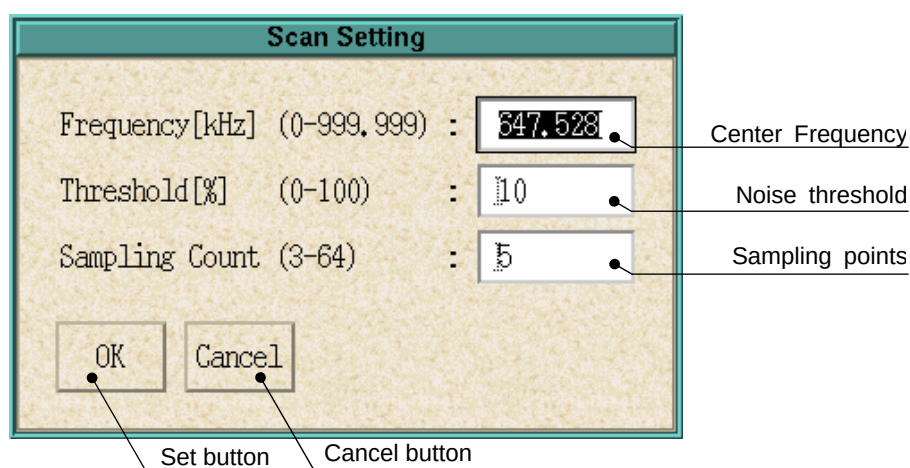


Fig. 6.38-5 Scan Setting Window Display Screen

6.38.4.6.2 Changing parameter

For procedure to change parameters, refer to Paragraph 6.38.4.1.

Table 6.38-2 Scan Setting Parameter Setting Range

Parameter Name	Setting range
Center Frequency	0.000 ~ 999.999
Noise Threshold	0 ~ 100
Sampling points	3 ~ 64

6.38.4.6.3 Determination of parameter change

By clicking [OK] button on the screen, the changed parameters can be determined and the protocol change screen is closed.

6.38.4.6.4 Cancellation of parameter change

By clicking [Cancel] on the screen, the change of parameter can be canceled and the protocol change screen is closed.

6.38.4.7 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.38.4.7.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.38.4.7.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

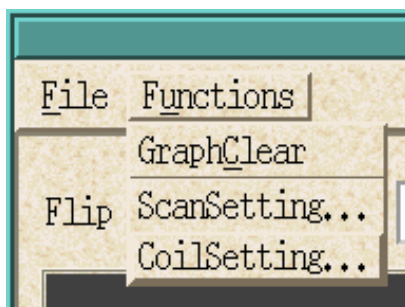


Fig. 6.38-6 Coil Setting Window Display Button

6.38.4.7.3 Display screen

The display screen and function names are shown in Fig. 6.38-7 and Fig. 6.38-8.

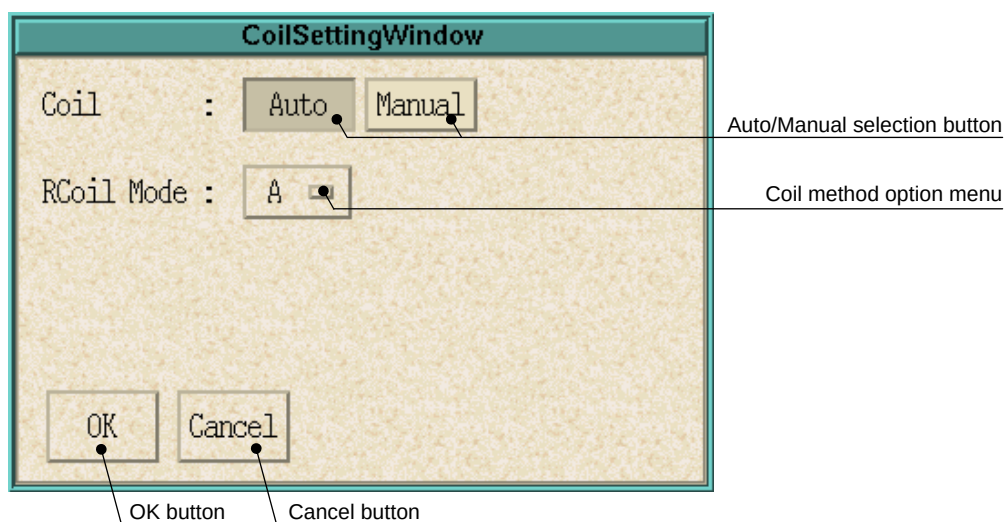


Fig. 6.38-7 Coil Setting Window Display Screen (Auto mode)

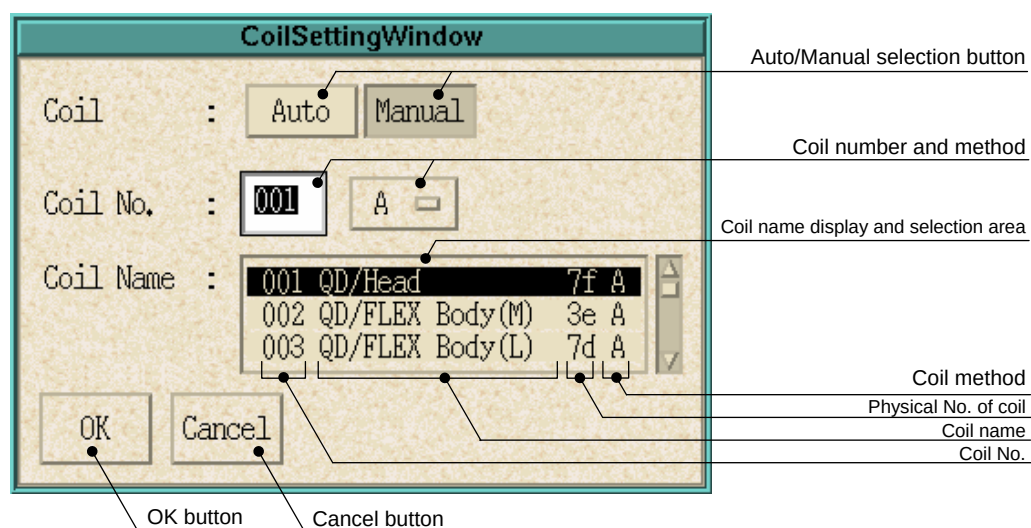


Fig. 6.38-8 Coil Setting Window Display Screen (Manual mode)

6.38.4.7.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the **Manual** screen in case of the **Auto** mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of **Manual** screen is [0], the coil method is [A], and the coil name is not selected.

6.38.4.7.5 Selection of coil application method

Select the method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in **Manual** mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.38.4.7.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area. The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.38.4.7.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.38.4.7.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.38.4.7.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.38.5 End

Selecting [Quit] button from "File Menu" in the Menu Bar pops up the dialog shown in Fig. 6.38-10. Clicking [Yes] button on this dialog exits from the application, while clicking [No] button returns to the previous screen without exiting from the process.

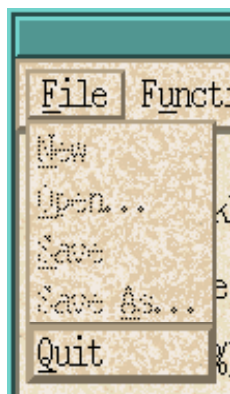


Fig. 6.38-9 Quit Button

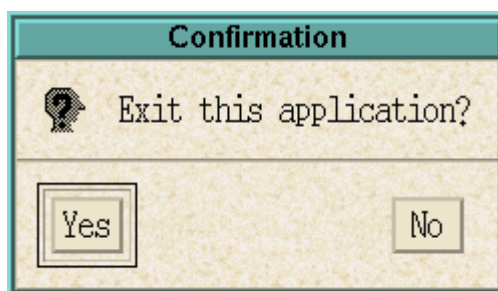


Fig. 6.38-10 Exit Confirmation Dialog

6.38.6 Error specifications

Error specifications are shown in Table 6.38-3.

Table 6.38-3 Error Specifications

Message	Status	Solution
The received signal is high. Please lower Flip Angle and perform once again.	The received signal is high. Abnormalities may be in a spectrum.	Change the value of Flip Angle, retry the measurement.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Finish the running task and retry the same operation procedure.
Can't read the SYSTEM_TABLE.	Reading the system table has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system status and retry the same operation procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the registered coils list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the registered coils list.
No coil is connected to the system.	The coil is not connected yet to the system.	Check the coil connection conditions.
Error occurred in the patient table.	Communication with the patient table failed.	Check the patient table condition.

6.38.7 Other cautions

Because this application is operated on the scan card, the buttons on the card are used to start and stop a scan.

Accordingly normal scan cannot be performed while this application is running, therefore close this application to start a normal scan.

Do not change the card while this application is running, otherwise the connection between this application and scan card may be broken. Before changing the card, be sure to close this application.

Should the card be changed without closing this application, immediately close this application. In this case, the data can be saved into the system table.

6.39 FatSepF Auto Adjustment

6.39.1 General

This is a tool to automatically adjust the gradient field pulse intensity for correction in the slice, phase and frequency axes in FatSepF sequence.

Carrying out measurements by changing correction value according to the given parameters, the optimum value among the obtained values is used as the adjustment value.

6.39.2 Start-up

Register the T/M license with the Key MOD/DVD (refer to Section 3), and select T/M Manager from "HMC Service Diagnostics Menu" in the System Tool. Then, T/M Manager is started (refer to Paragraph 1.3.6). Either click FatSepF Auto Adjustment in the displayed items and click [OK] or double-click the selected item.

6.39.3 Display screen

The screen displayed after start-up and function names are shown in Fig. 6.39-1.

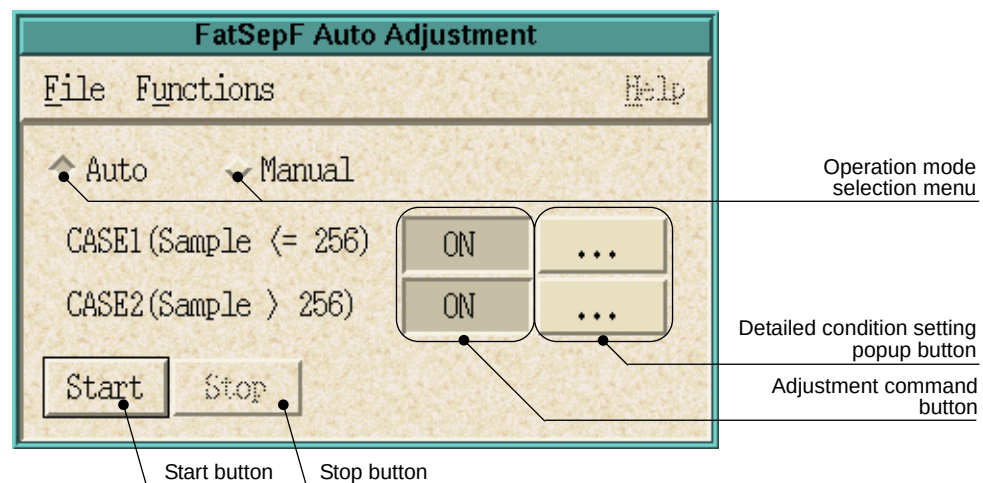


Fig. 6.39-1 FatSepF Auto Adjustment Display Screen

6.39.4 Operation procedure

Follow the next steps to execute the process.

6.39.4.1 Setting adjustment parameters

On the FatSepF auto adjustment main screen (hereafter called Main screen) shown in Fig. 6.39-1, set the adjustment command button to ON for the case to be adjusted, while set it to OFF for the case not be adjusted.

This ON/OFF setting can be done by moving the mouse cursor on the desired button and clicking the left mouse button.

For the case set to ON, the default adjustment item is set. To change the adjustment item, the detailed condition setting popup button is pushed. The detailed condition setting screen as shown in Fig. 6.39-2 is displayed. Then, click the left mouse button on the item to be adjusted to set ON/OFF.

To close the setting, click [OK] button.

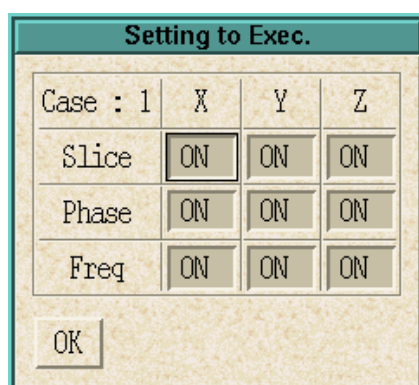


Fig. 6.39-2 Detailed Condition setting Screen

6.39.4.2 Selection of Auto/Manual mode

Select Auto or Manual on the mode selection menu of the Main screen. In Auto mode, a result of each measurement is not checked, but in Manual mode, a measurement result after each measurement is checked (Fig. 6.39-3).

6.39.4.3 Starting measurement

Click [Start] button to start the measurement.

After measurement is completed, to go to the measurement of the next adjustment item, click [OK] button. After the measurement of the last adjustment item, the adjustment result display screen as shown in Fig. 6.39-4 is displayed. To suspend the measurement before completing all the specified adjustment items, click [Cancel] button. In this case, the parameter entry screen (Fig. 6.39-11) is displayed, so follow the operation procedure described in Paragraph 6.39.4.8.

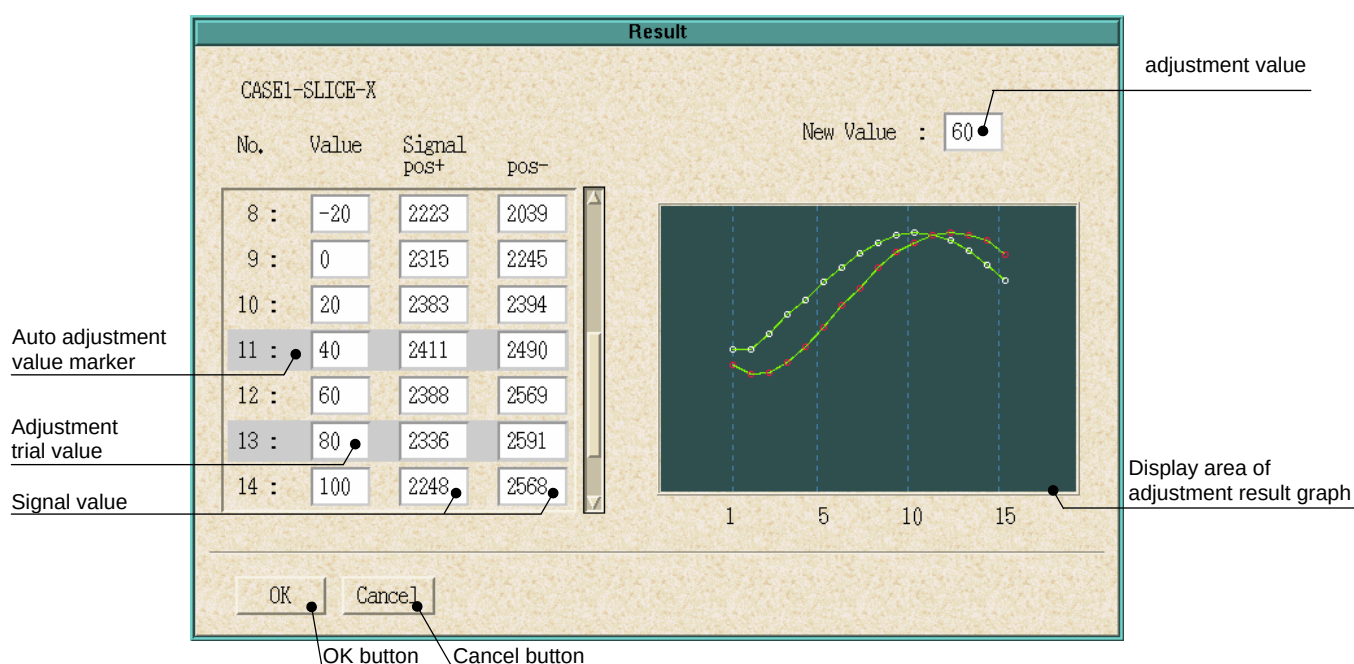


Fig. 6.39-3 Example of Measurement Result (Slice)

The adjustment value is the one obtained by the latest measurement if the result is OK, but is the previous one if the result is NG. This value is not written in the H/D but is registered in the system table memory.

The position of the peak value of the measurement result graph is displayed in the peak position, and the position is automatically marked as the optimum value among the adjustment trial values. When any peak value is not found, no marker is put on.

6.39.4.4 Stopping measurement

Click [Stop] button. In any of Auto and Manual mode, the measurement is stopped and the adjustment results by that time are displayed as shown in Fig. 6.39-4.

6.39.4.5 Checking adjustment results

Upon completion of adjustments of the specified items, the adjustment result screen as shown in Fig. 6.39-4 is displayed. ○ mark indicates the item been adjusted, while × mark indicates the item not been adjusted. The adjustment command button for the adjusted item is automatically turned OFF. But the button turns to the initial status by restarting the application.

This screen is also displayed when any measurement error occurs during measurement. Reset the measurement parameter (refer to Paragraph 6.39.4.8) and retry the adjustment.

Clicking [Close] button closes this screen and the step goes to Paragraph 6.39.4.6 Checking adjustment values.

Case No.	s-X	s-Y	s-Z	p-X	p-Y	p-Z	f-X	f-Y	f-Z
CASE1	●	●	●	●	●	●	×	●	●
CASE2	●	●	●	●	●	●	●	●	●

Close button Clear button

- ① Adjustment items in slice direction (X,Y,Z)
- ② Adjustment items in phase direction (X,Y,Z)
- ③ Adjustment items in frequency direction (X,Y,Z)

Fig. 6.39-4 Adjustment Result Screen

Clicking [Clear] button displays the dialog shown in Fig. 6.39-5. Clicking [Yes] on this dialog clears the resultant data and returns the screen to the initial status (but not to clear the adjustment value). Clicking [No] does not clear the data.

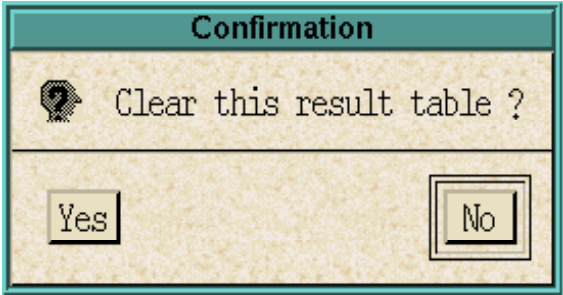


Fig. 6.39-5 Confirmation Screen to clear Adjustment Result

It is also possible to display this adjustment result screen by selecting [Result] on "Functions Menu".

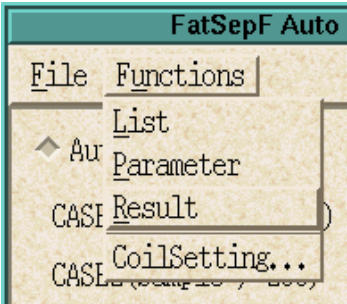


Fig. 6.39-6 Button to display Adjustment Result Screen

6.39.4.6 Checking adjustment value

After completion of the adjustment, select [List] on "Functions Menu" and check the adjustment value on the adjustment value screen shown in Fig. 6.39-8.

The automatically adjusted item is registered in the system table memory.

The adjustment value list display screen can be closed by clicking [Close] button on the screen.

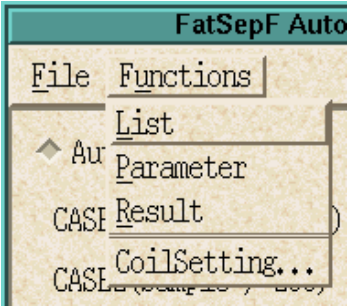
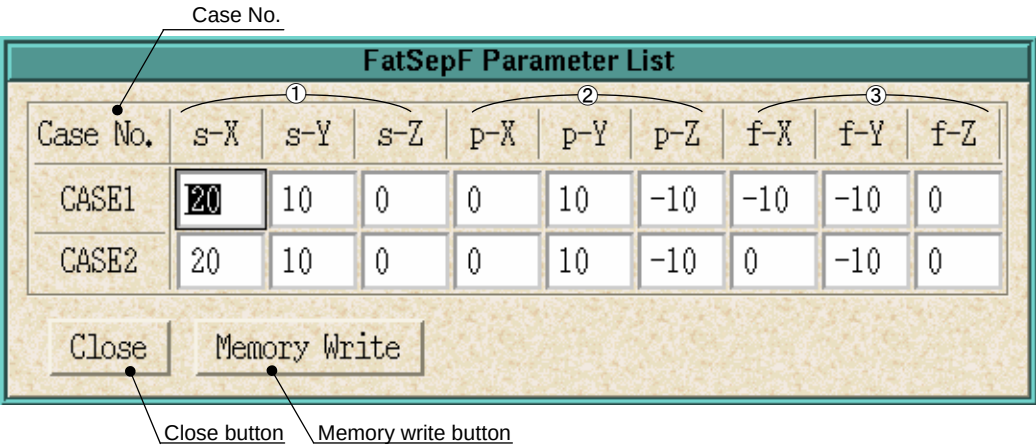


Fig. 6.39-7 Button to display Adjustment Value



- ① Adjustment items in slice direction (X,Y,Z)
- ② Adjustment items in phase direction (X,Y,Z)
- ③ Adjustment items in frequency direction (X,Y,Z)

Fig. 6.39-8 Screen to display Adjustment Values List

6.39.4.7 Direct entry of adjustment value

Adjustment values can be changed by direct input without using measurement values. Directly input adjustment values on the screen shown in Fig. 6.39-8. To set numerical parameter, place the mouse cursor on the entry area of the parameter to be changed and select the entry area. Input value from the keyboard and change the parameter value by either pressing <Enter> key or removing the focus from the entry area.

Table 6.39-1 Parameter Setting Range (Adjustment value)

Parameter Name	Setting range
Adjustment items in slice direction (X, Y, Z)	-999 ~ 999
Adjustment items in phase direction (X, Y, Z)	-999 ~ 999
Adjustment items in frequency direction (X,Y,Z)	-999 ~ 999

To register the changed adjustment value in the system memory, click [Memory Write] button.

By clicking [Close] button on the screen, the screen to display adjustment values list can be closed. When trying to close the screen without registering the values in the system table memory, the screen shown in Fig. 6.39-9 is displayed. Clicking [Yes] determines the change, while clicking [No] can cancel the change.

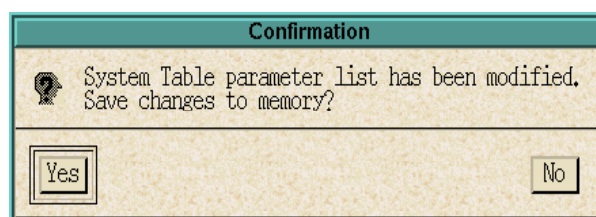


Fig. 6.39-9 Confirmation Screen for System Table Memory Change

6.39.4.8 Setting parameter

Select [Parameter] on "Functions Menu" of the Menu Bar and the parameter entry screen shown in Fig. 6.39-11 is displayed. Input measurement parameters as required. If no change is required, this operation can be omitted.

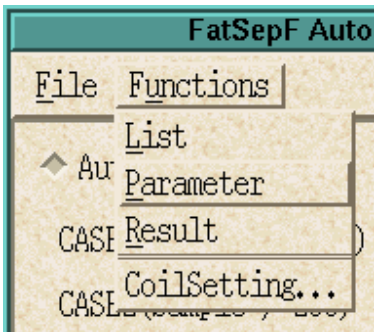


Fig. 6.39-10 Parameter Button

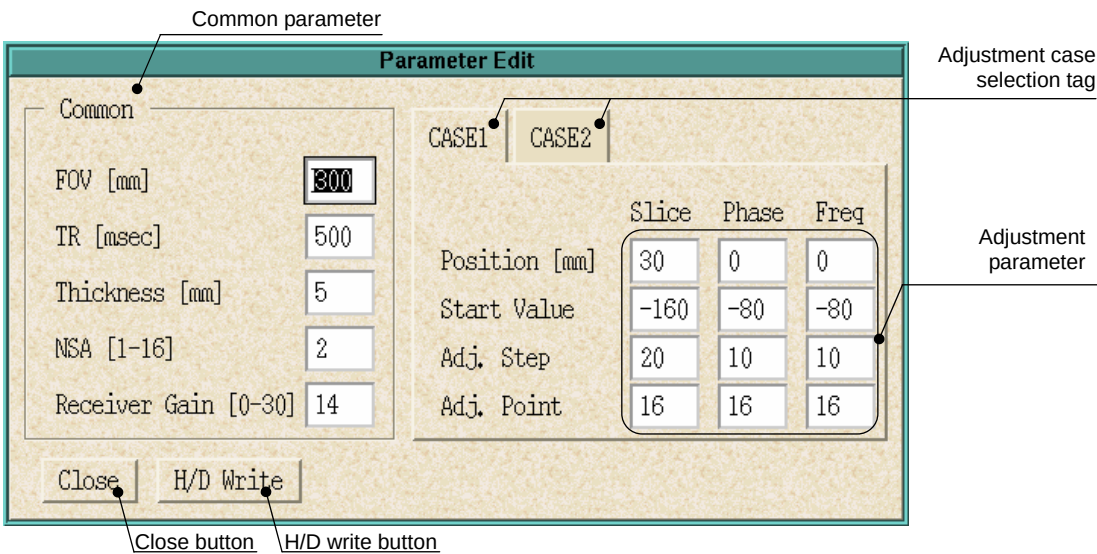


Fig. 6.39-11 Parameter Entry Screen

The changed parameter is stored in the H/D by clicking [H/D Write] button and is used as default parameter for the next measurement.

Clicking [Close] button closes this screen. If the screen is tried to close without storing the changed parameter, the H/D Write Confirmation screen as shown in Fig. 6.39-12 is displayed. To store it in the H/D, click [Yes], while not to store it, click [No].

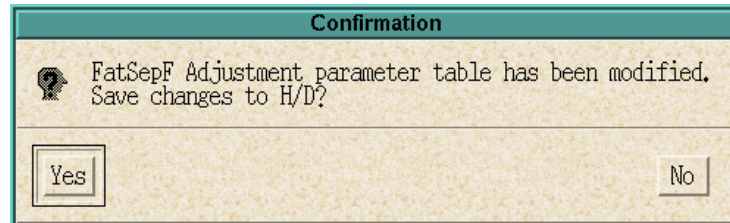


Fig. 6.39-12 H/D Write Confirmation Screen

Common Parameters on the parameter entry screen is common to all the cases. All of other parameters must be set for each case. To set numerical parameter, place the mouse cursor on the entry area of the parameter to be changed and select the entry area. Input a numerical value from the keyboard and change the parameter by either pressing <Enter> key on the keyboard or removing the focus from the entry area. Each parameter has a setting range as shown in Table 6.39-2, 6.39-3.

Table 6.39-2 Parameter Setting Range (Common Parameter)

Parameter Name	Setting range	Remarks
FOV	50 ~ 500	
TR	200 ~ 9999	
Thickness	2 ~ 100	
NSA	1 ~ 16	
Receiver Gain	0 ~ 30	Only even figures can be input.

Table 6.39-3 Parameter Setting Range (Adjustment Parameter)

Parameter Name	Parameter description	Setting range
Position	Adjustment position	0 ~ 999
Start Value	Initial value of adjustment value	-999 ~ 999
Adj. Step	Incremental value of adjustment value	1 ~ 100
Adj. Point	Number of measurement points for addition of incremental adjustment value	1 ~ 16

6.39.4.9 Coil setting

Either automatic recognition of coil or manual setting can be selected in the coil setting window.

6.39.4.9.1 Default

Default at the start time is **Auto**. Each coil is recognized automatically. However, the application causes an error, if the bed communication is impossible under the system condition. In such a case, set the coil manually.

6.39.4.9.2 Coil setting window display

Select [CoilSetting ...] from "Functions Menu". The coil setting window pops up. The starting screen shows default.

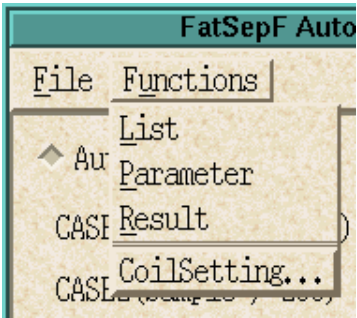


Fig. 6.39-13 Coil Setting Window Display Button

6.39.4.9.3 Display screen

The display screen and function names are shown in Fig. 6.39-14 and Fig. 6.39-15.

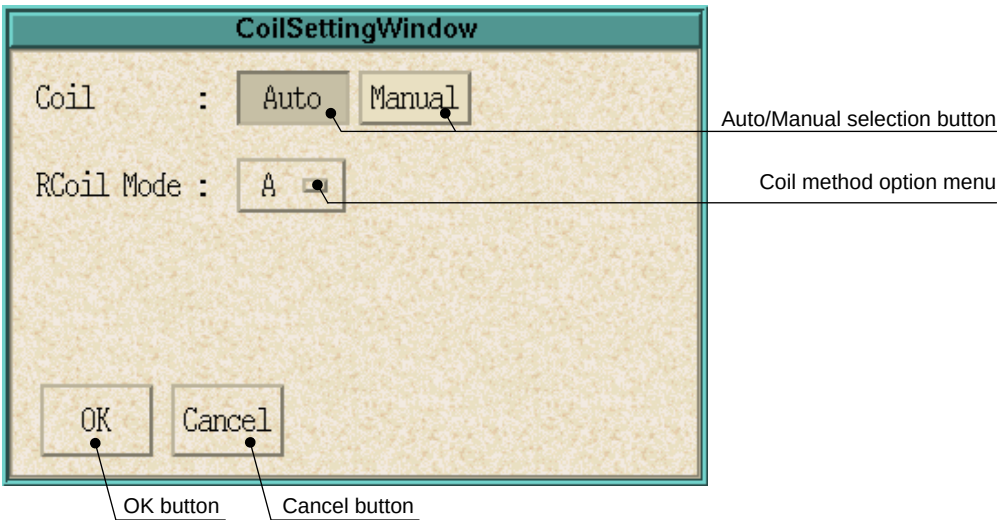


Fig. 6.39-14 Coil Setting Window Display Screen (Auto mode)

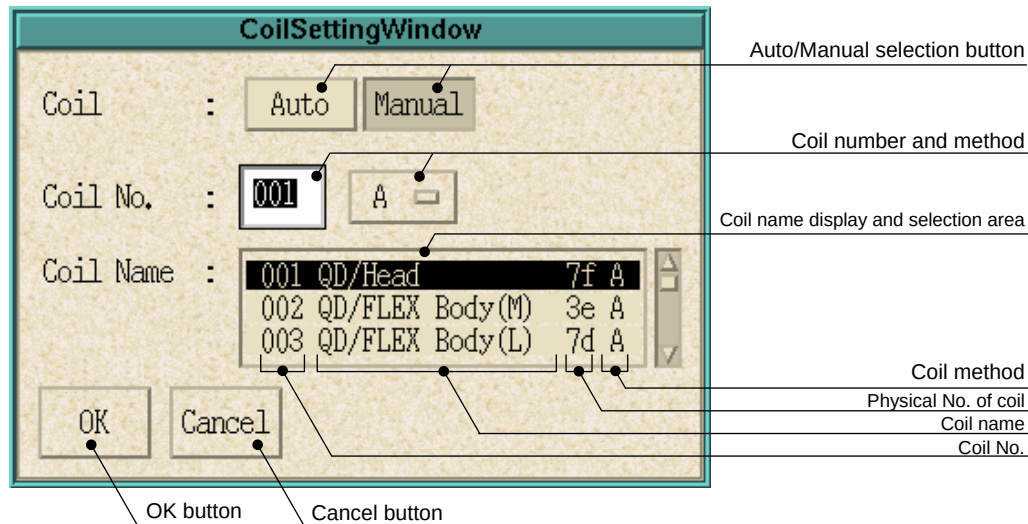


Fig. 6.39-15 Coil Setting Window Display Screen (Manual mode)

6.39.4.9.4 Auto/Manual selection

The push-button group displayed sideways is a single selection type, and it can be changed by clicking it with the mouse cursor. When [Auto] is selected, the receiving coil is recognized automatically. When [Manual] is selected, the designated coil information is used. The first display is started in the default set mode, and the second and subsequent display are started under the last set condition.

The default being displayed in the Manual screen in case of the Auto mode start shows the information obtained by automatic recognition of coil. If the coil information could not be obtained, coil number of Manual screen is [0], the coil method is [A], and the coil name is not selected.

6.39.4.9.5 Selection of coil method

Select the application method from the open menu by clicking [RCoil Mode] of the option menu. In case coil number and coil method have not been entered in Manual mode, an alarm sounds and the corresponding coil name is displayed in the coil name display area when it is used.

The range possible to set is [A]~[H], [J] and [K].

6.39.4.9.6 Numerical input to coil number input area

Select the input area by putting the mouse cursor to the coil number input area

The input area can be updated by pressing <Enter> key on the keyboard after inputting a numeric value from the keyboard or by taking the focus out of the input area.

If the combination of the coil number and coil method is not entered yet, a warning tone sounds. If the method only is different in an entered coil, the coil name is displayed in the coil name display area.

If the coil is not entered yet, the coil number is reset to the last value.

The corresponding coil name is displayed in the coil name display area in case of application. This setting is effective in **Manual** mode only.

The settable range is limited to coil entered numbers only out of [1~128].

6.39.4.9.7 Coil selection in coil name display area

The coil name display area is a list display area, and coil numbers can be set by utilizing this area. After putting the mouse cursor to the list display area, scroll the list by the scroll bar, and click a coil name to be changed. The black reversed display shows the selected name. The selected information is reflected on the coil number and coil method instantly. This setting is effective in **Manual** mode only.

6.39.4.9.8 Application of coil setting

After the end of coil setting, click [OK]. The coil set information becomes applicable and the coil setting window is closed.

6.39.4.9.9 Cancellation of coil setting

For canceling a coil setting change or closing the window without changing the coil setting, click [Cancel]. The changed contents are discarded and the coil setting window is closed.

6.39.5 End

By selecting [Quit] on "File Menu" of the Menu bar, the dialog box shown in Fig. 6.39-17 is popped up. Clicking [Yes] on this box exits from the application, while clicking [No] returns to the previous screen without exiting from the application.

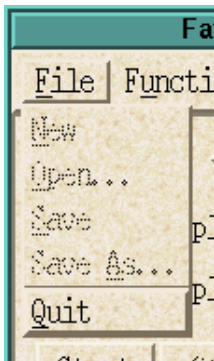


Fig. 6.39-16 Quit Button



Fig. 6.39-17 Exit Confirmation Dialog

When the adjustment value on the system table memory is changed, the dialog box as shown in Fig. 6.39-18 is popped up. Clicking [Yes] on this box writes the changed system table adjustment value in the H/D and closes the application, while clicking [No] closes the application without writing the value. Clicking [Cancel] returns to the previous screen without closing the application.

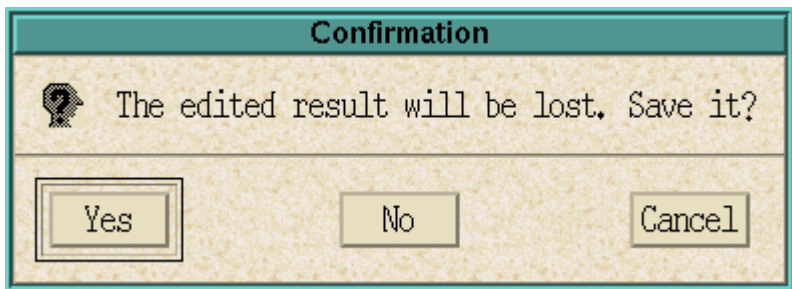


Fig. 6.39-18 System Table Storage and Exit Confirmation Dialog

6.39.6 Error specifications

Error specifications are shown in Table 6.39-4.

Table 6.39-4 Error Specifications

Message	Status	Solution
Signal has underflowed	Signal intensity is too low.	Adjust Gain.
Signal has overflowed	Signal intensity is too high.	Adjust Gain.
Parameter Error	Measurement error	Adjust FOV, TR, Thickness, etc.
T/M license is not registered.	T/M license has not been registered.	Register the license by the Key MOD/DVD.
Another task is running.	Another task is running.	Exit from the running task and retry the operation in the same procedure.
Can't start task.	The application has started unsuccessfully.	Check the system condition and retry the operation in the same procedure.
Can't end task.	The application has ended unsuccessfully.	Check the system condition and retry the operation in the same procedure.
The translation of parameter failed.	Parameter transfer has failed.	Check the system condition and retry the operation in the same procedure.
Can't read the SYSTEM_TABLE.	Reading SYSTEM_TABLE has failed.	[Retry] button: Read again [Exit] button: Closing application
Event Invalid.	Request is not accepted.	Check the system condition and retry the operation in the same procedure.
Can't obtain the coil information.	The coil information could not be obtained.	Check the coil registration list.
Connected coil is not registered to the system.	The connected coil has not been registered in the system.	Check the coil registration list.
No coil is connected to the system.	The coil is not yet connected.	Check the coil for connection.
Error occurred in the patient table.	Communication with the patient table has failed.	Check the condition of the patient table.
The system failed in the control of the patient table.	The control of the patient table has failed.	Retry the operation in the same procedure.

6.39.7 Other cautions

To obtain correct measurement values, a prescan must be performed at least once before executing this tool.